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CHAPTER

ATA 32

LANDING GEAR

THIS MANUAL IS INTENDED FOR TRAINING PURPOSE ONLY

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Landing Gear - General

Introduction

The landing gear supports the weight of the aircraft on the ground. It is also used to maneuver the aircraft on the ground.

General Description

[Refer Figure 32-1](#)

The landing gear is electrically controlled, hydraulically operated and mechanically locked. The hydraulically operated nose wheel steering gives directional control during taxiing, take-off and landing. Each main wheel has hydraulically powered anti-skid brakes. There is also an emergency/park brake system.

The tricycle gear is a retractable dual wheel installation. The main gears retract aft into the nacelles and the nose gear retracts forward into the nose section and has steerable nosewheels. Gear doors completely enclose the landing gear when it is retracted and partially enclose the gear when it is extended. Flight compartment advisory lights show position of gear doors and downlocks. An aural warning sounds if the gear is not extended and the aircraft is in a landing configuration.

LEGEND

1. Landing Gear Selector Lever and Control Panel
2. Alternate Release Handles.
3. MLG Lock Actuator.
4. MLG Retraction Actuator.
5. Alternate Extension Actuator.
6. Bypass Valve.
7. Proximity Sensor Electronic Unit.
8. NLG Lock Actuator.
9. NLG Retraction Actuator.

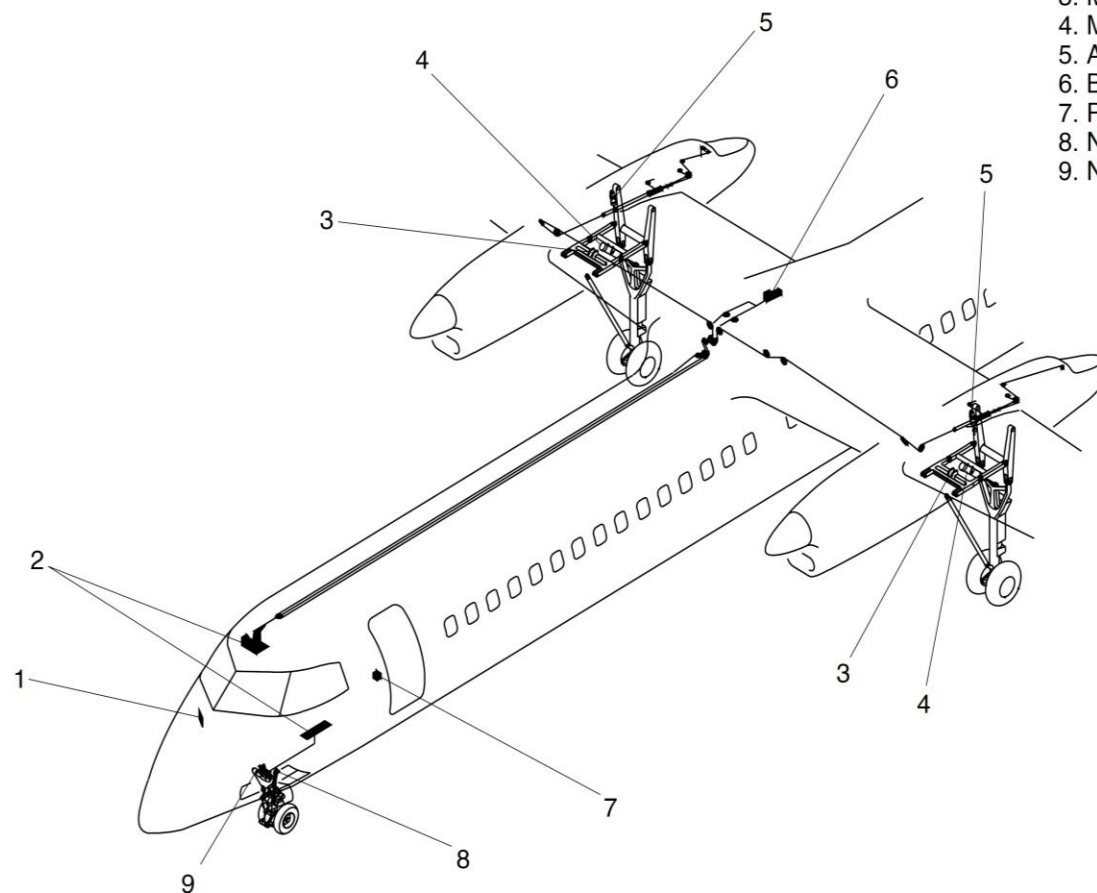


Figure 32-1 Landing Gear Locator

[Refer Figure 32-2](#)

A Proximity Switch Electronics Unit (PSEU) monitors and controls the operation of the landing gear components.

An alternate landing gear extension method can be used to extend the gear if the primary extension method fails. There is also an alternate downlock verification system.

The nosewheel is steerable by a flight compartment hand control or by the rudder pedals. The main wheels are equipped with anti-skid multiple disc brakes. The brakes can be controlled by the brake pedals or the EMERG BRAKE lever.

The landing gear is operated by the No. 2 hydraulic system and is controlled by the landing gear selector lever on the LANDING GEAR control panel. There is an alternate means of extension for the landing gear in the event that the primary system is not functioning.

The Landing Gear System has the following subsystems:

- Main Landing Gear and Doors
- Nose Landing Gear and Doors
- Landing Gear Extension and Retraction
- Wheels and Brakes
- Steering System
- Position and Warning

The main gear retracts aft and the nose gear retracts forward and has steerable nosewheels. The landing gear is operated by the No. 2 hydraulic system and is controlled by the landing gear selector lever on the LANDING GEAR control panel. There is an alternate means of extension for the landing gear in the event that the primary system is not functioning. Advisory lights give extension and retraction fail/safe information.

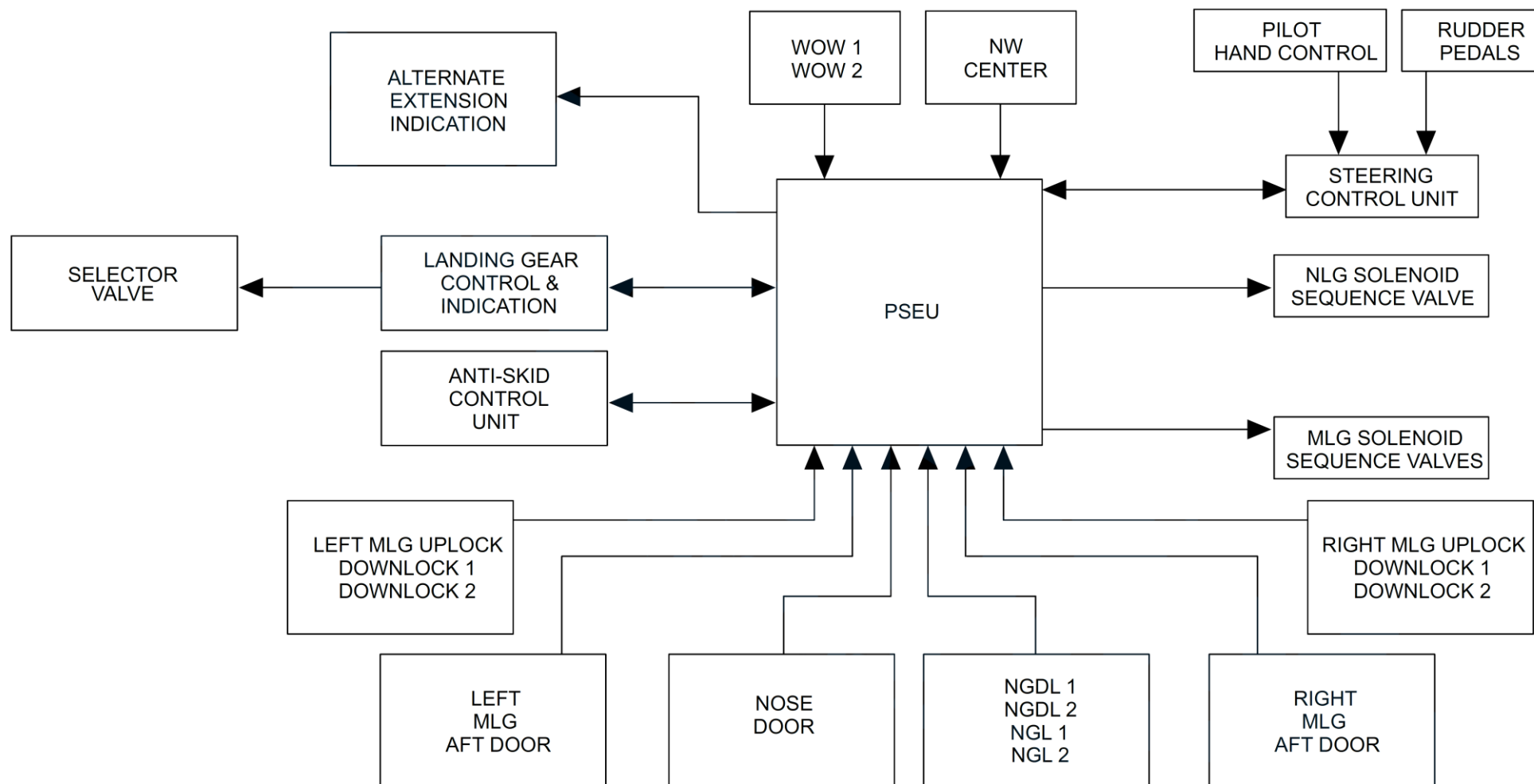


Figure 32-2 Landing Gear Funcional Block Diagram

Landing Gear Control Panel & Operation

[Refer Figure 32-3](#)

The landing gear position is controlled by the LANDING GEAR selector lever. The selector lever has a LOCK RELEASE button which must be pushed down before moving the LANDING GEAR selector lever. Moving the lever to the UP or DN position completes the electrical circuit to the hydraulic selector valve to retract or extend the gear. Gear proximity switches send signals to the Proximity Switch Electronic Unit (PSEU) that prevent gear retraction when the aircraft is on the ground.

Moving the LANDING GEAR selector lever to the UP position starts landing gear retraction sequence. Moving the LANDING GEAR selector lever to the DN position starts the landing gear extension sequence. This also causes the NO SMOKING signs to come on automatically. An amber light in the selector lever comes on when the LANDING GEAR selector handle does not agree with position of landing gear.

Landing Gear Advisory Lights

Gear and door position is shown by nine advisory lights on the LANDING GEAR control panel, and an amber light in the selector lever. Three green lights on the LANDING GEAR ALTERNATE EXTENSION panel, on the flight compartment floor, give backup visual verification of gear downlocking.

The L. DOOR, N. DOOR, or R. DOOR amber lights come on when the related hydraulic gear door is open. The LEFT, NOSE, RIGHT green lights come on when the related gear door is down and locked. The LEFT, NOSE, RIGHT red lights come on when the related gear is not locked in selected position. When the gear is up and locked all lights on the LANDING GEAR panel go off.



Figure 32-3 Landing Gear Control panel

Landing Gear Downlock Verification Lights

[Refer Figure 32-4](#)

Three green lights on the LANDING GEAR ALTERNATE EXTENSION panel, on the flight compartment floor, give backup visual verification of gear downlocking. The downlock verification toggle switch must be set aft for the lights to operate. The NG, LH or RH green light will come on when the related gear is locked down.

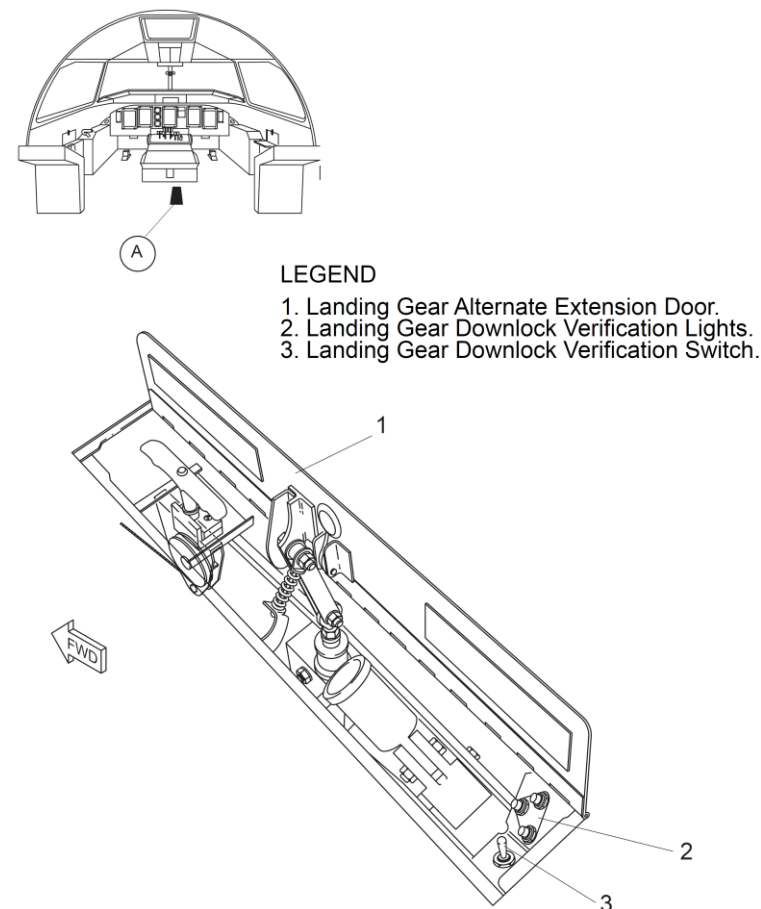


Figure 32-4 Landing Gear Alternate Downlock Verification Lights

MLG Ground Lock Pins and Door Lock Pins

[Refer Figures 32-5, 32-6 & 32-7](#)

Ground lock pins are supplied for the main gears. The main gear lock-pins may be kept in the forward compartment of the forward passenger door. With the gear extended, the pins are inserted into the main gear stabilizer brace assemblies.

There are also landing gear door lock pins for the nose and main hydraulic doors. These prevent the hydraulic gear doors from closing.

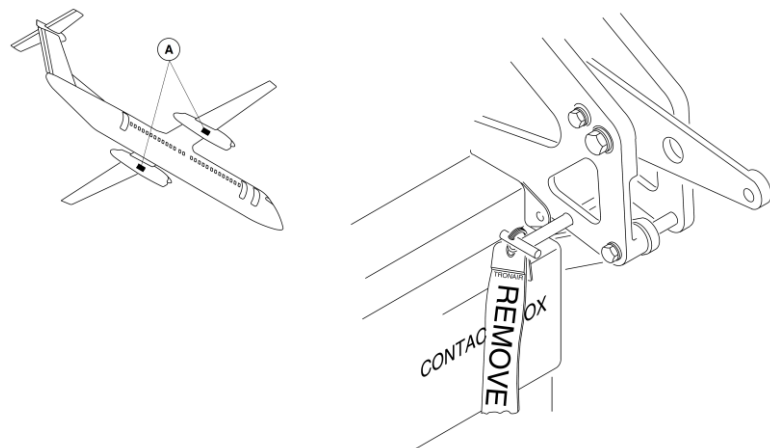


Figure 32-6 MLG Door Lock Pin

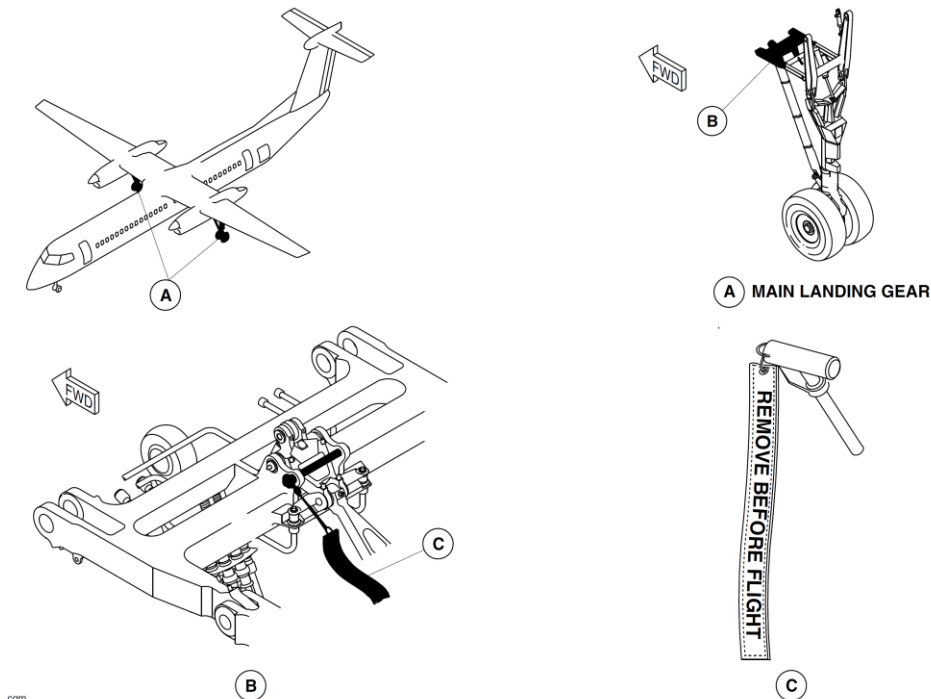


Figure 32-5 MLG Ground Lock Pin

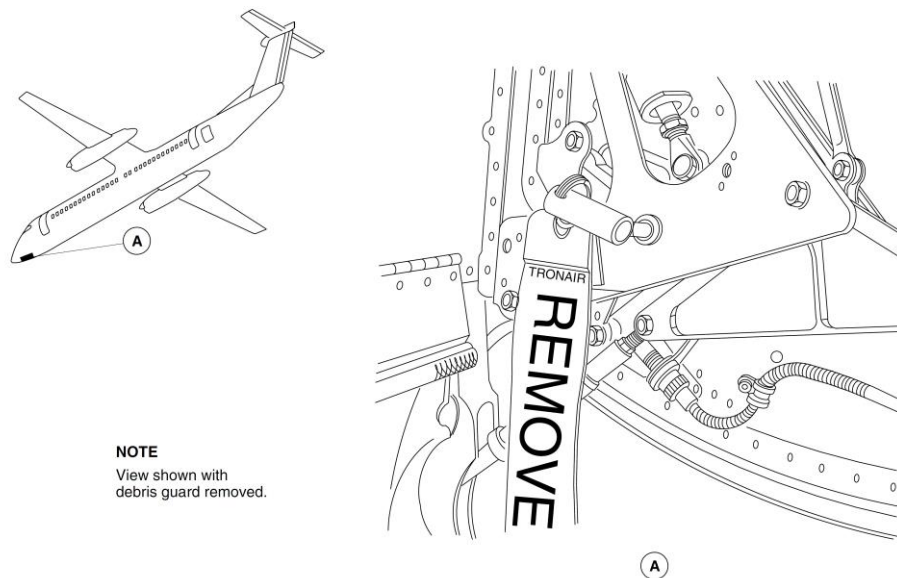


Figure 32-7 NLG Door Lock Pin

Nose Gear Ground Lock Control Handle

[Refer figure 32-8](#)

An integral ground lock mechanism is for nose gear which is controlled from outside the aircraft for locking the nose gear. You can find the ground lock control on the left side of the external nose fuselage. The nose gear ground lock control handle has a push-button. When you push the push-button, the nose gear control handle releases from the fuselage, and the control handle moves out. You must pull the control handle out fully from the sprung position for it to turn.

When the control handle is in and flush with fuselage, the nose gear unlocked.

When the control handle is fully out and rotated clockwise, the nose gear is locked.

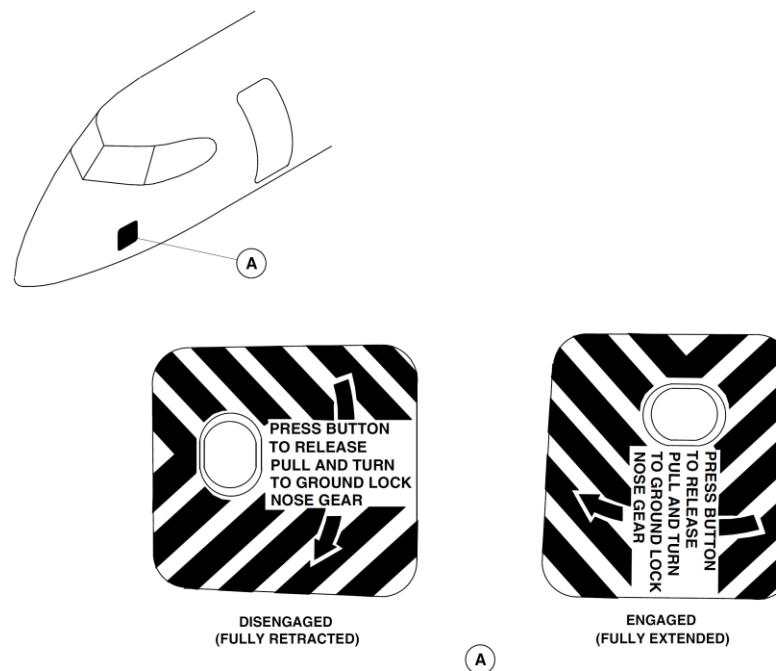


Figure 32-8 NLG Ground Lock Mechanism

Nosewheel Steering

[Refer Figures 32-9 & 32-10](#)

Nosewheel steering is by the steering tiller or the rudder pedals when the steering switch is set to STEERING. The steering tiller is a self-centering unit that lets the pilot steer the nosewheels for forward taxi at low speed. The steering tiller is installed in the flight compartment on the pilot's left side console. The steering tiller gives full authority steering command signals of the nose wheel position to the Steering Control Unit (SCU). It can steer the nosewheels ± 70 degrees to the left and right of the center position. The rudder pedals are self-centering and allow the crew to steer the nosewheels for high speed taxi, take off, and landing purposes. The rudder pedals can steer the nosewheels to a maximum of ± 8 degrees left and right of the center position. The potentiometer for the rudder pedals is installed on the support structure of the control quadrant of the rudder system. A linkage assembly connects the potentiometer to the control quadrant of the rudder system. The potentiometer gives limited authority steering command signals of the nosewheel position to the SCU.

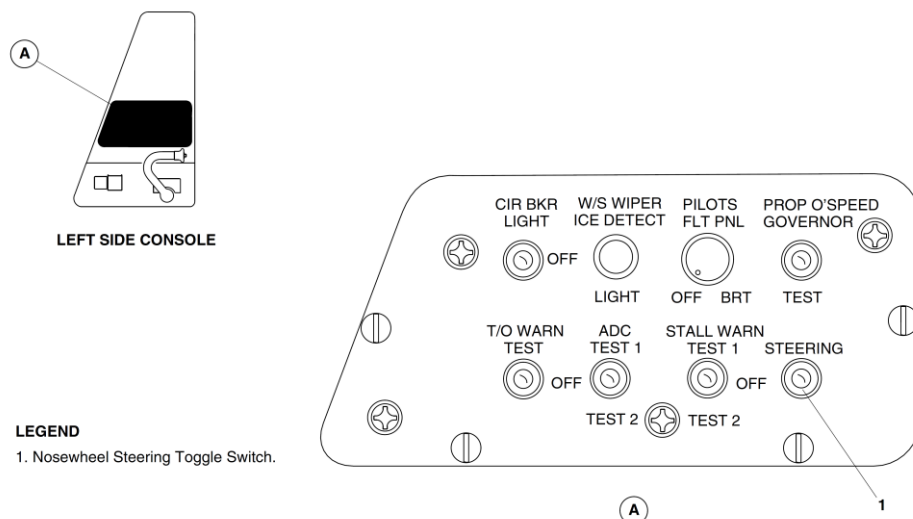


Figure 32-9 Nose Wheel Steering Switch

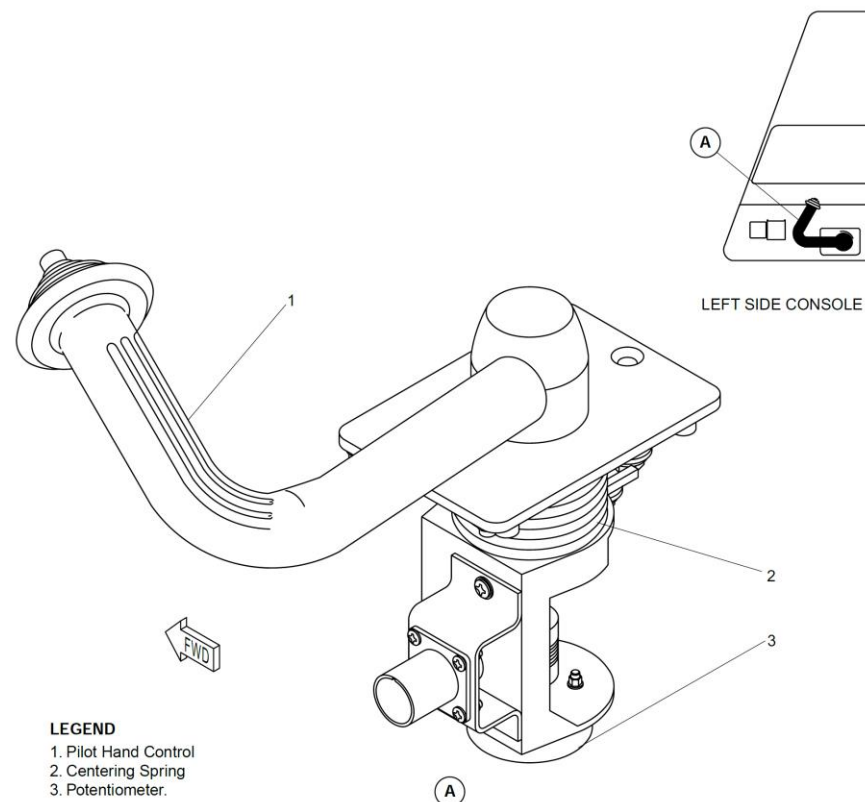


Figure 32-10 Nose Wheel Steering Hand Control

Landing Gear Alternate Release Door

[Refer Figure 32-11](#)

If the LANDING GEAR INHIBIT switch is set to the INHIBIT position, the gear will not retract or extend with the LANDING GEAR selector. If the LANDING GEAR ALTERNATE RELEASE door, located in the ceiling of the flight compartment, is opened, a bypass valve isolates the No. 2 hydraulic system. Pulling the LANDING GEAR ALTERNATE RELEASE handle opens the MLG doors and releases the MLG uplocks. Tension springs in the MLG door mechanism pull the doors open and the MLG free-falls into the down position.

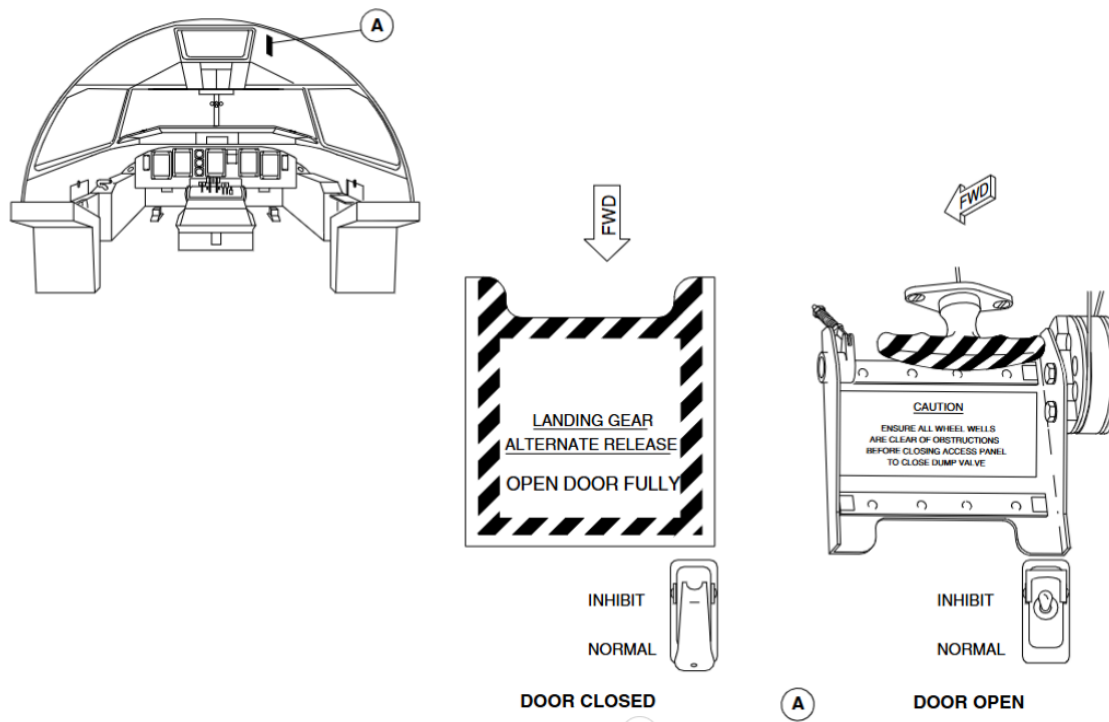


Figure 32-11 Landing Gear Alternate Release Door

Landing Gear Alternate Extension Door

Refer Figure 32-12

Opening the LANDING GEAR ALTERNATE EXTENSION door, located in the flight compartment floor, operates the NLG alternate selector valve and reveals the handpump socket. If the MLG does not reach the down and locked position, the handpump is used to fully extend the MLG. This pumps hydraulic fluid from the emergency hydraulic reservoir to the auxiliary main landing gear actuators. The pump handle is located behind copilot's seat.

Pulling the LANDING GEAR ALTERNATE EXTENSION handle unlocks the NLG forward doors. Tension springs in the NLG door mechanism pull the doors open. When the doors are fully open, the NLG uplock is released by further pulling the handle. The NLG free-falls to the fully extended position. Springs installed on the NLG drag strut move the lock links into the down and locked position.

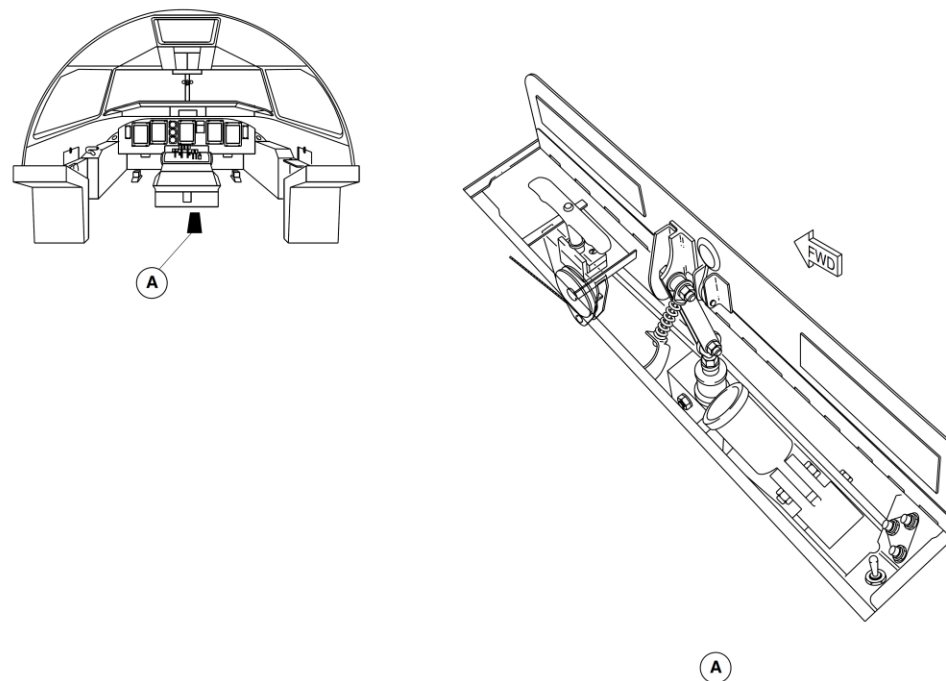


Figure 32-12 Landing Gear Alternate Extension Door

Anti Skid Switch

Refer Figure 32-13

If the ANTI SKID switch is in the OFF position, the antiskid system is deactivated. The INBD ANTISKID and OUTBD ANTISKID caution lights come on. If the ANTI SKID switch is in the ON position, the antiskid system is activated. The INBD ANTISKID and OUTBD ANTISKID caution lights will go off if there is no system failure. If the ANTI SKID switch is held in the TEST position while in the air, with gear down and locked, the INBD ANTISKID and OUTBD ANTISKID caution lights come on for 3 seconds then go off. If the ANTI SKID switch is held in the TEST position while on the ground, the INBD ANTISKID and OUTBD ANTISKID caution lights come on for 6 seconds then go off. The INBD ANTISKID and OUTBD ANTISKID caution lights will come on if there is a system failure.

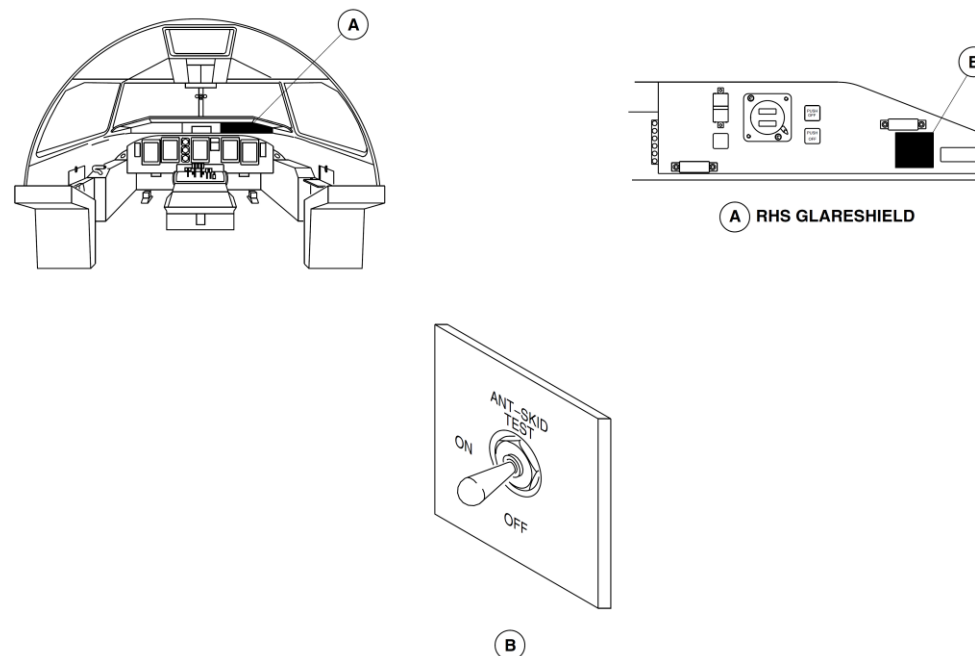


Figure 32-13 Anti Skid Switch

Emergency/Parking Brake Lever

Refer Figure 32-14

If the EMERGENCY/PARKING BRAKE LEVER is in the PARK detent, the parking brake is set if parking brake hydraulic pressure is more than 1500 psi (10342.5 KPa). The PARKING BRAKE caution light also comes on. The emergency brake system uses No. 2 hydraulic or accumulator pressure.

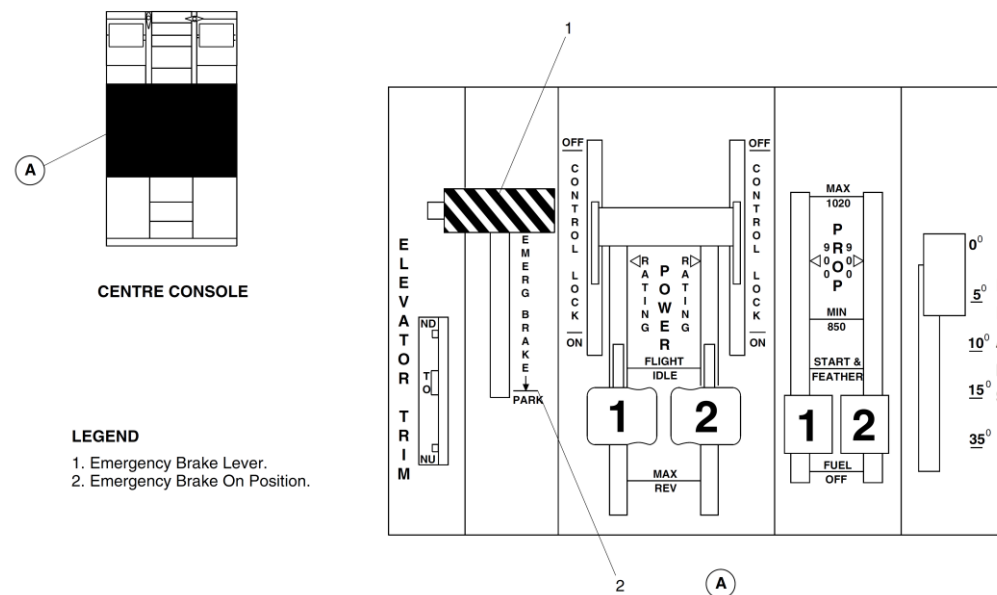


Figure 32-14 Emergency/Parking Brake Lever

Main Landing Gear

Introduction

The Main Landing Gear (MLG) absorbs and distributes the landing loads. The MLG and wheels support the aircraft on the ground. The MLG doors supply the fairing for the engine nacelle with the MLG in the retracted position.

General Description

[Refer Figure 32-15](#)

Each MLG assembly is installed on the airframe structure in the wheelwell of the related engine nacelle. Each MLG has two wheel and tire assemblies and retracts rearwards into the aft section of the wheelwell.

A retraction actuator is attached to the yoke and to the shock strut to extend and retract the MLG. A lock actuator on the stabilizer brace locks the MLG in the down position. An auxiliary extension actuator is attached to the airframe structure in the aft section of the wheelwell and to the arm of the yoke. The auxiliary extension actuator extends the MLG during an alternate extension sequence. An uplock assembly is attached to the top of the aft section of the wheelwell and locks the MLG in the retracted position.

A proximity sensor on the uplock assembly gives a signal to the Proximity Sensor Electronic Unit (PSEU) of the status of the MLG. Proximity sensors on the stabilizer brace send signals to the PSEU when the MLG is down and locked. Proximity sensors on the torque links give an indication to the PSEU of a Weight-On-wheels or a Weight-Off-Wheels condition.

The Main Landing Gear includes the components that follow:

- Yoke
- Shock Strut
- Stabilizer Brace
- Drag Strut
- Uplock Assembly
- Downlock Release Actuator
- Retraction Actuator

- Auxiliary Extension Actuator

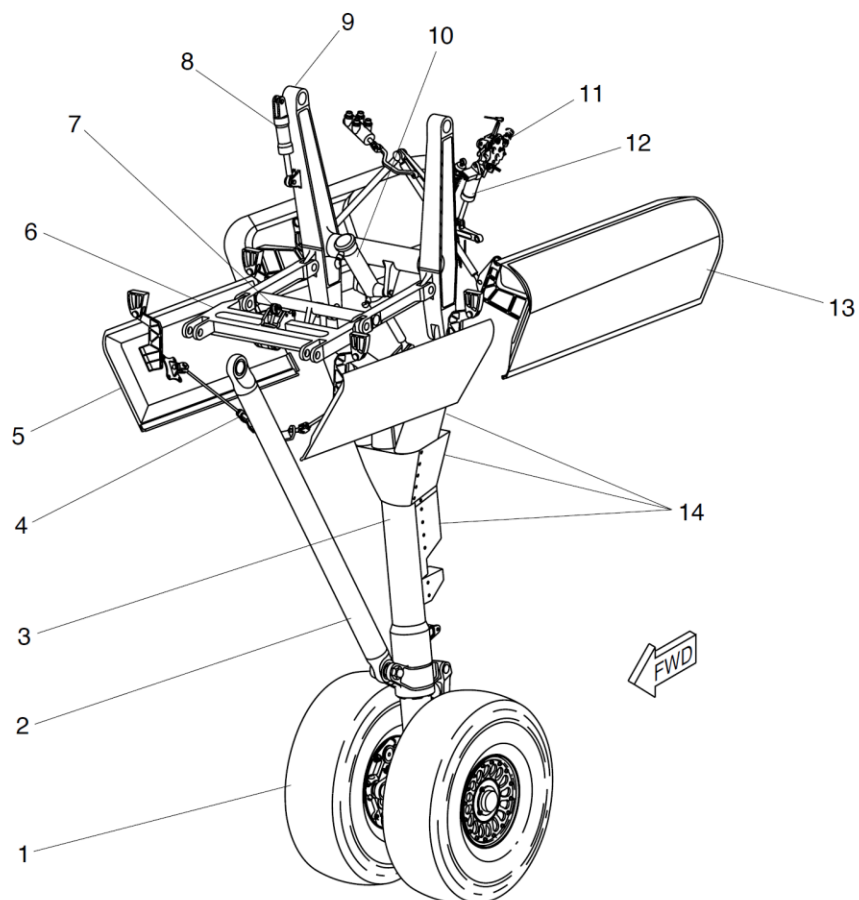


Figure 32-15 Main Landing Gear Details

Detailed Description

[Refer Figure 32-16](#)

The yoke is attached to the top of the aft section of the wheelwell. The shock strut is attached to the bottom of the yoke. A stabilizer brace keeps the yoke in position in the wheelwell. The stabilizer brace is attached to the front of the yoke and to the forward frame of the forward section of the wheelwell.

The shock strut is held in position, in the wheelwell, by the drag strut. The drag strut is attached to the bottom of the shock strut cylinder and to the forward frame of the forward section of the wheelwell. Fairings attached to the shock strut streamline the shock strut when the MLG is extended in flight. The shock strut has provision for the wheels, brake units, and anti-skid devices to attach to the axles. A jack pad on the bottom of the shock strut is for the aircraft to be lifted on jacks to change wheels.

The retraction actuator pulls the shock strut up and rearwards into the aft section of the MLG wheelwell. The stabilizer brace folds upwards and pulls the yoke forward. The shock strut hinges on the bottom of the yoke and on the bottom of the drag strut. The MLG uplock roller engages the uplock assembly and locks the MLG in the retracted position. The aft door actuator extends to close the aft wheelwell doors. The forward doors of the MLG wheelwell are closed by the upward movement of the drag strut when the shock strut moves backwards and up.

Proximity sensors located on the outer cylinder indicate a Weight-Off-Wheels condition to let the MLG retract. A proximity sensor on the aft doors mechanism gives an indication when the doors move and when the doors are closed and locked. Proximity sensors on the stabilizer brace

overcenter lock mechanism give an indication that the MLG downlock is not in the downlock position. A proximity sensor on the uplock assembly gives an indication that the MLG is locked in the retracted position.

Extension of the MLG is controlled by the components operation of the retraction and extension system. Hydraulic pressure is supplied to the aft MLG doors actuator to extend and open the doors. The uplock assembly actuator extends to unlock and release the MLG uplock roller. The retraction actuator extends to lower the shock strut.

The lock actuator on the stabilizer brace unfolds the stabilizer brace. The bottom of the yoke moves backwards and the shock strut pivots around the yoke bottom hinge points and the drag strut hinge points.

When the stabilizer brace has unfolded, the lock actuator pushes the stabilizer brace into an overcenter locked position. The actuator of the MLG wheelwell aft doors is sequenced to extend and close the doors. The forward doors of the MLG wheelwell are opened by the downward movement of the drag strut when the extension sequence occurs.

A proximity sensor on the aft doors mechanism gives an indication when the doors move and when the doors are closed and locked. A proximity sensor on the uplock assembly gives an indication that the MLG is unlocked from the uplock assembly. Proximity sensors on the stabilizer brace overcenter lock mechanism give an indication that the MLG is in a down and locked condition. Proximity sensors on the outer cylinder indicate a Weight-On-Wheels condition and will stop the MLG from retracting.

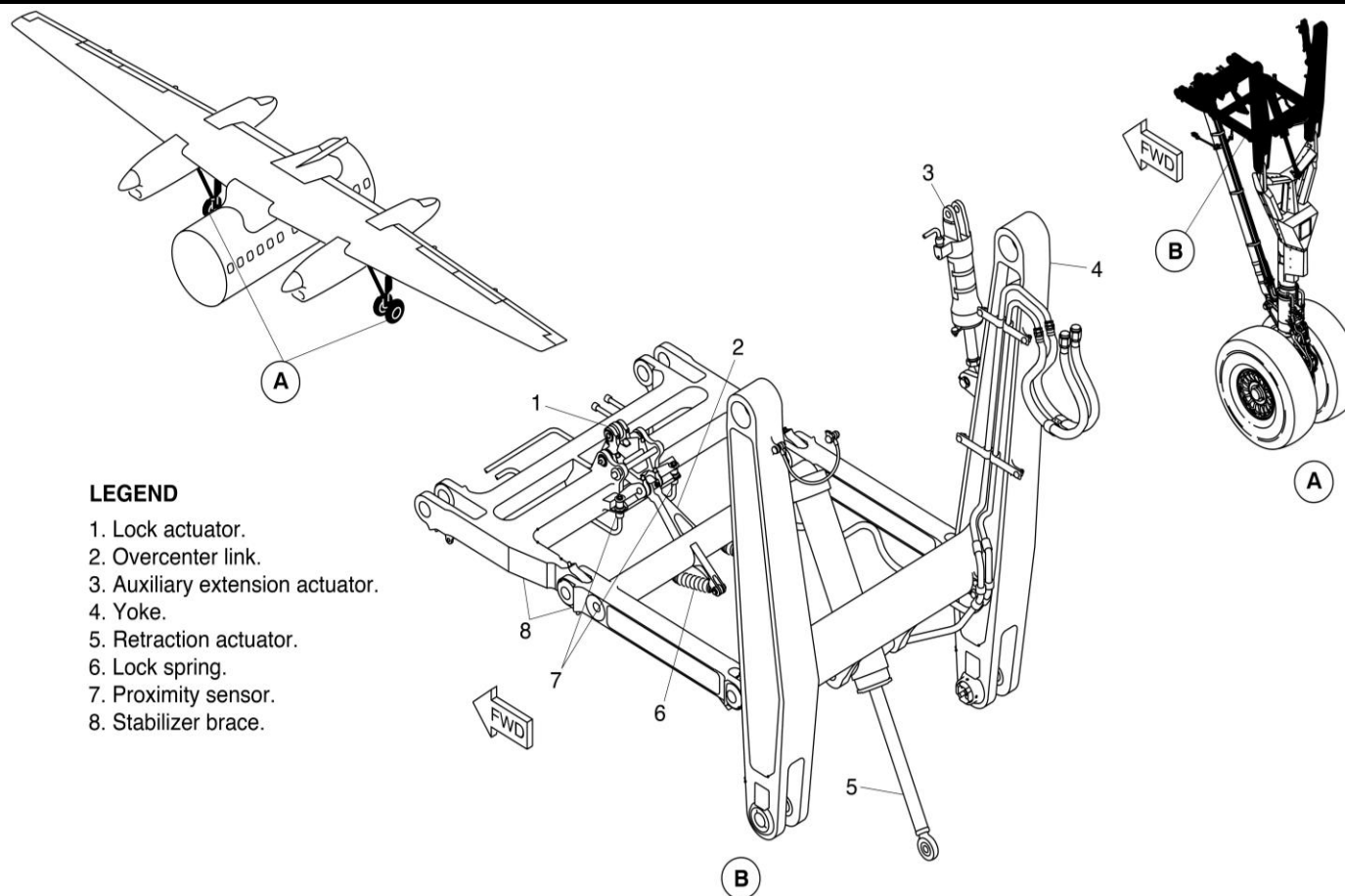


Figure 32-16 Main Landing Gear Component Details

Main Landing Gear - Component Description

Yoke

Refer Figure 32-16

The yoke assembly is an H-shaped structural component that connects the upper end of the shock strut to the airframe structure in the aft section of the wheelwell. The top of the yoke is attached to the airframe by two lubricated hinge points. The bottom of the yoke is attached to the top of the shock strut by two lubricated hinge points. The yoke attaches to the stabilizer brace at the cross beam with two lubricated hinge points. The retraction actuator cylinder attaches to the cross beam of the yoke with two lubricated hinge points. The rod end of the auxiliary retraction actuator is attached to lugs at the top of the yoke with lubricated hinge points. The yoke rotates about the top hinge points and the bottom of the yoke moves forward in the wheelwell during the retraction sequence.

Shock Strut

Refer Figure 32-17

The shock strut outer cylinder is a Y-shaped component with a single stage nitrogen and oil filled shock absorbing piston fitted inside the outer cylinder. Fixed fairings are attached to the forward and aft sides of the shock strut Y-shaped top section and to the outer cylinder. The fairings have an aerodynamic shape and are made from composite material. The anti-skid devices, brake units, and wheel assemblies are attached to a removable and interchangeable axle at the bottom of the shock strut piston. An uplock roller on the back of the outer cylinder engages the uplock assembly in the aft section of the wheelwell to lock the MLG in the retracted position.

Rotation of the shock strut piston and axle is prevented by two connected torque links. The lower torque link is attached to two lugs on the rear of the shock strut piston. The upper torque link is attached to two lugs on the bottom of the outer cylinder.

The top end of the shock strut is attached to the yoke with lubricated hinge points. The lower end of the shock strut outer cylinder is attached to the lower

end of the drag strut with a lubricated hinge point. The rod end of the retraction actuator is attached to the top of the shock strut with a lubricated hinge point.

Proximity sensor targets on the torque links move as the piston moves to give a Weight-On-Wheels or a Weight-Off-Wheels indication to the PSEU. The proximity sensor and anti-skid wheel speed transducer electrical harnesses and brake hydraulic lines are fitted to the bottom section of the shock strut cylinder. Nitrogen and oil charging and bleeding valves are located on the outer cylinder.

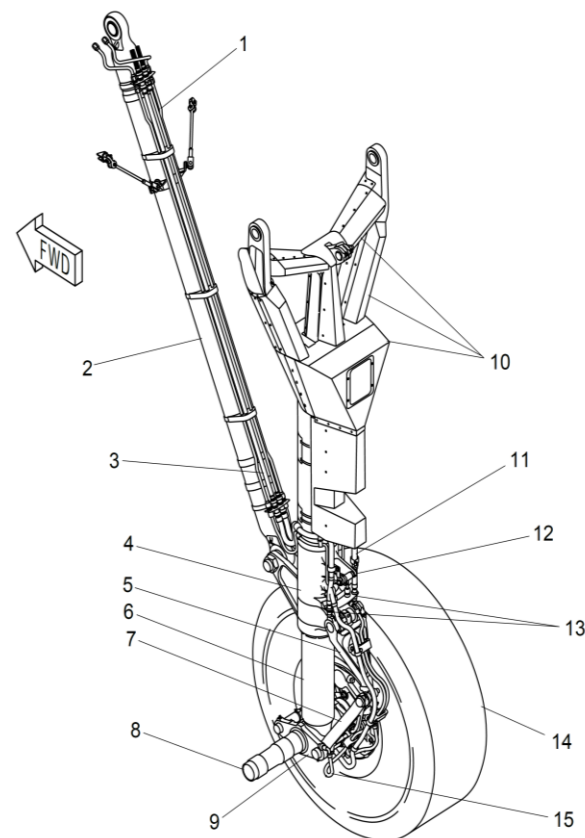


Figure 32-17 MLG Shock Strut Detail

LEGEND

1. Electrical Harness
2. Drag Strut
3. Hydraulic Brake Line
4. Shock Strut
5. Upper Torque Link
6. Shock Strut Piston
7. Lower Torque Link
8. Axle
9. Jack Pad
10. Fairings
11. Electrical Harness
12. Uplock Roller
13. Proximity Sensors
14. Wheel and Brake Assembly
15. Hydraulic Brake Line.

Stabilizer Brace

[Refer Figure 32-16](#)

The stabilizer brace is a two piece folding structural component. The stabilizer brace keeps the yoke and the shock strut in position when the MLG is in the extended or retracted position. The forward section of the stabilizer brace is attached to the airframe structure in the forward section of the wheelwell with two lubricated hinge points. The aft section of the stabilizer brace is attached to the yoke with two lubricated hinge points.

The stabilizer brace has an overcenter link subassembly attached between the two sections of the stabilizer brace. The overcenter link subassembly supplies a mechanical lock for the MLG in the down position. The mechanical lock is broken by the lock actuator. The two sections of the link subassembly fold upward to move the bottom of the yoke forward during the MLG retraction sequence. The links are moved into a mechanical lock position by the lock actuator during the MLG extension sequence. Two lock springs keep the links in the mechanical lock position.

The lock springs lock the stabilizer brace in the down position when the alternate extension system is in operation. The stabilizer brace has provision for the installation of a ground lock pin when the MLG is extended.

Drag Strut

[Refer Figure 32-18](#)

The drag strut is a fixed length tubular component. The drag strut is attached at the lower end to the shock strut outer cylinder with a lubricated hinge point. The drag strut is attached at the upper end to the airframe structure of the forward section of the wheelwell with a lubricated hinge point. A bracket assembly near the top end of the drag strut is to attach to the linkages of the forward doors of the wheelwell.

Brackets attach the electrical harnesses for the proximity sensors and wheel speed transducers, and hydraulic pipes for the brake units along the rear of the drag strut. A grounding point to ground the aircraft during maintenance procedures is fitted to the top of the drag strut.

The landing gear pivots about the drag strut upper end hinge during MLG retraction and extension.

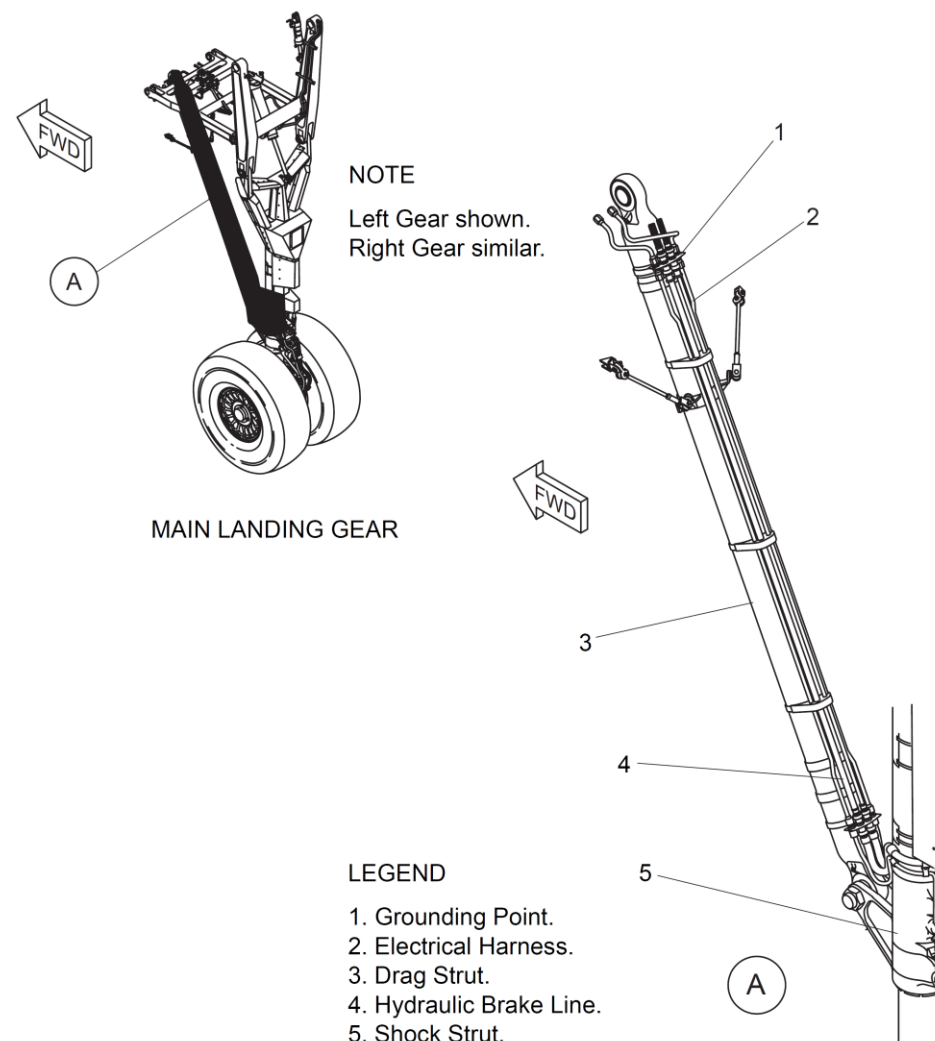


Figure 32-18 MLG Drag Strut

Uplock Assembly

Refer Figure 32-19

The uplock assembly is made up of two side plates to form the body of the uplock assembly. The uplock assembly is made up of a latch assembly, a lock link, an actuator assembly, an alternate release arm, internal extension springs, and a torsion spring.

The uplock assembly is attached to a frame in the top of the aft section of the wheelwell. The latch assembly and the lock link are attached to the body with lubricated hinge points. The latch assembly and the lock link pivot in the body of the uplock assembly. This action locks the uplock roller of the MLG in the retracted position.

The uplock roller on the MLG engages the inner face of the latch assembly and the latch assembly pivots into the locked position. The force of the two extension springs keeps the latch assembly fully rotated **to lock the MLG in the retracted position. The lock link prevents the latch assembly from rotating to the unlocked position.**

The lock link is rotated to the unlocked position when the piston of the uplock actuator extends under pressure from the No. 2 hydraulic system. The latch assembly returns to the locked position when hydraulic pressure is removed from the uplock actuator. The external compression spring on the uplock actuator returns the piston to the retracted position when hydraulic pressure is removed from the uplock actuator.

When the landing gear alternate release handle in the flight compartment is pulled, a cable rotates the alternate release arm. The lower portion of the alternate release arm touches the latch assembly and rotates the latch assembly to release the MLG uplock roller. The MLG is partially lowered by gravity. The torsion spring returns the alternate release arm to the normal position when the alternate release handle in the flight compartment is reset.

A proximity sensor mounted on the uplock assembly gives an indication to the PSEU on the status of the uplock assembly. The target on the lock link moves

away from the proximity sensor when the lock link is rotated during the MLG extension sequence. This gives an indication to the PSEU that the uplock assembly is in the unlocked position. The target on the lock link moves to the proximity sensor when the lock link is rotated during the MLG retraction sequence. This gives an indication to the PSEU that the uplock assembly is in the locked position.

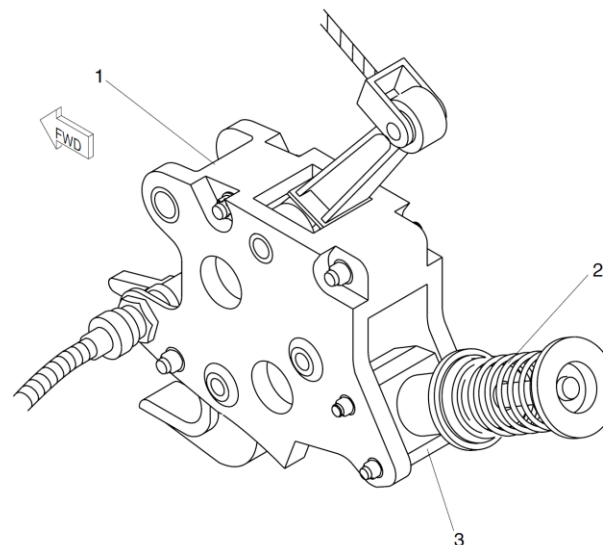


Figure 32-19 MLG Uplock Assembly

LEGEND

1. MLG Uplock Assembly.
2. External Spring.
3. MLG Uplock Actuator.

Downlock Release Actuator

[Refer Figure 32-20](#)

The downlock release actuator is of a simple two port design with hydraulic pressure supplied by the No.2 hydraulic system. The downlock release actuator is attached to the stabilizer brace forward section with lubricated hinge points. The rod end of the downlock release actuator is attached to the forward lock link of the stabilizer brace assembly with a lubricated hinge point.

The piston of the downlock release actuator retracts to break the lock links overcenter position. Further movement folds the stabilizer brace and pulls the yoke forward during the retraction sequence. The piston of the downlock release actuator extends to bring the lock links into an overcenter position to lock the stabilizer brace.

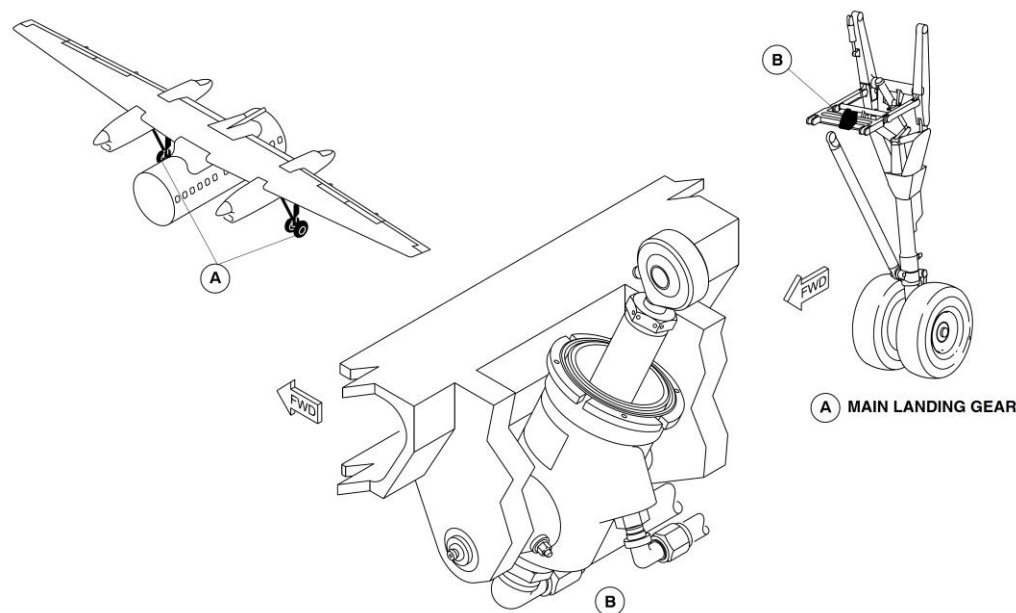


Figure 32-20 MLG Downlock Release Actuator

Retraction Actuator

[Refer Figure 32-21](#)

The retraction actuator is of a simple two port design with hydraulic pressure supplied by the No.2 hydraulic system. The retraction actuator is attached to the yoke cross beam lugs with lubricated hinge points. The rod end of the retraction actuator is attached to the top of the shock strut with a lubricated hinge point. The piston of the retraction actuator retracts to retract the MLG. The piston extends to lower the MLG.

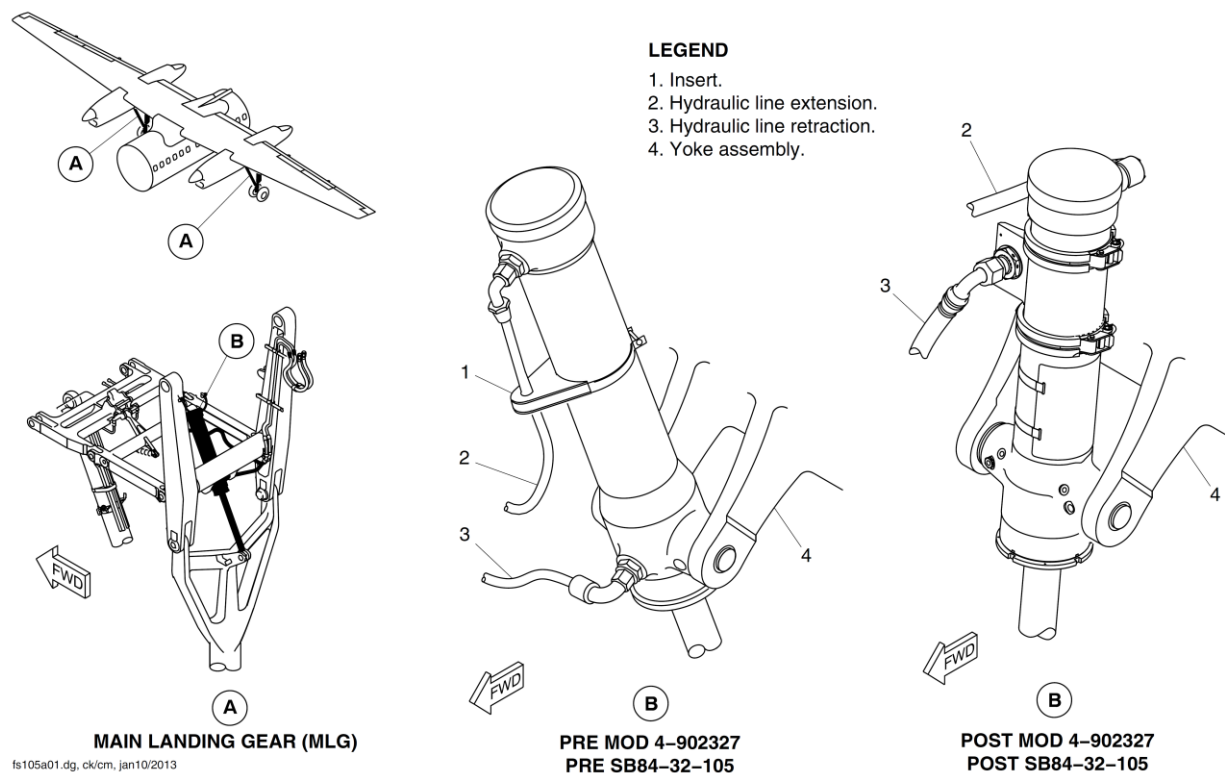


Figure 32-21 MLG Retraction Actuator

Auxiliary Extension Actuator

[Refer to Figure 32-22](#)

The auxiliary extension actuator is of a one-port design with hydraulic pressure supplied by the handpump hydraulic system. The auxiliary extension actuator attaches to the airframe structure at the top of the wheelwell. The rod end of the auxiliary extension actuator attaches to the forward side of the yoke with a lubricated hinge point.

The piston of the auxiliary extension actuator extends to lower the MLG during the alternate extension sequence.

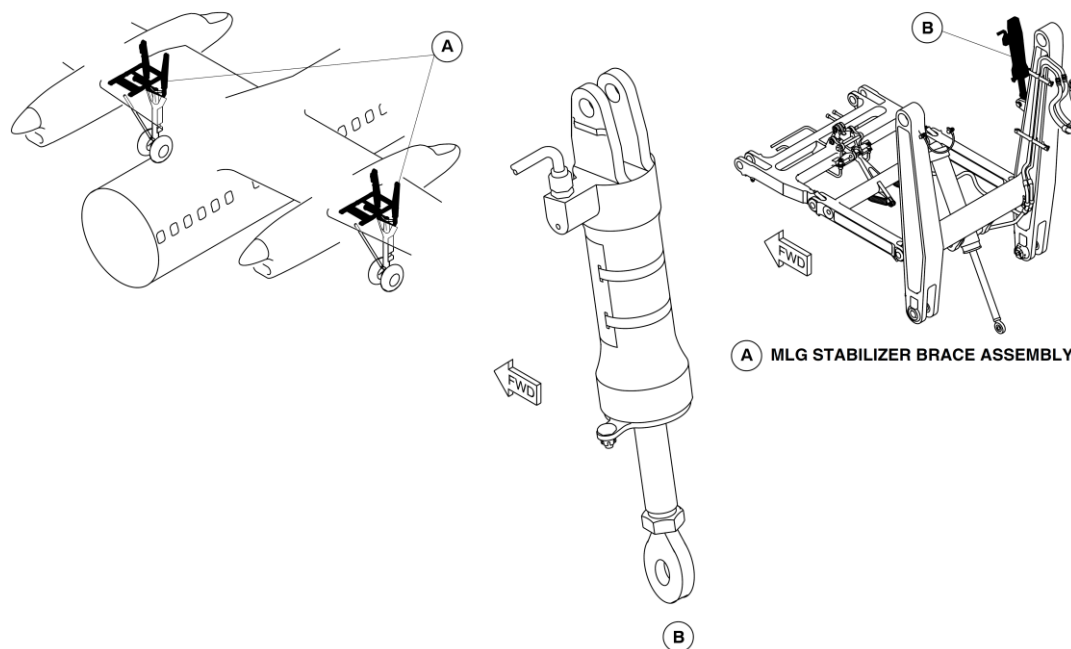


Figure 32-22 MLG Alternate Extension Actuator

Main Landing Gear Doors

Introduction

The Main Landing Gear (MLG) doors enclose the retracted MLG in the engine nacelle. The four doors on each engine nacelle make up the fairings for the MLG in the retracted position.

General Description

[Refer Figure 32-23 & 32-24](#)

Two doors close off the forward section of the MLG wheelwell. The forward doors are open when the MLG is extended and closed when the MLG is retracted. The doors are opened and closed by adjustable links connected to the drag strut.

Two doors close off the aft section of the MLG wheelwell. The aft doors are closed when the MLG is retracted and closed when the MLG is extended.

An alternate release arm is used to open the doors during alternate extension of the MLG.

The status of the aft doors is shown on the LANDING CONTROL panel in the flight compartment.

The MLG include the components that follow:

- Forward Doors
- Aft Doors
- Forward Doors Linkages
- Aft Doors Linkage Mechanism
- Aft Doors Actuator
- Detailed Description

Detailed Description

[Refer Figure 32-24](#)

The forward doors are operated by mechanical links attached to the drag strut of the MLG. When the MLG is extended the forward doors are opened mechanically. When the MLG is retracted the doors are closed mechanically

The aft doors are opened and closed by a single hydraulic actuator and a linkage mechanism. The linkage mechanism has a mechanical sequence valve that is connected to the pivot arm by a rod. The door pushrods, lock links, and the tension spring are attached to the pivot arm. The pushrods connect the doors to the pivot arm. The lock links hinge point supplies an overcenter lock for the doors in the closed position.

An alternate release arm breaks the overcenter lock of the lock links hinge point to open the doors during alternate extension of the MLG.

A proximity sensor sends a signal to the Proximity Sensor Electronic Unit (PSEU). The PSEU sends a signal to the LANDING GEAR Control Panel in the flight compartment, to show the status of the aft doors.

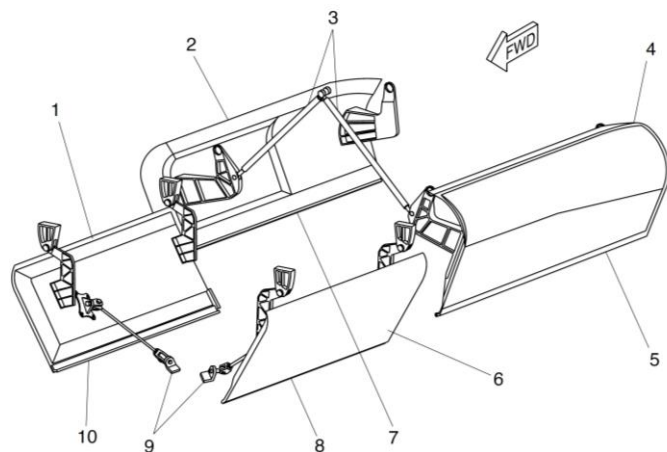
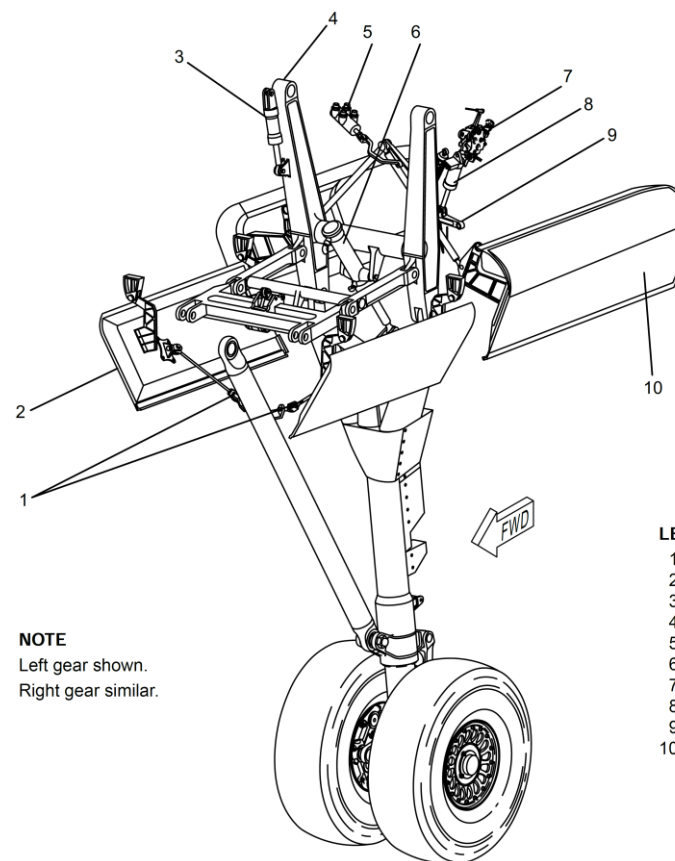


Figure 32-23 Main Landing Gear Doors Detail

LEGEND

1. Forward Door.
2. Aft Door.
3. Push Rods.
4. Aft Door.
5. Door Seal.
6. Forward Door.
7. Door Seal.
8. Door Seal.
9. Forward Door Connecting Rods.
10. Door Seal.



NOTE

Left gear shown.
Right gear similar.

LEGEND

1. Forward Doors Connecting Rods
2. Forward Door
3. Alternate Extension Actuator
4. Yoke
5. Mechanical Sequence Valve
6. Retraction Actuator
7. Uplock Assembly
8. Aft Doors Mechanism
9. Alternate Release Arm
10. Aft Door.

Figure 32-24 Main Landing Gear Doors Mechanism

Main Landing Gear Doors – Component Description

Forward doors

[Refer Figure 32-25](#)

The forward doors are double skinned and are made in the same shape as the bottom contour of the center section of the engine nacelle. The doors are connected to the bottom of the center section of the engine nacelle by hinges. Each door has a hinge on the forward and aft corner. The doors are symmetrical and meet at the engine nacelle centerline when closed. The door operating links, are attached to fittings on the forward section of the doors and to the MLG drag struts.

Forward Doors Linkages

[Refer Figure 32-25](#)

The forward door linkages have articulated, adjustable links. The links are attached to a bracket assembly on the upper part of the drag strut and to fittings on the forward section of the doors. The links open the doors as the drag strut moves when the MLG extends. The links close the doors as the drag strut moves when the MLG retracts.

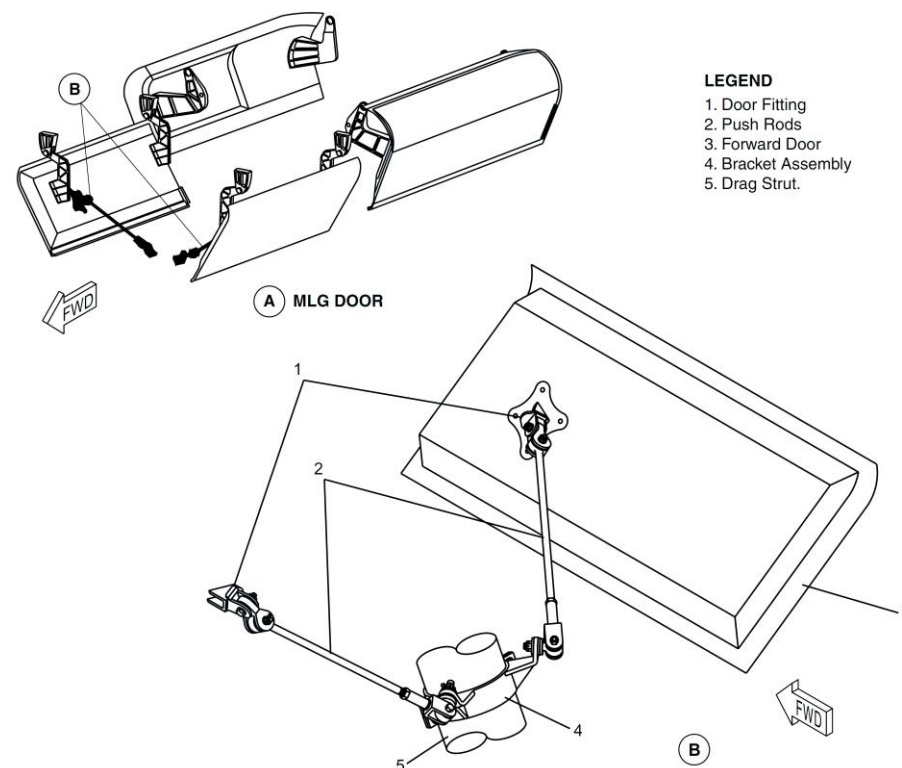
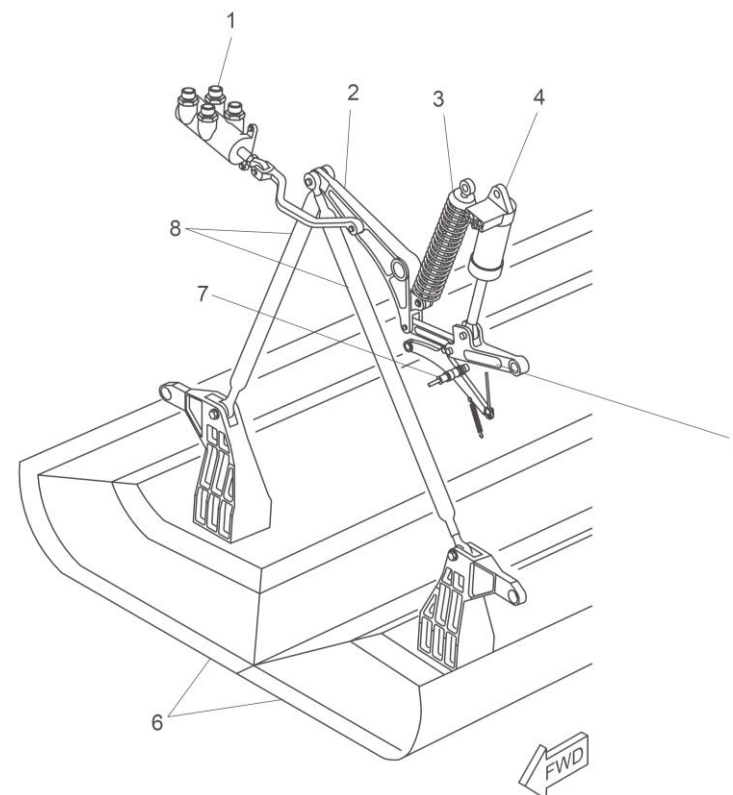


Figure 32-25 MLG Forward doors and connecting rods

Aft Doors

Refer Figure 32-26

The aft doors are double skinned and are made in the same shape as the bottom contour of the aft section of the engine nacelle. The doors are connected to the bottom of the aft section of the engine nacelle by hinges. Each door has a hinge on the forward corner and a hinge on the aft corner. The doors are symmetrical and meet at the engine nacelle centerline when closed. The door operating linkage mechanism is attached to the forward hinges of the doors.



LEGEND

1. Mechanical Sequence Valve.
2. Pivot Arm.
3. Tension Spring.
4. Actuator.
5. Lock Link.
6. Aft Door.
7. Proximity Sensor.
8. Push Rods.

Figure 32-26 MLG Aft doors & connecting rods

Aft Doors Linkage Mechanism

[Refer Figure 32-27](#)

The door linkage has two push rods that are attached to the forward hinges of the aft doors and to the pivot arm. The pivot arm is attached to a frame assembly on the left side of the aft section of the wheelwell. The pivot arm has hinge points to form a bellcrank between the push rods and the lock links. The lock links are attached to the bottom of the pivot arm and to the side of the aft section of the wheelwell.

The lock links hinge point is pushed into an overcenter position by the actuator to lock the doors in the closed position. The lock links hinge point is pulled from an overcenter position by the actuator to open the doors. The lock links fold and rotate the pivot arm to push open the doors with the push rods. During the alternate extension of the MLG, the alternate release arm rotates to move the lock links from an overcenter position. A tension spring is attached to the pivot arm and to the side of the aft section of the wheelwell. The tension spring turns the pivot arm for the push rods to push open the doors.

A proximity sensor is installed on the left side of the aft section of the wheelwell. The proximity sensor sends a signal to the Proximity Sensor Electronic Unit (PSEU). The PSEU sends a signal to the LANDING GEAR control panel, located in the flight compartment, to show the status of the aft doors. When the doors are open the amber L DOOR and R DOOR lights come on. The lock links in an overcenter position will show that the doors are closed and locked, and the lights will go off. When the lock links are moved from the overcenter position, the sensor target on the lock links is moved away from the proximity sensor. This will give an indication that the doors are not closed and locked.

Aft Doors Actuator

[Refer Figure 32-27](#)

The hydraulic actuator is a two port design connected to the No.2 hydraulic system of the aircraft. The actuator is located on the left side of the aft section

of the wheelwell. Lubricated hinge points attach the body of the actuator to the side of the wheelwell and the rod end of the piston to the pivot arm. The hydraulic actuator is sequenced by a solenoid sequence valve. The hydraulic actuator retracts to open the doors and extends to close the doors..

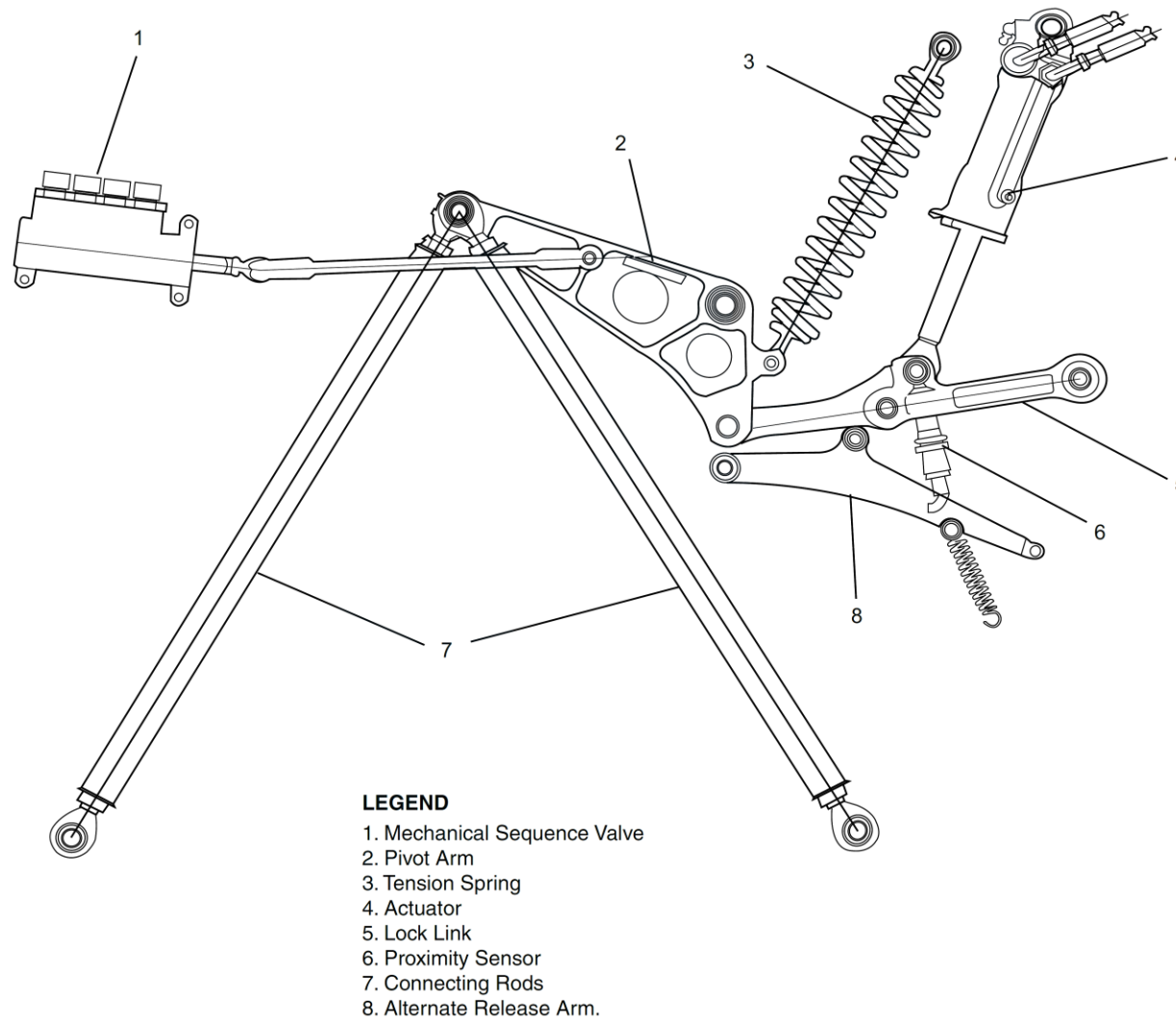


Figure 32-27 MLG Aft Doors Linkage Mechanism

Nose Landing Gear

Introduction

The nose landing gear (NLG) lets the forward fuselage absorb the shock of landing and gives stability and direction during aircraft taxi. The NLG doors enclose the retracted NLG in the forward fuselage and add to the aerodynamic efficiency of the aircraft

General Description

[Refer to Figures 32-28 & 32-29](#)

The NLG is installed in the wheelwell in the nose fuselage, located forward of the forward pressure bulkhead. When the NLG retracts the forward and aft doors enclose it in the wheelwell.

A taxi light is installed on the lower right side of the outer cylinder of the shock strut.

The GROUND LOCK control handle operates a cable system to engage the downlock safety lever.

The nose landing gear assembly has the components that follow:

- Shock Strut Assembly
- Drag Strut Assembly
- Harness Electrical

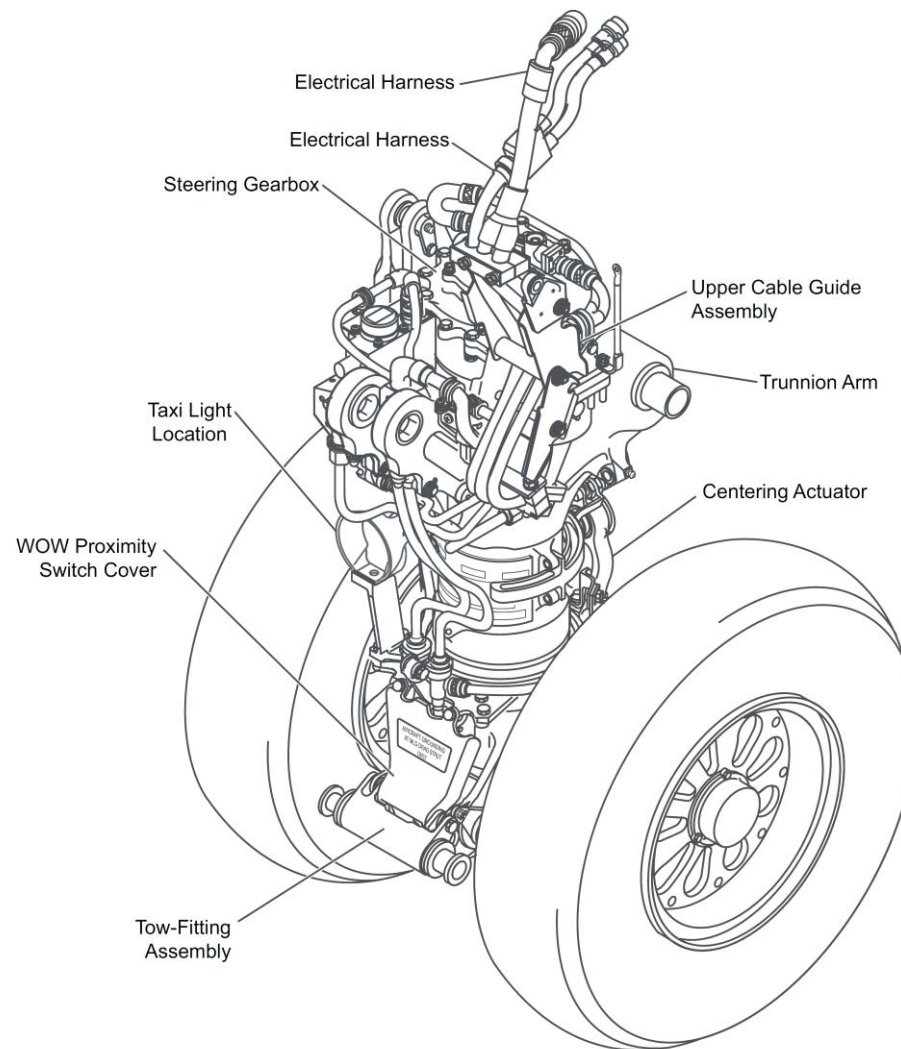
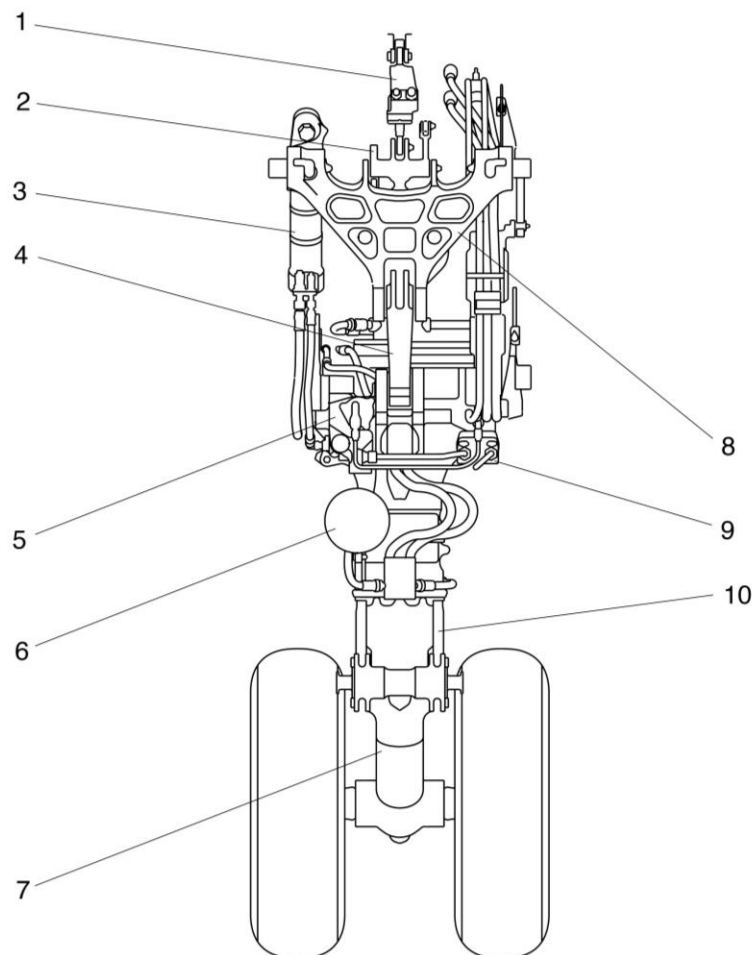


Figure 32-28 Nose Landing Gear Assembly



LEGEND

1. Downlock Actuator.
2. Pivot Tube Assembly.
3. Retraction Actuator.
4. Lower Drag Strut.
5. Nosewheel Steering Manifold.
6. Taxi Light.
7. Trailing Arm.
8. Upper Drag Strut.
9. Steering Hydraulic Motor.
10. Inner Cylinder.

Figure 32-29 Nose Landing Gear Aft View Detail

Detailed Description

Refer Figure 32-30 & 30-31

The NLG has two hydraulically operated forward doors and two mechanically operated aft doors, actuated by the NLG when the landing gear extends and retracts. When the NLG retracts into the wheelwell, the two sets of doors close around it.

A taxi light is installed on the lower right side of the outer cylinder of the shock strut.

The ground lock control handle operates a cable to engage the downlock safety lever. The lever is engaged when the downlock arm of the pivot tube assembly is on the drag strut of the NLG.

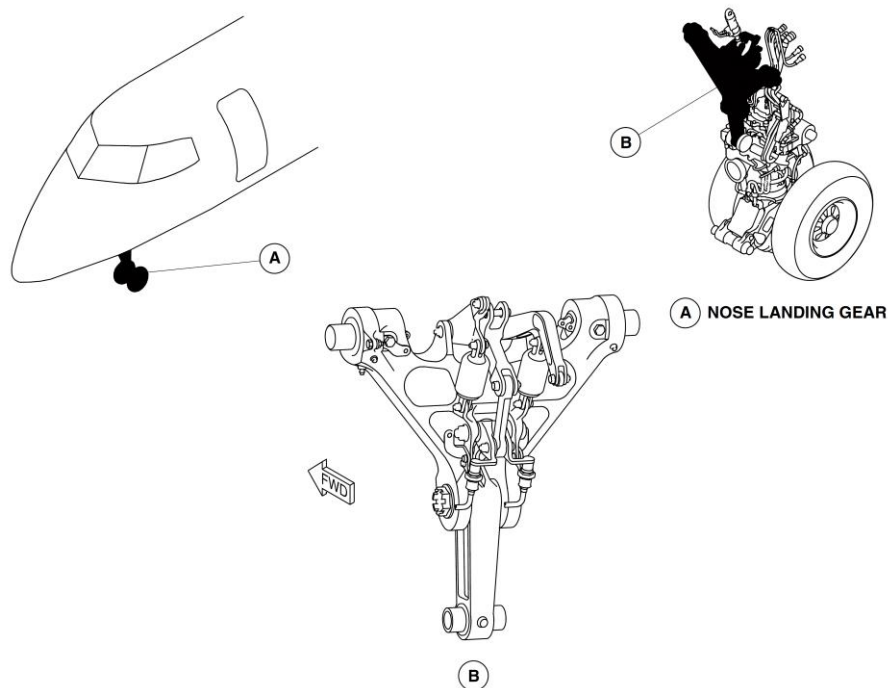


Figure 32-30 Nose Gear Strut/Steering Locator

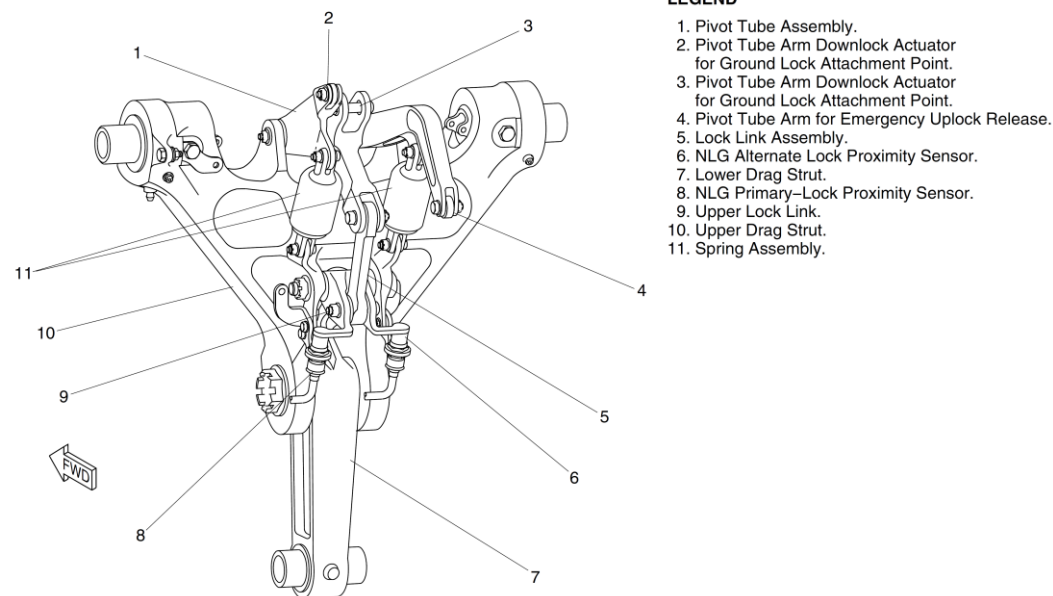


Figure 32-31 Nose Gear Strut/Steering Details

Nose Landing Gear – Component Description

Shock Strut Assembly

[Refer Figures 32-31 & 32-32](#)

The one stage, nitrogen/oil filled shock strut assembly uses a lever suspension geometry. The shock strut has an outer cylinder, an inner cylinder, a piston assembly, a link, and a trailing arm. **The trailing arm has a jacking point to retract and extend the NLG.**

Two trunnion arms are installed at the upper rear portion of the outer cylinder, where the trunnion bearings are attached and the NLG is installed. The trunnion bearings are the NLG pivot points during extension and retraction. The retraction actuator is attached to the mounting lugs on the top of the right trunnion arm. At the top forward side of the outer cylinder are two mounting lugs. The lower drag strut is attached to the two mounting lugs. A taxi light is installed on the bottom right side of the outer cylinder.

The power steering hydraulic motor and gearbox assembly are installed between the trunnion arms. The NLG steering centering actuator assembly, the steering sensor, and the centering cam are installed on the aft lower side of the outer cylinder.

The outer cylinder has attachment points for the electrical harnesses. The Weight On Wheels (WOW) proximity sensors are installed on the inner cylinder above the tow fitting assembly.

The piston assembly in the inner cylinder is attached to a link. The link is installed between two lugs on the trailing arm. The link transmits ground loads from the trailing arm to the piston assembly.

A low pressure valve is on the top surface of the inner cylinder. The valve is used to service the piston assembly with hydraulic fluid and dry nitrogen.

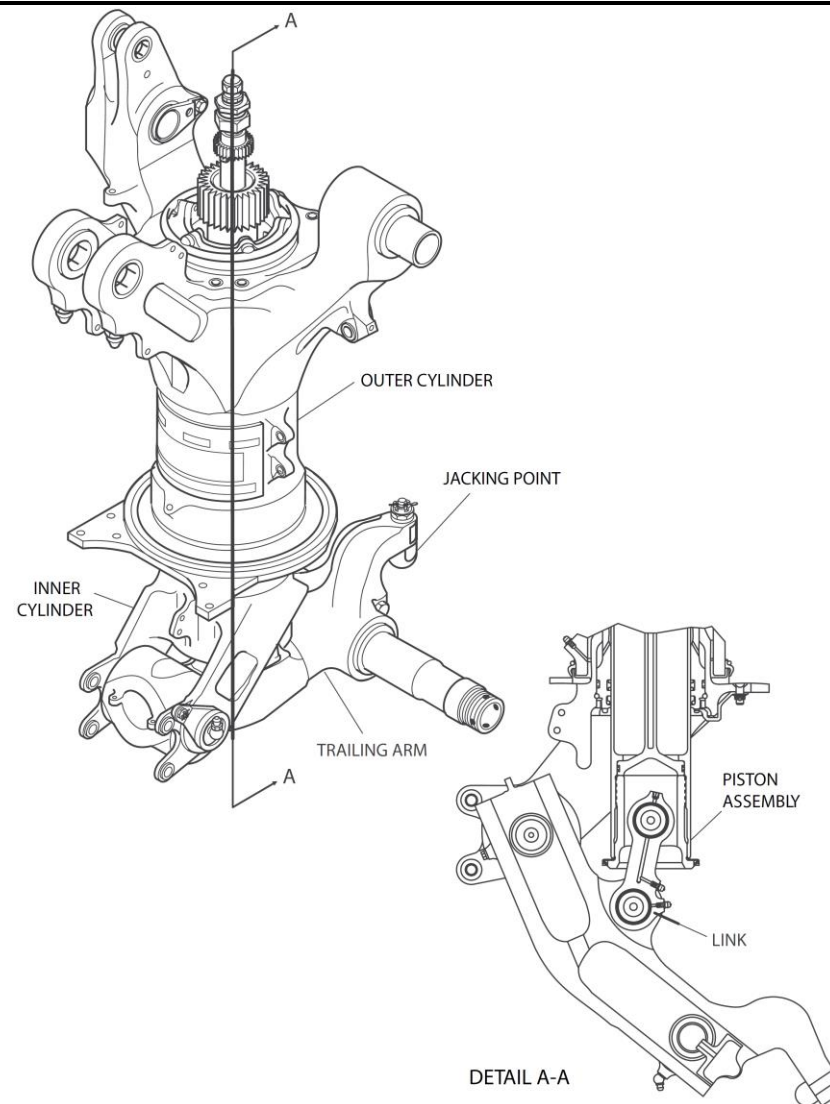


Figure 32-31 NLG Shock Strut Assembly

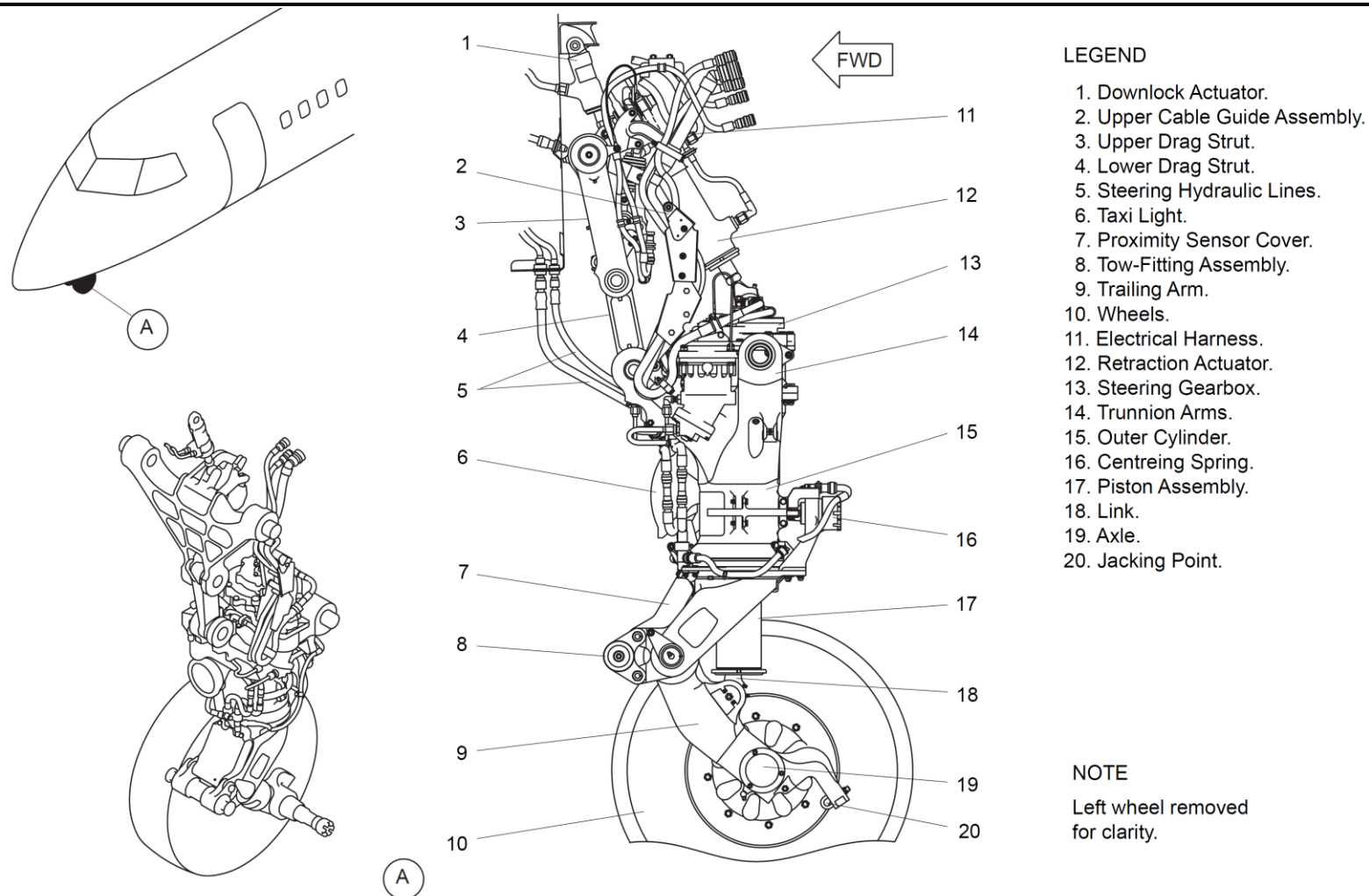


Figure 32-32 NLG Shock Strut Assembly - Detail

Drag Strut Assembly

[Refer Figure 32-33](#)

The drag strut assembly stabilizes the shock strut in either the retracted or the extended position.

The drag strut assembly has the components that follow:

- An upper drag strut
- A lower drag strut
- A pivot tube assembly
- Two spring assemblies
- A link assembly
- An upper lock link
- A lower lock link.

The two spring assemblies and the pivot tube assembly are installed on the upper drag strut. The pivot tube assembly has six arms:

- Two for the spring assemblies
- One for the downlock actuator
- One for the link assembly
- One for the emergency uplock release mechanism
- One for the ground lock mechanism.

The lower lock link is installed on the lower drag strut and is connected to the upper lock link. The upper lock link is installed on the upper drag strut and is connected to the lock link assembly. The over centering action of the upper and lower lock links makes a mechanical lock when the gear is in the retracted and the extended positions.

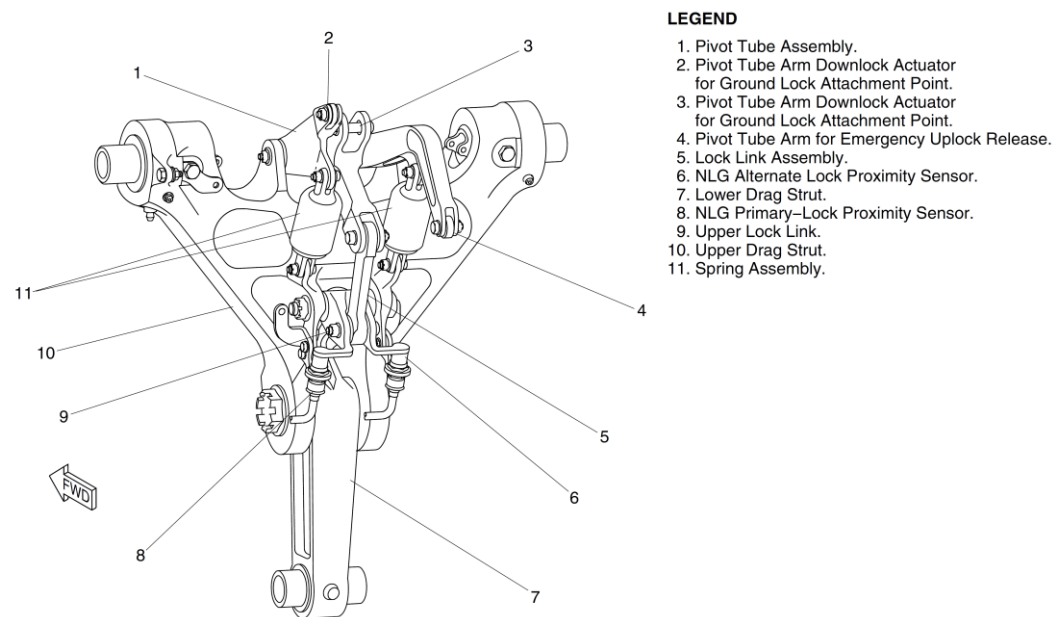


Figure 32-33 NLG Drag Strut Detail

NLG Electrical Harness

Four electrical harnesses are attached at different places on the shock strut. Two additional electrical harnesses are attached to the drag strut. The four harnesses on the shock strut give power/signal input for the components that follow:

- The taxi light
- The WOW proximity sensors
- The steering proximity sensor
- The steering manifold pressure switches
- The Rotary Variable Differential Transformers (RVDTs).

The two harnesses on the drag strut supply electrical power/signal input for the primary and alternate lock proximity sensors.

Training Information Points

When you tow the aircraft or move the NLG you must obey the turn limits. Do not move the NLG more than 120° right and left of center. If you do not do this, you can cause damage to the equipment. There are labels on the two aft NLG doors that show these limits.

When you do maintenance on the nose landing gear make sure that the nose landing gear ground lock is engaged, the door lock pin is installed and that the hydraulic pressure is shut off. If you do not do this, you can cause injuries to persons and/or damage to the equipment.

Make sure that the aircraft is well clear of the hangar structure and ground equipment before you release the pressure from the landing gear shock struts. If you do not do this, damage to the aircraft and/or equipment can occur.

Use only nitrogen when you charge the landing gear shock struts. Do not use compressed air. If you use compressed air, damage to the shock struts can occur.

Use the approved hydraulic fluid when you service the landing gear. The use

of phosphate ester-base fluids is strictly prohibited. If you use unapproved fluids, severe deterioration of the shock strut seals will occur.

Nose Landing Gear Doors

Introduction

The Nose Landing Gear (NLG) doors enclose the retracted NLG in the forward fuselage wheelwell and add to the aerodynamic efficiency of the aircraft.

General Description

[Refer Figure 32-34](#) & [32-35](#)

The four NLG doors consist of two hydraulically operated forward doors, and two mechanically operated aft doors.

The NLG doors have the components that follow:

- The NLG Forward Doors
- The NLG Forward Doors Connecting Rods
- The NLG Aft Doors
- The NLG Aft Doors Connecting Rods

Detailed Description

[Refer Figure 32-34](#) & [32-35](#)

The two NLG forward doors are hydraulically operated to open before the NLG extends or retracts, and close after the NLG has extended or retracted. The two NLG aft doors are connected to the NLG shock strut and are actuated by the NLG.

NLG Aft Doors

[Refer Figure 32-35](#)

The two NLG aft doors are connected to the NLG shock strut by two adjustable door linkage assemblies. These doors are open when the NLG is extended and close when the NLG is retracted. Each NLG aft door is hinged to the fuselage at two places.

NLG Aft Doors Linkage Assemblies

[Refer Figure 32-35](#)

The two adjustable connecting rods, connect the NLG shock strut to the doors for mechanical operation and let the doors be rigged correctly, in the closed position.

NLG Forward Doors

[Refer Figure 32-34](#)

The two NLG forward doors are hinged to the fuselage at two places. The forward hinges form the attachment brackets for the operating mechanism. The operating mechanism has two adjustable connecting rods, an extension spring, a bellcrank, a toggle link assembly, a link assembly, a mechanical sequence valve, and a hydraulic actuator. This operating mechanism is attached to the forward wall of the nosewheel bay and is protected by a removable debris guard. Each door is made up of an inner skin and an outer skin. The inner skin is contoured to give clearance for the nosewheels in the retracted position.

NLG Forward Doors Connecting Rods

[Refer Figure 32-34](#)

The two adjustable connecting rods connect the bellcrank to the doors and let the doors be rigged correctly in the closed position.

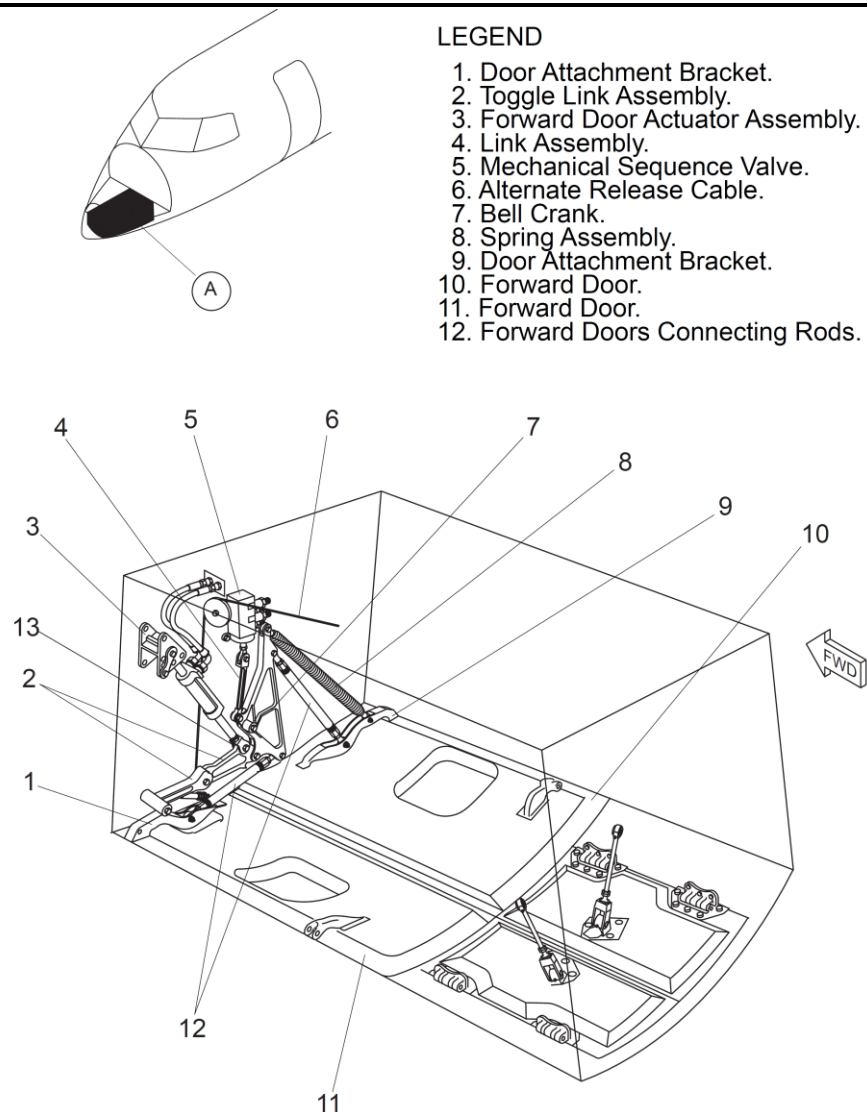


Figure 32-34 NLG Forward Doors and Connecting Rods

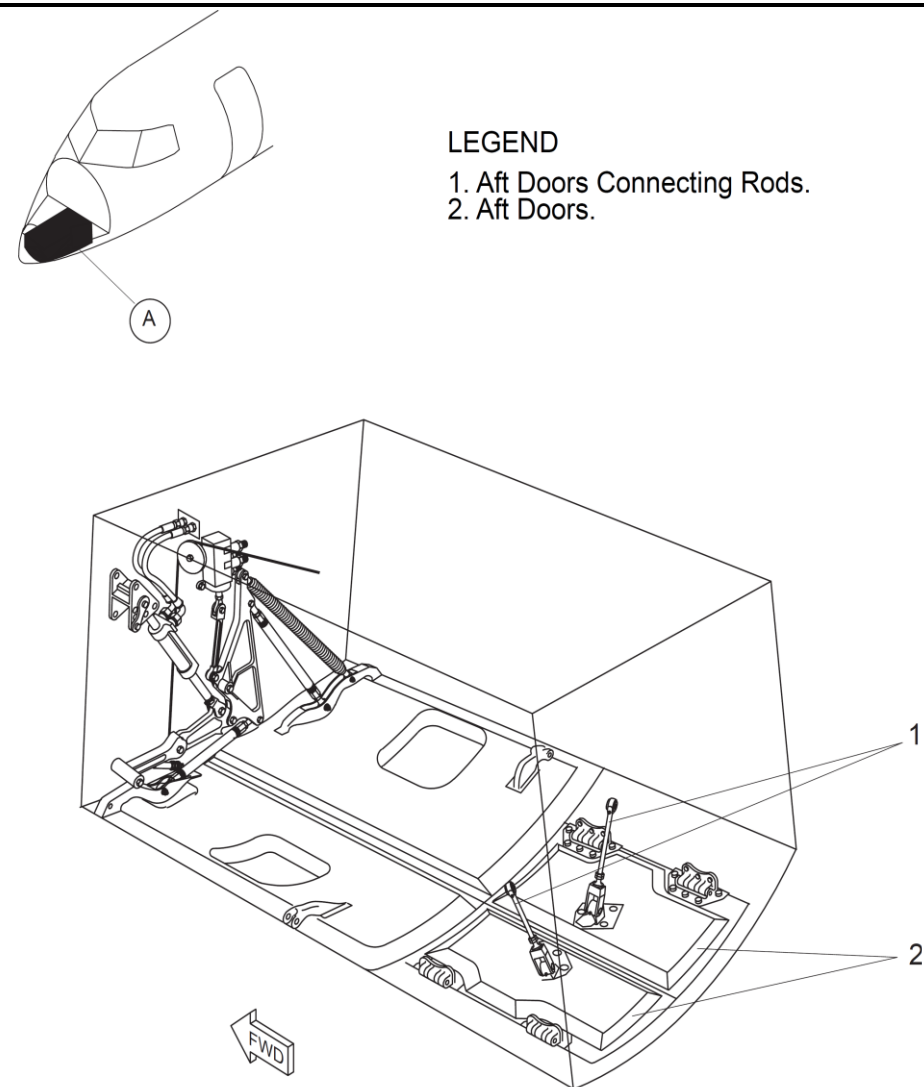


Figure 32-35 NLG Aft Doors & Connecting Rods

Nose Landing Gear Ground Lock

Introduction

A remotely operated nose gear ground lock mechanism is controlled from the outside of the aircraft.

When you engage the ground-lock system, an advisory indication shows on the pilot's side panel.

General Description

The ground lock control is on the left side nose fuselage.

The ground lock mechanically prevents NLG retraction.

Detailed Description

[Refer Figure 32-36](#)

The nose gear ground-lock control handle engages the downlock safety lever.

The ground-lock control handle has a momentary action pushbutton.

Pushing the pushbutton releases the nose gear control handle from the fuselage. This will cause the control handle to spring partly out.

The control handle must be pulled fully out and rotated 90° clockwise to point the chevrons towards the ground. This will lock the nose gear.

The nose gear ground-lock is engaged when the downlock arm of the pivot tube assembly is on the drag strut of the NLG.

When the control handle is in and flush with the fuselage, the nose gear is unlocked.

Nose Gear Ground-Lock Position Indication System (Optional)

[Refer Figure 32-37](#)

This system is optional for aircraft with Modsum 4Q428586. The switch and, cable assembly are located in the nose landing gear wheel well.

The ground-lock position indication system is connected to the ground lock control system.

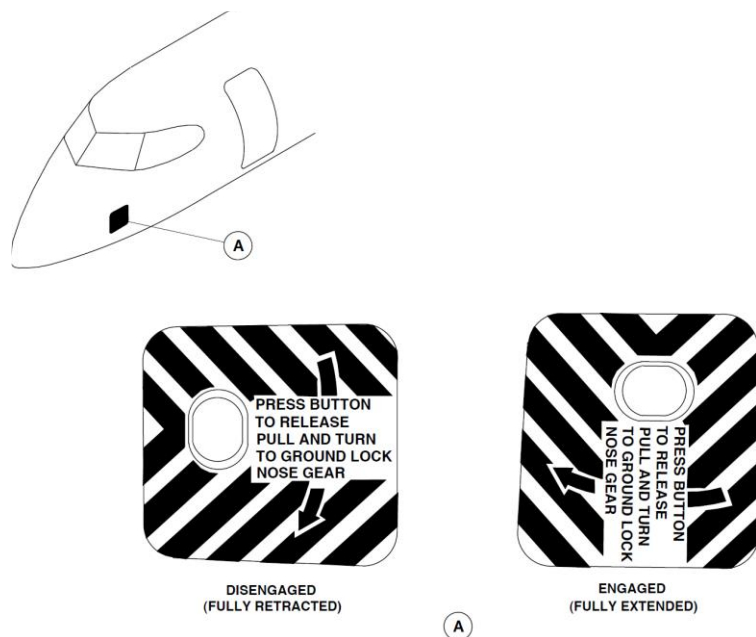


Figure 32-36 NLG Ground-Lock Control Handle

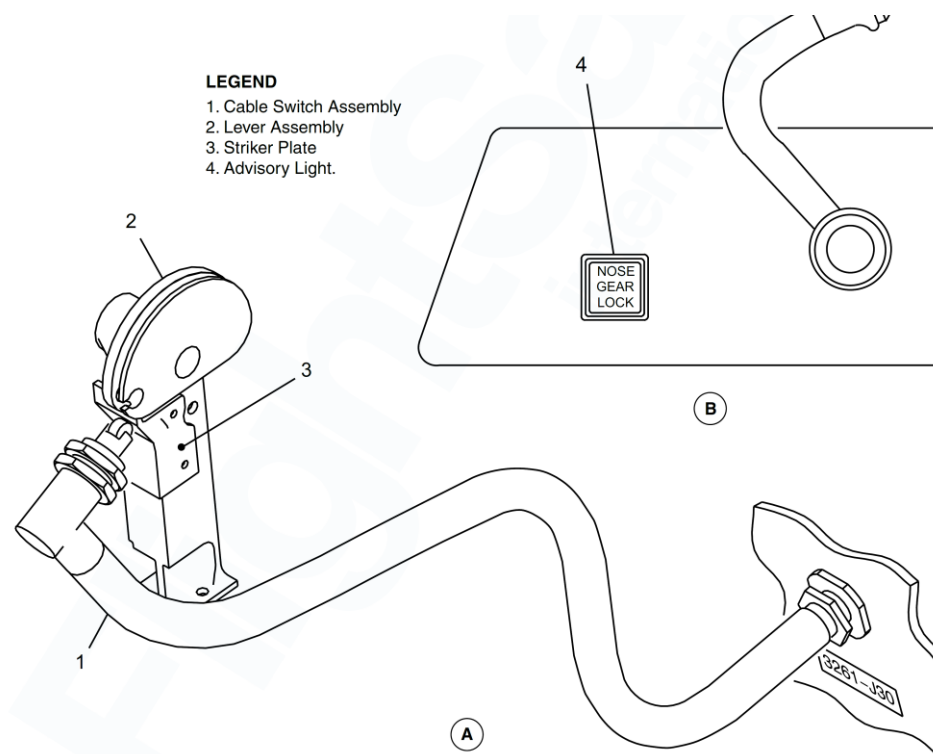


Figure 32-37 Nose Gear Lock Position Indication System

Landing Gear Extension and Retraction

Introduction

The landing gear Retraction Control System (RCS) controls the hydraulic retraction and extension of the landing gear. It also controls the alternative manual extension of the main landing gear (MLG) and the nose landing gear (NLG). The RCS shows the landing gear and landing gear door status to the flight crew in the flight compartment.

The Primary Extension and Retraction system controls the extension and retraction of the landing gear and the landing gear doors operation.

General Description

[Refer Figure 32-38 & 32-39](#)

The LANDING GEAR selector lever and the Proximity Sensor Electronic Unit (PSEU) control the operation of the landing gear. The No. 2 hydraulic system supplies the power to move the landing gear.

Hydraulic retraction and extension starts when the landing gear selector lever is unlocked and moved to the desired command position. The Proximity Sensor Electronic Unit (PSEU) checks the status of the landing gear and the landing gear doors, and compares it to the command position selected. The PSEU then controls the hydraulic sequences to either fully extend or fully retract the landing gear. The indicator lights installed on the landing gear control panel assembly show the landing gear and landing gear door status. A LDG GEAR INOP caution light is shown on the flight compartment Caution and Warning panel in the event of a system malfunction.

Each MLG wheelwell has four doors. The two aft doors are hydraulically operated and the two forward doors are mechanically operated.

The NLG wheelwell has four doors. The two forward doors are hydraulically operated and the two aft doors are mechanically operated.

The alternate extension system is an alternate means to manually extend the landing gear when the No. 2 hydraulic system or the PSEU itself fails.

Detailed Description

Normal landing gear operation is controlled by the PSEU as a result of input signals from the landing gear selector lever. The No. 2 hydraulic system supplies the power to move the landing gear.

The PSEU does sequencing logic for the RCS and gives indications for the landing gear and the landing gear door position. The PSEU interfaces with the landing gear selector lever to turn on the indicator lights on the landing gear control panel.

The Landing Gear Control Panel indicator lights are as follows:

- LEFT, NOSE, RIGHT green lights indicate the gear is down and locked.
- LEFT, NOSE, RIGHT red lights indicate the gear is not locked up, or down
- L DOOR, N DOOR, R DOOR amber lights indicate landing gear doors are open.

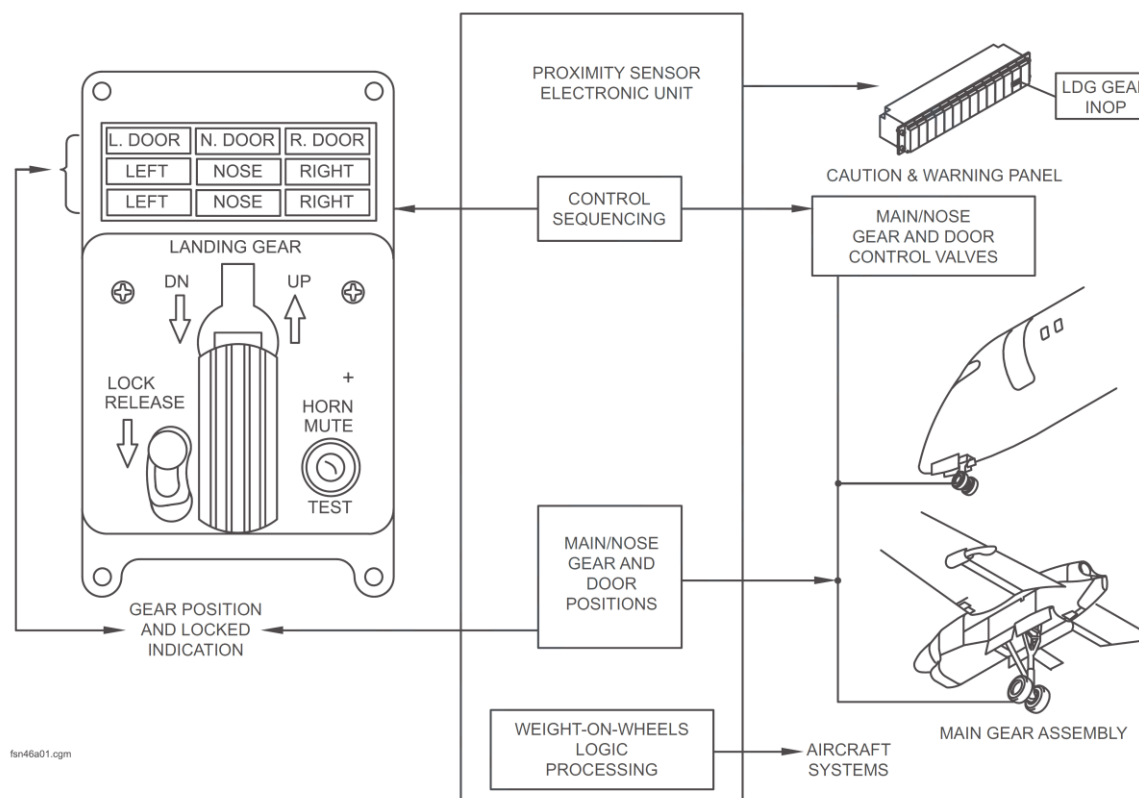


Figure 32-38 Landing Gear Block Diagram

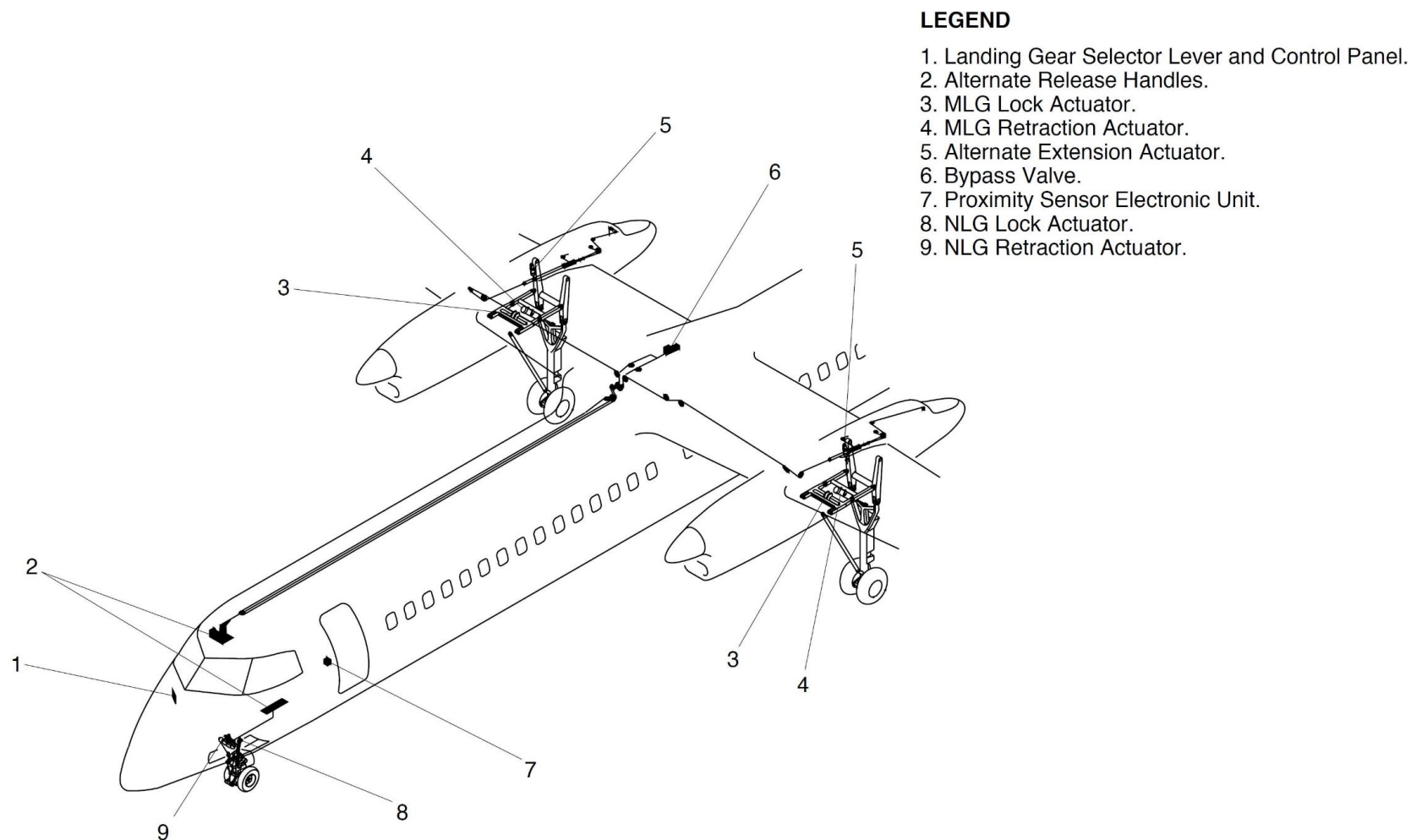


Figure 32-39 Landing Gear Extension and Retraction – Component Details

Refer Figure 32-40

If the primary downlock indication fails, an alternate downlock indication system is available. There are individual green indicator lights for each landing gear, to indicate a down and locked condition. These indicator lights are installed in the flight compartment floor, under the ALTERNATE EXTENSION door.

The MLG is designed to retract rearward into the wheelwell and extends toward the nose of the aircraft. The NLG retracts forward into the nose wheelwell and extends toward the aft of the aircraft. The four doors of each landing gear hydraulically and mechanically open in sequence with the landing gear movement. The two aft MLG doors and the two forward NLG doors are operated hydraulically by the PSEU.

Mechanical sequence valves check the time of landing gear extension and retraction in relation to the hydraulic landing gear doors movement. The two forward MLG and the two aft NLG doors are mechanically linked to the landing gear. The doors are opened and closed by the movement of the landing gear. The mechanically actuated landing gear doors stay open while the landing gear are in the extended position. Hydraulically actuated doors are closed at the end of the extension and retraction cycle.

There are three conditions for the PSEU to activate the selector valve. These conditions are as follows:

- The landing gear selector lever is moved to the UP position
- The PSEU receives a signal that the aircraft is airborne
- The NLG is centered.

The PSEU electrically signals the selector valve to supply hydraulic power to the retract circuit of the landing gear hydraulic system. When the landing gear selector lever is moved to the DOWN position, the PSEU signals the selector valve to supply hydraulic power to the extend circuit of the landing gear hydraulic system.

The alternate extension system is a self resetting, cable actuated design. The system is accessible in the flight compartment. The alternate extension system has a bypass valve and a manual handpump hydraulic system. The bypass valve isolates the landing gear hydraulics from the No.2 hydraulic system. The manual handpump hydraulic system helps move the MLG into the down and locked position.

For an alternate extension of the landing gear to occur, the NLG forward doors, the MLG aft doors, and the landing gear uplocks open mechanically. The NLG free falls to the down and locked position with the help of the airflow over the fuselage. The MLGs free fall and are moved to the down and locked position by the alternate extension actuators. A LANDING GEAR SELECTOR INHIBIT switch installed in the flight compartment ceiling is used to isolate the normal landing gear selector valve during flight. The LANDING GEAR SELECTOR INHIBIT switch must be set from NORMAL to INHIBIT before the alternate extension system can be used.

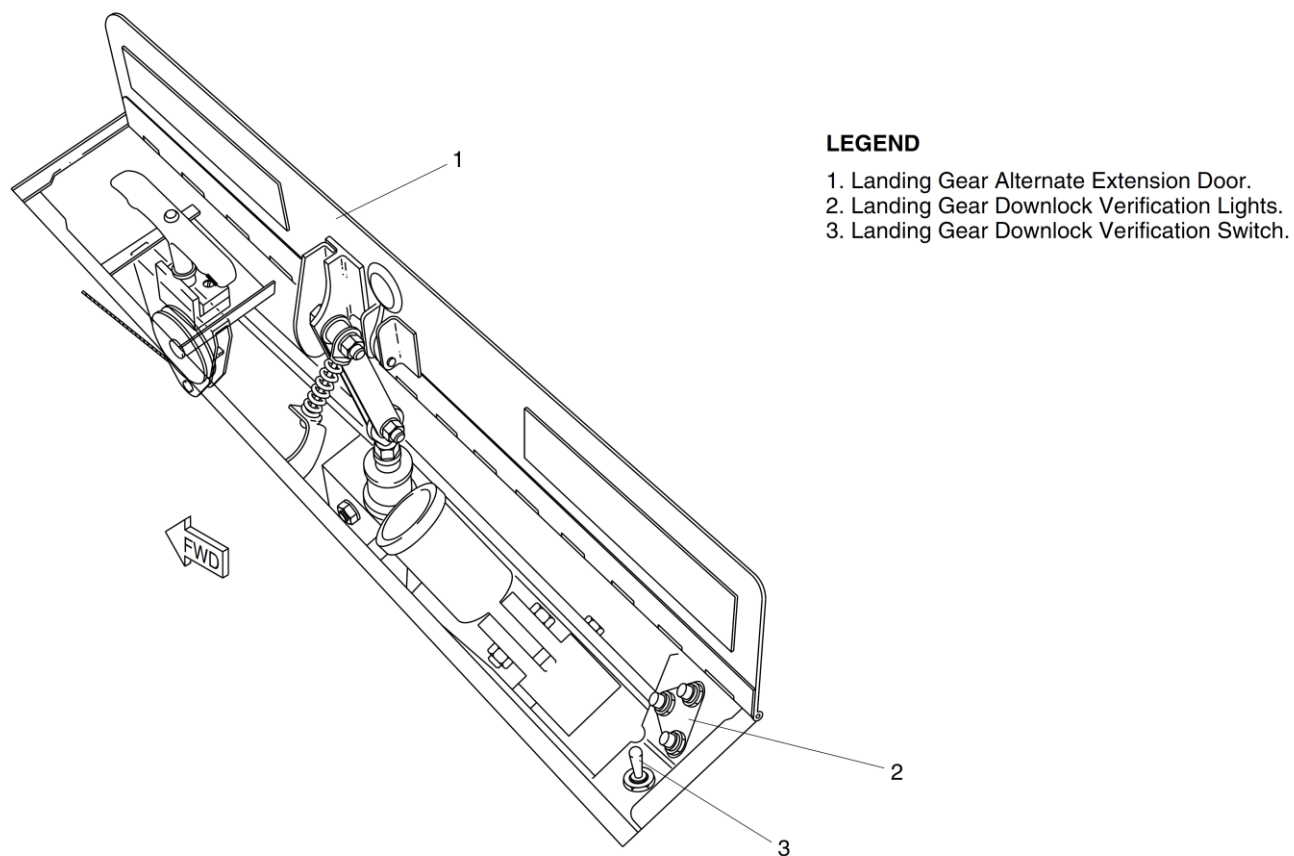


Figure 32-40 Landing Gear Alternate Extension Controls

Primary Extension and Retraction

- Valve Solenoid Sequence NLG Doors
- A Power Transfer Unit

Introduction

The Primary Extension and Retraction system controls the extension and retraction of the landing gear and the landing gear doors operation.

General Description

The LANDING GEAR selector lever and the Proximity Sensor Electronic Unit (PSEU) control the operation of the landing gear. The No. 2 hydraulic system supplies the power to move the landing gear.

The primary extension and retraction system has the components that follow:

- Panel Control, Landing Gear
- Valve Selector, Landing Gear
- Actuator Retraction MLG
- Actuators Downlock MLG
- Actuators Uplock MLG
- Actuator Doors MLG
- Mechanism Doors MLG
- Valves Mechanical Sequence MLG Doors
- Restrictors Inline MLG Doors
- Valve Solenoid Sequence MLG Doors
- Actuator Retraction NLG
- Actuator NLG Lock
- Actuator NLG Doors
- Mechanism NLG Doors
- Valve Mechanical Sequence NLG Doors
- Restrictor Inline NLG

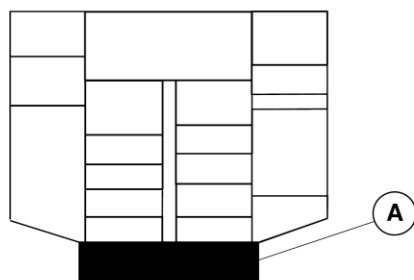
Detailed Description

[Refer Figure 32-41](#)

Hydraulic retraction and extension starts when the LANDING GEAR selector lever is moved to the desired position. The PSEU checks the status of the landing gear and the landing gear doors, and compares it to the command selected. The PSEU controls the hydraulic sequences to either fully extend or fully retract the landing gear. The status of the landing gear and the landing gear doors is shown by the indicator lights on the LANDING GEAR control panel. A LDG GEAR INOP caution light on the Caution and Warning panel shows a fault in the landing gear retraction and extension system..

The following is an in depth description of the Extension and Retraction systems in the order that follows:

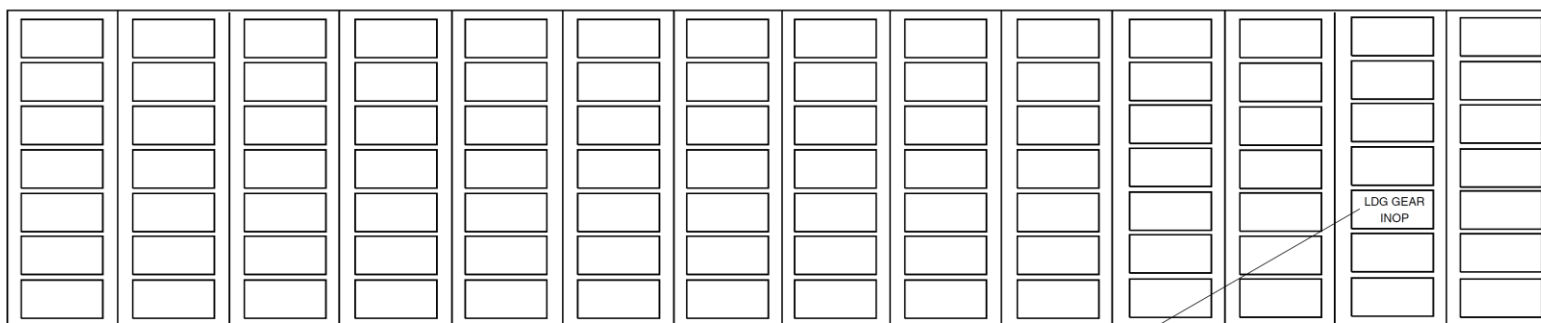
- Retraction sequence
- Retraction sequence configuration
- Main landing gear retraction
- Nose landing gear retraction
- Retraction sequence completion
- Extension sequence configuration
- Main landing gear extension
- Nose landing gear extension



OVERHEAD CONSOLE

LEGEND

1. Landing Gear Inoperative (Amber).



A

1

Figure 32-41 LDG GEAR INOP Caution Light

[Refer Figure 32-42](#)

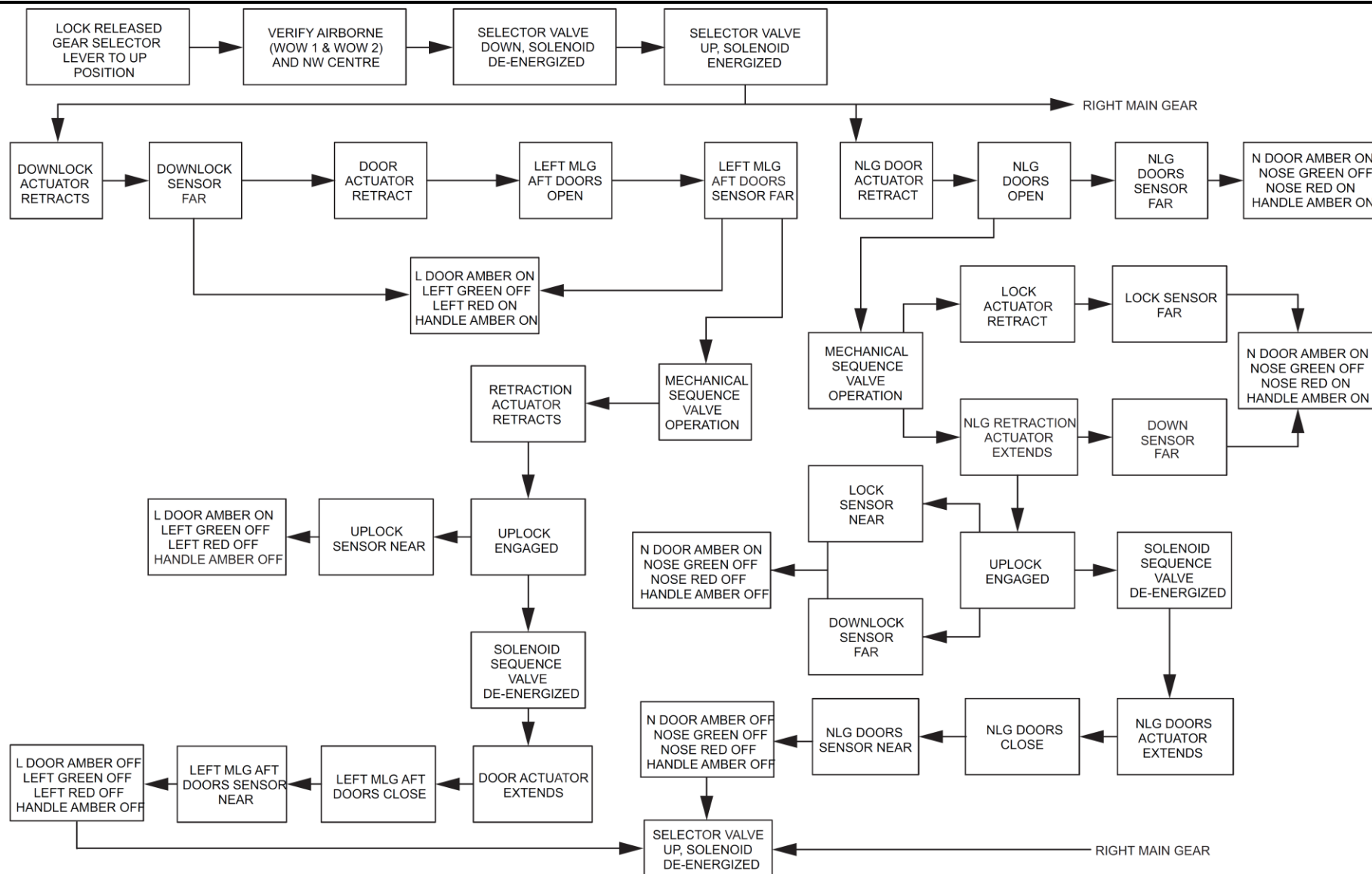
Retraction Sequence

When the landing gear is down and locked, the solenoid sequence valves and the down solenoid of the selector valve are kept in an energized condition.

This keeps hydraulic pressure on the down side of the retraction actuators. The lock actuators are also used to downlock the landing gear and to keep the hydraulically actuated wheelwell doors closed. Retraction of the landing gear starts when the LANDING GEAR selector lever is unlocked and moved to the UP position.

Retraction Sequence Configuration

When the LANDING GEAR selector lever is moved to the UP position, the down solenoid of the landing gear selector valve is de-energized. The PSEU makes sure that the aircraft is airborne and that the Nose Landing Gear (NLG) is centered. The up solenoid of the landing gear selector valve, now receives electrical power. When the landing gear selector valve receives electrical power, the up solenoid configures the selector valve. This lets hydraulic system pressure flow into the retract side of the landing gear hydraulic system.



[Refer Figure 32-43](#)

Main Landing Gear Retraction

Hydraulic pressure is supplied to each Main Landing Gear (MLG) downlock release actuator to release the downlock. Hydraulic pressure is supplied through an energized solenoid sequence valve to the open side of the MLG aft door actuators.

NOTE: The solenoid sequence valves stay energized from the time of closing the doors during the landing gear extension sequence. They will not change the state.

This causes the MLG aft doors to open. The operation of the downlock release actuator and the MLG aft door actuators is monitored by the proximity sensors.

The MLG door mechanism operates a mechanical sequence valve to interlock the retraction/extension part of the hydraulic system. This continues until the doors are open enough for the landing gear to retract so that it does not touch the doors. At approximately 93 percent travel of the MLG aft doors, the MLG aft doors linkage operates the mechanical sequence valve. When the mechanical sequence valve is in operation, hydraulic pressure is supplied to the up side of the MLG retraction actuator.

The MLG starts to travel to the fully retracted position where it is locked in place by the mechanical uplock. The proximity sensors monitor the gear and door positions. When the PSEU receives the input signals that the MLG is up and locked, the PSEU de-energizes the solenoid sequence valves. This causes the solenoid sequence valves to supply pressure to the close side of the MLG door actuators and close the doors.

At approximately 7 percent reverse travel of the MLG doors, the mechanical sequence valves stop their operation. This action removes pressure from the up side of the MLG retraction actuators. Inline restrictors bypass the mechanical sequence valves and keep the MLG retraction actuators pressurized to 3000 psi (20684 kPa). Pressure is kept at 3000 psi (20684 kPa)

until the landing gear hydraulic system is depressurized upon completion of the retraction cycle.

Nose Landing Gear Retraction

Hydraulic pressure is supplied to the NLG uplock/downlock release actuator to release the downlock. Hydraulic pressure is supplied through an energized solenoid sequence valve to the open side of the NLG forward doors actuator.

NOTE: The solenoid sequence valve stays energized from the time of closing the doors during the landing gear extension sequence. It will not change the state.

This causes the NLG forward doors to start to open. The operation of the NLG uplock/downlock release actuator and the NLG forward doors actuator is checked by the proximity sensors.

At approximately 92 percent travel of the NLG forward doors, the door linkage operates the mechanical sequence valve. When the mechanical sequence valve is in operation, it stops to interlock the retraction/extension part of the NLG hydraulic system. Hydraulic pressure is supplied to the up side of the NLG retraction actuator. The NLG starts to travel to the fully retracted position where it is locked in place by the mechanical uplock.

The proximity sensors make sure that these procedures have been completed. When the PSEU receives the input signals that the NLG is up and locked, the PSEU de-energizes the solenoid sequence valves. This causes the solenoid sequence valves to pressurize the NLG forward doors actuator to close the doors.

At approximately 8 percent reverse travel of the NLG forward doors, the mechanical sequence valve stops its operation. This action removes pressure from the up side of the NLG retraction actuator. Inline restrictors bypass the mechanical sequence valve and keep the NLG retraction actuator pressurized to 3000 psi (20684 kPa). Hydraulic pressure is kept at 3000 psi (20684 kPa) until the landing gear hydraulic system is depressurized upon completion of the retraction cycle. A proximity sensor senses that the NLG forward doors are closed and sends an input signal to the PSEU.

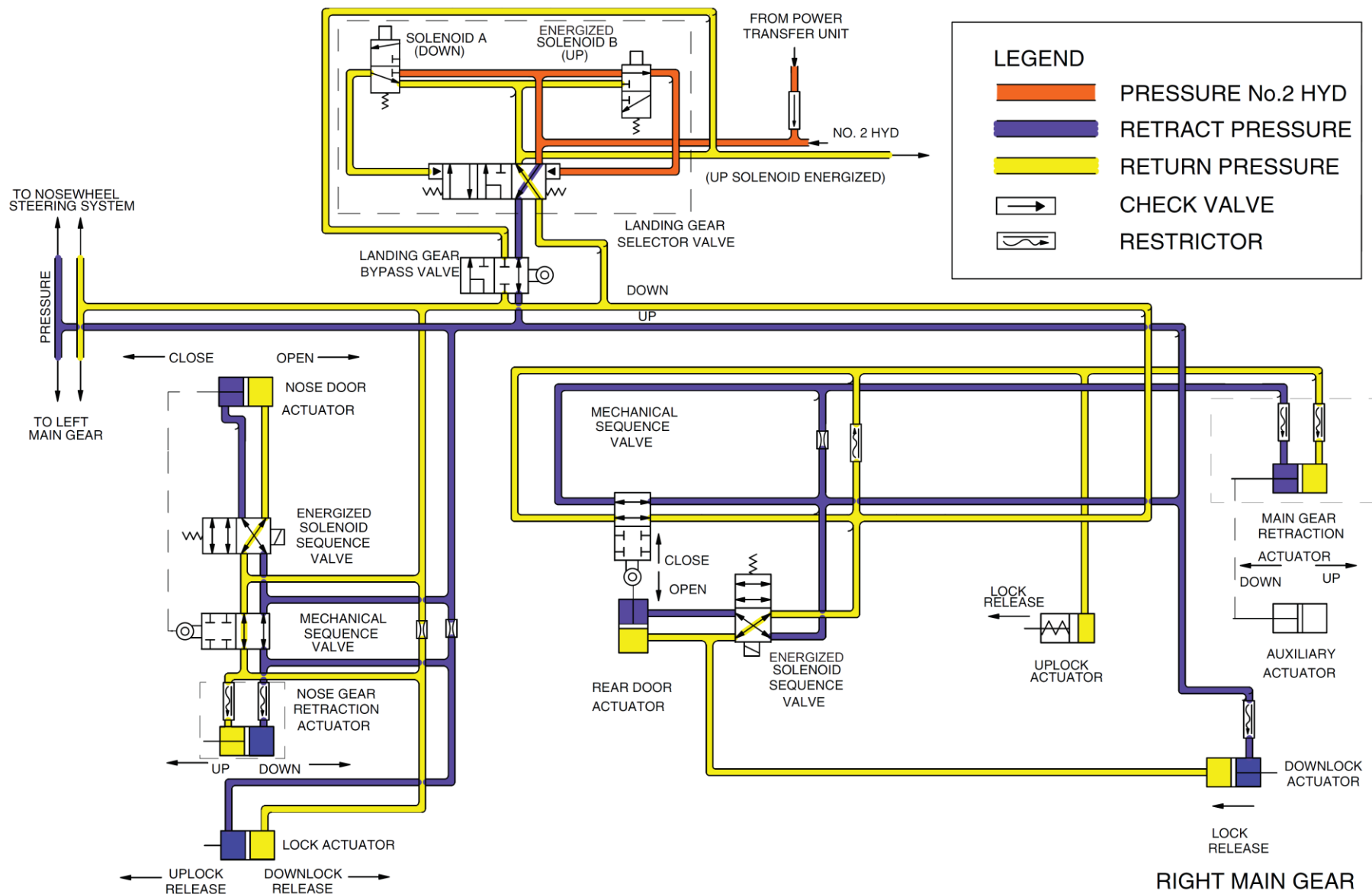


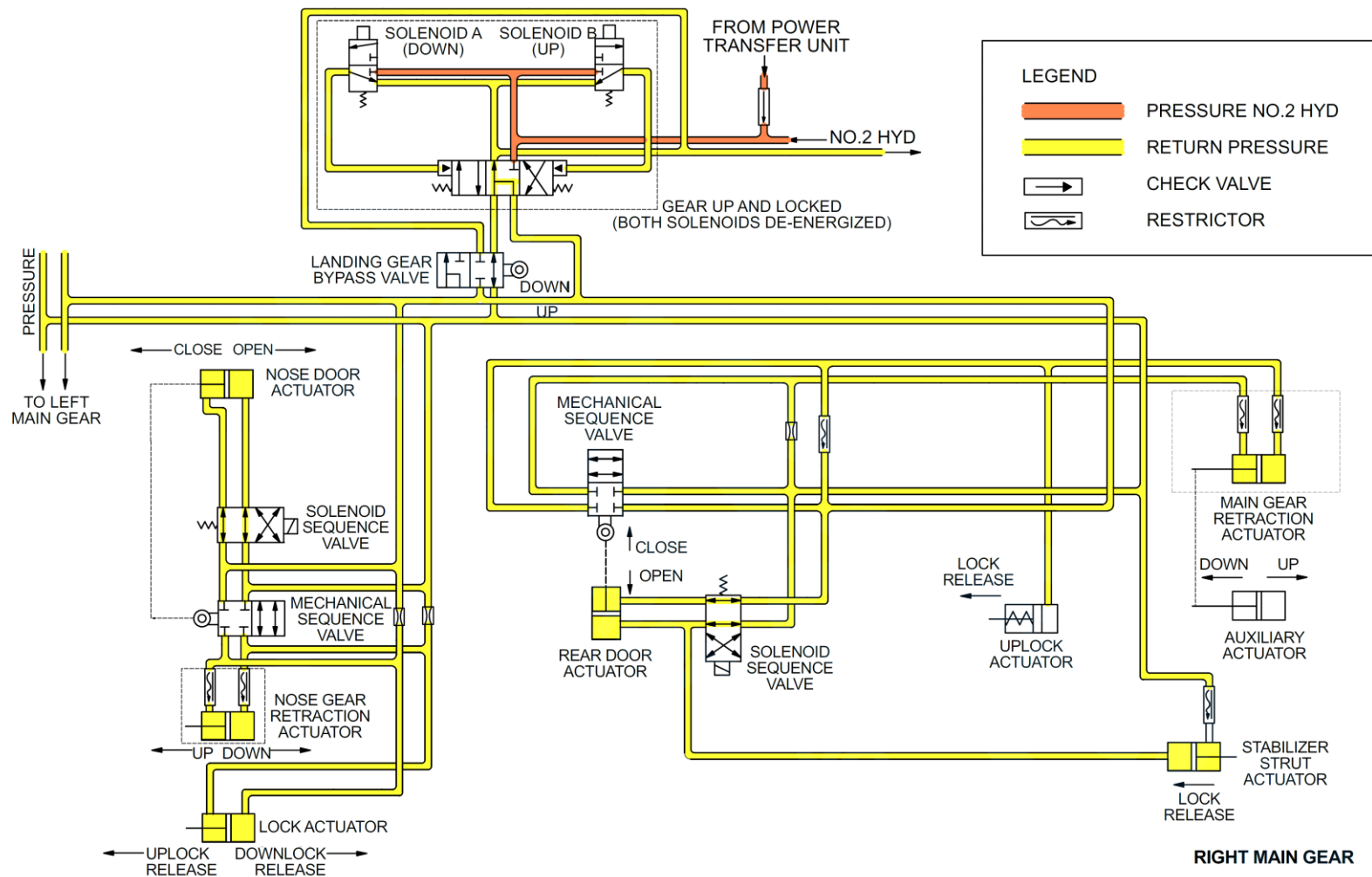
Figure 32-43 Initial Up Selection Doors Open

Retraction Sequence Completion

Refer Figure 32-44

The MLG and NLG proximity sensors sense that the gear is up and locked and

that the MLG and NLG doors are closed. The PSEU gives a three second interval before it de-energizes the up and the down solenoid in the selector valve. The PSEU then depressurizes the landing gear hydraulic circuit.



*ALL SOLENOIDS DE-ENERGIZED

Figure 32-44 Gear Up and Doors Closed

Landing Gear Retraction -Electrical Schematic

[Refer Figure 32-45 \(Sheet 1 of 3\)](#)

Before landing gear retraction can start, the PSEU must receive the signals as follows:

- LG control handle is in the RETRACT position
- WOW sensors show that the aircraft is in the air as reported
- LG control is NOT inhibited
- The nosewheel is centered

When the above conditions are achieved:

The PSEU de-energizes the EXTEND solenoid and at the same time outputs 28 VDC to Pin L of the LG control panel. Because the LG control handle is in the retract position. The LG selector switch is positioned to output the 28 VDC through Pin R to the RETRACT solenoid of the LG selector valve.

- Hydraulic pressure through the retract solenoid acts on one end of the hydraulic spool valve. The other end of the hydraulic spool is open to return through the de-energized EXTEND solenoid.
- The hydraulic spool is moved to the position that allows the hydraulic pressure to be applied to the retract manifold through the bypass valve.
- At the same time the PSEU energizes the solenoid sequence valves by outputting 28 VDC to each valve, the hydraulic pressure then passes through the valves to the door actuators to open the landing gear doors.

[Refer Figure 32-46 \(Sheet 2 of 3\)](#)

The PSEU outputs 28 VDC through Pins 12E, 11D and 11E to the LG Control and Indication Panel to bring on the door OPEN amber lights.

- Hydraulic pressure is applied through a restrictor to the MLG downlock actuator, the actuator retracts and unlocks the overcenter lock in the stabilizer brace assembly.
- Pressure is also applied to the NLG lock release actuator to unlock the overcenter lock condition of the upper and lower drag strut assemblies.

The PSEU outputs 28 VDC through the advisory control unit for the following:

- MLG unsafe red lights through Pin 4C and Pin 3B to Pins H and D
- NLG unsafe red light through Pin 3A to Pin F
- Light in the control handle through Pin 2A to Pin A on the LG Control and Indication Panel.

[Refer Figure 32-47 \(Sheet 3 of 3\)](#)

When the PSEU gets all the signals listed below:

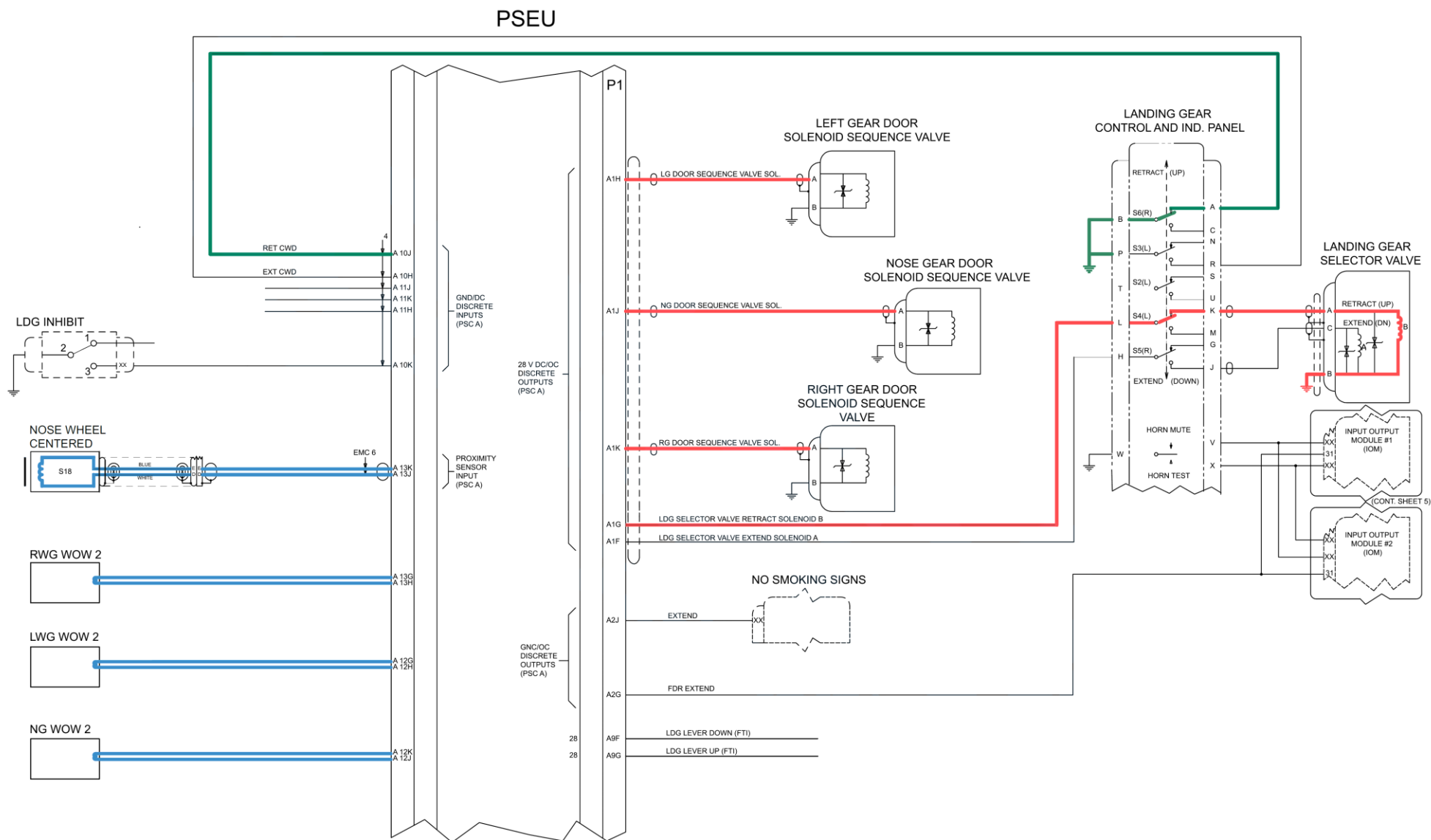
- RMG uplock sensor at Pin D14A and D14B
- LMG uplock sensor at Pin D14C and D14D
- NG lock 1 at Pins D13C and D13D
- NG lock 2 at Pins B8G and B8H
- The solenoid sequence valves are de-energized.

Hydraulic pressure will now be supplied to the close side of the door actuators closing the doors.

When the PSEU gets all the signals that follow:

- RMG Door Closed at Pins B9A and B9B
- LMG Door Closed at Pins B8A and B8B
- NG Door Closed at Pins E9C and E9D.

The RETRACT solenoid in the LG selector valve is de-energized and the door OPEN amber lights extinguish.



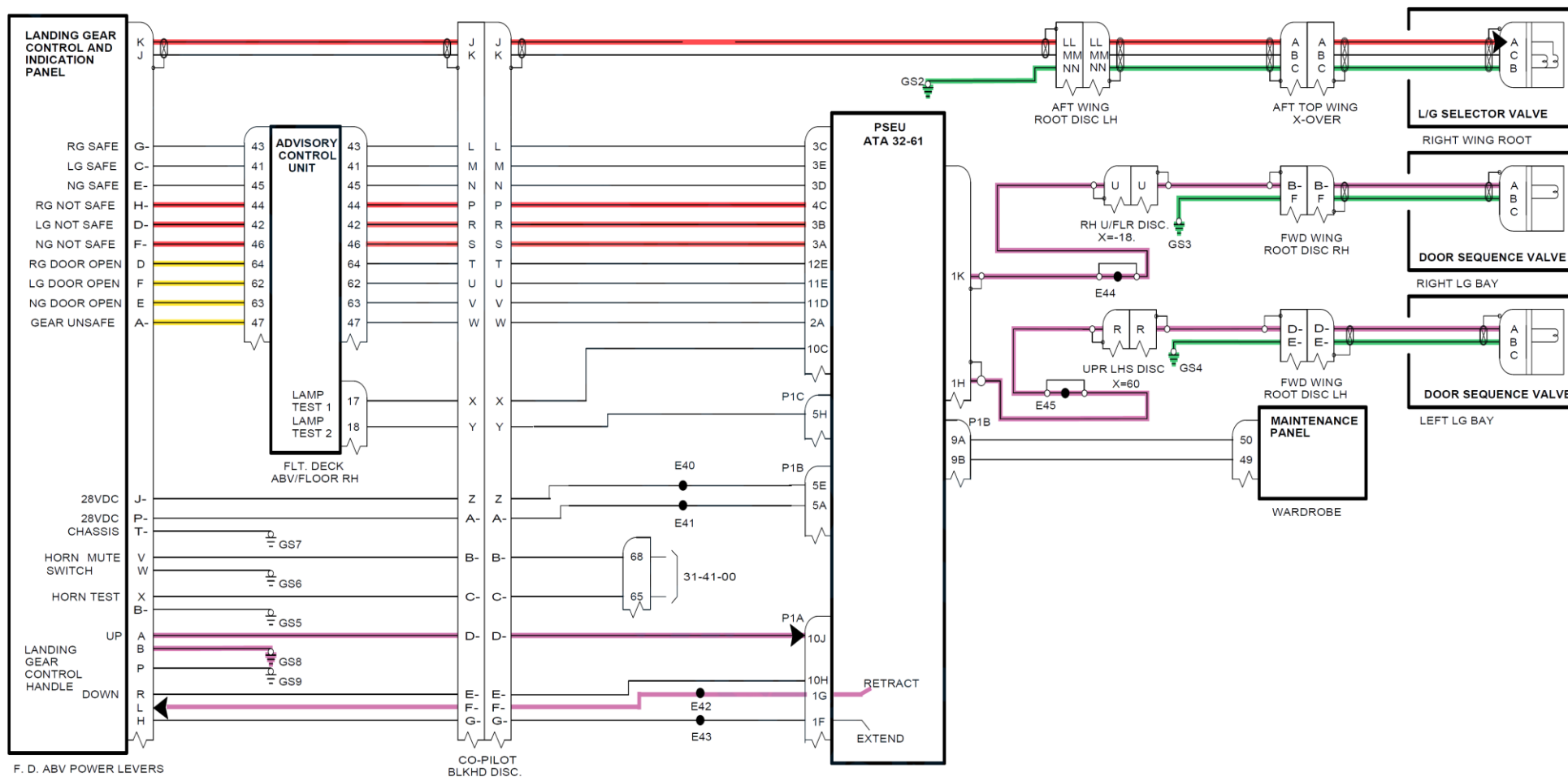


Figure 32-46 Landing Gear Retraction (Sheet 2 of 3)

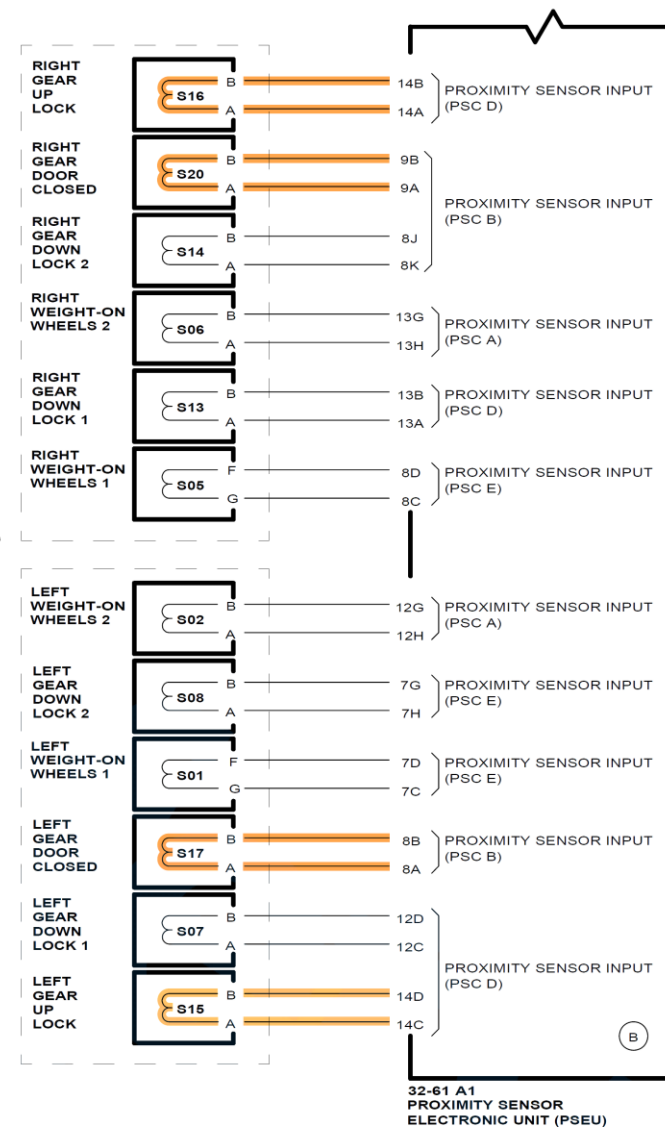
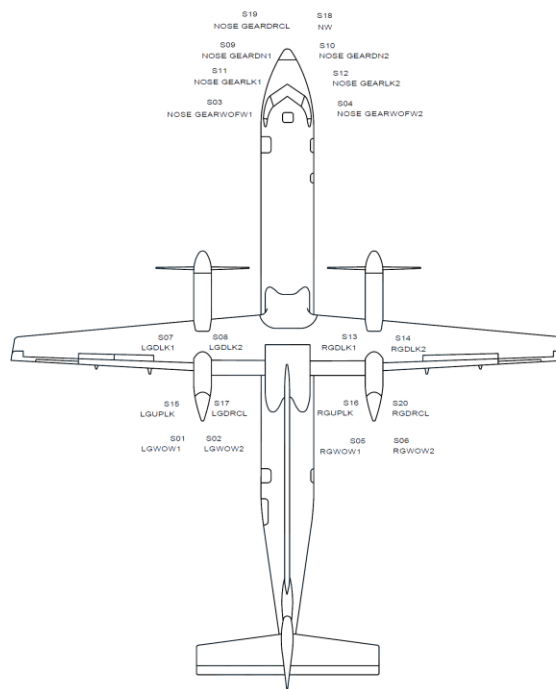
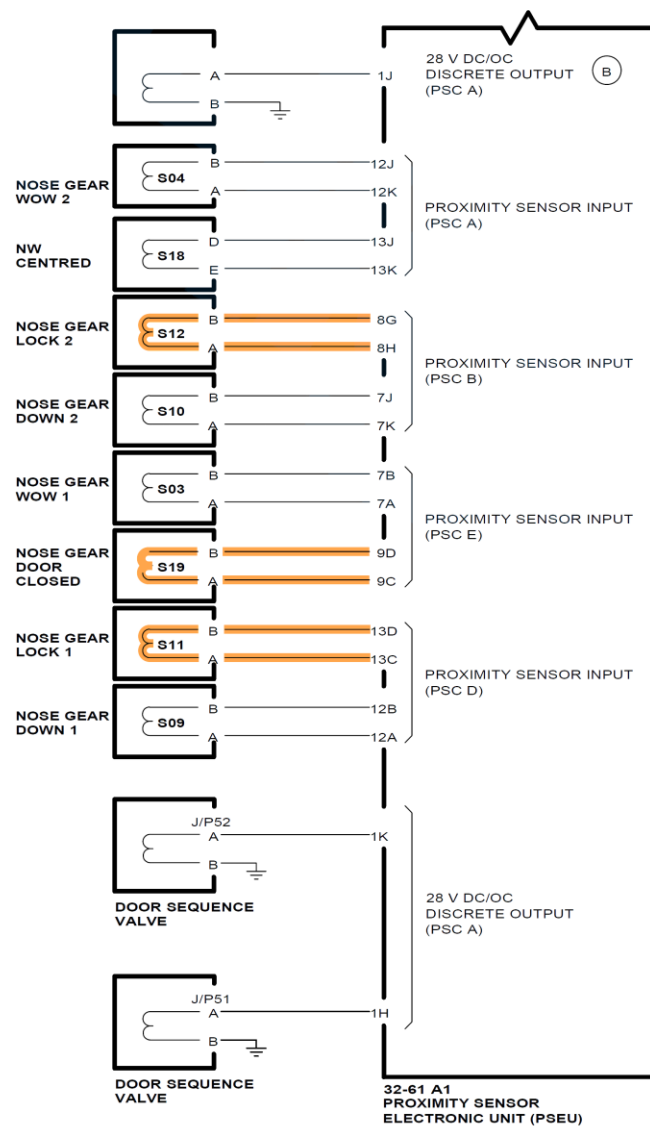


Figure 32-47 Landing Gear Retraction (Sheet 3 of 3)

Extension Sequence Configuration

The landing gear starts to extend when the LANDING GEAR selector lever is unlocked and moved to the DN position. The down solenoid of the selector valve receives electrical power. The selector valve now supplies aircraft hydraulic system pressure and flow into the extend side of the landing gear hydraulic system.

[Refer Figure 32-48 & 32-49](#)

Main Landing Gear Extension

When the LANDING GEAR selector lever is moved to the DN position, the solenoid sequence valves stay de-energized. The de-energized solenoid sequence valves supply hydraulic pressure to the open side of the MLG aft doors actuators to open the MLG aft doors.

NOTE: The solenoid sequence valves stay de-energized from the time of closing the doors during the landing gear retraction sequence. They will not change the state.

At approximately 93 percent travel of the MLG aft doors, the MLG aft doors linkage operates the mechanical sequence valve. The valve supplies hydraulic pressure to the uplock release actuators and to the down side of the MLG retraction actuators. The MLG starts to travel to the down and locked position.

There are three proximity sensors used to monitor the MLG extension sequence. Each MLG has two down and locked sensors and one MLG aft doors, closed sensor. When the PSEU receives input signals that the MLG is down and locked, the PSEU energizes the solenoid sequence valves.

Pressure is supplied to the MLG aft doors actuators to close the MLG aft doors. At approximately 7 percent reverse travel of the MLG doors, the mechanical sequence valves stop their operation. This action removes pressure from the up side of the MLG retraction actuators. Inline restrictors keep the MLG retraction actuators pressurized to 3000 psi (20684 kPa) at the end of the extension sequence.

Nose Landing Gear Extension

When the LANDING GEAR selector lever is moved to the DN position, hydraulic pressure is supplied to the open side of the NLG doors actuator through the de-energized solenoid valve. This causes the NLG forward doors to open.

NOTE: The solenoid sequence valve stays de-energized from the time of closing the doors during the landing gear retraction sequence. It will not change the state.

At approximately 92 percent travel of the NLG forward doors, a mechanical sequence valve is actuated by the door linkage. When the mechanical sequence valve operates, hydraulic pressure is sent to the lock actuator and the down side of the NLG retraction actuator. The NLG then starts to travel to the down and locked position.

There are five proximity sensors that monitor the NLG extension sequence. The NLG has two down sensors on the NLG trunnion arms, two down and locked sensors on the NLG drag strut, and one NLG doors closed sensor. When the PSEU receives the input signals that the NLG is down and locked, the solenoid sequence valve is energized by the PSEU. Hydraulic pressure is supplied to the NLG forward doors actuator to close the doors.

At approximately 8 percent reverse travel of the NLG forward doors, the mechanical sequence valve stops its operation. This action removes pressure from the uplock/downlock actuator and the NLG retraction actuator. An inline restrictor keeps the NLG retraction actuator and the lock actuator pressurized to 3000 psi (20684 kPa) at the end of the extension cycle. A mechanical downlock, locks the NLG in the fully extended position. The PSEU keeps power on the down solenoid of the selector valve as long as the LANDING GEAR selector lever is in the DN position.

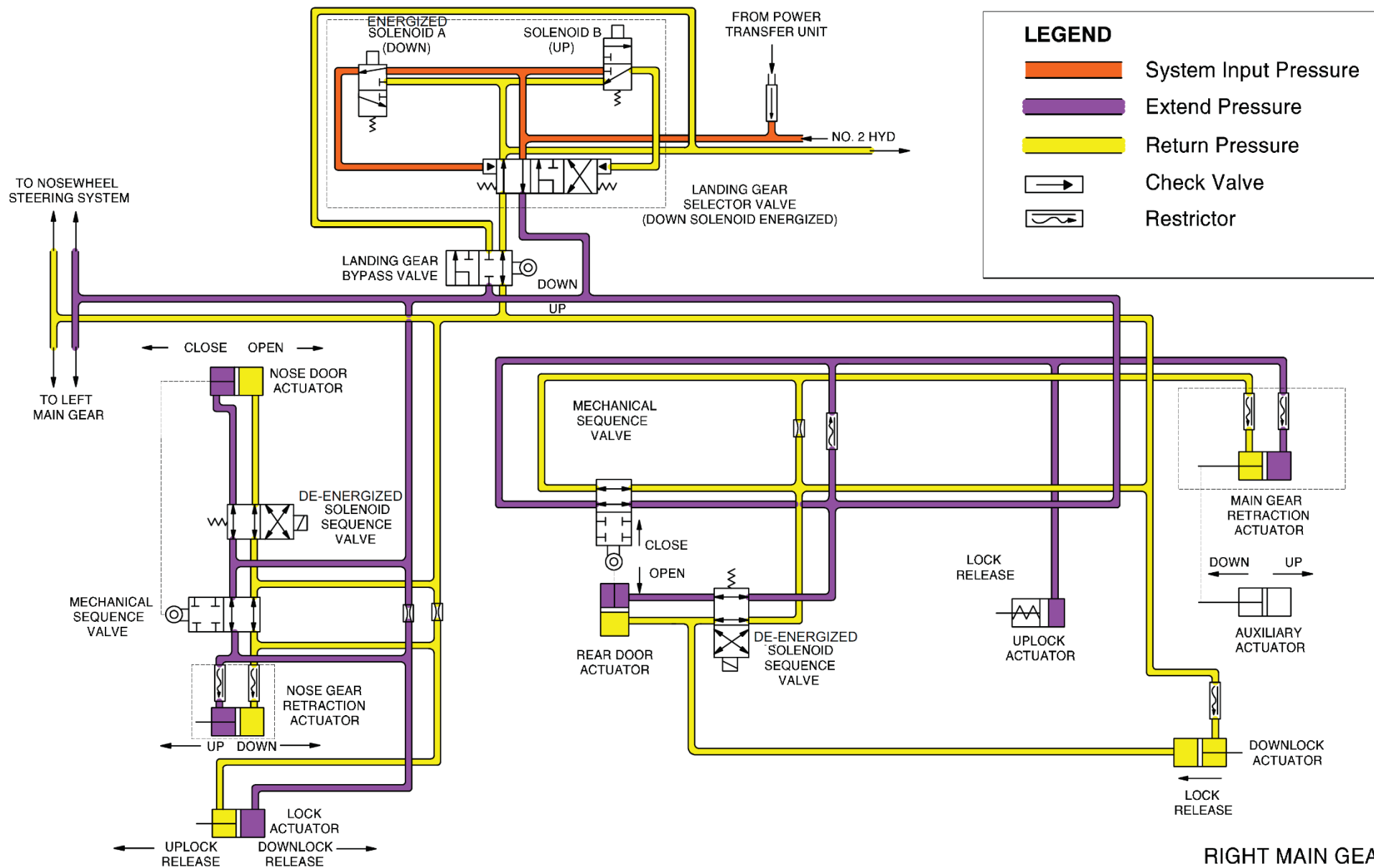


Figure 32-48 Initial DOWN selection Doors opening

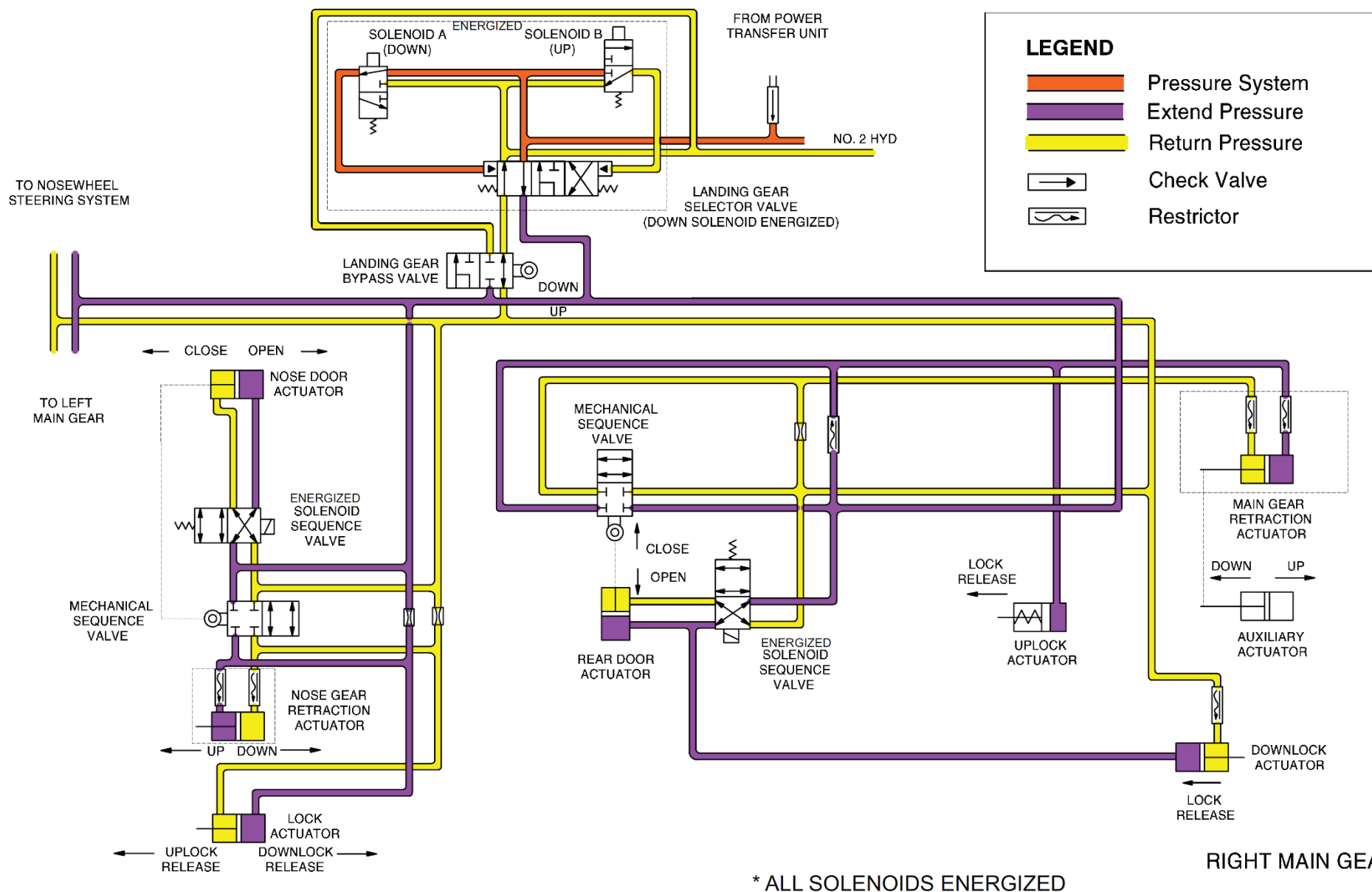


Figure 32-49 Extension Completed & Doors closed

Landing Gear Normal Extension – Electrical Schematic

[Refer Figure 32-50 \(Sheet 1 of 2\)](#)

Before the landing gear normal extension can start the PSEU must receive the signals that follow:

- EXTCMD discrete input on Pin A10H
- No LDGINH signal at Pin A10K.

The PSEU outputs 28 VDC to Pin H of the LG Control and Indication panel.

Because the LG control lever is selected DOWN 28 VDC is output on Pin J to energize the EXTEND solenoid in the LG selector valve.

Hydraulic pressure is directed through the EXTEND solenoid to one end of the hydraulic spool, the other end is open to return through the de-energised RETRACT solenoid. The hydraulic spool is positioned to supply hydraulic pressure directly to the EXTEND manifold.

The pressure is supplied through the de-energized solenoid sequence valves to the retract side of the door actuators to OPEN the hydraulically operated doors.

Pressure is also supplied to the following:

- The extend side of the MLG uplock actuators to release the uplock.
- The extend side of the NLG lock release actuator to unlock the overcentre lock in the upper and lower drag strut assembly.
- Through the inline restrictors to the retraction/extension actuators.

The PSEU outputs 28 VDC through the Advisory Control Unit for the following:

- The MLG unsafe red lights through Pin 4C and Pin 3B to Pins H and D
- NLG unsafe red light through Pin 3A to Pin F
- The light in the control handle through Pin 2A to Pin A on the LG Control and Indication Panel.

[Refer Figure 32-51 \(Sheet 2 of 2\)](#)

When the doors are approximately 93% open, full hydraulic pressure and flow is supplied to the actuators to extend the gear.

When the main gear is down and locked, the PSEU receives signals from the MLG sensors that follow:

- RMG Downlock 1 on Pins E8C and E8D
- RMG Downlock 2 on Pins B8J and B8K
- LMG Downlock 1 on Pins D12C and D12D
- LMG Downlock 2 on Pins E7G and E7H.

When the nose gear is down and locked, the PSEU receives signals from the NLG that follow:

- NG Down 1 on Pins D12A and D12B
- NG Lock 1 on Pins D13C and D13D
- NG Down 2 on Pins B7J and B7K
- NG Lock 2 on Pins B8G and B8H.

When the PSEU has received the down and locked signals from the LG sensors, it energizes the solenoid sequence valves.

Hydraulic pressure is supplied to the extend side of the door actuators closing the doors.

The PSEU receives the following signals from the door sensors:

- RG Door Closed on Pins B9A and B9B
- LG Door Closed on Pins B8A and B8B
- NG Door Closed on Pins E9C and E9D.

The PSEU turns OFF the amber door lights on the LG control and indication panel.

The EXTEND solenoid and the solenoid selector valves remain energized.

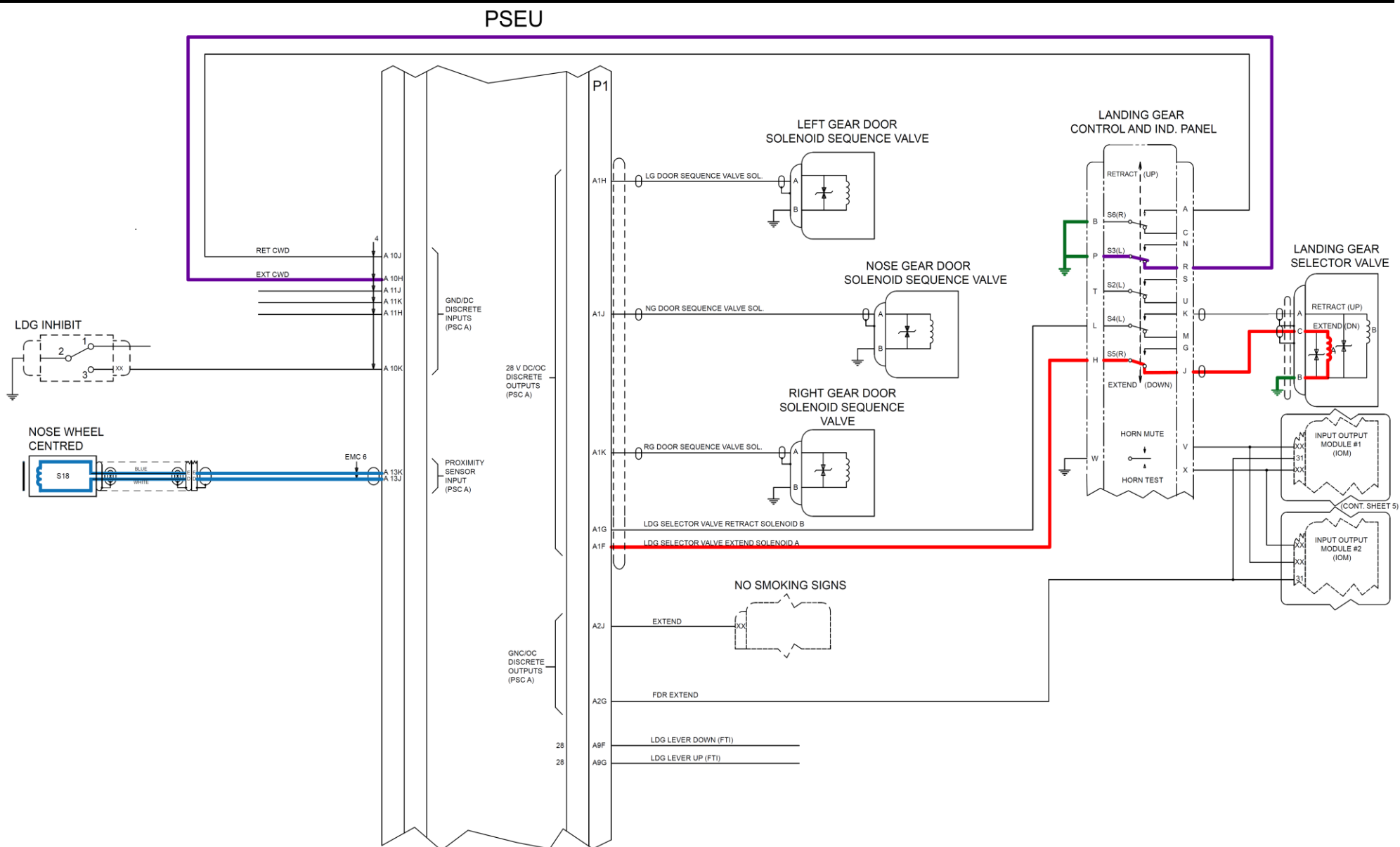


Figure 32-50 (Sheet 1 of 2) Landing Gear Normal Extension

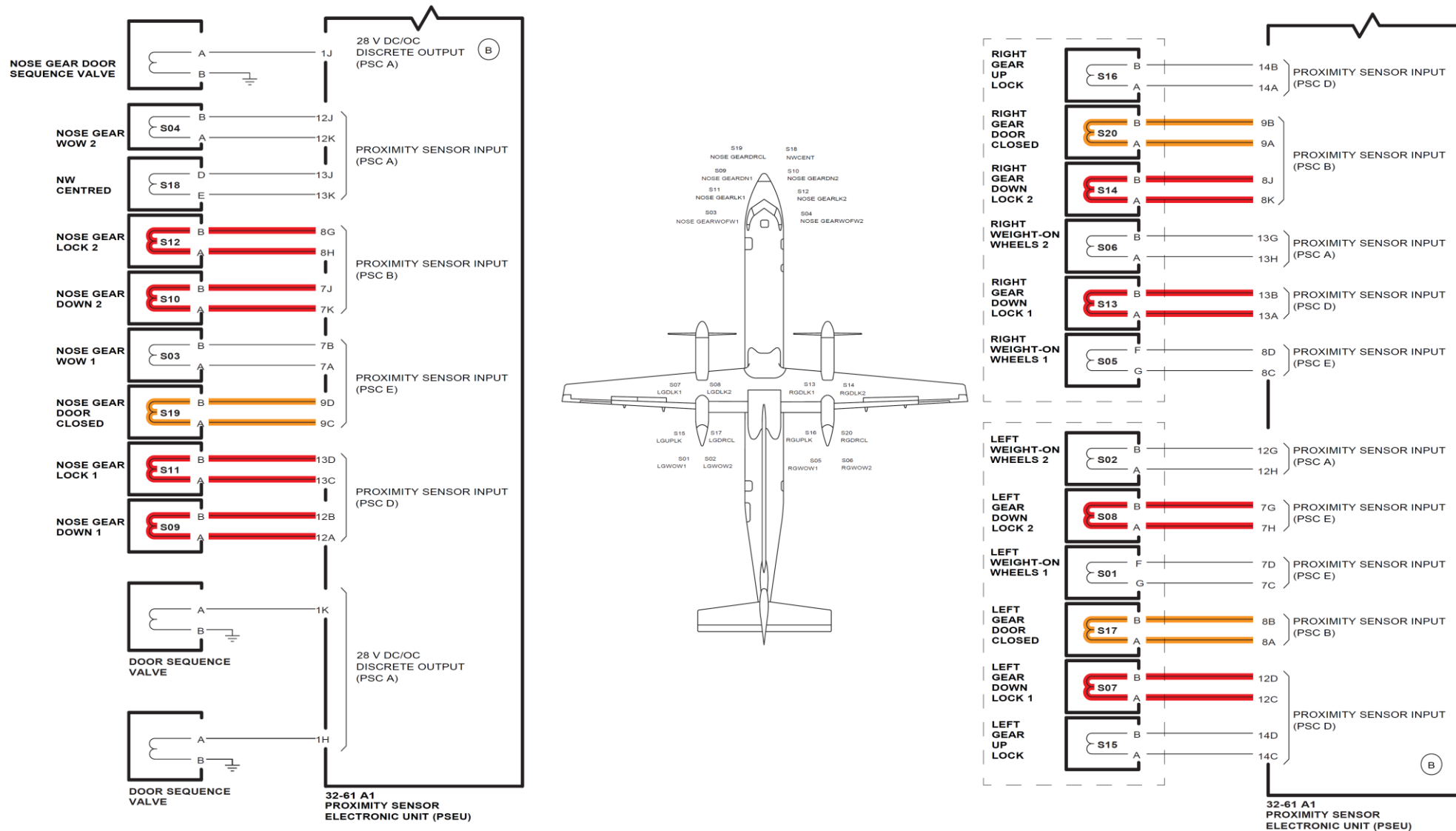


Figure 32-51 (Sheet 2 of 2) Landing Gear Normal Extension – Electrical Schematic

Primary Extension and Retraction – Component Description

Landing Gear Control Panel

[Refer Figure 32-52](#)

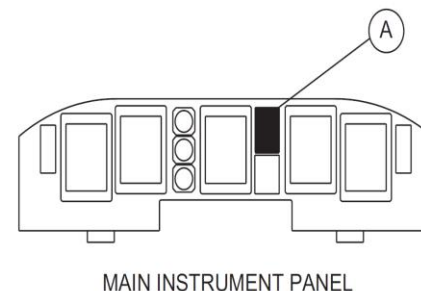
The landing gear is controlled and monitored from the LANDING GEAR control panel located on the right side of the Engine Display (ED). The panel has a LANDING GEAR selector lever, a LOCK RELEASE selector lever, landing gear and landing gear door advisory lights, and a landing gear warning HORN MUTE/TEST switch. The landing gear is commanded to the UP or DN position with the LANDING GEAR selector lever. The LANDING GEAR selector lever LOCK RELEASE must be held down before the selector lever can be moved to either the UP or the DN position. An amber light in the LANDING GEAR selector lever comes on to show that the landing gear position does not agree with the LANDING GEAR selector handle position.

Nine advisory lights are located at the top of the landing gear control panel. Three amber lights L DOOR, N DOOR, and R DOOR show that the landing gear doors are open during extension and retraction, or after the sequence is completed. Three green lights show that the LEFT, NOSE, and RIGHT landing gear are down and locked. Three red lights show that the LEFT, NOSE, and RIGHT landing gear are unlocked.

When the LANDING GEAR selector lever is unlocked and moved to the UP position, the landing gear downlocks are released. Hydraulic pressure is supplied to the landing gear actuators and the landing gear is retracted. When the LANDING GEAR selector lever is unlocked and moved to the DN position, the landing gear uplocks are released. Hydraulic pressure is supplied to the landing gear actuators and the landing gear is extended.

A landing gear configuration warning tone sounds when the aircraft is in the landing configuration and not all the landing gear are down and locked. The two position landing gear warning HORN MUTE/TEST switch is a momentary type toggle switch. When the switch is set to the TEST position, the landing gear configuration warning tone will sound at a constant 800 Hz. The MUTE

position is used for training purposes.



LEGEND

1. "Green" Gear Downlocked.
2. "Red" Gear Not Uplocked Or Downlocked.
3. "Amber" Door Open.
4. Handle "Amber" (Concurrent W/"Red" Unsafe Lights) Gear In-Transit.
5. Warning Horn Mute/Test Switch.

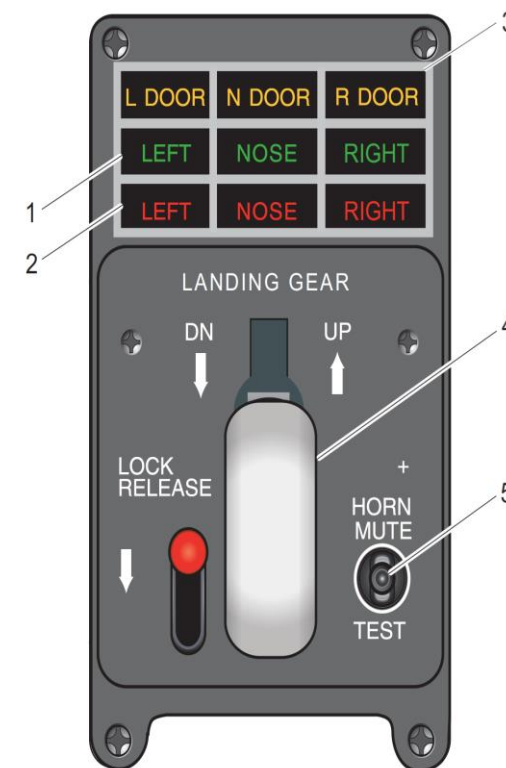
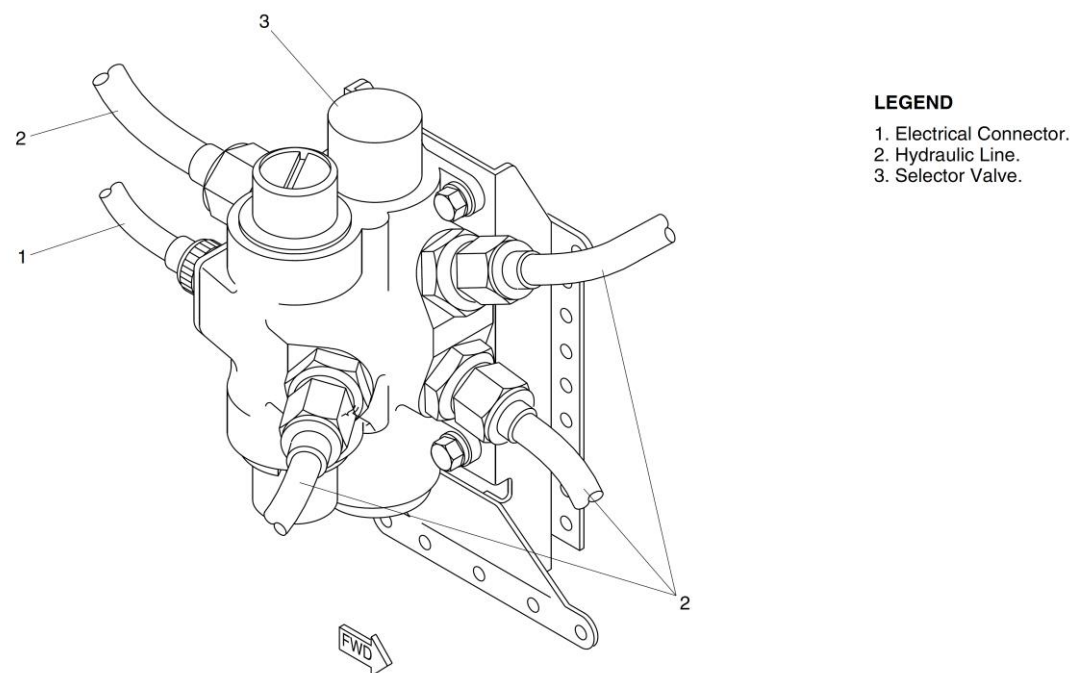


Figure 32-52 Landing Gear Control Panel

Landing Gear Selector Valve

[Refer Figure 32-53](#)

The landing gear selector valve is a self-contained assembly which has two solenoid valves. The landing gear selector valve controls hydraulic pressure to position a directional control valve that is spring centered. The position of the valve controls the supply of hydraulic pressure to either the UP or DOWN hydraulic circuits of the landing gear system. The landing gear system will be configured for either normal retraction or extension.



LEGEND

1. Electrical Connector.
2. Hydraulic Line.
3. Selector Valve.

Figure 32-53 Landing Gear Selector Valve

Main Landing Gear Retraction Actuator

[Refer Figure 32-54](#)

The main landing gear retraction actuator is a hydraulic device that has two ports. There are restrictors in the retract and extend ports. The rod end of the actuator piston has a ball and race with a lubrication fitting. The MLG retraction actuator cylinder is attached to the lower front of the MLG yoke cross beam. The rod end is attached to the center top of the MLG shock strut.

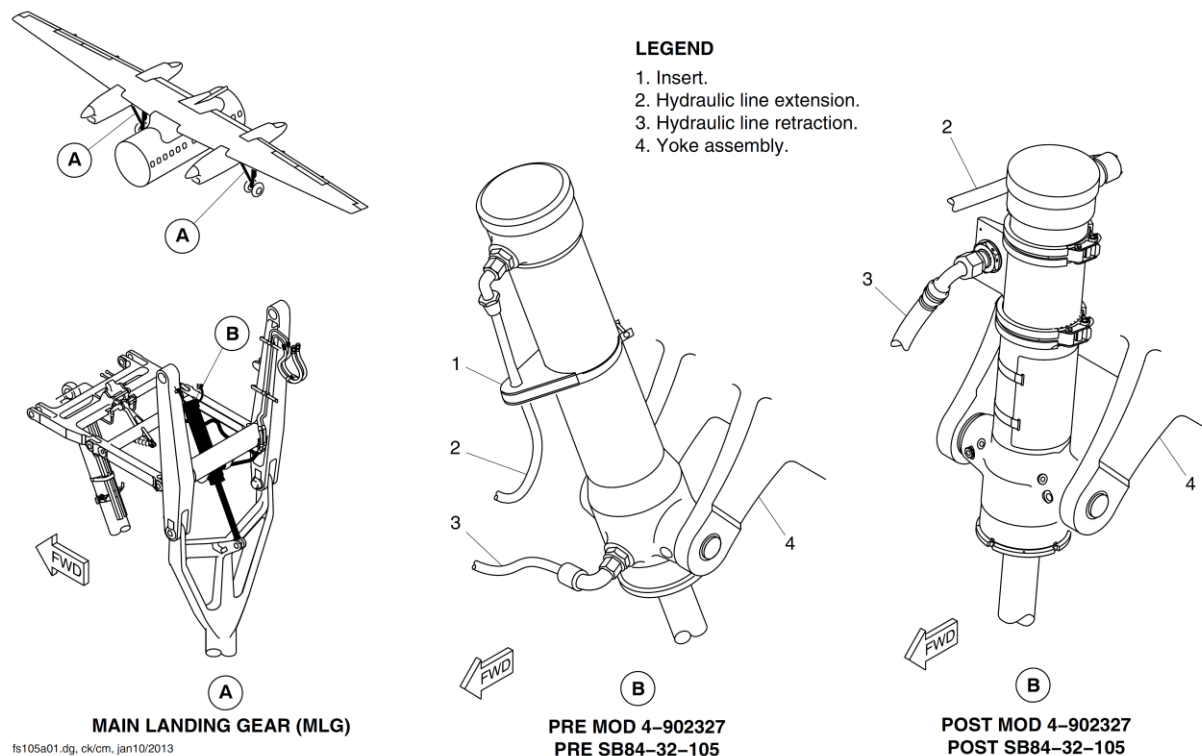


Figure 32-54 Main Landing Gear Retraction Actuator

MLG Downlock Actuator

[Refer Figure 32-55](#)

The MLG downlock actuator has two ports. The rod end of the actuator has a ball and race with a lubrication fitting. Hydraulic lines are attached to the two ports and are sealed with O-rings. The MLG downlock actuator is attached to the MLG stabilizer brace assembly.

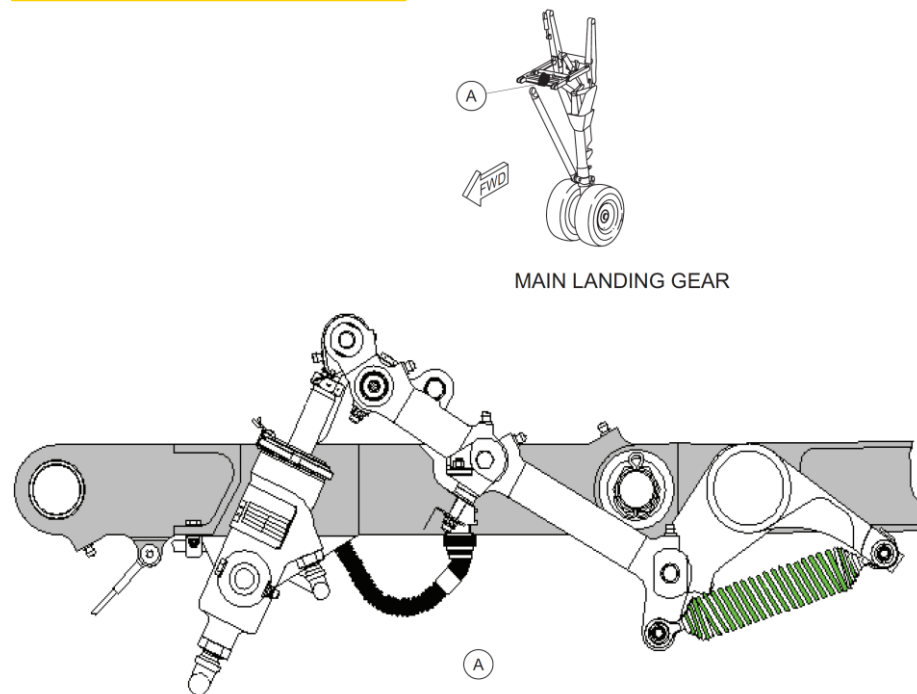


Figure 32-55 Main Landing Gear Downlock Actuator

MLG Uplock Actuator

[Refer Figure 32-56](#)

The MLG uplock actuator is a part of the MLG uplock assembly. The MLG uplock actuator has an external spring that helps to return it to its retracted position when hydraulic pressure is removed.

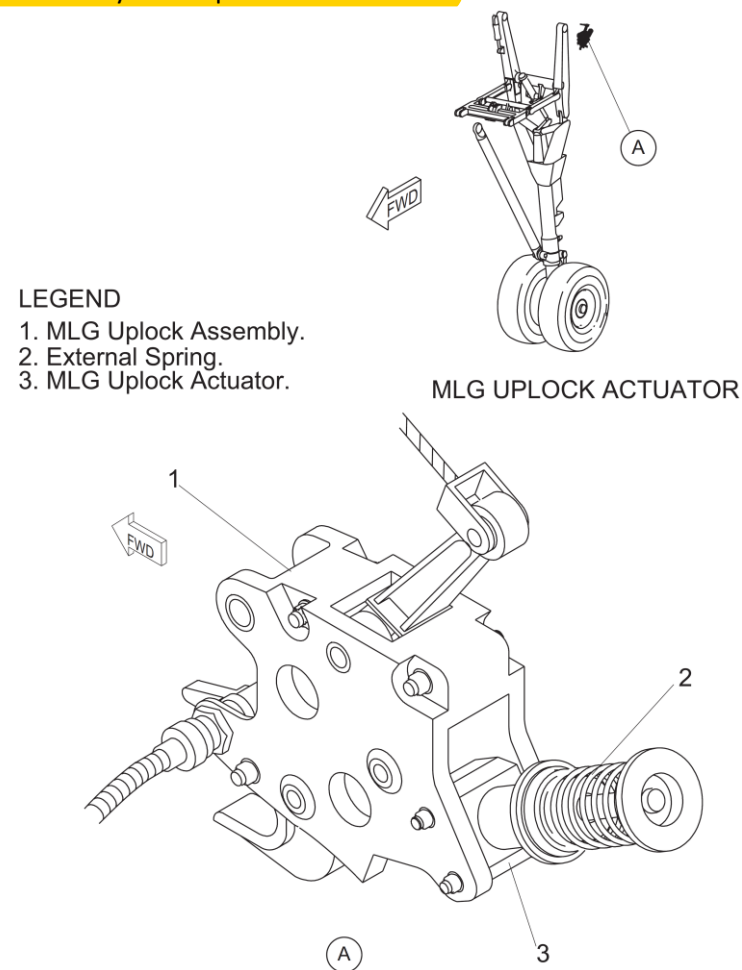
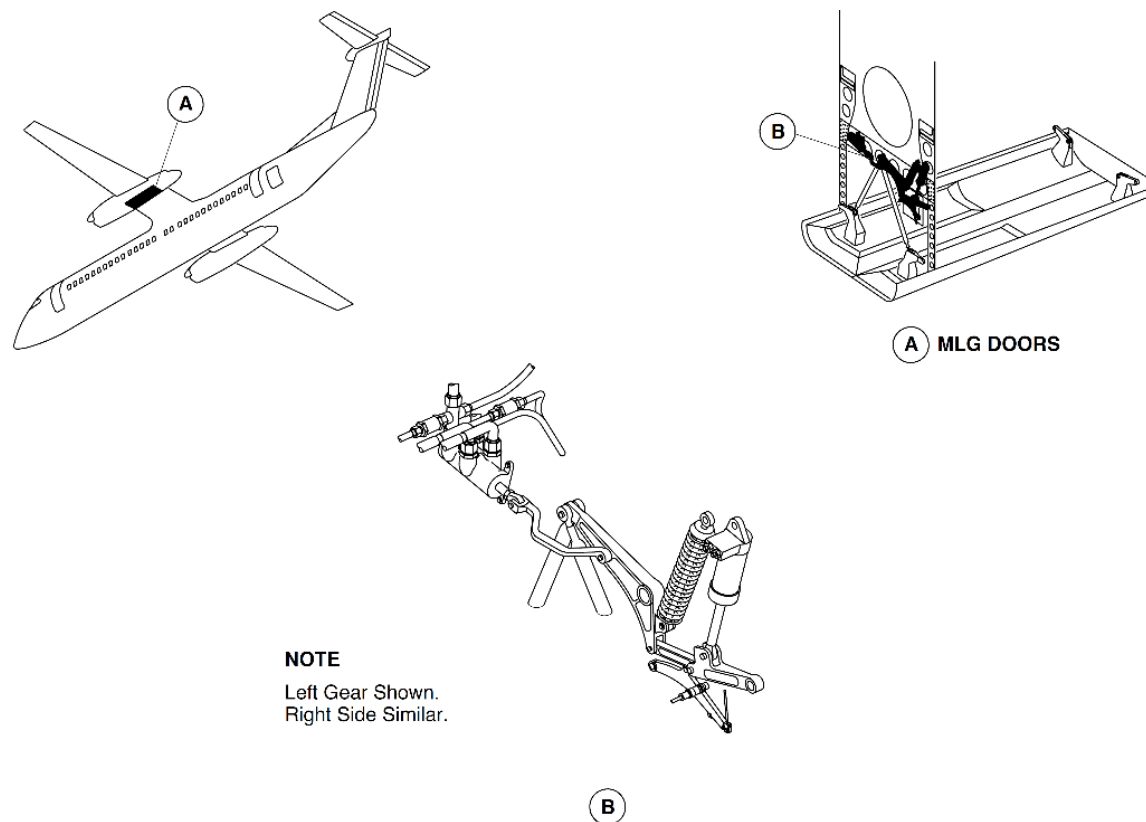


Figure 32-56 Main Landing Gear Uplock Assembly & Actuator

MLG Doors Actuator

[Refer Figure 32-57](#)

The MLG doors actuator is a hydraulic device that has two ports. A restrictor valve is installed at the extension port and an internal restrictor is located under the retraction port in the cylinder wall. These restrictors are used to monitor the time for extension and retraction. Lubrication fittings are located at the attachment point of the actuator cylinder and on the piston rod end.



NOTE

Left Gear Shown.
Right Side Similar.

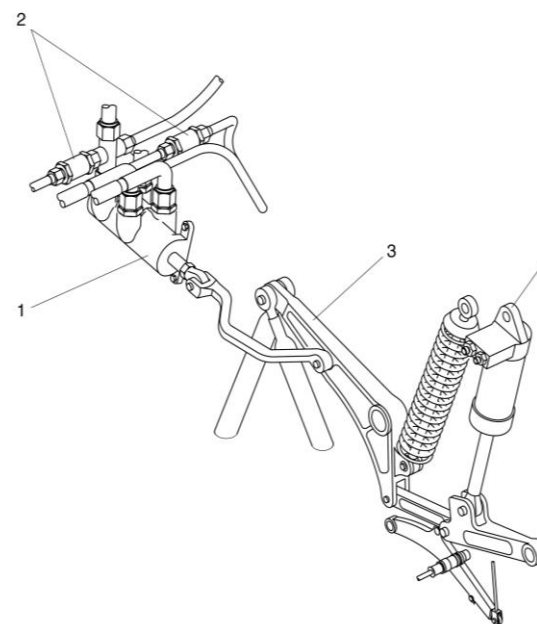
Figure 32-57 MLG Door Actuator

MLG Doors Mechanism

[Refer Figure 32-58](#)

The MLG door actuator powers the MLG doors mechanism to open or close the aft MLG doors. When the MLG doors actuator retracts, it unlocks the over-center lock link. As the actuator continues to retract, it folds the lock link and turns the bellcrank clockwise. The bellcrank acting on the two push rods, pushes the aft doors to the open position.

When the MLG doors actuator extends, it unfolds the lock link to the over-center position. As the actuator continues to extend, it unfolds the lock link and turns the bellcrank anti-clockwise. The bellcrank, acting on the two push rods, pulls the aft doors to the closed position. The aft doors are closed when the MLG is in the extended or the retracted position. The forward doors are connected to and operated by the upper part of the shock strut of the MLG through two links. The forward doors are open when the MLG is extended and closed when the MLG is retracted.



LEGEND

1. MLG Doors Mechanical Sequence Valve.
2. MLG Doors Inline Restrictors.
3. MLG Doors Mechanism.
4. MLG Doors Actuator.

Figure 32-58 MLG Doors Mechanism

MLG Doors Mechanical Sequence Valve

[Refer Figure 32-59](#)

The MLG mechanical sequence valve is a bolt-on assembly. During landing gear retraction and extension, the mechanical sequence valve operates to correctly sequence the MLG uplock release and the MLG retraction actuators. This prevents the landing gear from traveling through partially opened MLG landing gear doors.

MLG Doors Inline Restrictor

[Refer Figure 32-59](#)

The MLG doors inline restrictors are unions that give a specific hydraulic restriction, to help in the door timing sequence. The MLG doors inline restrictors bypass the MLG mechanical sequence valves to keep the MLG retraction actuators pressurized to 3000 psi (20684 kPa). The pressure is kept at 3000 psi (20684 kPa) until the retraction cycle is completed.

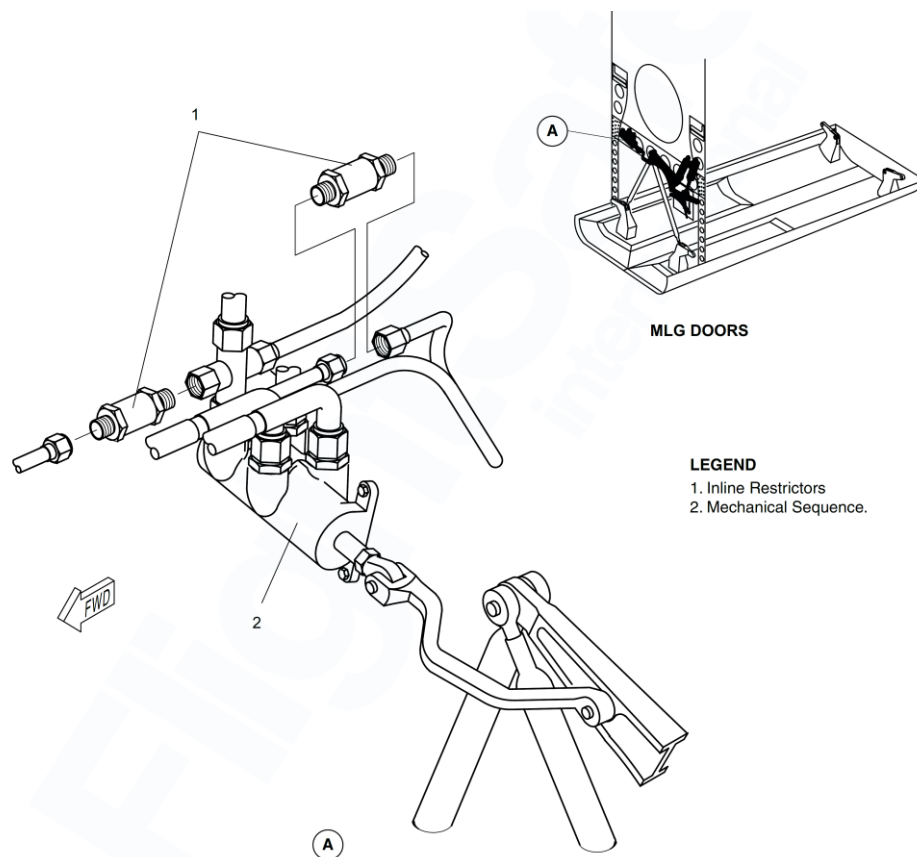


Figure 32-59 MLG Doors Mechanical Sequence Valve and Doors Inline Restrictors

MLG and NLG Doors Solenoid Sequence Valve

[Refer Figure 32-60](#)

The MLG and NLG doors solenoid sequence valves are located in each landing gear wheelwell. The doors solenoid sequence valve is a self-contained, bolt-on assembly. This valve lets the hydraulically operated wheelwell doors open and close.

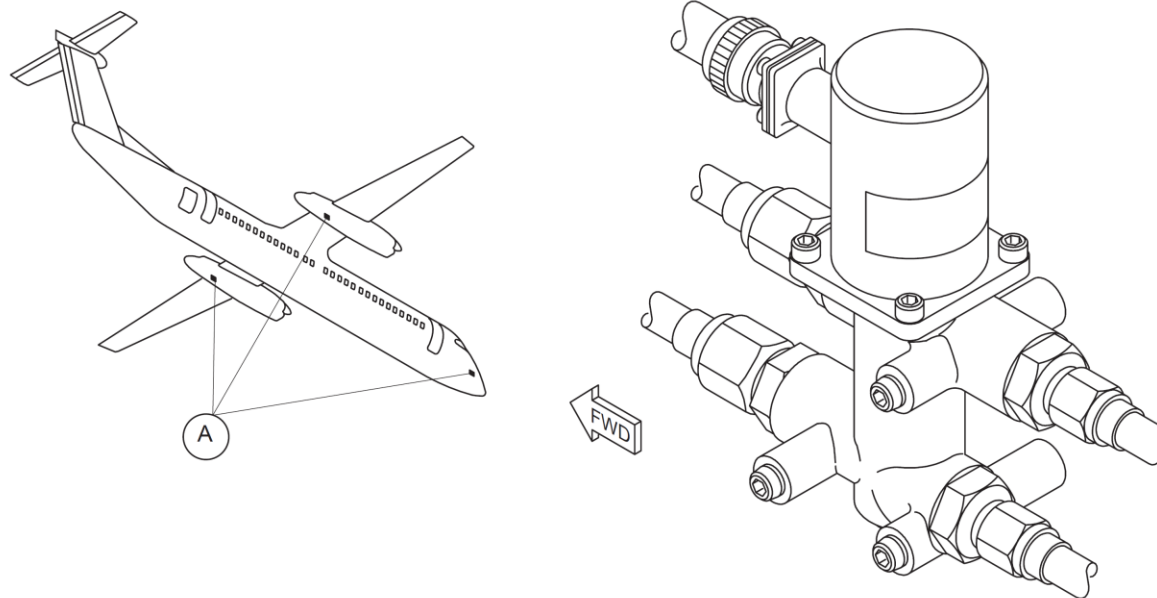


Figure 32-60 MLG & NLG Doors Solenoid Sequence Valve

Nose Landing Gear Retraction Actuator

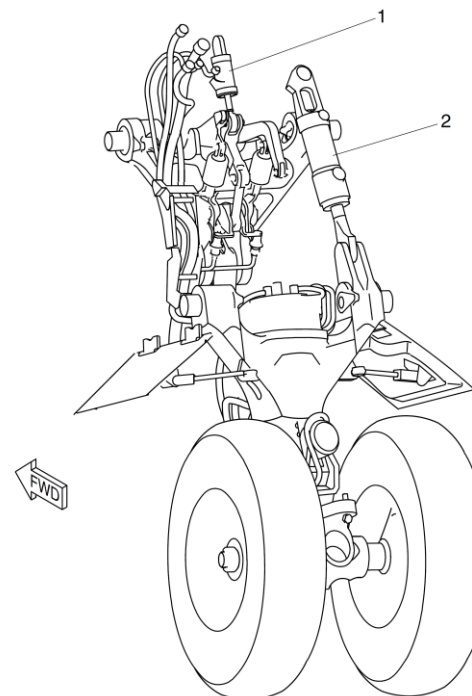
[Refer Figure 32-61](#)

The nose landing gear retraction actuator has two ports. The rod end of the piston is installed with bushings and a lubrication fitting. The cylinder end of the actuator has a ball and race with a lubrication fitting. Hydraulic lines are attached to the two ports and are sealed with O-rings. The NLG retraction actuator piston is attached at the top right hand of the NLG shock strut and the cylinder is attached to the aircraft structure.

NLG Lock Actuator

[Refer Figure 32-61](#)

The NLG lock actuator has two ports. The piston is held in the cylinder by a gland nut and a cup washer. Both ends of the actuator have a ball and race with a lubrication fitting. The NLG lock actuator is attached to the NLG drag strut near the lock links.



LEGEND

1. NLG Lock Actuator.
2. NLG Retraction Actuator.

Figure 32-61 NLG Lock & Retraction Actuator

NLG Doors Actuator

[Refer Figure 32-62](#)

The NLG door actuator has two ports. The piston is held in the cylinder by a gland nut. The NLG door actuator extends to break the linkage over-center and open the doors. The NLG door actuator retracts to break the linkage over-center and close the doors.

NLG Doors Mechanism

[Refer Figure 32-62](#)

The NLG doors mechanism is actuated by the NLG door actuator to open or close the forward NLG doors. When the NLG doors actuator retracts, it unlocks the over centered lock link. As the actuator continues to retract, it folds the lock link and turns the bellcrank clockwise. The bellcrank acting on the two push rods pushes the forward doors to the open position.

When the NLG doors actuator extends, it unfolds the lock link to the over-center position. As the actuator continues to extend, it unfolds the lock link and turns the bellcrank anti-clockwise. The bellcrank acting on the two push rods, pulls the forward doors to the closed position. The forward doors are closed when the NLG is in the extended or the retracted position. The aft doors are connected to and operated by the upper part of the shock strut of the NLG through two articulated links. The aft doors are open when the NLG is extended and closed when the NLG is retracted.

NLG Doors Mechanical Sequence Valve

[Refer Figure 32-62](#)

The NLG mechanical sequence valve is a bolt-on assembly. It correctly sequences the NLG uplock/downlock release and the NLG retraction actuators. This prevents the landing gear from traveling through partially opened NLG landing gear doors. The NLG mechanical sequence valve is open when the landing gear is moving between either the fully extended or the fully retracted position. The NLG mechanical sequence valve is closed when the landing gear is locked in either the fully extended or the fully retracted position.

LEGEND

1. Door Attachment Bracket
2. Toggle Link Assembly
3. Forward Door Actuator Assembly
4. Link Assembly
5. Mechanical Sequence Valve
6. Alternate Release Cable
7. Bell Crank
8. Spring Assembly
9. Door Attachment Bracket
10. Forward Door
11. Forward Door
12. Forward Doors Connecting Rods.

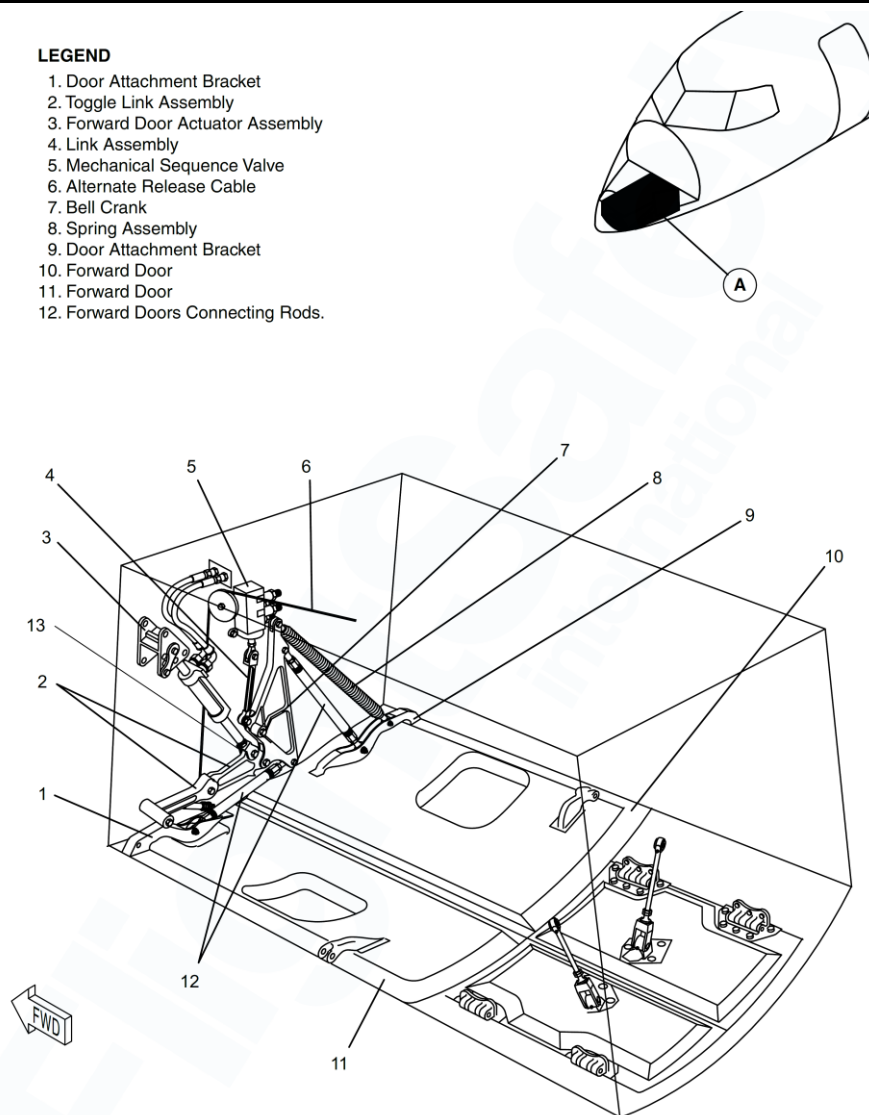


Figure 32-62 NLG Door Mechanism & Doors Mechanical Sequence Valve

NLG Inline Restrictors

[Refer Figure 32-63](#)

The NLG inline restrictors are unions that provide a specific hydraulic restriction. The NLG inline restrictors bypass the NLG mechanical sequence valves to keep the NLG retraction actuators pressurized to 3000 psi (20684 kPa). Hydraulic pressure is kept at 3000 psi (20684 kPa) upon completion of the retraction cycle.

NLG Doors Solenoid Sequence Valve

[Refer Figure 32-62](#)

The NLG doors solenoid sequence valve is a bolt-on assembly with four ports. The valve is directly operated by a single coil solenoid. This valve lets the hydraulically operated NLG wheelwell doors open and close.

Training Information Points

Make sure that all hydraulic power and electrical power is off before any maintenance is performed on the landing gear. The landing gear can accidentally retract. This can cause injuries or death to persons and/or damage to equipment.

Make sure that all persons and equipment are clear of the landing gear and door movement area otherwise injuries to persons and/or damage to equipment can occur.

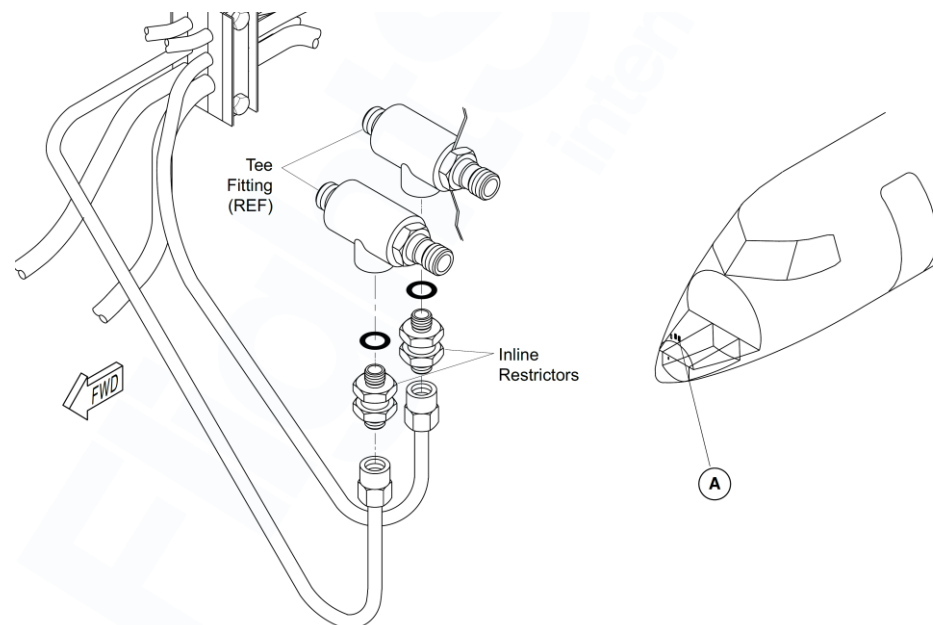


Figure 32-63 NLG Inline Restrictors

Alternate Extension System

Introduction

The alternate extension system can be used to extend the landing gear when the No. 2 hydraulic system, or the Proximity Sensor Unit (PSEU) are not serviceable. The system can also be used, if the normal extension system fails to lock the landing gear in the down position.

General Description

[Refer Figure 32-64](#)

The alternate extension system is a self-resetting, cable actuated design. Access to the alternate extension system is through the flight compartment. The alternate extension system includes a bypass valve to isolate the landing gear hydraulics from the No. 2 hydraulic system. A manual handpump hydraulic system is available to assist in lowering the Main Landing Gear (MLG) into the down and locked position.

The alternate extension of the landing gear occurs when the Nose Landing Gear (NLG) forward doors, the MLG aft doors, and the landing gear uplocks are mechanically opened. The NLG free-falls to the down and locked position, assisted by the airflow. The MLGs free-fall and, if required, are pumped to the down and locked position by the alternate extension actuators.

To isolate the normal landing gear selector valve during flight, a LANDING GEAR INHIBIT switch is installed in the flight compartment ceiling. To use the alternate extension system, the LANDING GEAR INHIBIT switch must be set from NORMAL to INHIBIT.

The alternate extension system has the components that follow:

- Two alternate release handles
- One alternate release door
- One alternate extension door
- Two alternate extension actuators
- A bypass valve
- Control Cables

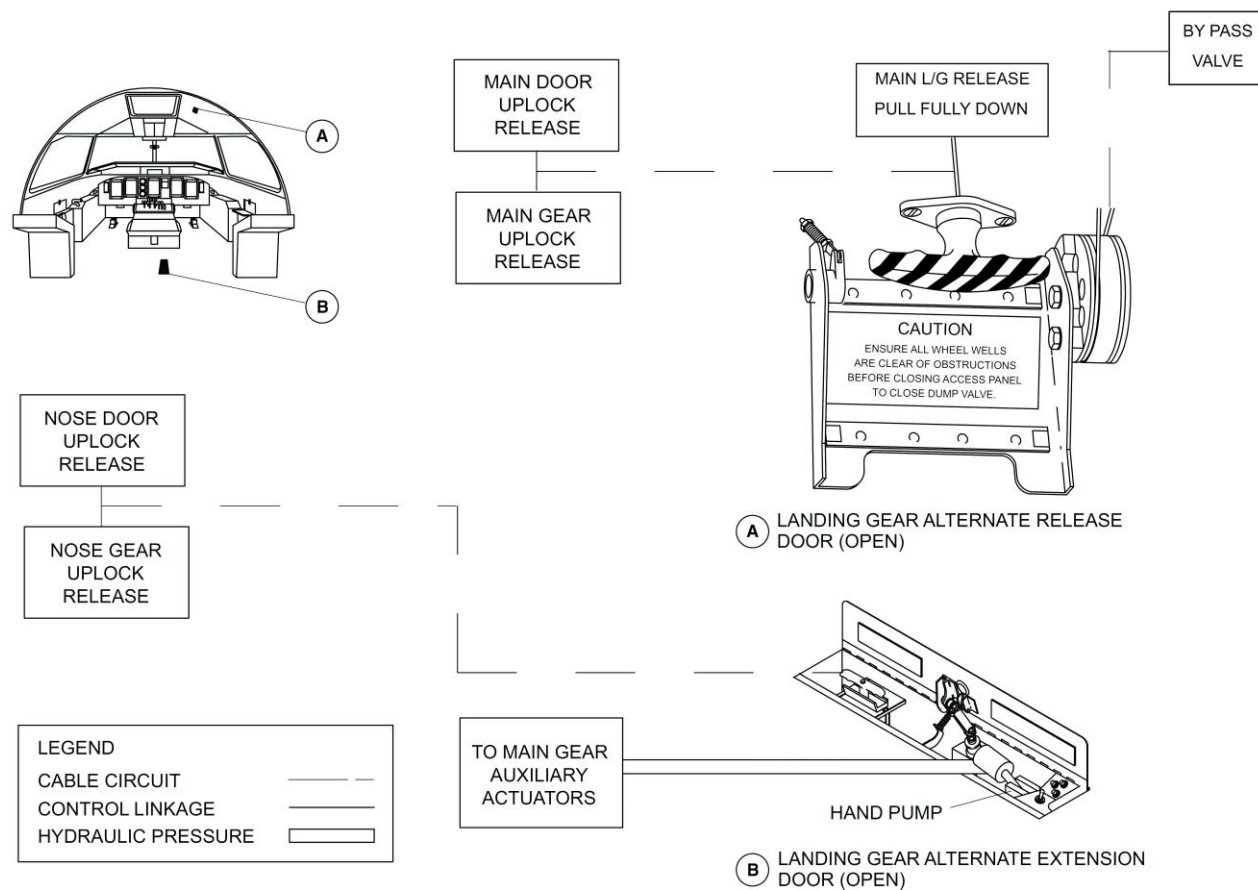


Figure 32-64 Alternate Extension System Block Diagram

Detailed Description

Refer to Figures 32-65 & 32-66

Alternate manual extension of the landing gear is accomplished in four stages.

The LANDING GEAR INHIBIT switch is set to the INHIBIT position to disable the PSEU.

The LANDING GEAR ALTERNATE RELEASE door, located in the ceiling of the flight compartment, is opened. This actuates the bypass valve to isolate the No. 2 hydraulic system.

The LANDING GEAR ALTERNATE RELEASE handle is pulled to open the MLG doors and release the MLG uplocks.

Tension springs in the MLG door mechanism pull the doors open and the MLG free-falls into the down position.

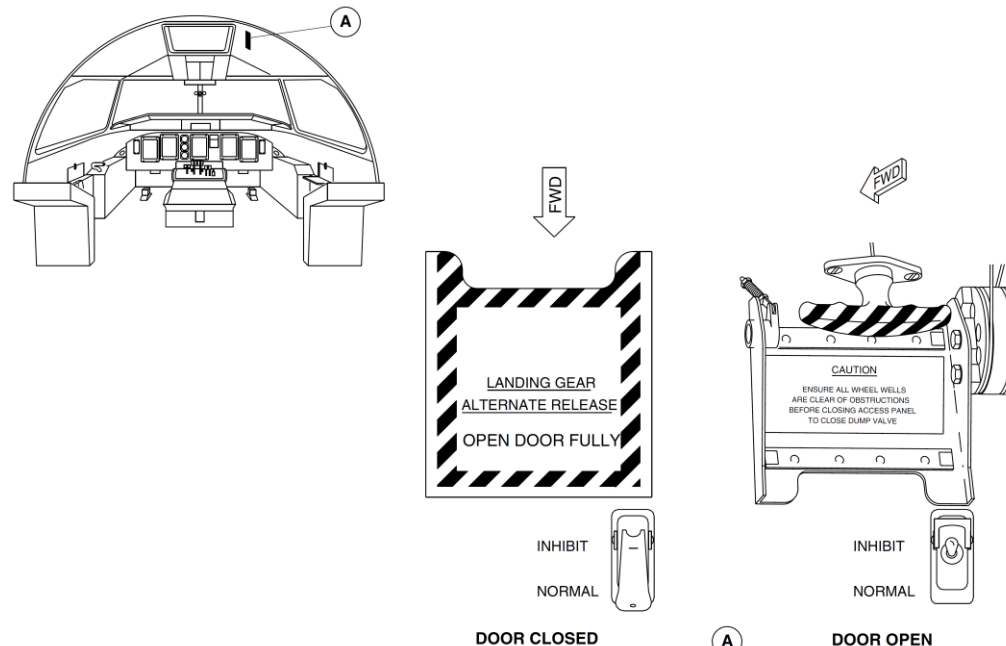


Figure 32-65 Landing Gear Alternate Release Controls Locator

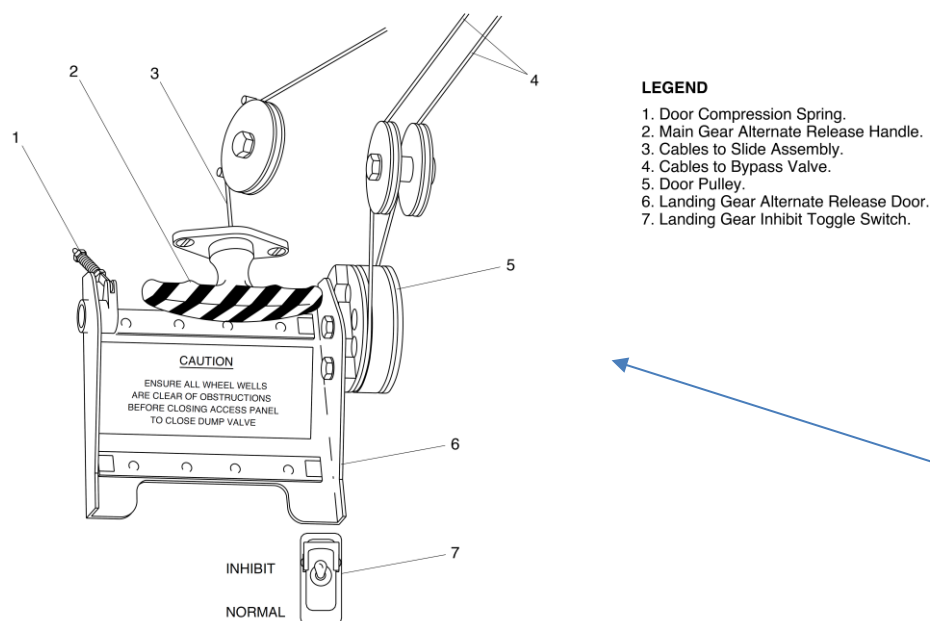


Figure 32-66 Landing Gear Alternate Release Details

[Refer Figure 32-67](#) & [32-68](#)

The LANDING GEAR ALTERNATE EXTENSION door located in the flight compartment floor is opened. The door operates the alternate landing gear selector valve and reveals the handpump socket. If the MLG does not reach the down and locked position, the handpump is used to fully extend the MLG. Springs installed on the MLG stabilizer brace move the lock links into the down and locked position. The LANDING GEAR ALTERNATE EXTENSION handle is pulled to unlock the NLG forward doors. Tension springs in the NLG door mechanism pull the doors open. When the doors are fully open, the NLG uplock is released by a further pull on the handle. The NLG free-falls to the fully extended position. Springs installed on the NLG drag strut move the lock links into the down and locked position.

An alternate landing gear down and locked verification system is located in the in the flight compartment, in the aft section of the LANDING GEAR ALTERNATE EXTENSION compartment. The system has a two position, momentary type toggle switch, a light panel that contains three green lights, and four proximity sensors. Two of the proximity sensors are used in series and are attached to the NLG. One proximity sensor is installed on each of the MLG. The system elements on each landing gear are activated when the downlock verification switch is moved to the ON position. The system indication lights come on as each landing gear is locked down.

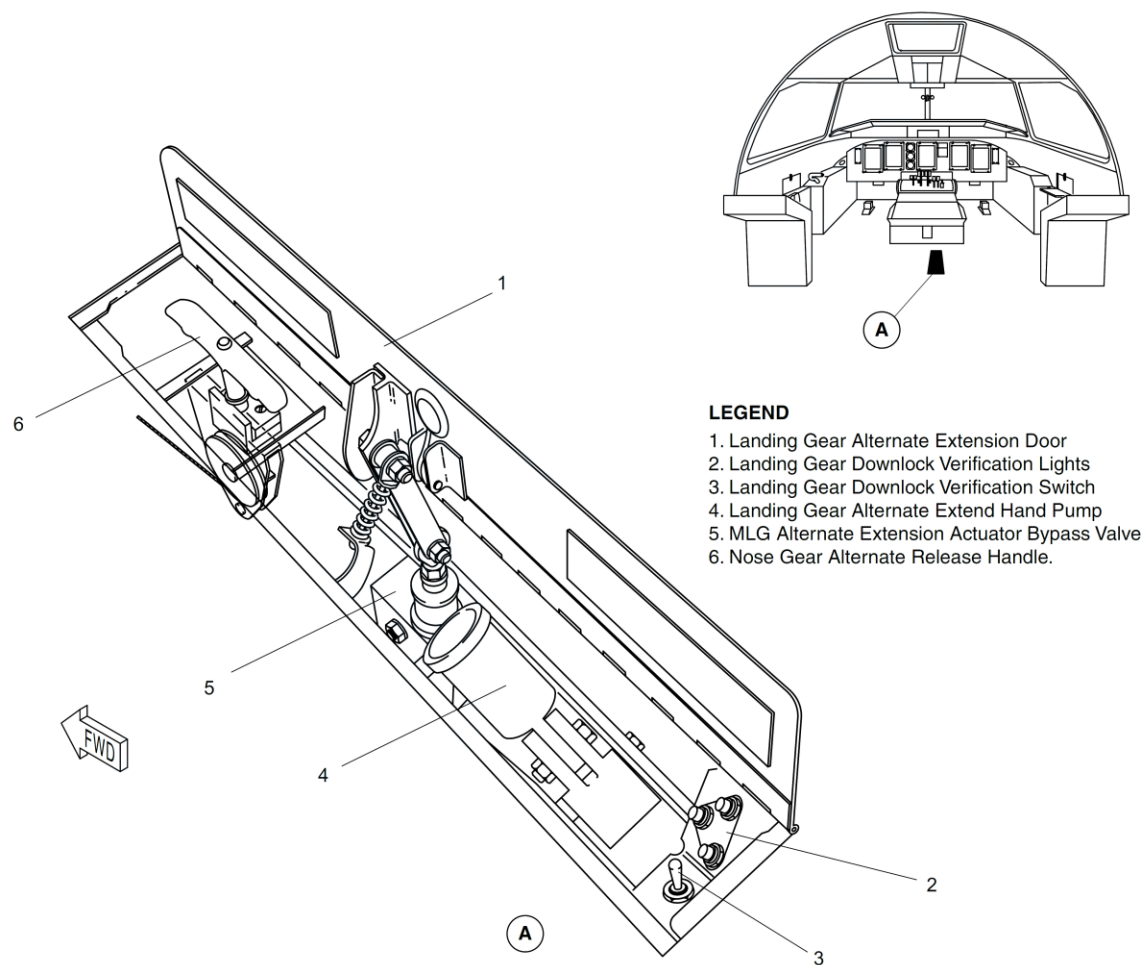


Figure 32-67 Landing Gear Alternate Extension Door Details

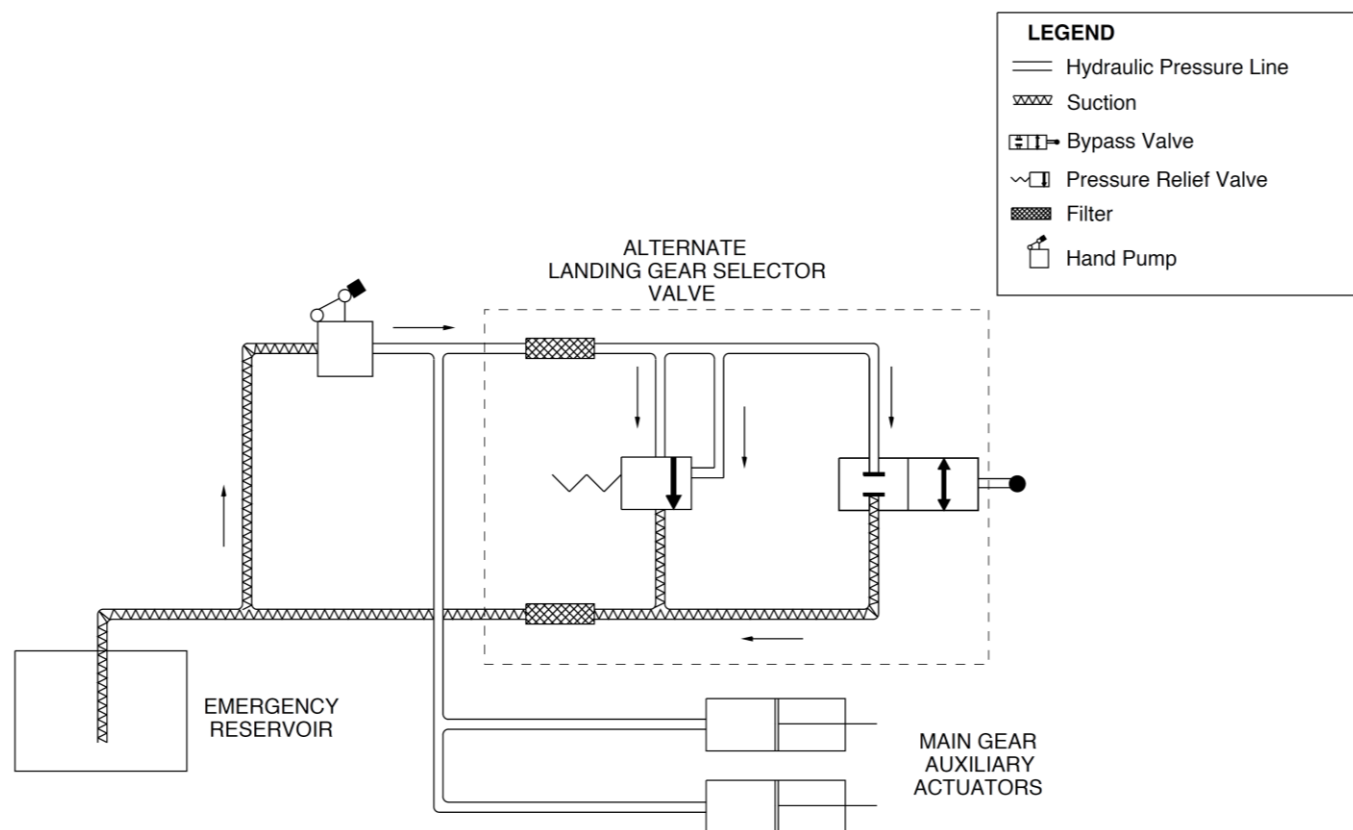


Figure 32-68 Alternate Extension System Hydraulic Synoptic

Alternate Extension System – Component Description

Alternate Release Handles

[Refer to Figure 32-69](#)

The alternate extension of the NLG uses the NLG alternate release handle located in the floor of the flight compartment under the LANDING GEAR ALTERNATE EXTENSION door. The handle is connected by cables, turnbuckles and pulleys to the NLG uplock release arm, a pulley cam mechanism and to the NLG door release operating arm. The pulley cam mechanism makes sure that the NLG forward doors open before the NLG uplock is opened to release the NLG.

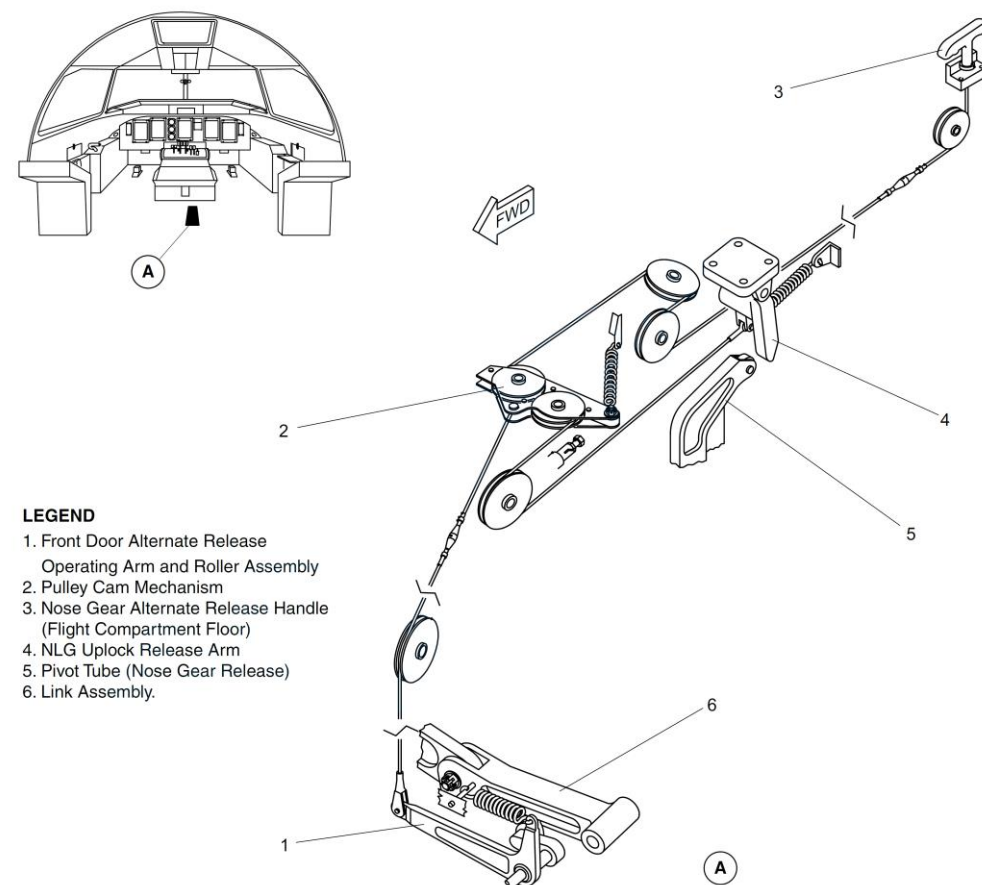


Figure 32-69 NLG Alternate Release Cable

[Refer Figures 32-70, 71-A & 71-B](#)

The alternate extension of the MLG uses the MLG alternate release handle, the handpump, and the auxiliary actuators. The alternate release handle is located in the ceiling of the flight compartment behind the LANDING GEAR ALTERNATE RELEASE door. To actuate the MLG bypass valve, open the LANDING GEAR ALTERNATE RELEASE door. The handle is connected by cables, turnbuckles, pulleys, a slide assembly, and a cam mechanism to the MLG uplocks and the MLG aft doors release mechanisms. The cam mechanism makes sure that the MLG aft doors open before the MLG uplocks are opened to release the MLGs.

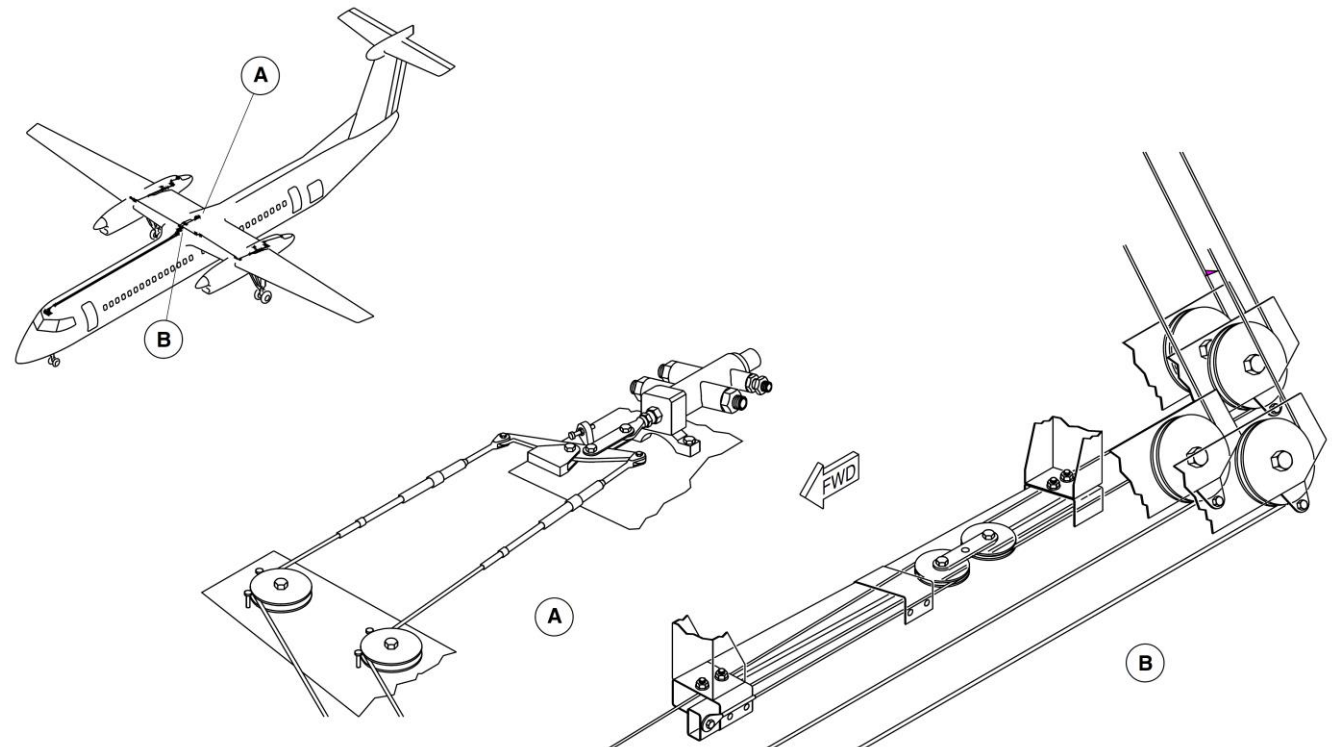


Figure 32-70 Alternate Release Cables Locator

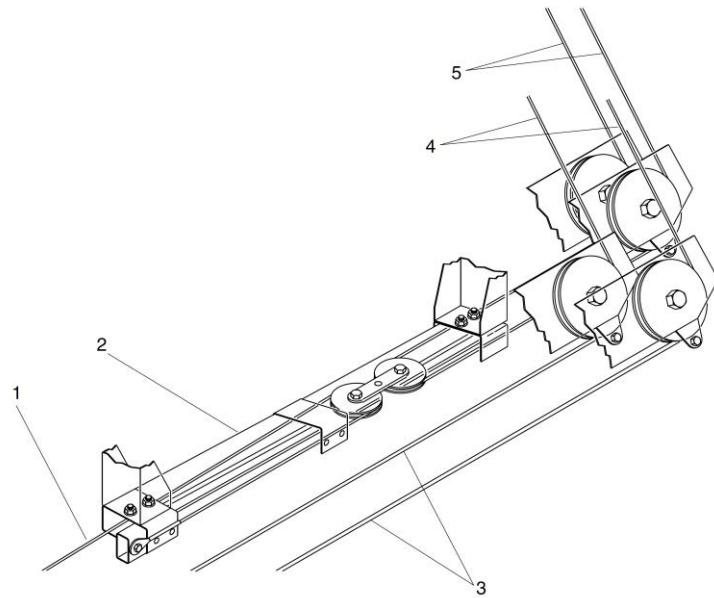


Figure 32-71A Alternate Release Cables Details 1

LEGEND

1. Cable from Alternate Release Handle.
2. Slide Assembly.
3. Cables from Door Pulley.
4. Cables to Bypass Valve Operating Lever.
5. Cables to Left and Right Nacelles.

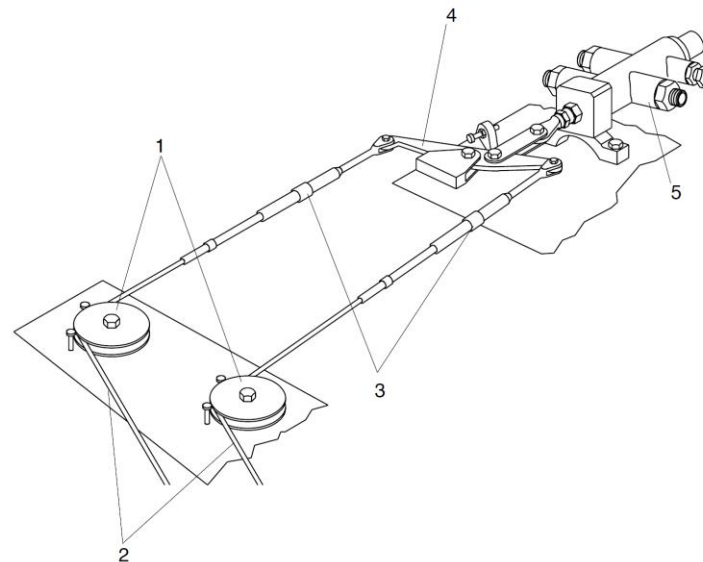


Figure 32-71B Alternate Release Cable Details 2

LEGEND

1. Pulleys.
2. Cables from Door Pulley.
3. Turnbuckles.
4. Operating Lever.
5. Bypass Valve.

Alternate Extension Actuator

[Refer Figure 32-72](#)

Alternate extension actuators are located on the right side of each MLG yoke. Hydraulic pressure required to operate the actuators is supplied by a handpump located in the flight compartment floor. The hydraulic fluid is supplied from the emergency hydraulic reservoir.

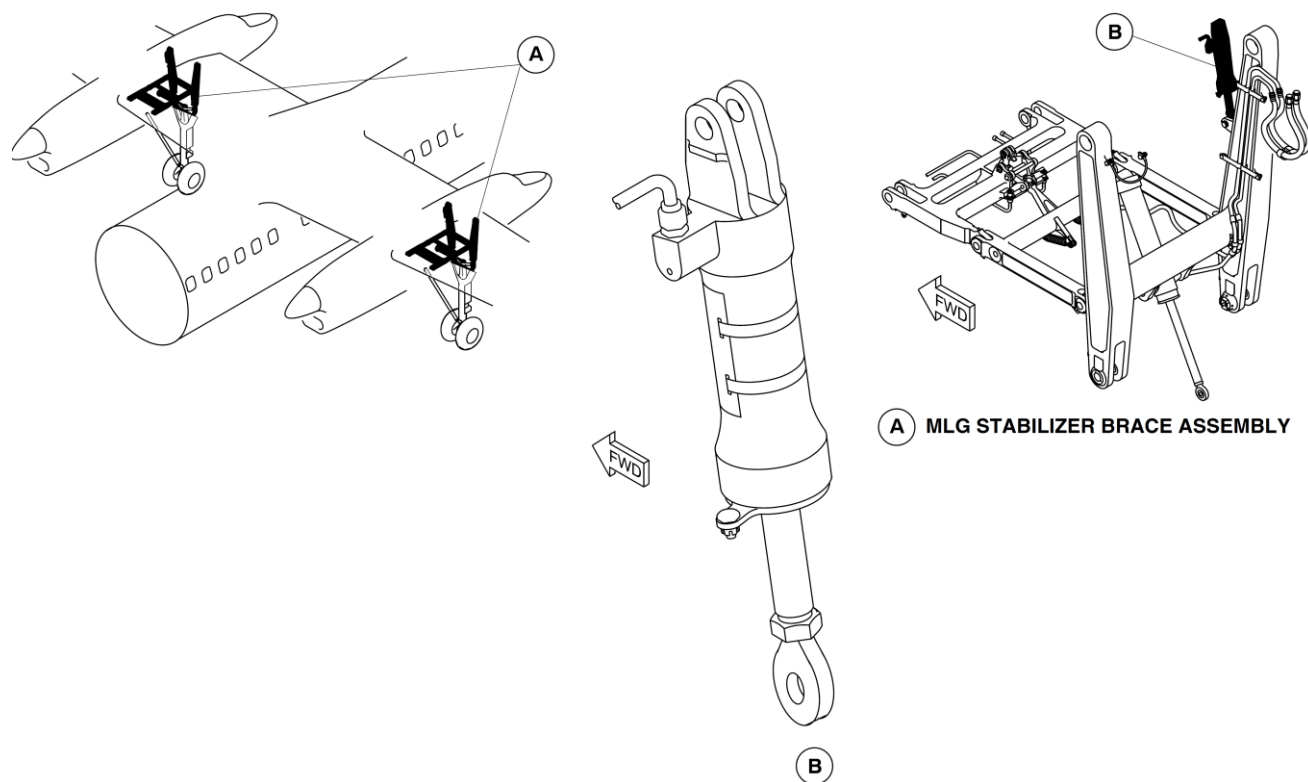
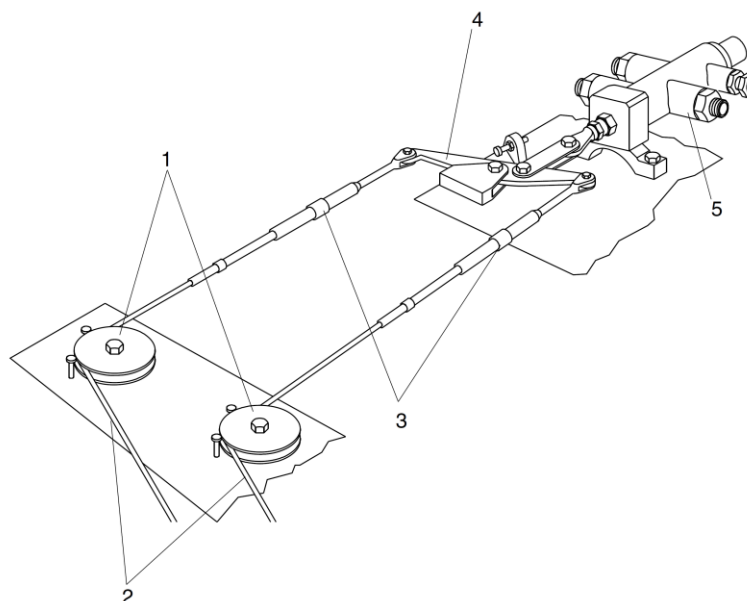


Figure 32-72 Alternate Extension Actuator

Bypass Valve

[Refer Figure 32-73](#)

The bypass valve is installed in the forward right wing root area, adjacent to the landing gear selector valve. The bypass valve isolates the No. 2 hydraulic supply lines from the main and nose gear retraction actuators, and connects the lines to the return lines. The bypass valve is actuated when the LANDING GEAR ALTERNATE RELEASE door is opened. Cables extend from the LANDING GEAR ALTERNATE RELEASE door and attach to a lever at the bypass valve. The lever operates the bypass valve.



LEGEND

1. Pulleys.
2. Cables from Door Pulley.
3. Turnbuckles.
4. Operating Lever.
5. Bypass Valve.

Figure 32-73 Alternate Extension Actuator

Control Cables

[Refer Figure 32-74](#)

The MLG ALTERNATE RELEASE door installed in the flight compartment ceiling has a closed loop cable system. When the door is opened, it actuates the bypass valve to the BYPASS position. This isolates the main landing gear hydraulic system and depressurizes the extension actuators. It also lets the MLG be pumped to the down and locked position. When the door is closed, the bypass valve returns to the NORMAL position. Control cables are routed from the MLG alternate release handle through a slider mechanism located in the ceiling of the aircraft forward of the wing root. The slider mechanism lets the full force of the pull be concentrated first on the release of one MLG and then on the other. From here, the control cables are routed through to the engine nacelles to the cam mechanisms. The cam mechanism allows the force of the pull to be concentrated first on the release of the MLG aft doors and then on the MLG. Control cables are routed from the NLG alternate release handle through a pulley cam mechanism located in the forward fuselage above the nose wheel well. The pulley cam mechanism lets the full force of the pull be concentrated first on the release of the NLG forward doors and then on the NLG.

Training Information Points

Make sure that all hydraulic power and electrical power is off before you do maintenance on the landing gear. The landing gear can accidentally retract. This can cause injuries or death to persons and/or damage to equipment.

Make sure that all persons and equipment are clear of the landing gear and door movement area. If you do not do this, injuries to persons and/or damage to equipment can occur.

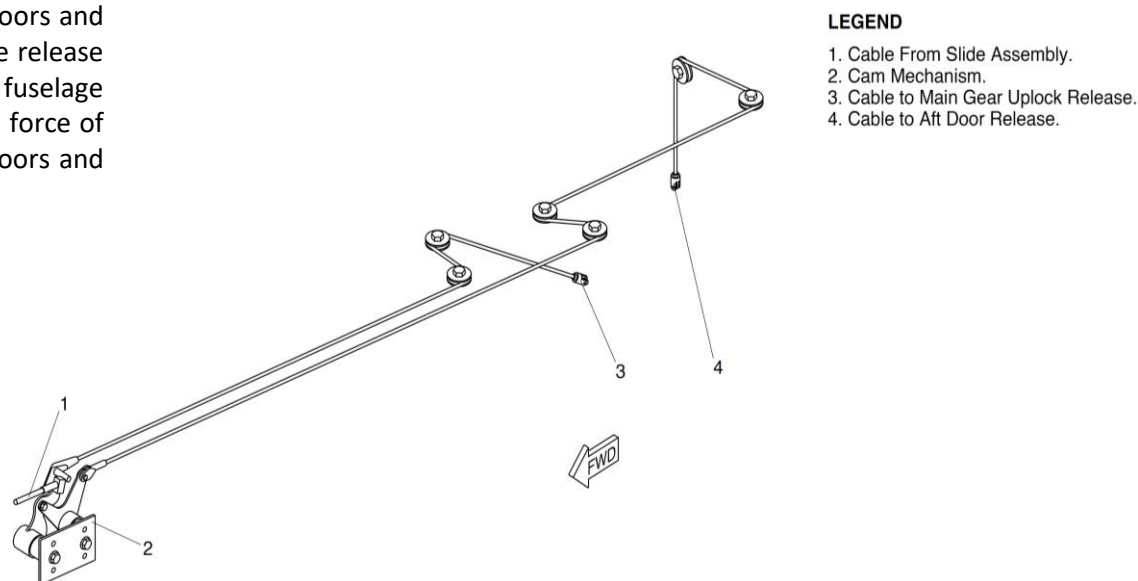


Figure 32-74 Landing Gear Alternate Release Cables and Uplock Detail

Wheels

Introduction

The wheels support the weight of the aircraft and let the aircraft move while on the ground.

General Description

The aircraft has six wheels. Two wheels are installed on the nose landing gear axle and two wheels are installed on each main landing gear axle.

Detailed Description

[Refer Figure 32-75](#)

Two wheels are installed on the axle of each main landing gear. The main wheels are a split-web type assembly made of two forged aluminum wheel halves. There are two different wheel assemblies that can be used on the main landing gear. One main wheel assembly has a 32×8.8–16, 18 ply tubeless tire. The other main wheel assembly has a 34×10.75–16, 16 ply tubeless tire.

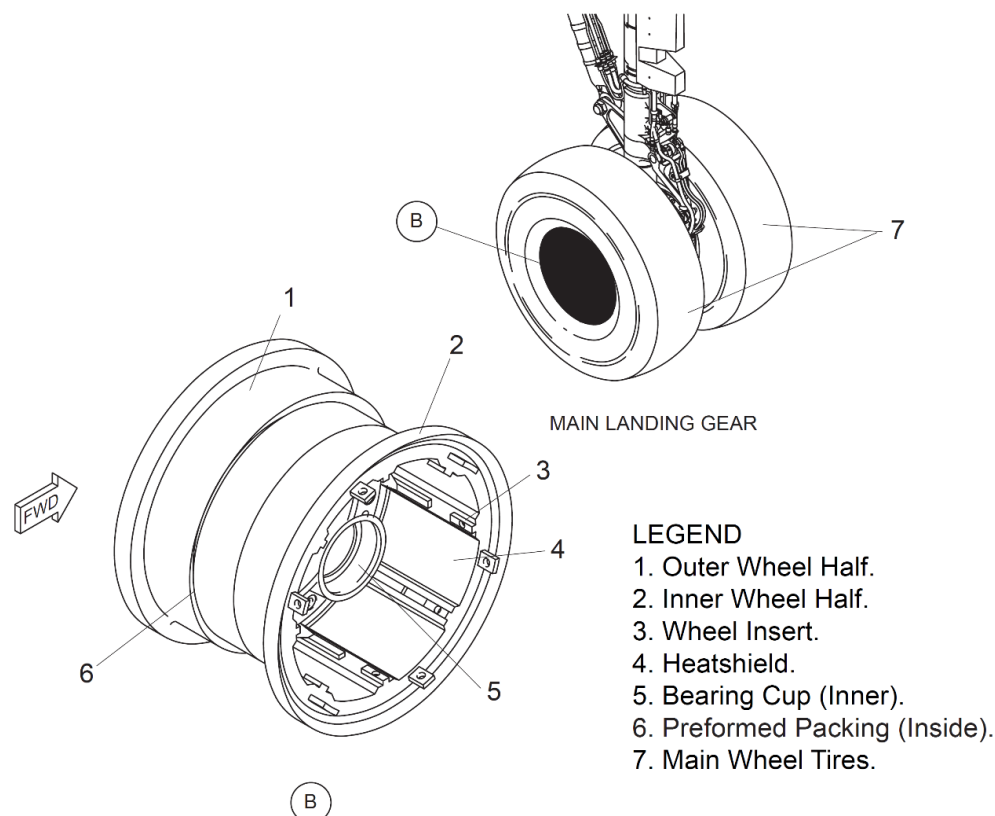


Figure 32-75 Main Wheel Details

Refer Figure 32-76

The main wheels are tubeless, split-web type assemblies made of forged aluminum wheel halves. The two halves are held together with equally spaced tie bolts, washers, and nuts. A preformed packing is installed on the inner wheel half assembly to prevent air leakage through the wheel mating surfaces.

Tapered wheel bearings and bearing cups are installed in the inner and outer wheel halves. The bearings and cups are installed in pairs and lubricated with grease. Bearing seals are installed at the outer surface of the bearings to prevent contamination. A retaining ring holds the outer bearing in place. Grease retainers are installed next to each bearing cup to prevent the flow of grease into the center of the wheel. A hubcap and gasket is installed on the outer wheel half to protect the bearing from dirt and moisture.

An inflation valve assembly is installed in the outer wheel half assembly. **An over inflation plug is installed 180 degrees from the inflation valve to prevent over inflation of the tire.** The over inflation plug contains a disk that ruptures, and deflates the tire to zero pressure. Four thermal relief plugs are installed in the web of the inner wheel half. Each thermal relief plug is made of fusible metal that melts at a specified temperature. This will release the pressure in the tire and prevent an explosion if the wheel gets too hot.

Wheel inserts are installed on the inner wheel. The inserts engage the drive slots in the brake disks of the main wheel brake units, which turn the disks when the wheel turns. **A one piece heatshield is installed in the inner wheel half. The heatshield decreases the transfer of heat from the brake assembly to the wheel and tire assembly.**

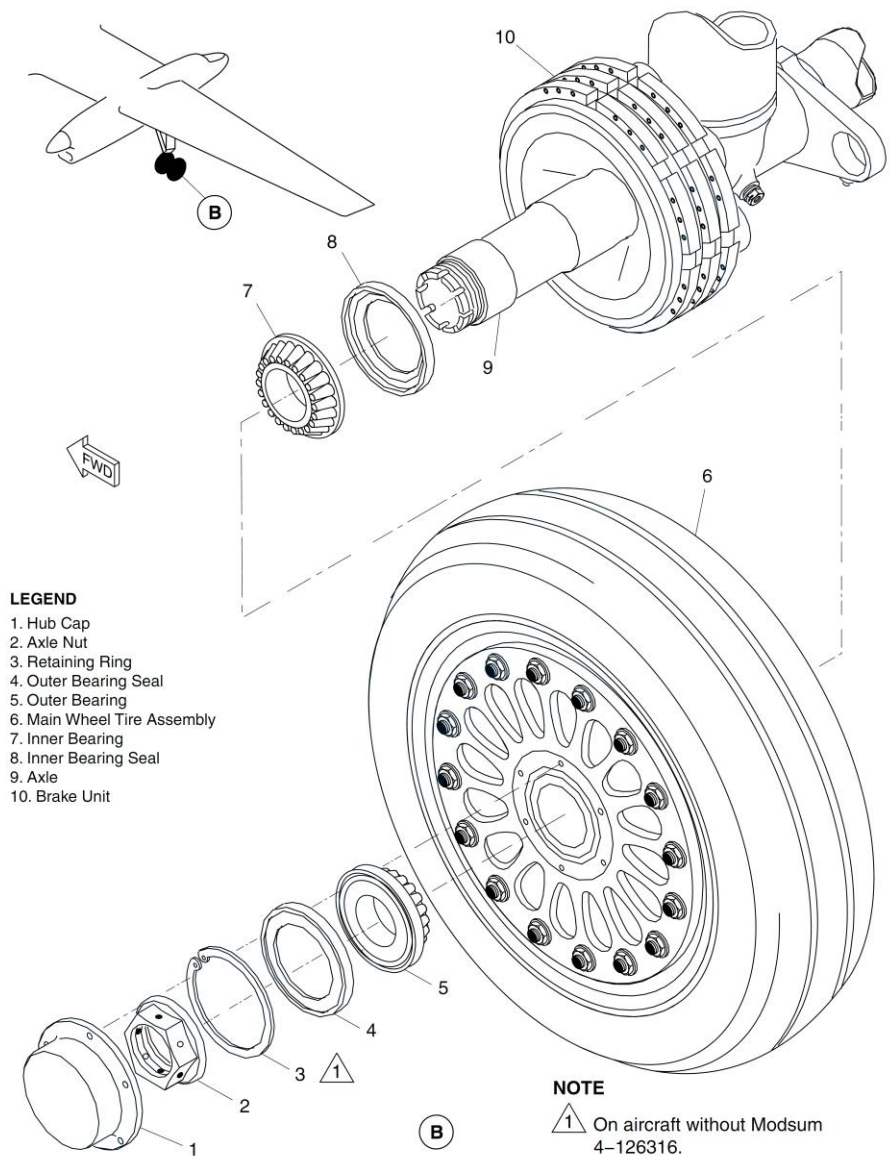


Figure 32-76 Main Wheel & Tire Assembly

Nose Wheels

[Refer to Figure 32-77 & 32-78](#)

Two wheels are installed on the axle of the nose landing gear. The nosewheels are a split-web type assembly made of two forged aluminum wheel halves. Each nosewheel has a 22×6.50–10, 8 ply tubeless tire.

The nose wheels are tubeless, split-web type assemblies made of forged aluminum wheel halves. The two halves are held together with equally spaced tie bolts, washers, and nuts. A preformed packing is installed on the inner wheel half assembly to prevent air leakage through the wheel mating surfaces.

Tapered wheel bearings and bearing cups are installed in the inner and outer wheel halves. The bearings and cups are installed in pairs and lubricated with grease. Bearing seals are installed at the outer surface of the bearings to prevent contamination. A water deflector shield is installed on the inner wheel half assembly to keep water spray away from the bearing seals and bearing. A hubcap and gasket are installed on the outer wheel half to protect the bearing from dirt and moisture.

An inflation valve assembly is installed in the outer wheel half assembly. An overinflation plug is installed in the outer wheel half assembly to prevent overinflation of the tire. The plug contains a disk that ruptures, and deflates the tire to zero pressure.

Training Information Points

Do not use compressed air to charge tires. Use only compressed dry nitrogen. If you use compressed air, you can cause damage to the tires.

Make sure the aircraft is correctly jacked and supported before you remove the wheels.

Fully deflate the tires before you disassemble the wheel.

The valve core can be forcefully ejected if removed before the tire is completely deflated. Use an approved deflating tool.

Lubrication of the wheel bearings by mechanical or pressure procedures is recommended. These methods are more efficient and decrease possible contamination.

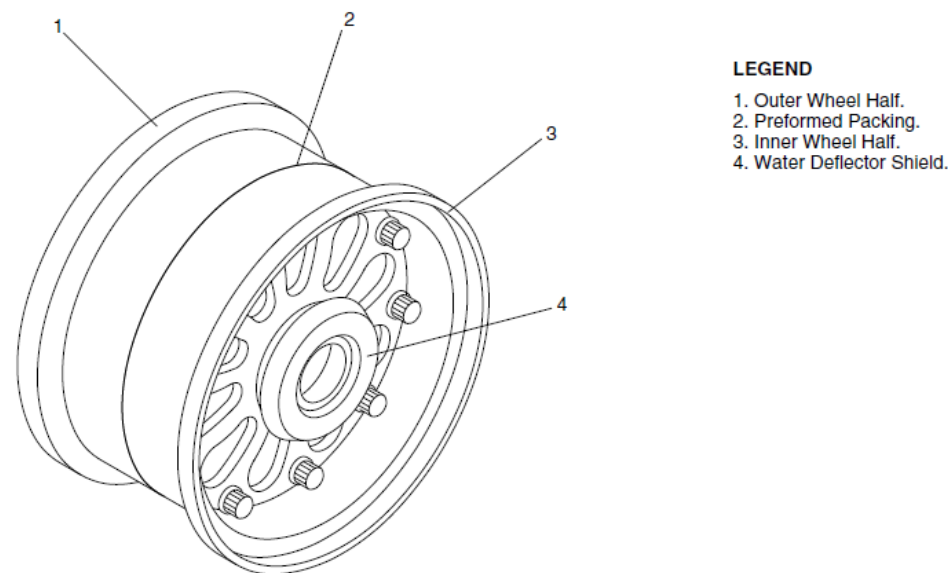
Bearings and cones are matched sets and must be kept together. Do not mix bearing and cone sets.

Do not spin bearings with compressed air. To prevent wheel bearing failure, be careful with wheel bearings.

Do not use impact or power wrenches to tighten or torque wheel tie bolts and nuts.

Do not inflate the tires until the tie bolts have been torqued. Inflate tires in a metal cage to prevent injury to personnel.

When you install the water shield on the axle, make sure the words THIS SIDE OUT point to the nose wheel strut.



LEGEND

1. Outer Wheel Half.
2. Preformed Packing.
3. Inner Wheel Half.
4. Water Deflector Shield.

Figure 32-77 Nose Wheel Detail

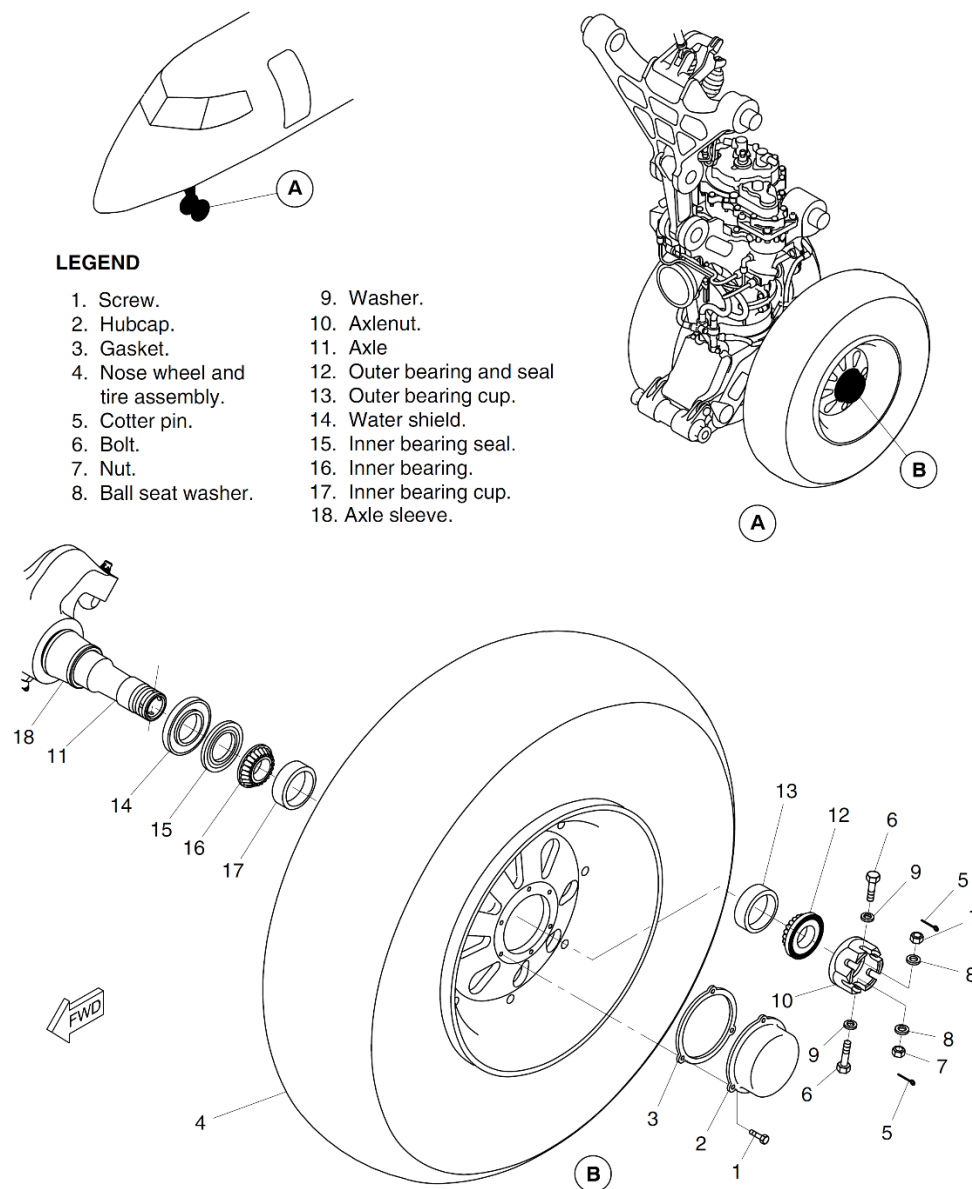


Figure 32-78 Nose Wheel & Tire Assembly

Main Landing Gear Brake System

Introduction

The brake system for the Main Landing Gear (MLG) lets the aircraft brake normally to decrease its speed during ground operations. The brakes convert energy necessary to bring the aircraft to a stop.

General Description

[Refer Figure 32-79](#)

Two brake units are installed on each axle of the main landing gear. The MLG brake system is controlled by the pilot's and co-pilot's brake pedals.

The brake system has the components that follow:

- Main wheel brake units
- Brake control valve
- Fuse/shuttle valves
- Pilots' brake pedals
- Brake control cables

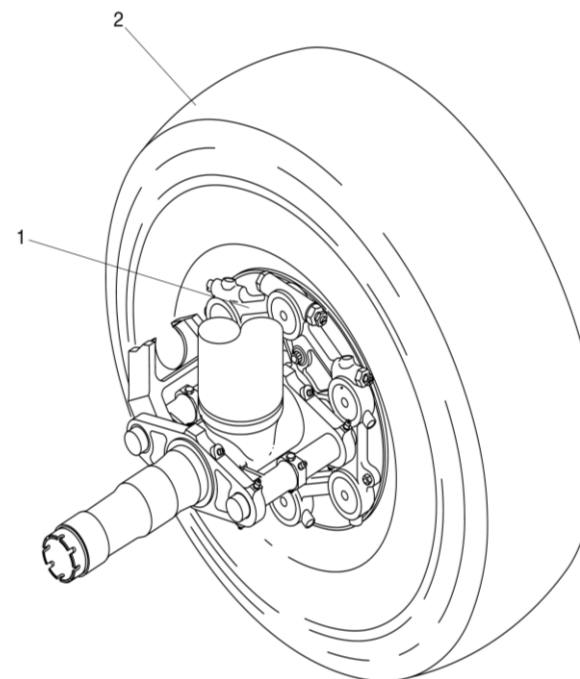
Detailed Description

[Refer Figure 32-80](#)

When a pilot's brake pedal is pushed, the attached tie rod moves a pushrod to operate the brake control lever. Movement of the brake control lever extends the lever spring and pulls the control cable to operate the applicable lever on the brake control valve. Hydraulic pressure from the No. 1 system is supplied to the skid control valve when the brake control valve lever is operated. The brake control valve will supply hydraulic pressure in proportion to the amount of brake pedal travel. Hydraulic pressure is supplied from the skid control valve through the fuse/shuttle valves to the applicable MLG brake units.

When the brake pedal is released, the lever spring returns the brake pedal to the off position. A spring in the brake control valve returns the valve to the off position. The brake pressure is released to return and the valve inlet port is closed.

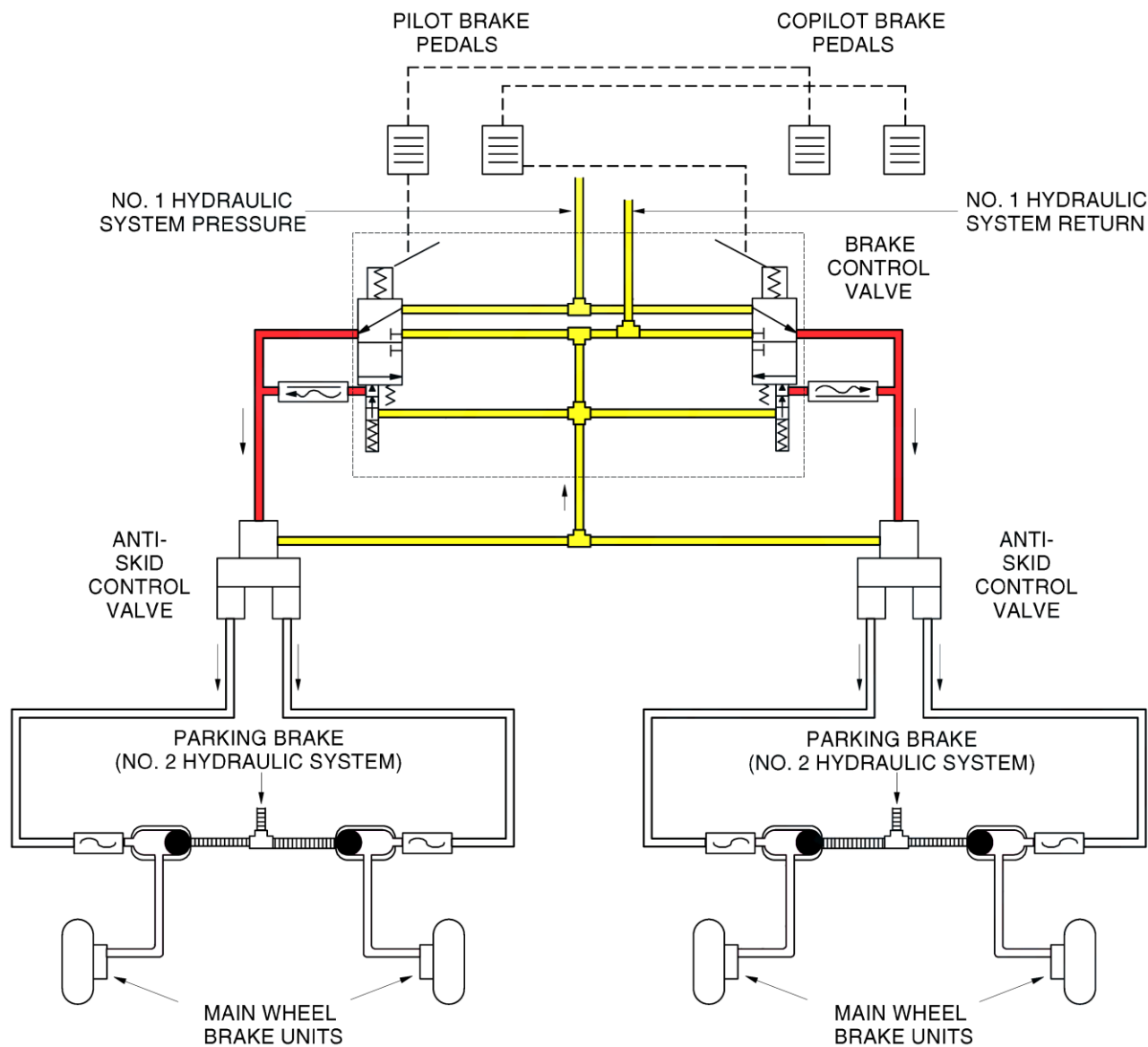
When the MLG brake unit is actuated, pressurized hydraulic fluid from the MLG brake system moves the pistons to push the pressure plate assembly. The pressure plate assembly pushes the rotor disks, the stator disks, and the end plate together. As the stator disks and the rotor disks are pushed against each other, the friction between the disks causes resistance. The resistance is transmitted from the MLG brake unit to the wheel and tire assemblies causing the wheel to slow down.



LEGEND

1. Main Wheel Brake Unit.
2. Main Wheel and Tire Assembly.

Figure 32-79 Main Wheel & Brake Assembly



LEGEND

	SUPPLY PRESSURE
	METERED PRESSURE
	ANTI SKID PRESSURE
	RETURN
	STATIC
	FUSE
	SHUTTLE VALVE
	RESTRICTOR

NOTE

System shown with brakes applied.

Figure 32-80 Brakes System Hydraulic Synoptic

Main Landing Gear Brake System – Component Description

Main Wheel Brake Units

[Refer Figure 32-81 & 32-82](#)

Two brake units are installed on each axle of the main landing gear. The brake pins on the MLG shock strut engage with the bushings in the piston housing assembly to transfer the braking loads to the landing gear.

The main wheel brake unit is a piston operated, carbon heat sink, multiple carbon disk brake.

The brake assembly has three primary components:

- The piston housing assembly
- The carbon heat sink stack
- The torque plate assembly

The piston housing is made of aluminum. It contains six adjuster assemblies, two hydraulic bleeder valves and adapters, and one bleeder plug. The adapters and bleeder plugs are interchangeable to let the brake assembly be installed on the left or right side of each main landing gear axle. Two wear indicator pins are attached to the pressure plate assembly and extend out of the piston housing through sleeves to give an indication of brake wear. When the wear pin is level with the wear indicator sleeve, replacement of the MLG brake unit is required. The piston housing with its integral parts performs the actuation, retraction and clearance adjustment functions of the brake. The piston housing is installed to the torque plate assembly with bolts.

The carbon heat sink stack includes the pressure plate, the three rotor disk assemblies, the two stator disk assemblies, and the end plate assembly that is attached to the torque plate assembly. All of these parts have friction surfaces that cause the brake assembly to stop the wheel when the brake is actuated. The pressure plate assembly has drive slots on the inner diameter that engage with the lugs on the torque plate assembly. The pressure plate does not turn, but slides as necessary on the torque plate assembly when the

brake is operated.

The rotor disk assemblies have drive slots on the outer diameter that engage with the inserts on the inner wheel half. The rotor disk assemblies turn with the wheel, and slide as necessary on the inserts on the inner diameter of the wheel when the brake is operated.

The stator disks have drive slots on the inner diameter that engage with the lugs on the torque plate assembly. The stator disks do not turn, but slide on the torque plate as necessary when the brake is actuated. The stator disks are installed between the three rotor disks.

The end plate assembly has nine torque buttons that engage with holes in the torque plate assembly. The end plate assembly does not turn or slide when the brake operates. The torque plate assembly ensures that the pressure plate assembly, the stator disks, and the end plate assembly maintain alignment. Lugs on the torque plate transfer brake torque from the pressure plate assembly and the stator disks to the torque plate and the piston housing.

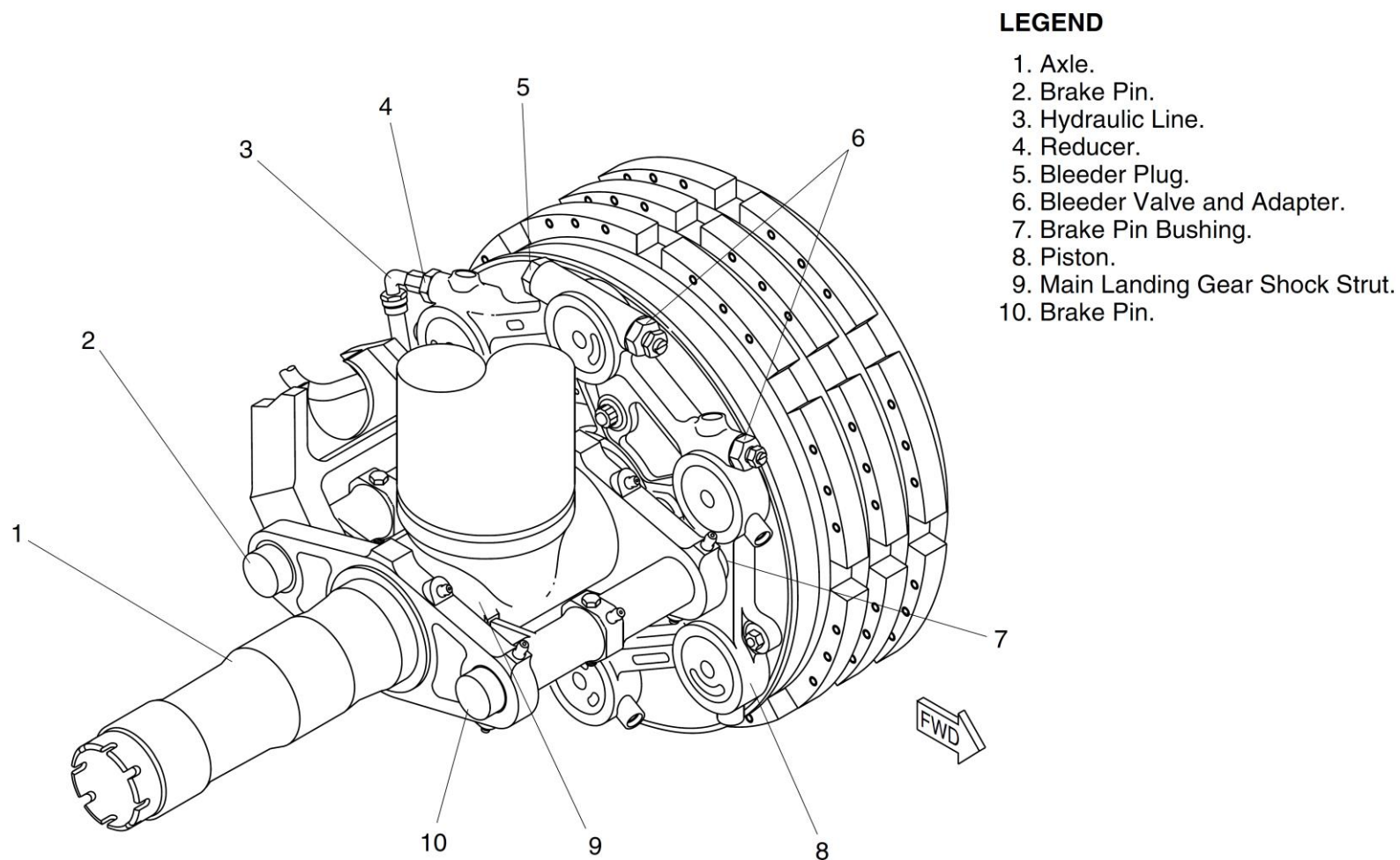
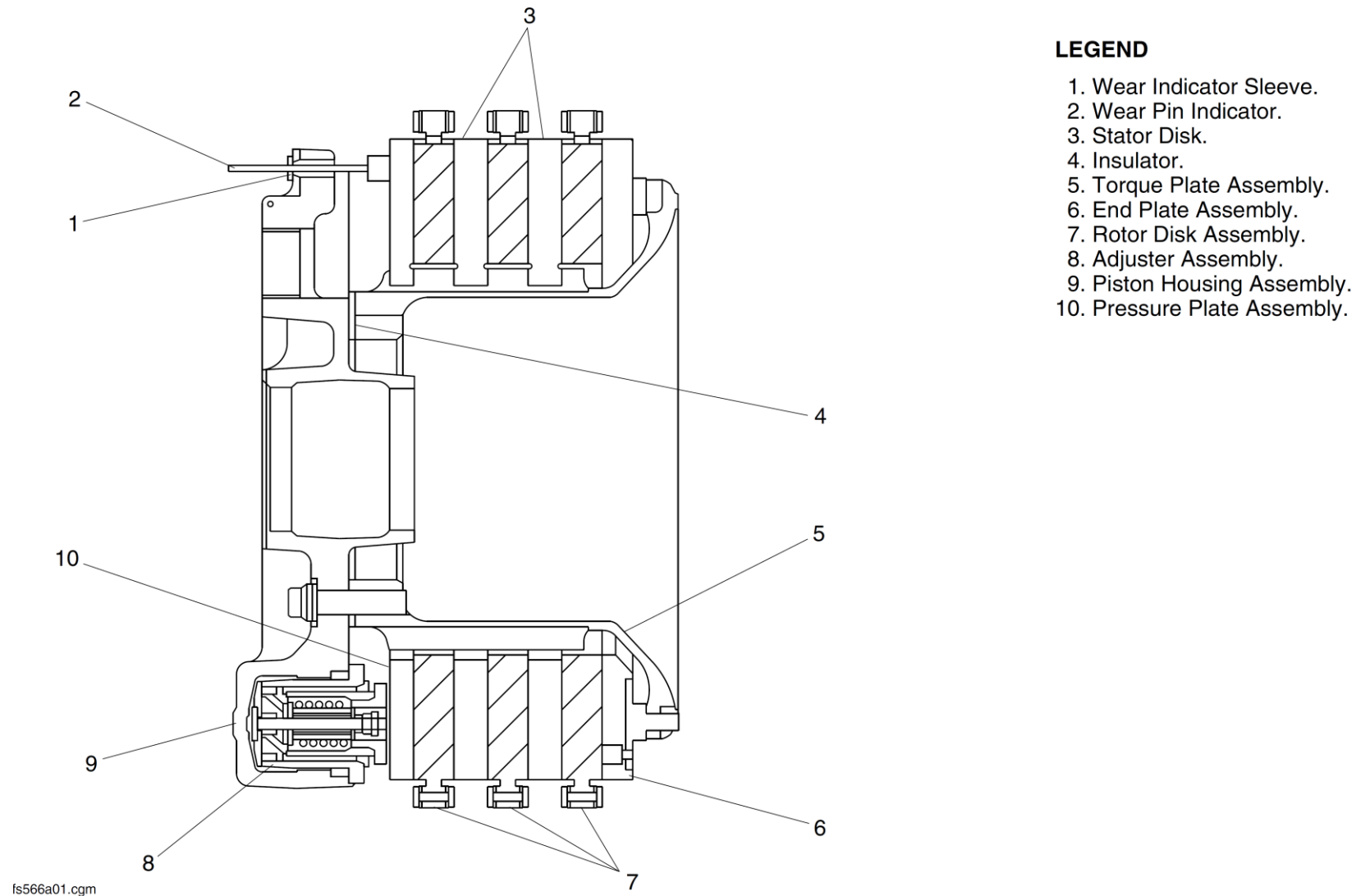


Figure 32-81 Axles & Brake Assembly Detail



LEGEND

1. Wear Indicator Sleeve.
2. Wear Pin Indicator.
3. Stator Disk.
4. Insulator.
5. Torque Plate Assembly.
6. End Plate Assembly.
7. Rotor Disk Assembly.
8. Adjuster Assembly.
9. Piston Housing Assembly.
10. Pressure Plate Assembly.

Figure 32-82 Brake Assembly Detail

Brake Control Valve

[Refer Figure 32-83](#)

The brake control valve is installed in the left wing root at Sta. X401.00. The pilot's and copilot's left brake pedals are connected to the left lever of the brake control valve. The pilot's and copilot's right brake pedals are connected to the right lever of the brake control valve. The brake control valve has two internal metering sections in a common housing. Each metering section is actuated progressively by an integral lever mechanism assembly.

The housing also supplies common system pressure and return ports. Hydraulic pressure and return lines connect the brake control valve to the No.1 hydraulic system. Hydraulic lines connect the brake control valve to the anti-skid valves.

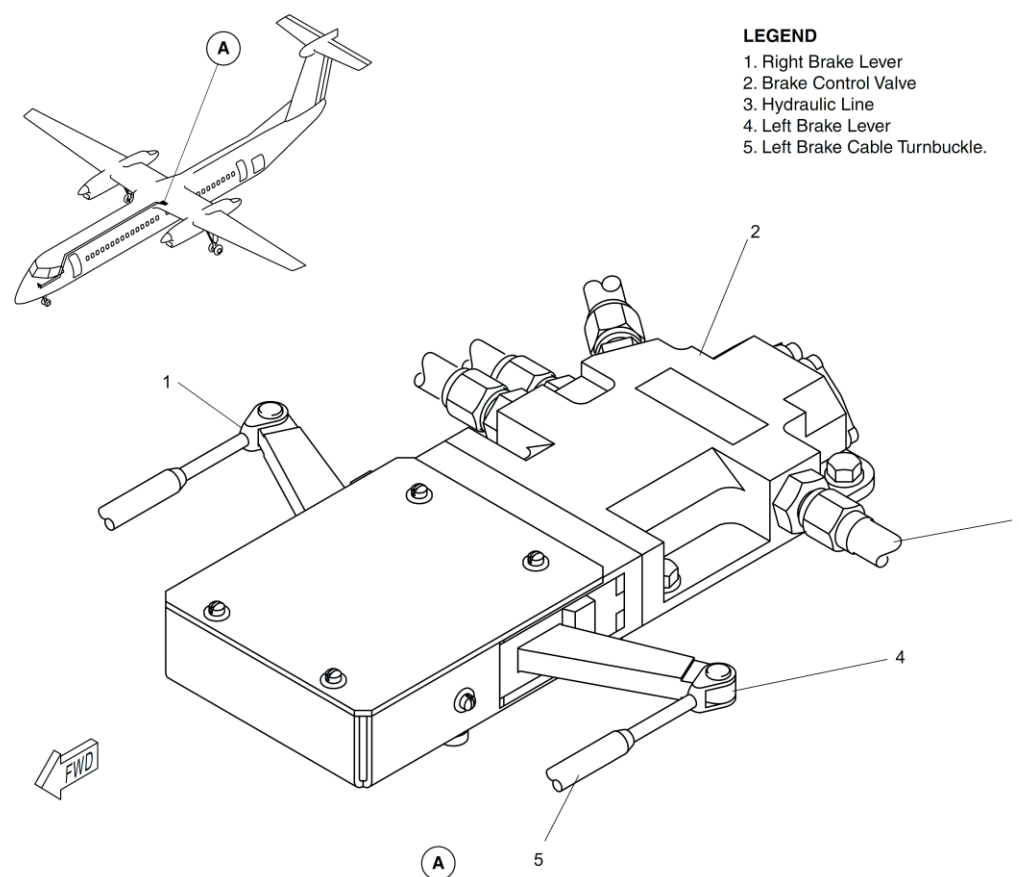


Figure 32-83 Brake Control Valve

Fuse & Shuttle Valve

[Refer Figure 32-84](#)

There are four fuse/shuttle valves installed on the left wall of the left and the right nacelles. Hydraulic pressure is supplied from the No.1 hydraulic system to the brake units through the skid control valves and the fuse/shuttle valves.

The fuse component of the fuse/shuttle valve closes the hydraulic line to the brake unit, if the fluid flow is more than a specified quantity. This will limit the leakage of hydraulic fluid if a brake line on the main landing gear is cut or if a brake unit develops a leak. The shuttle component of the fuse/shuttle valve allows either normal brake pressure or parking/emergency brake pressure to be used to actuate the brake units.

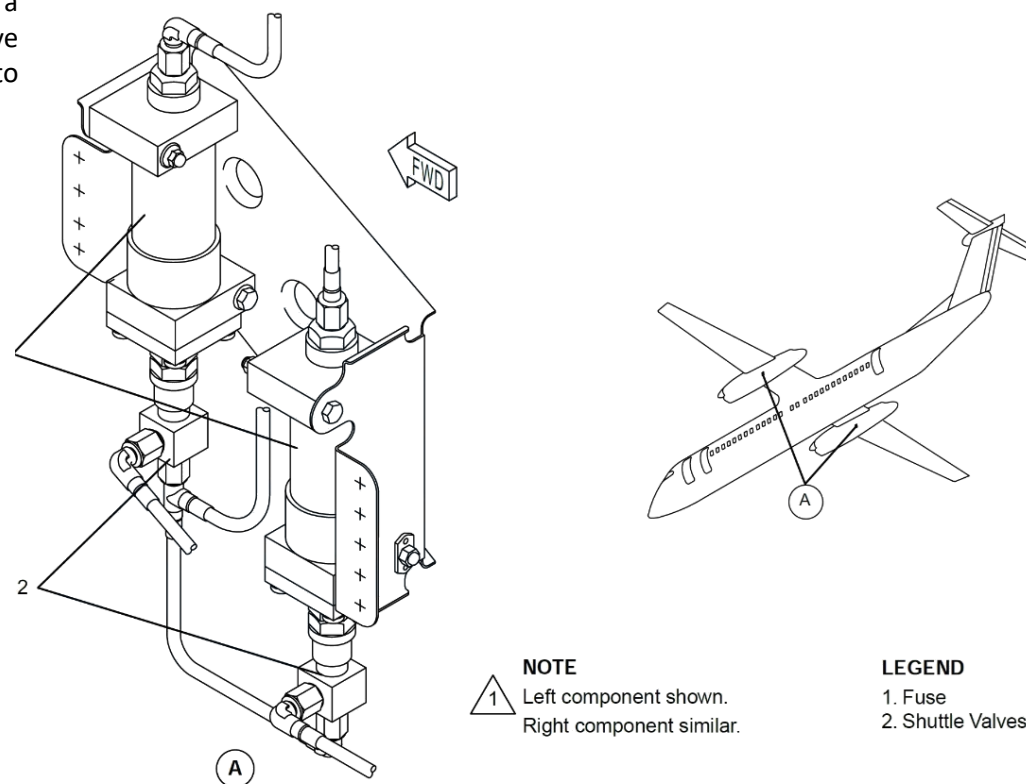


Figure 32-84 Fuse & Shuttle Valve

Brake Pedals

Refer Figure 32-85

The pilot's and copilot's brake pedals are installed in the flight compartment and are connected to the brake control valve. The pilots' inputs are applied to the brake control levers through pushrods. Movement of the pilot's brake pedals does not operate the copilot's brake pedals. The copilot's inputs are transmitted by a cable system to the spring loaded lever arm. Movement of the copilot's brake pedals will move the pilot's brake pedals through the interconnect cables from the copilot's brake pedal.

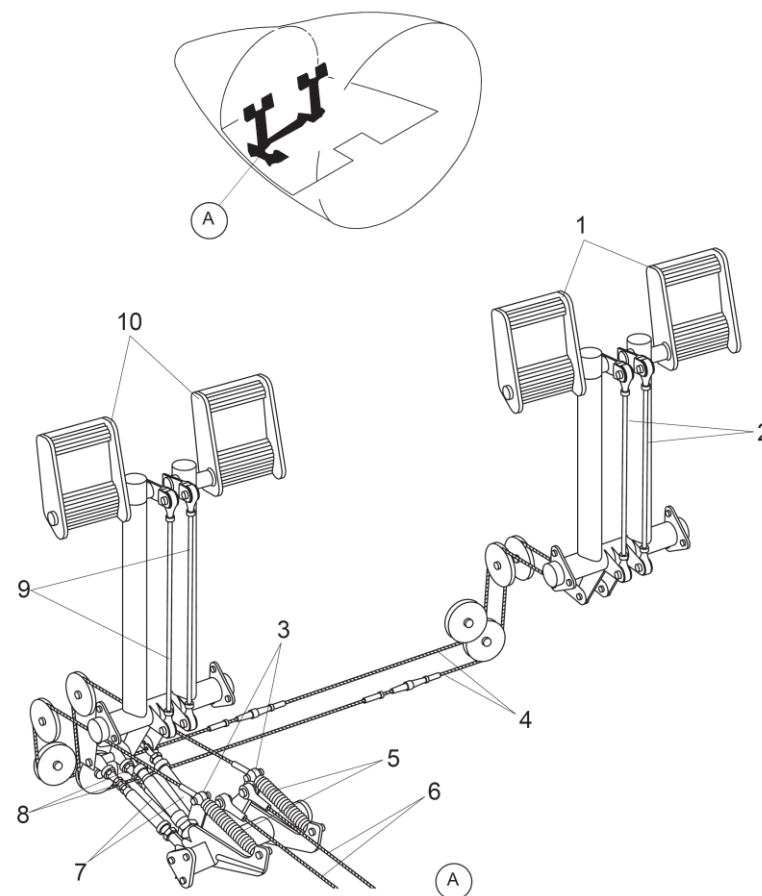
The linkage attached to the base of the pedals is spring loaded, to return the pedals to the off position when the pedals are released. The springs in the cable system return the copilot's pedals to off. Two spring rods remove the tension in the braking loads.

Training Information Points

Before you do maintenance on or around the landing gear, make sure that the main gear lock pins are installed, the nose gear downlock is engaged and hydraulic and electric power are not connected.

Make sure the aircraft is correctly jacked and supported before you remove the wheels.

When the brake unit wear pin is level with the wear indicator sleeve, replace the brake assembly.



LEGEND

- | | |
|-------------------------------|-----------------------------------|
| 1. Co-pilot's Brake Pedals. | 6. Cables to Brake Control Valve. |
| 2. Tie Rods. | 7. Push Rods. |
| 3. Brake Control Levers. | 8. Spring Rods. |
| 4. Brake Interconnect Cables. | 9. Tie Rods. |
| 5. Lever Springs. | 10. Pilot's Brake Pedal. |

Figure 32-85 Brake Pedals Assembly

Anti-Skid System

Introduction

The anti-skid system controls hydraulic pressure to the four main wheel brake units to supply individual wheel anti-skid protection.

General Description

[Refer Figure 32-86](#)

The brake pedals control hydraulic pressure from the No. 1 hydraulic system through the brake control valve to the anti-skid control valves. The anti-skid control valves control hydraulic pressure to the fuse/shuttle valves and the brake units. When the anti-skid system senses a skid condition, it sends a signal to the applicable skid control valve. The skid control valve releases the brake pressure through the anti-skid return line. The anti-skid system supplies individual wheel anti-skid control for each main wheel. The anti-skid system operates with the normal brake system.

The anti-skid system supplies the functions that follow:

- Individual wheel anti-skid control
- Touchdown protection
- Locked wheel protection
- System monitoring.

The anti-skid system has the components that follow:

- Anti-Skid Control Unit
- Wheel Speed Transducers
- Skid Control Valves
- Anti-Skid Switch

Detailed Description

The Anti Skid Control Unit (ASCU) receives wheel speed information from the wheel speed transducers to sense wheel deceleration. When a skid is sensed, the ASCU sends a signal to the skid control valve to release the brake pressure

to the non-rotating wheel. The skid control valve reacts to the signal and releases a portion of the brake pressure through the anti-skid return line. This decreases the brake pressure to the affected wheel so that it can recover from the skid. When the wheel has recovered from the skid, the ASCU removes the signal from the skid control valve. This lets the metered brake pressure from the brake control valve through to the brake unit.

Touchdown protection prevents brake pressure from being applied to the brake units before touchdown or before wheel spin-up has occurred.

The wheel control boards in the ASCU receive Weight-On-Wheel (WOW) and gear up and locked signals from the PSEU, as well as, wheel speed information from the wheel speed transducers to determine when touchdown protection is to be provided.

The ASCU supplies a full brake release signal to the skid control valve when: .

- The wheel speed is less than 30 knots (55.6 km/h)
- The WOW signals from the PSEU indicates "AIR"
- The gear "Up and Locked" signal indicates the gear is up and locked.

When the skid control valve receives the signal from the ASCU, it will release the brake pressure through the anti-skid return line

Touchdown protection (full brake release signal) will be removed or inhibited to the skid control valve when the wheel speed exceeds 35 knots (64.86 km/h) or 5 seconds after WOW signals from the PSEU indicates "GROUND".

Locked wheel protection lets the affected wheel recover from deep skids.

The outboard wheels (1 and 4) and the inboard wheels (2 and 3) are paired for locked wheel protection. The higher of the two wheel speeds is the used as the reference velocity. The reference velocity is allowed to decelerate by 25 ft/sec² (7.62 m/sec²). A locked wheel condition exists when the wheel

speed of either wheel decreases to less than 30% of the reference velocity.

When a locked wheel condition occurs, the ASCU sends a signal to the skid control valve to release brake pressure to the wheel that is locked. The skid control valve will release the brake pressure through the anti-skid return line until the wheel recovers from the deep skid. To recover from a locked wheel

condition, the wheel speed must be more than 35% of the reference velocity. The ASCU will remove the full brake release signal to the skid control valve to permit wheel braking.

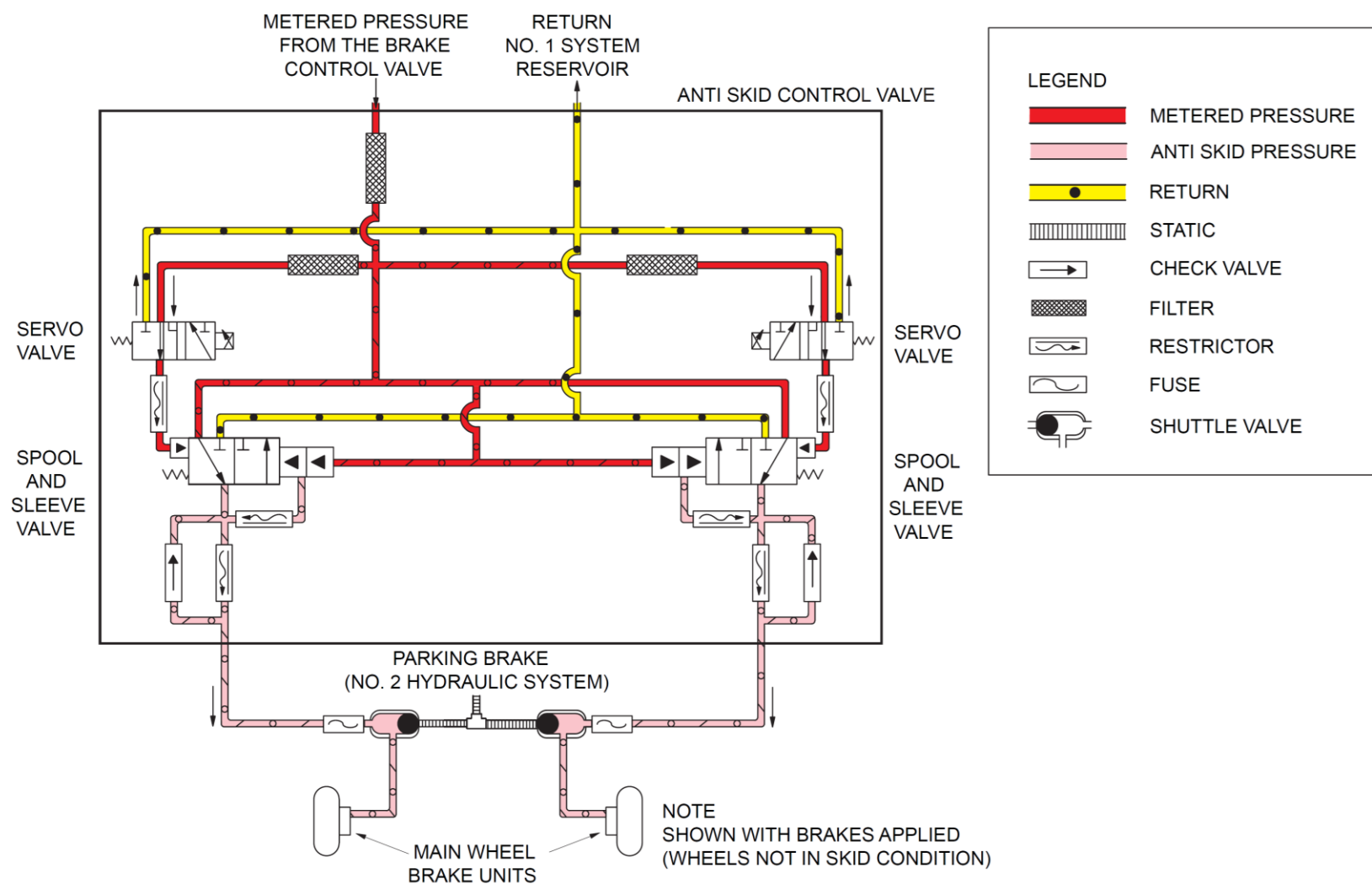


Figure 32-86 Anti-Skid System – Hydraulic Schematic

Anti-Skid System – Component Description

Anti-Skid Control Unit

[Refer Figure 32-87](#)

The ASCU is installed on the right side of the aircraft, below the passenger cabin floor at the mid-wing area, on the equipment rails. The ASCU is an electronic box that has two wheel control circuit boards, one for the inboard wheels and one for the outboard wheels. The functions of the wheel control circuit boards are for anti-skid control and Built-In-Test (BIT). Each wheel board uses a micro-controller and related circuitry to do the digital processing of the control and built-in-test software. Each wheel board also contains analog circuits for filters, wave shaping, test signal generation, valve drivers, and monitor functions.

The ASCU receives 28 Vdc power from the right main DC bus through ANTISKID INBD and ANTISKID OUTBD circuit breakers, through the anti-skid switch. Both power sources are needed for the ASCU to supply full anti-skid control. The fault data recorded by the ASCU is transmitted over the ARINC 429 data bus as a series of pages and is shown on the Audio Radio Control Display Unit (ARCDU). The number of pages transmitted depends on the number of faults. Current and past faults can show on the display unit.

The ASCU sends signals to the caution and warning light and advisory system and the Central Diagnostic System (CDS) to show and record faults in the system.

The ASCU receives signals from the:

- Anti-skid switch
- Four wheel speed transducers
- Four skid control valves.

The ASCU receives signals from the Proximity Sensor Electronic Unit (PSEU) for landing protection functions.

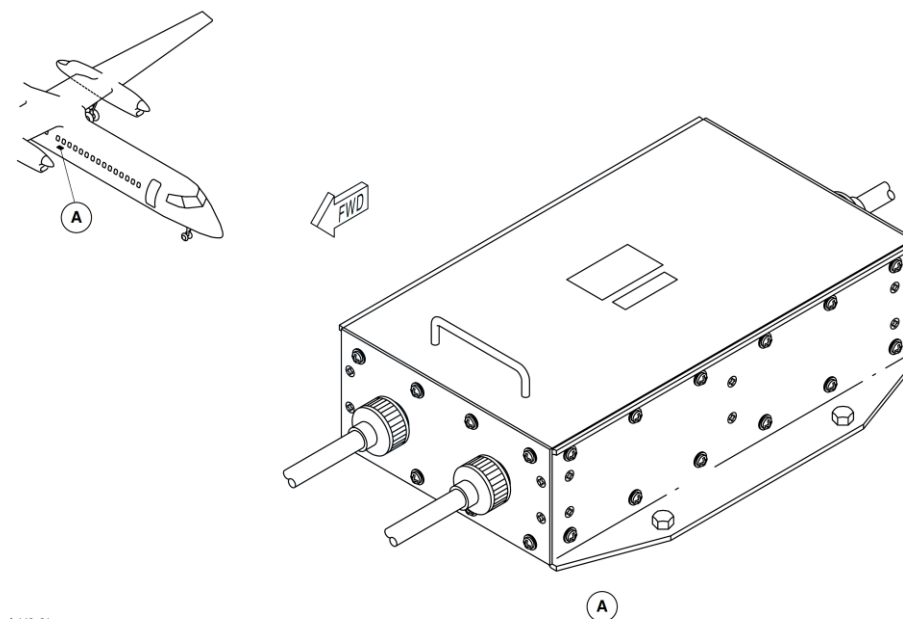


Figure 32-87 Anti-Skid Control Unit

Skid Control Valves

[Refer Figure 32-88](#)

The skid control valves are two separate valves in one common manifold. Each skid control valve contains a screen on the inlet port. The screen protects the skid control valve components from contaminants larger than 40 microns. The skid control valves receive metered pressure from the brake control valve.

The right control valve gives skid control for the inboard and outboard brake units on the right main landing gear. The left control valve gives skid control for the inboard and outboard brake units on the left main landing gear. The skid control valves are installed on the left wall of the left and right nacelle.

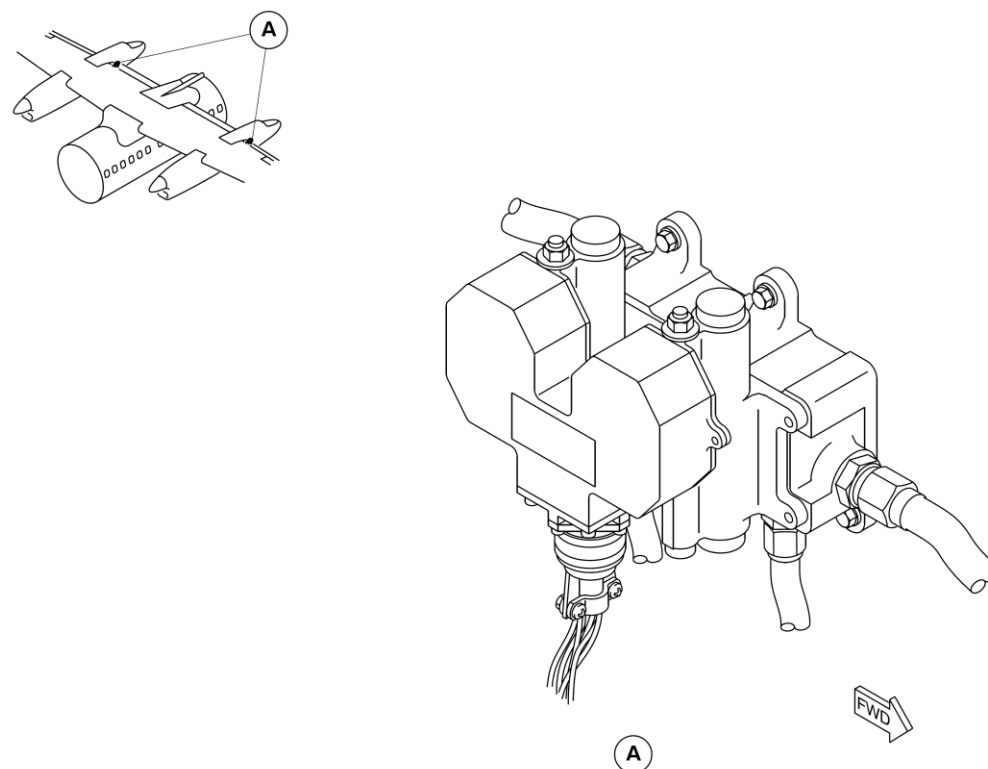


Figure 32-88 Anti-Skid Control Valve

Wheel Speed Transducer

[Refer Figure 32-89](#)

The wheel speed transducer is an electrical generating device that supplies wheel speed information to the ASCU. A wheel speed transducer is installed in each end of the main landing gear axles.

The hubcap on the wheel engages with a clip on the wheel speed transducer. When the wheel turns, the wheel speed transducer supplies an AC signal with a frequency of 100 cycles per wheel revolution. This signal is changed to a digital signal in the ASCU for anti-skid control use.

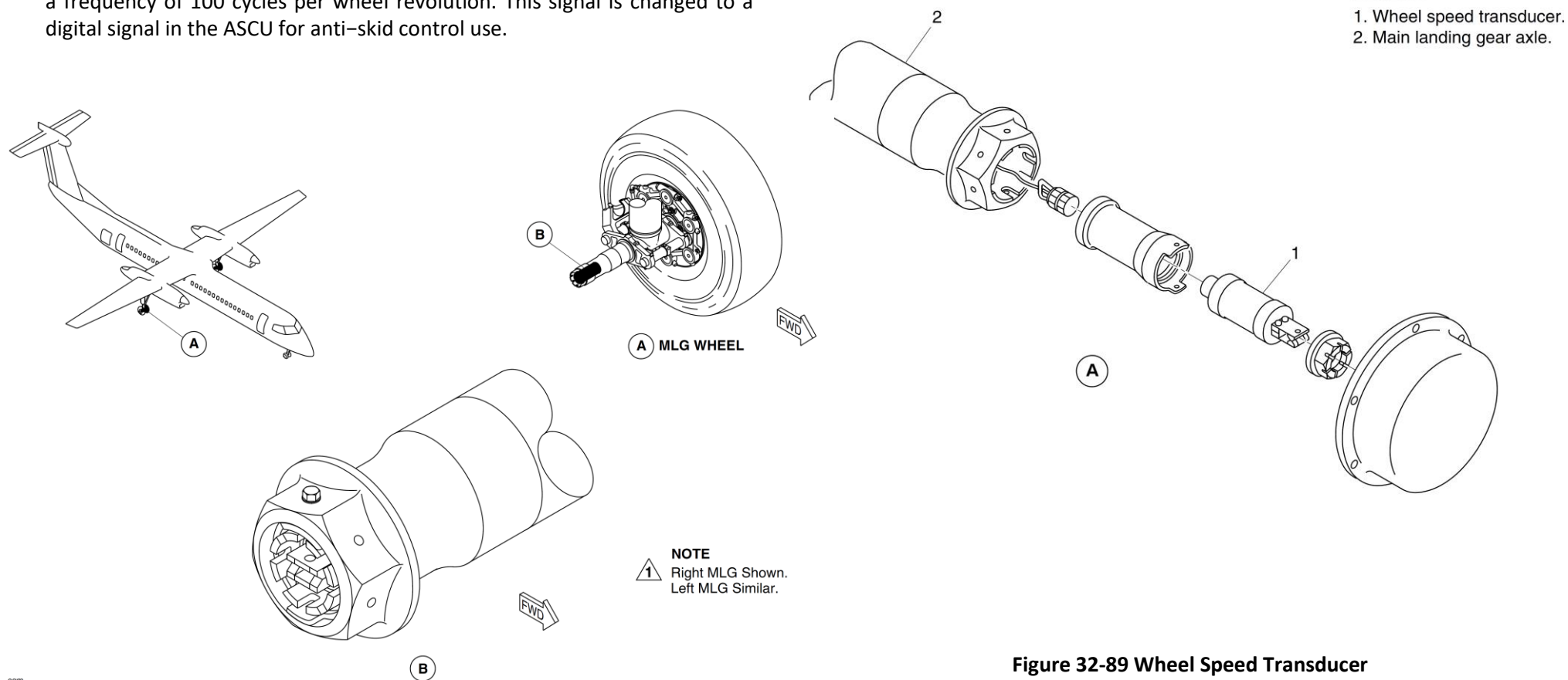


Figure 32-89 Wheel Speed Transducer

Anti-Skid Switch

[Refer Figure 32-90](#)

The ANTI-SKID switch is located on the copilot's glareshield panel. The ANTI-SKID switch is a three position OFF/ON/TEST switch. When the ANTI-SKID switch is set to the ON position, 28 Vdc is supplied to the inboard and outboard wheel control circuit boards in the ASCU. When set to the TEST position, 28 Vdc is supplied to the wheel control circuit boards and the system Built-In-Test function. When the ANTI-SKID switch is set to OFF, all electrical power is removed from the ASCU.

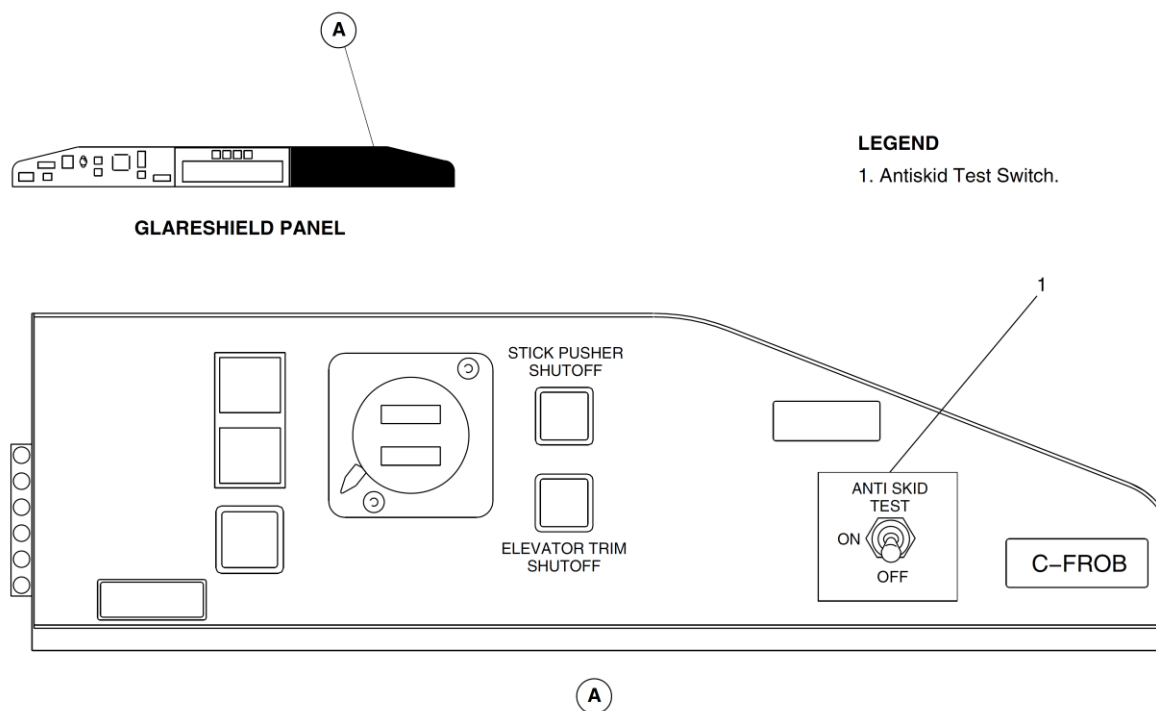
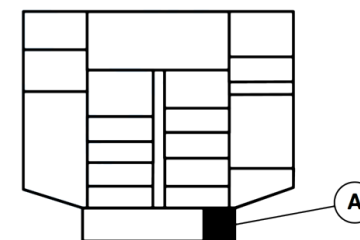
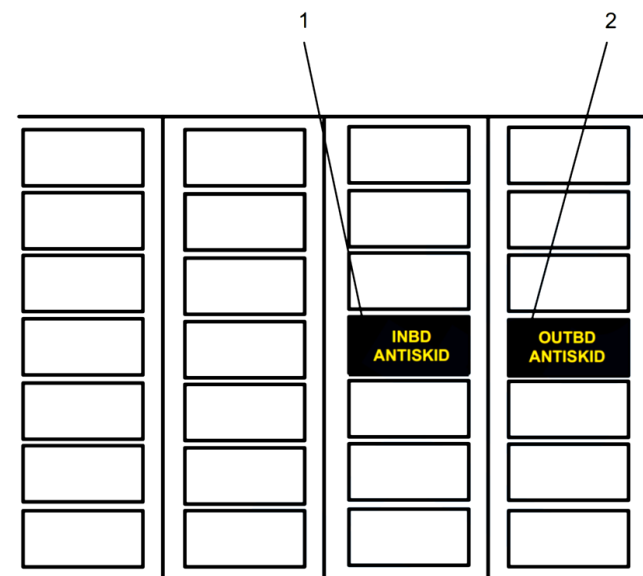


Figure 32-90 Anti-Skid Switch

[Refer Figure 32-91](#)

If the ANTI SKID switch is in the OFF position, the anti-skid system is deactivated. The INBD ANTISKID and OUTBD ANTISKID caution lights come on.

If the ANTI SKID switch is in the ON position, the anti-skid system is activated. The INBD ANTISKID and OUTBD ANTISKID caution lights will go off if there is no system failure. If the ANTI SKID switch is held in the TEST position while in the air, with gear down and locked, the INBD ANTISKID and OUTBD ANTISKID caution lights come on for 3 seconds then go off. If the ANTI SKID switch is held in the TEST position while on the ground, the INBD ANTISKID and OUTBD ANTISKID caution lights come on for 6 seconds then go off. The INBD ANTISKID and OUTBD ANTISKID caution lights will come on if there is a system failure.



OVERHEAD CONSOLE

LEGEND

- 1. Inboard Antiskid (Amber)
- 2. Outboard Antiskid (Amber)

Figure 32-91 Anti-Skid System Caution Lights

Anti-Skid System Operation – Electrical Schematic

[Refer Figure 32-92](#)

When the Anti-Skid Switch is selected ON, Power is supplied from the 28 VDC right main bus through Pins F and D of the switch and out of the switch through Pins C and E to Pin 17 of the Anti-Skid Control Unit.

When the aircraft is on the approach for landing the PSEU sends signals to the Anti-Skid Control Unit. The signals are as follows:

WOFFW to Pins 13

Gear NOT Up and Locked to Pins 28.

The ASCU gets the wheel speed < 35 knots signals from:

- Pins 32 and 33 for right inboard and right outboard transducers.
- Pins 8 and 9 for the left inboard and outboard transducers.

Touchdown Protection Mode

It causes a full brake release signal sent from:

- ASCU Pins 23 to right dual skid control valve at Pins 1 and 3.
- ASCU Pins 11 to left dual skid control valve at Pins 1 and 3.

NOTE: All the pins mentioned are used throughout all modes of anti-skid operation.

This causes the servo valve to release all hydraulic pressure into the return line. The brakes receive no pressure and the brakes will release.

Initialization

When the aircraft lands the anti-skid system goes into Initialization Mode when the following signals are received by the ASCU:

- MLGWOW for >5 seconds OR
- Wheel speed >75 knots.

Metered hydraulic pressure from the brake control valve is slightly reduced by the servo valves, causing the spool and sleeve valves to open partially,

preventing full braking force.

Individual Wheel Protection Mode

Individual wheel Anti-Skid Mode starts with the signals that follow:

- MLGWOW > 5seconds OR
- Wheel speed <70 knots.

Full metered pressure from the brake control valve passes through the servo valves. This pressure operates the brakes.

If a skid is detected, the ASCU commands decreased pressure to the affected wheel brake unit, until the wheel speeds up to match the speed of the other wheels.

Locked Wheel Protection Mode

If an inboard, or outboard wheel develops a skid producing a difference in speed of <30% of the reference speed, **Locked Wheel Protection** is now initiated.

All metered pressure, for that brake unit, is released to return by the servo valve until the locked wheel speed is >35% of reference speed. Reference speed is the speed of the matched inboard or outboard wheel, on the other side of the aircraft.

Interfaces

- The ASCU receives 28 VDC power from the right main DC circuit breakers (ANTISKID INBD CB Q5 and ANTISKID OUTBD CB R5). The ASCU requires power from the two CBS for correct operation.
- The fault data recorded by the ASCU is transmitted over the ARINC 429 data bus as a series of pages and is shown on the Audio Radio Control Display Unit (ARCDU) as a part of the Central Diagnostic System.

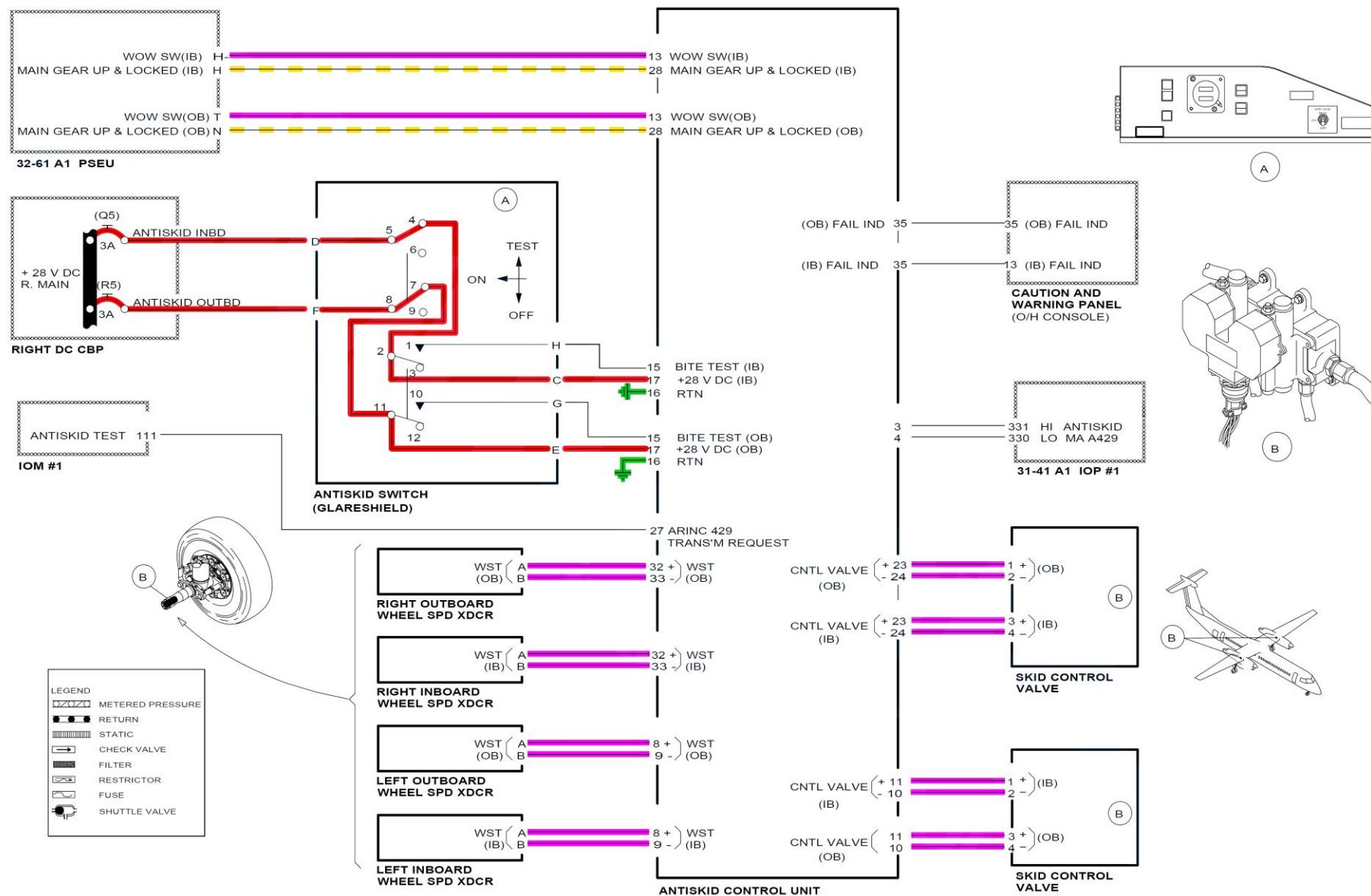


Figure 32-92 Anti-Skid System Operation – Electrical Schematic

System Monitoring

The ASCU monitors the system with a built-in-test function. The built-in-test operates in two modes, continuous test mode and start up self test mode.

Continuous tests are initiated during normal operation of the anti-skid system. The tests do not cause interference with anti-skid operation. If the anti-skid system fails, the ASCU sends a signal to the Caution and Warning Panel (CAWP). Failure data is recorded in non-volatile memory in the ASCU.

The start up self test is initiated when the anti-skid switch is held to the TEST position or whenever the ASCU is energized. The start up self test is a check on the proper response of the anti-skid control system functions and to determine the operational status of the ASCU, the skid control valves, and the wheel speed transducers. Normal anti-skid operation may be inhibited during the start up self test. .

If a start up self test is initiated while using the anti-skid switch on the ground, the anti-skid failure lights comes on in the flight compartment for 6 seconds. They will come on for 3 seconds if the test was done while in the air. The start up self test is inhibited when the wheel speed is more than 17 knots (31.5 km/h). If the anti-skid system passes the start up self test, normal anti-skid system operation will continue.

If the anti-skid system fails the start up self test, the fail indicators on the CAWP comes on.

Fault Detection

The ASCU monitors the anti-skid system and senses operational faults. If a fault is sensed, the ASCU turns on the INBD or OUTBD ANTISKID caution lights on the CAWP. Failure data is stored in non-volatile memory in the ASCU. Past failure data is recorded and stored in non-volatile memory in the ASCU. The failure data is available to maintenance personnel through the CDS. A maximum of fifteen past faults can be recorded.

The ASCU monitors the current and voltage of the skid control valve, and checks for open and short circuits. The ASCU monitors the wheel speed transducers for open and short circuits.

Anti-Skid Fault Display on the CDS

Fault data from the ASCU is transmitted at the ARINC 429 interface as a series of pages to be shown on the display. Depending on the number of faults, either two or three pages are transmitted. Two types of faults are included, current faults and past faults.

Current faults are always one page long. The faults are a list of names of line replaceable units shown in a fixed sequence. If a fault is corrected, it will be removed from the list when a new fault display is requested. The current faults page can show eleven names of line replaceable units.

Past faults may be one or two pages long. All past faults have a flight leg shown with the name of the line replaceable unit. Past faults are displayed in the reverse order that they occurred. The maximum number of past faults displayed is 15.

The oldest recorded faults are overwritten with new faults as they occur. For each flight leg where there was a fault recorded, one old fault will be overwritten.

The ASCU fault is always recorded in the non-volatile memory. The remaining LRU faults are recorded only when the wheel speed on both the wheels is more than 8 knots (14.82 km/h) or when the aircraft is in the air. Each faulty LRU will be recorded only once for each flight leg even if the fault has occurred multiple times. The same fault can be present in the present faults and the historical faults lists.

The flight leg zero always shows the present flight leg. The flight leg number for the stored faults (past faults) is increased by one when the sequence that follows occurs:

-
- Power-up or start-up test
 - The air/ground switch indicates ground
 - The air/ground switch indicates air
 - The air/ground switch indicates ground.

Training Information Points

The anti-skid system operates with the normal brake system. It does not operate with the parking/emergency brake system.

The wheel speed transducer is an electrical generating device that supplies wheel speed information to the ASCU. A wheel speed transducer is installed in each end of the main landing gear axles.

Parking Brake System

Introduction

The parking brake system is used to keep hydraulic pressure applied to the brake units on the Main Landing Gear (MLG) when the aircraft is parked. The parking brake system can also be used to stop the aircraft on the ground when the normal brake system is not available.

General Description

The parking brake system uses hydraulic pressure from the No. 2 hydraulic system. An accumulator in the brake system supplies pressure for parking brake operation when the engines are stopped. A handpump can be used to manually pressurize the accumulator when the aircraft is on the ground. Parking brake pressure is applied equally to all of the main wheel brake units.

A PARKING BRAKE caution light on the Caution and Warning panel in the flight compartment comes on when the parking brake lever is in the PARK position. A pressure indicator is installed in the flight compartment to indicate the parking brake pressure available.

The parking brake system has the components that follow:

- Parking Brake Control Lever
- Parking Brake Pressure Indication
- Control Cable Assembly
- Parking Brake Control Valve
- Handpump
- Accumulator
- Charge Valve and Gauge

Detailed Description

[Refer Figure 32-93](#) & [32-94](#)

When the PARK/EMERG control lever is moved from the stowed position toward the PARK position, the park brake control valve is actuated by a control cable. The parking brake control valve single metering section applies progressive hydraulic pressure to the fuse/shuttle valves. The hydraulic pressure moves the shuttle valve component of the fuse/shuttle valve to close

the inlet ports for the normal brake system. This lets the hydraulic pressure from the parking brake control valve go through the shuttle valves and operate the brake units.

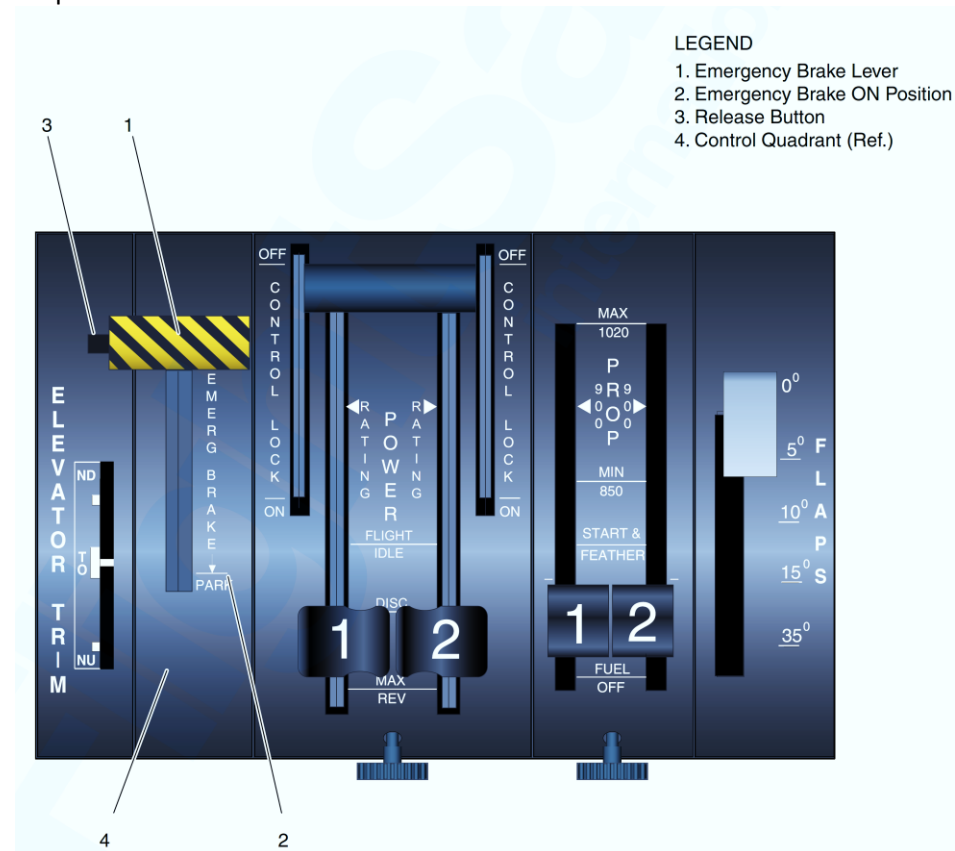


Figure 32-93 Parking Brake Control Quadrant

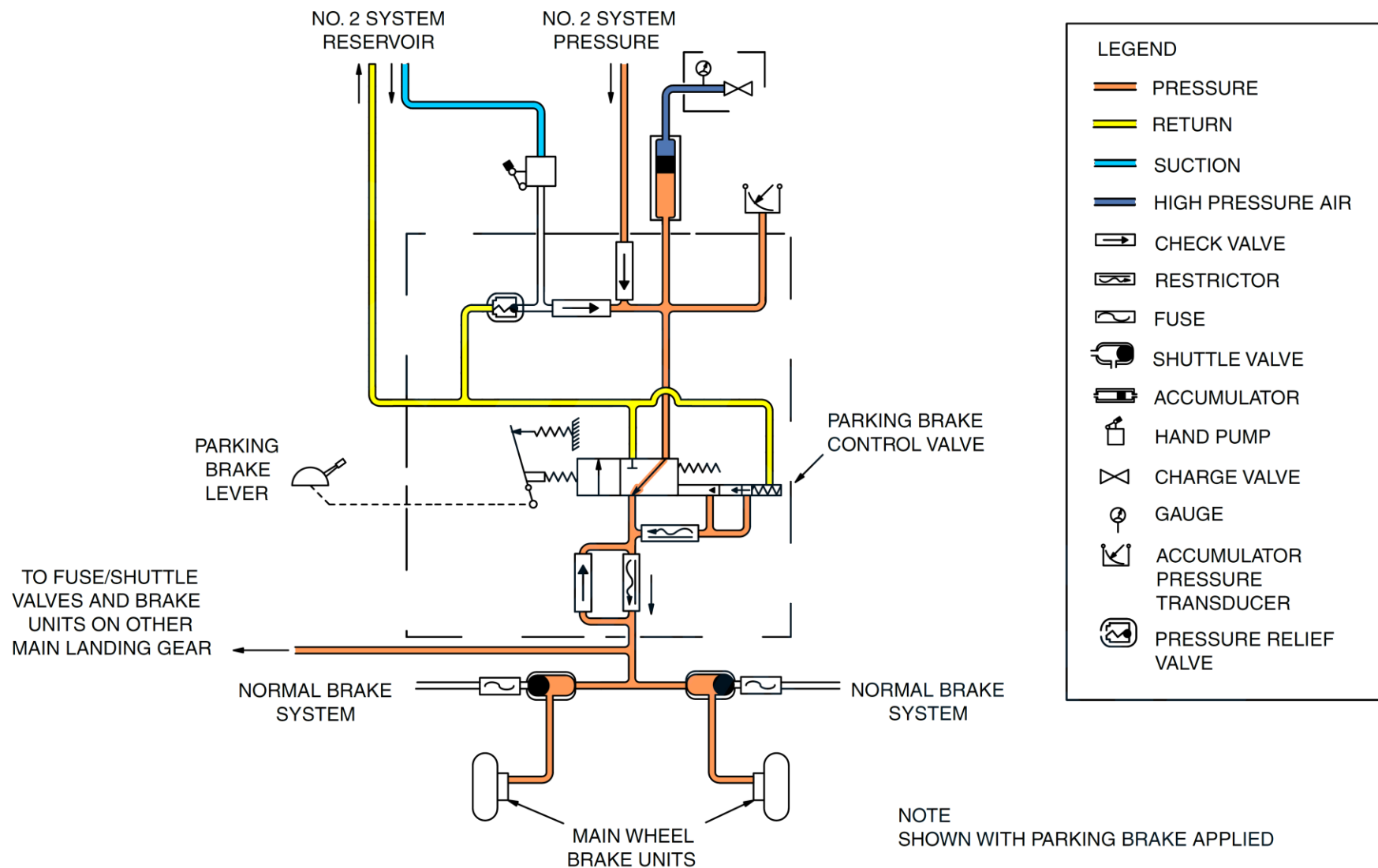


Figure 32-94 Parking Brake System Hydraulic Synoptic

Refer Figure 32-95

When the parking brake lever is in the PARK position, the PARKING BRAKE caution light on the Caution and Warning panel comes on. When No. 2 hydraulic pressure is not available, hydraulic accumulator pressure is shown on the PARK BRAKE pressure indicator in the flight compartment. A 28 Vdc electrical power source is needed for the PARK BRAKE pressure indicator and the pressure transmitter to operate.

Refer Figure 32-93

The parking brake is released when the release button on the parking brake lever is pushed in and the lever is moved forward to the stowed position. A

spring in the parking brake control valve closes the pressure port and opens the brake lines to the return side of the hydraulic system. This releases the brakes and the PARKING BRAKE caution light goes off.

The fluid side of the hydraulic accumulator is pressurized from the No. 2 hydraulic system when the No. 2 engine is operating. After the engine has stopped, a check valve closes and hydraulic pressure is stored in the accumulator to supply six full parking brake applications. The handpump can be used to pressurize the accumulator when the No. 2 hydraulic system is not available.

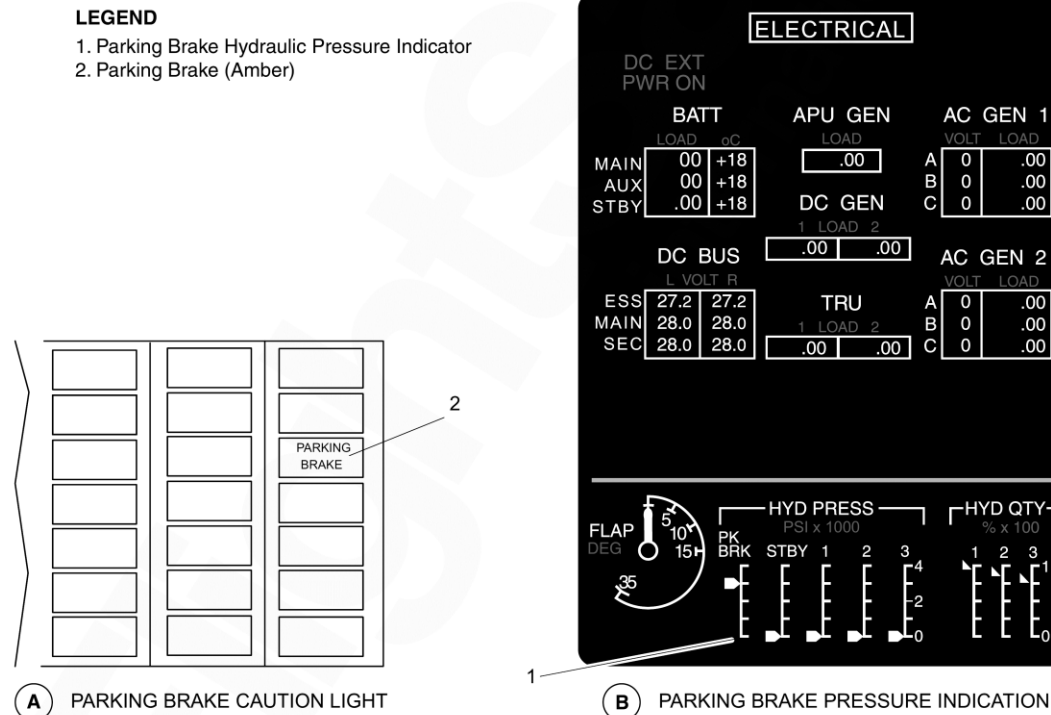


Figure 32-95 Parking Brake Pressure Indication

Parking Brake System – Component Description

Parking Brake Lever

The parking brake lever is installed in the flight compartment, in the parking brake control quadrant, on left side of the center console. A cable assembly connects the parking brake lever to the parking brake control valve. A release button on the parking brake lever unlocks the lever from the PARK position.

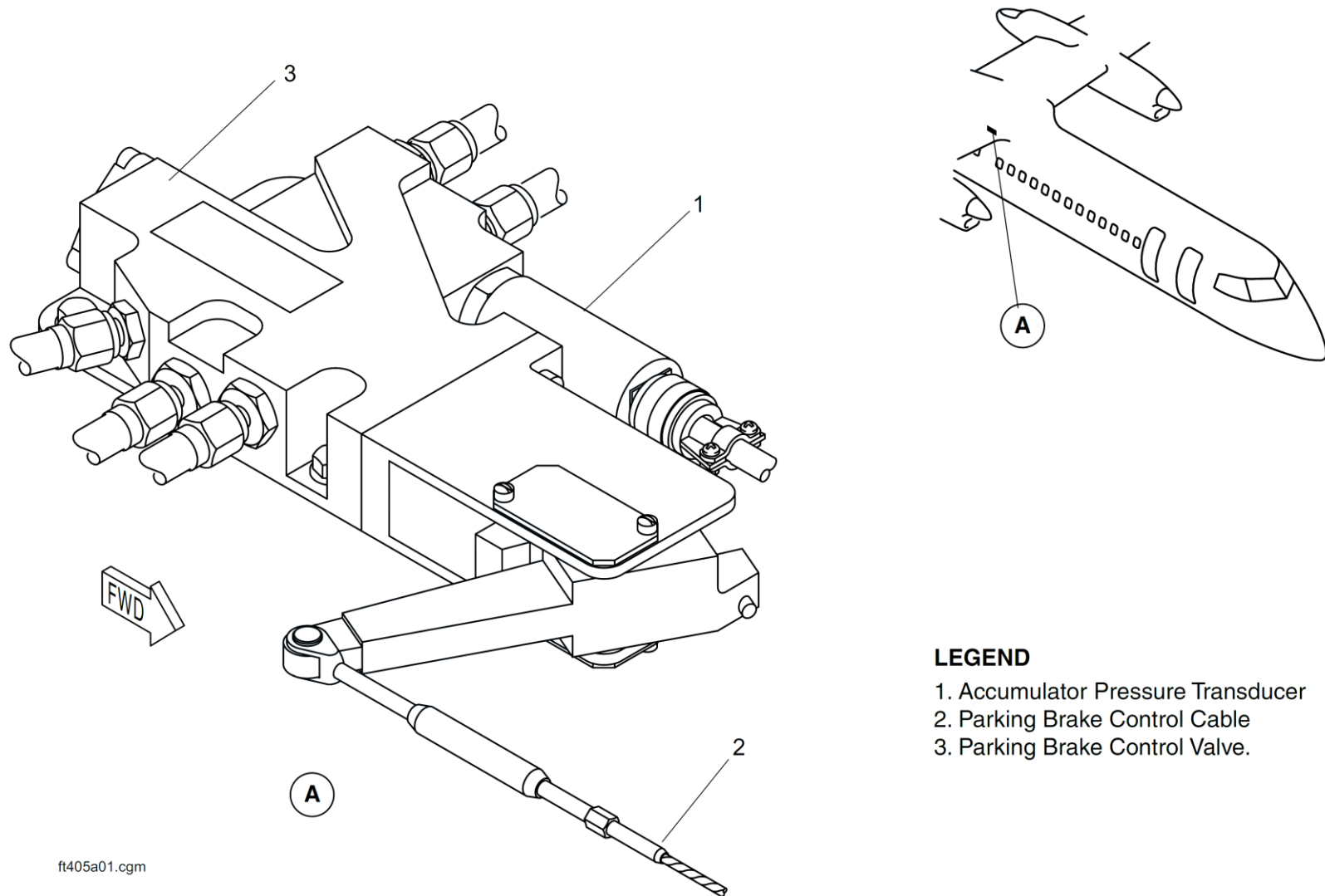
Parking Brake Control Valve

[Refer Figure 32-96](#)

The parking brake control valve is installed in the right wing root, aft of Sta. X388.00 and is operated by the parking brake control cable. The No. 2 hydraulic system pressure is supplied to the parking brake control valve. From

the control valve, pressure is metered to the fuse/shuttle valves and the brake units in proportion to the amount of parking brake lever movement. This lets the pilot apply proportional braking pressure through the park/emergency brake system when the normal braking system has failed.

When the parking brake lever is moved from the stowed to the PARK position, full hydraulic pressure is applied to the four brake units. A pressure transducer in the parking brake control valve supplies a signal to the flight compartment indication system to indicate the hydraulic pressure in the parking brake system. The parking brake control valve has a relief valve to prevent an over pressure condition when the handpump is used to pressurize the system. Two check valves are contained in the brake control valve housing. These valves are used to isolate the No. 2 hydraulic system and the handpump from the parking brake system.



LEGEND

- 1. Accumulator Pressure Transducer
- 2. Parking Brake Control Cable
- 3. Parking Brake Control Valve.

Figure 32-96 Parking Brake Control Valve

Handpump

[Refer Figure 32-97](#)

A lever operated handpump is installed in the right engine nacelle. It supplies an alternate method to pressurize the accumulator to 3000 psi (20684 kPa) when the aircraft is on the ground and the engines are not operating. The handpump handle is stowed adjacent to the handpump in the right nacelle. On aircraft with Modsum 4-113723 or SB84-32-99 incorporated, the handpump handle is stowed in the fuselage nose compartment.

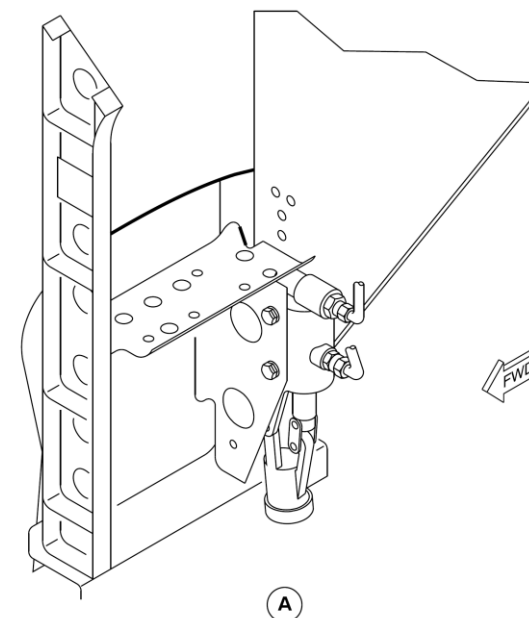
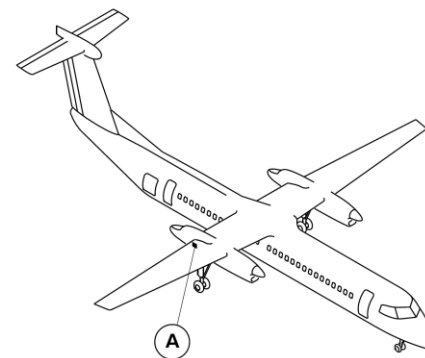


Figure 32-97 Parking Brake Hand Pump Location

Charge Valve and Gauge

[Refer Figure 32-98](#)

The charge valve and gauge are installed in the right wing root and are connected to the gas pressure side of the accumulator. The gauge supplies an indication of the accumulator pressure. The charge valve is used to charge the gas side of the accumulator with nitrogen.

Parking Brake Accumulator

[Refer Figure 32-98](#)

The accumulator is installed in the right wing root area. It supplies hydraulic pressure for the parking brake operation when the engines are stopped, or if No. 2 hydraulic pressure fails. The accumulator is a cylindrical piston type with a gross rated volume of 50 inches³ (820 centimeters³).

The accumulator is pressurized with nitrogen on one side of the internal piston and hydraulic fluid on the other side. The gas side of the accumulator is pre-charged to 500 psi (3447 kPa). The fluid side is pressurized by the No. 2 hydraulic system or the handpump. The fluid side will hold enough pressure to supply six full parking brake applications and releases.

Training Information Points

The parking brake system uses hydraulic pressure from the No. 2 hydraulic system.

The parking brake system can be used to stop the aircraft on the ground when the normal brake system is not available.

The handpump can be used to manually pressurize the accumulator when the aircraft is on the ground and the engines are not operating.

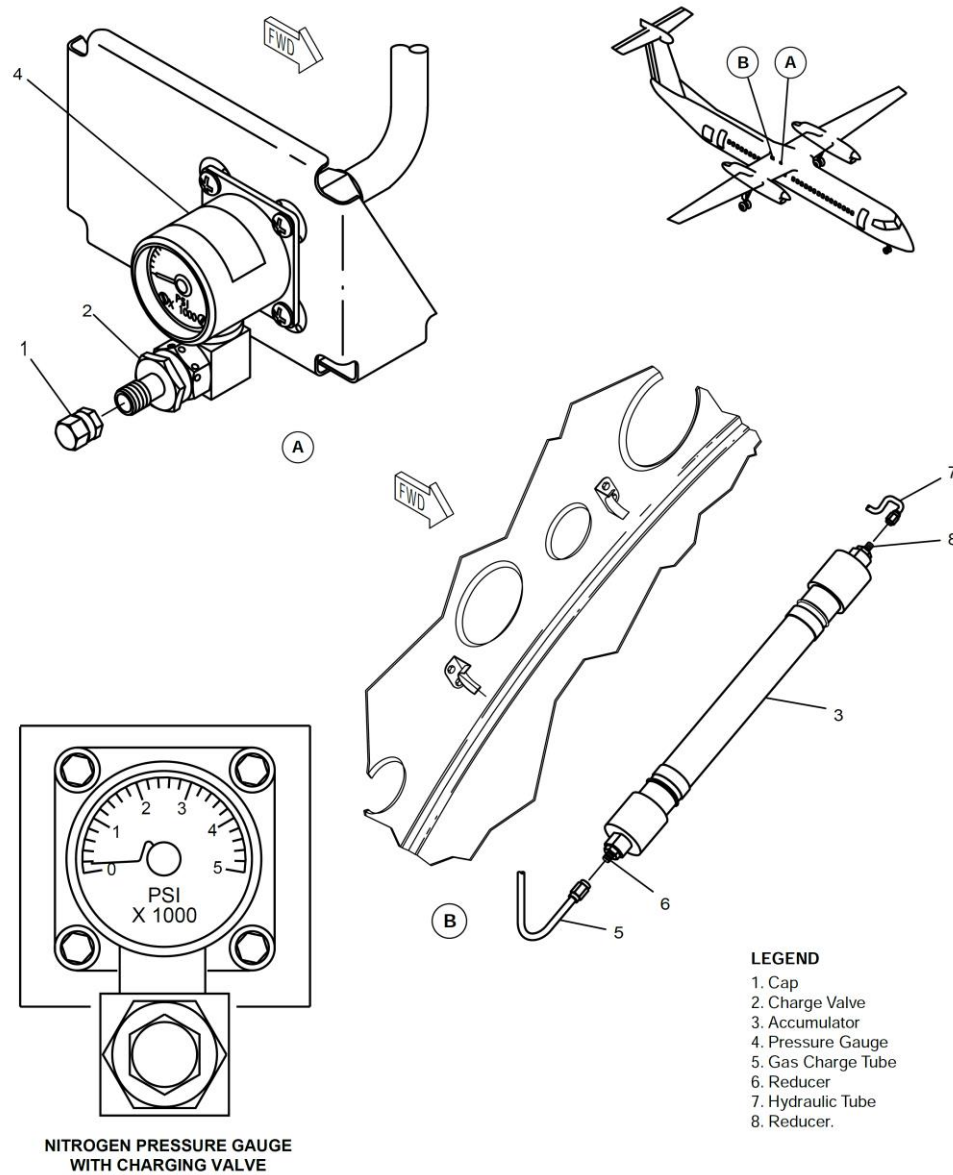


Figure 32-98 Parking Brake Accumulator, Charging Valve and Gauge

NOSEWHEEL STEERING SYSTEM

Introduction

The steering system for the nosewheels gives electronically controlled and hydraulically powered steering of the Nose Landing Gear (NLG). Steering of the nosewheels is for taxi, take-off, and landing.

General Description

[Refer Figure 32-99](#)

Hydraulically powered steering of the nosewheels is available when the NLG is down and locked and the aircraft is on the ground. The STEERING switch, located in the flight compartment on the pilot's side console, must be in the STEERING position for the steering system to operate.

The Steering Control Unit (SCU) electronically controls the steering system for the nosewheels. Electrical power for the SCU is supplied through the STEERING switch controlled by the pilot. The switch receives electrical power from the 28 VDC left and right main busses. The left main bus power is through C.B. C6, right main bus power is through C.B. P5. The SCU does three types of Built-In-Test (BIT) on the components of the NWS system:

- Power-Up BIT (PBIT)
- Continuous BIT (CBIT)
- Maintenance Diagnostic BIT (MBIT).

The power steering unit has a steering manifold, a hydraulic motor, and a reduction gearbox with Rotary Variable Differential Transformers (RVDT). Hydraulic pressure to operate the steering system for the nosewheels is supplied by the No.2 hydraulic system. The steering manifold controls the hydraulic pressure to the hydraulic motor. The hydraulic motor operates a reduction gearbox which steers the nosewheels.

The nosewheel steering system includes the components that follow:

- Electronic Steering Control Unit (32-51-01)
- Pilot Hand Control (32-51-06)

- Rudder Pedal Steering Potentiometer
- Nosewheel Steering Manifold (with the NWS Manifold Solenoid and NWS Pressure Switch)
- NLG Centering Mechanism
- Power Steering Unit
- Hydraulic Steering Motor
- Steering Feedback Assembly (Rotary Variable Differential Transformer)
- STEERING Switch

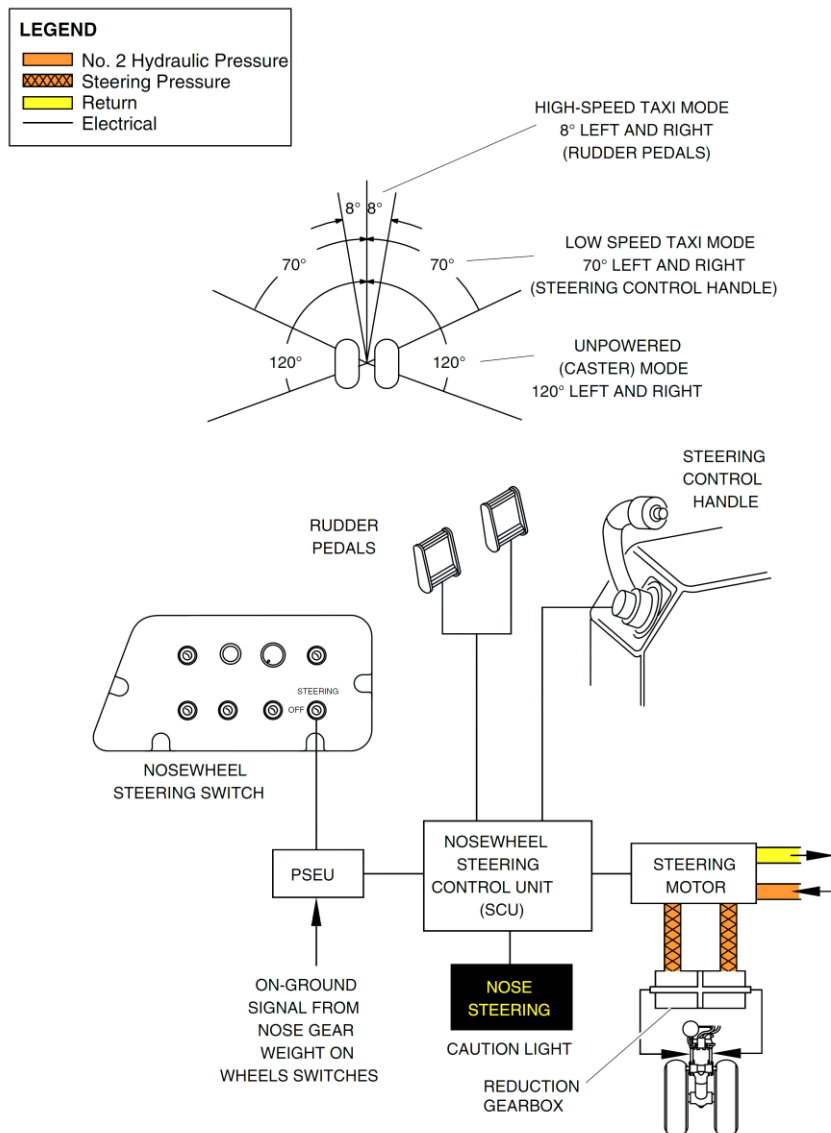


Figure 32-99 Nose Wheel Steering System Block Diagram

Detailed Description

Full authority steering of the nosewheels is by the pilot hand control located on the pilot's side console, in the flight compartment. The function of the pilot hand control is to steer the nosewheels for low speed forward taxiing only.

Limited authority steering of the nosewheels is by the rudder pedals in the flight compartment. The rudder pedals control steering of the nosewheels for high speed forward taxiing, take-off, and landing.

Hydraulic powered shimmy damping of the nosewheels is supplied by the steering manifold when the STEERING switch is in the STEERING position. Passive, free castor shimmy damping of the nosewheels is supplied by the nosewheel centering system when the STEERING switch is in the OFF position. If any component in the steering system fails, the nosewheel centering system will give passive, free caster shimmy damping of the nosewheels. The passive, free castor mode lets the nosewheels move ± 120 degrees to the left and to the right of the center position.

If any component in the steering system fails, a NOSE STEERING caution light on the Caution and Warning panel in the flight compartment comes on.

Hydraulic power is removed from the NWS system when there is no Weight-On-Wheels (WOW) input from the PSEU and the NLG selector valve begins NLG retraction. The SCU delays the closing of the Solenoid Operated Valve (SOV) of the steering manifold by 5 seconds after lift-off because of a possible Weight-Off-Wheels (WOFW) false signal that can occur during the lift-off phase of flight.

NOTE: The SCU will also delay the opening of the SOV by 1 second after it gets a WOW signal from the PSEU (after touch-down) because of a possible WOW false signal that can occur during the touch-down phase of flight.

Normal Modes of NWS System Operation

[Table 1](#) shows the different NWS modes of operation:

PHASE	SYSTEM REQUIREMENTS	FUNCTIONS
Ground	- SCU on and no faults detected - NLG is down and locked - WOW input from the PSEU - STEERING switch in the ON position.	After a 1 second delay after touch-down, power steering is available. Directional control of the aircraft from pilot inputs (PHC or rudder pedal input).
Lift Off	- SCU on and no faults detected - NLG is down and locked - No WOW input from the PSEU - STEERING switch in the ON position.	The SCU commands a straight-ahead signal to the NLG. Pilot inputs are disregarded. After a 5 second delay, the nosewheels are centered and the SOV is de-energized.
Air	- SCU on and no faults detected - NLG is not down and locked - No WOW input from the PSEU - STEERING switch in the ON position.	The SOV is de-energized. The SCU commands a straight-ahead reference signal to the NLG. Pilot inputs are disregarded.
Approach	- SCU on and no faults detected - NLG is down and locked - No WOW input from the PSEU - STEERING switch in the ON position.	The SOV is de-energized. The SCU commands a straight-ahead reference signal to the NLG. Pilot inputs are disregarded. The SCU does a validity check of the PHC and rudder pedals.

Table 1 – Nose Wheel Steering Modes of Operation

Nose Wheel Steering System – Component Description

Steering Control Unit

[Refer to Figure 32-100 & 32-101](#)

The Steering Control Unit is an electronic box installed on the outer side of the Nose Landing Gear wheelwell wall. A panel on the right side of the nose fuselage gives access to the SCU in the equipment bay of the forward nose fuselage.

The SCU uses input commands from the PSEU, rudder pedals and the PHC. Input signals are also received from the STEERING switch, the nosewheel steering manifold and the feedback system RVDTs. The SCU gives output commands to the steering manifold and failure signals to the Caution and Warning panel.

The WOW condition signal and the landing gear down and locked condition signal are necessary for the SCU to enable the powered steering mode of operation. The WOW condition signal will arm the STEERING switch when it is set to the STEERING position.

When the aircraft is in the air, the SCU sends a straight-ahead reference signal (zero-angle command). When the aircraft lands (and the WOW input signals and the landing gear down and locked condition signal are received from the PSEU), the straight-ahead reference signal is removed by the SCU. The SCU commands the SOV to energize (the SOV opens) and then starts to count to 1 second. The mode selector valve of the steering manifold is opened by the SOV. The hydraulic pressure increases to the system operating pressure in a fraction of a second and is monitored by the NWS pressure switch. At the same time, the SCU synchronizes the input commands from the PHC potentiometer and the rudder pedal potentiometer with the actual nosewheel position (by RVDT feedback). The difference between the input and feedback commands is then gradually reduced over the 1 second period to give a smooth transition to the powered steering mode.

NOTE: This synchronization feature will not start until the steering pressure switch detects that there is sufficient hydraulic pressure available to the NWS system.

NOTE: If the SCU detects that the PHC is not between ± 8 degrees of the straight-ahead position in the moment that the system operating pressure is reached and synchronization starts, the SCU will command the SOV to de-energize. The SOV and the mode selector valve of the steering manifold will close to put the NWS system in the passive, free castor mode. At the same time, the SCU sends a signal to the Caution and Warning panel. When the signal is received, the NOSE STEERING caution light comes on.

The SCU receives steering command inputs from the potentiometers of the PHC and rudder pedals. The SCU receives signals from the RVDTs on the reduction gearbox relative to the steered position of the nosewheels. The SCU cancels the steering signal when the steered position of the nosewheels matches the position commanded by the PHC or rudder pedals. The SCU keeps the nosewheel steering angle by the hand control to between ± 70 degrees and by the rudder pedals to between ± 8 degrees. The steering system can be set to STEERING or OFF with the STEERING switch in the flight compartment.

The steering system is tested by the SCU for faults on power up. The SCU continuously monitors the system for faults while it is in operation. The SCU can record and store up to 100 failures in a non-volatile memory (NVM) for maintenance functions. Maintenance personnel can use the push button on the SCU to review the last 18 failures. The complete failure history can be downloaded to a personal computer for analysis by maintenance personnel.

The SCU has a command path in the computer hardware that monitors input commands (from the PHC and rudder pedal potentiometers) and monitors the feedback from the RVDTs on the reduction gearbox. The logic to control the different phases of the powered steering mode are also in the command path. A separate monitor path in the computer hardware duplicates the

command path. If there is a difference between the command path and the monitor path, the SCU system will prevent the operation of the NWS system and a fault will be logged in the NVM.

When a component in the steering system fails, the SCU sends a signal to the Caution and Warning panel. When the signal is received, the NOSE STEERING caution light comes on. The SCU monitors the SOV for mechanical failures and monitors the potentiometer of the PHC for proper function. The SCU monitors the Electro Hydraulic Servo Valve (EHSV) torque motor to make sure it responds correctly to input commands. The SCU monitors the Linear Variable Differential Transducer (LVDT) to make sure that it responds correctly to input commands.

A computer can be connected to terminals on the SCU for steering system rigging. The computer measures output signals from components that follow to rig the steering system:

- RVDTs
- Pilot hand control potentiometer
- Rudder potentiometer

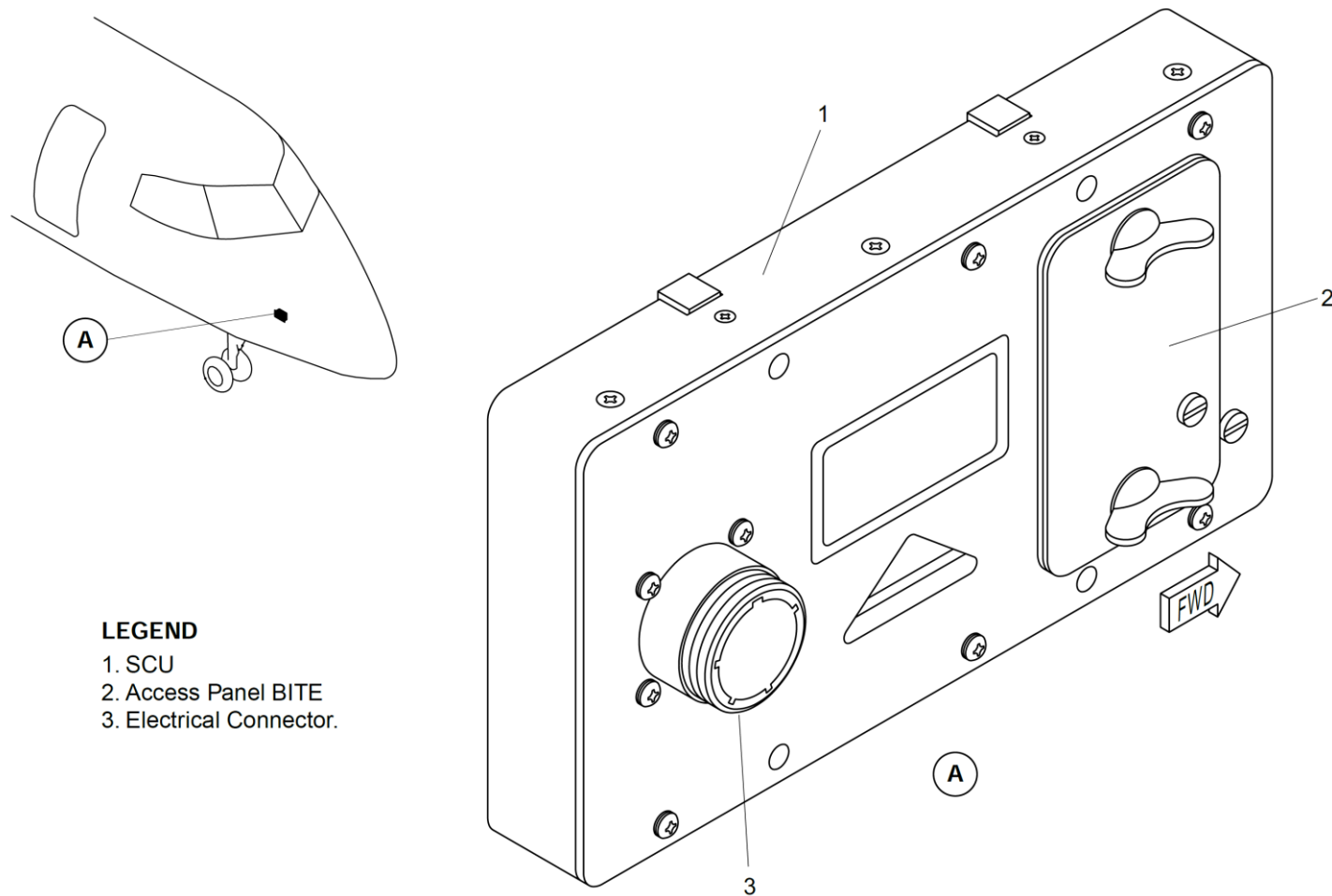


Figure 32-100 Steering Control Unit

Nose Wheel Steering System – Component Description

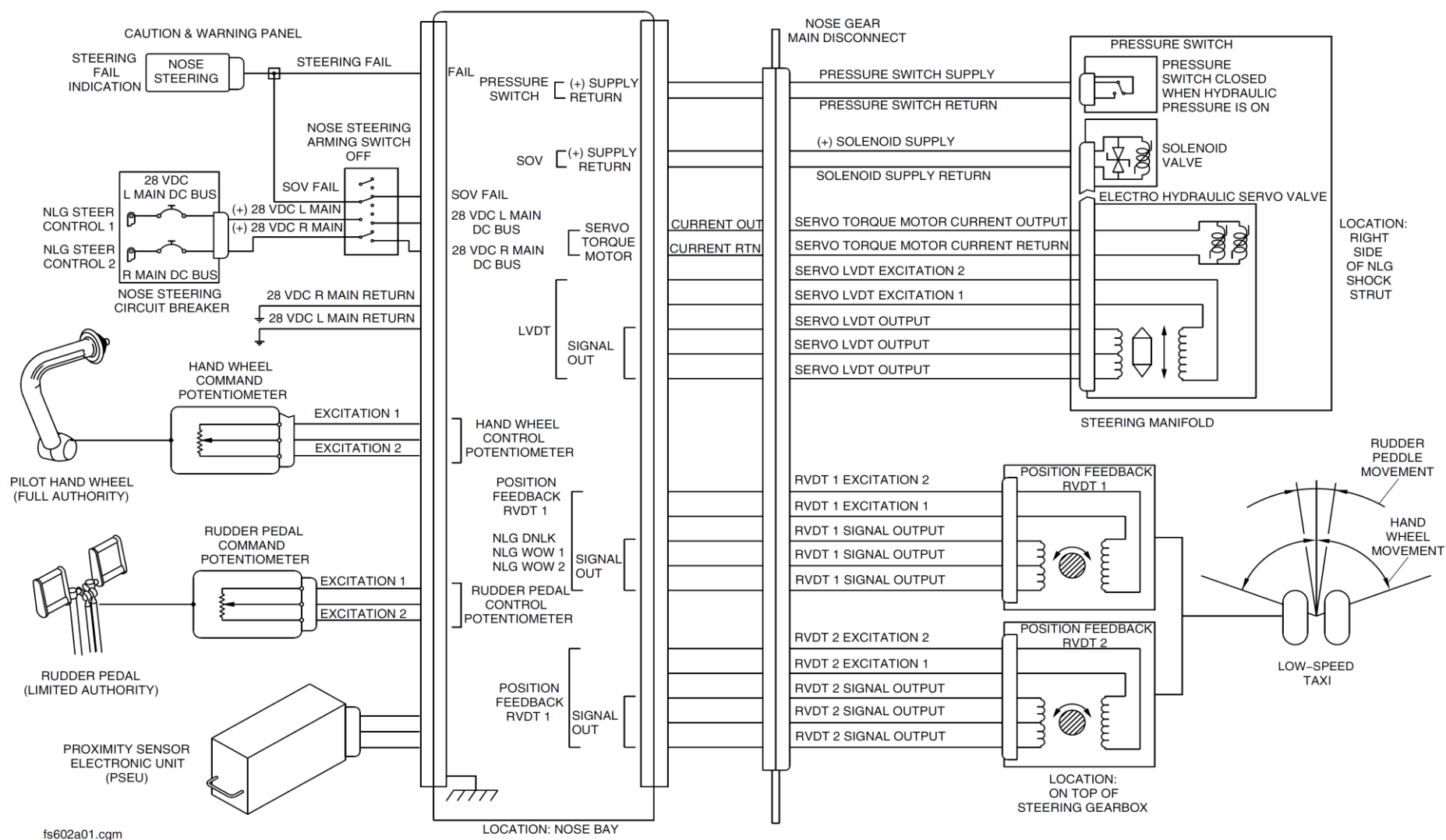


Figure 32-101 Nose Wheel Steering Electrical Synoptic

Nosewheel Steering Manifold

[Refer Figures 32-102 & 32-103](#)

The nosewheel steering manifold is a self-contained assembly which is installed on the right front side of the NLG outer cylinder. The steering manifold controls hydraulic pressure to the steering hydraulic motor for powered steering of the nosewheels. Hydraulic power for the steering manifold is supplied by the No. 2 hydraulic system through the landing gear selector valve extension tubing. Hydraulic fluid is returned to the No. 2 hydraulic system through the retraction tubing of the landing gear selector valve.

Two electrical connectors supply electric power to the steering manifold. One connector supplies electrical power to the SOV, the EHSV, and the EHSV LVDT. The second connector supplies electrical power to the NWS pressure switch.

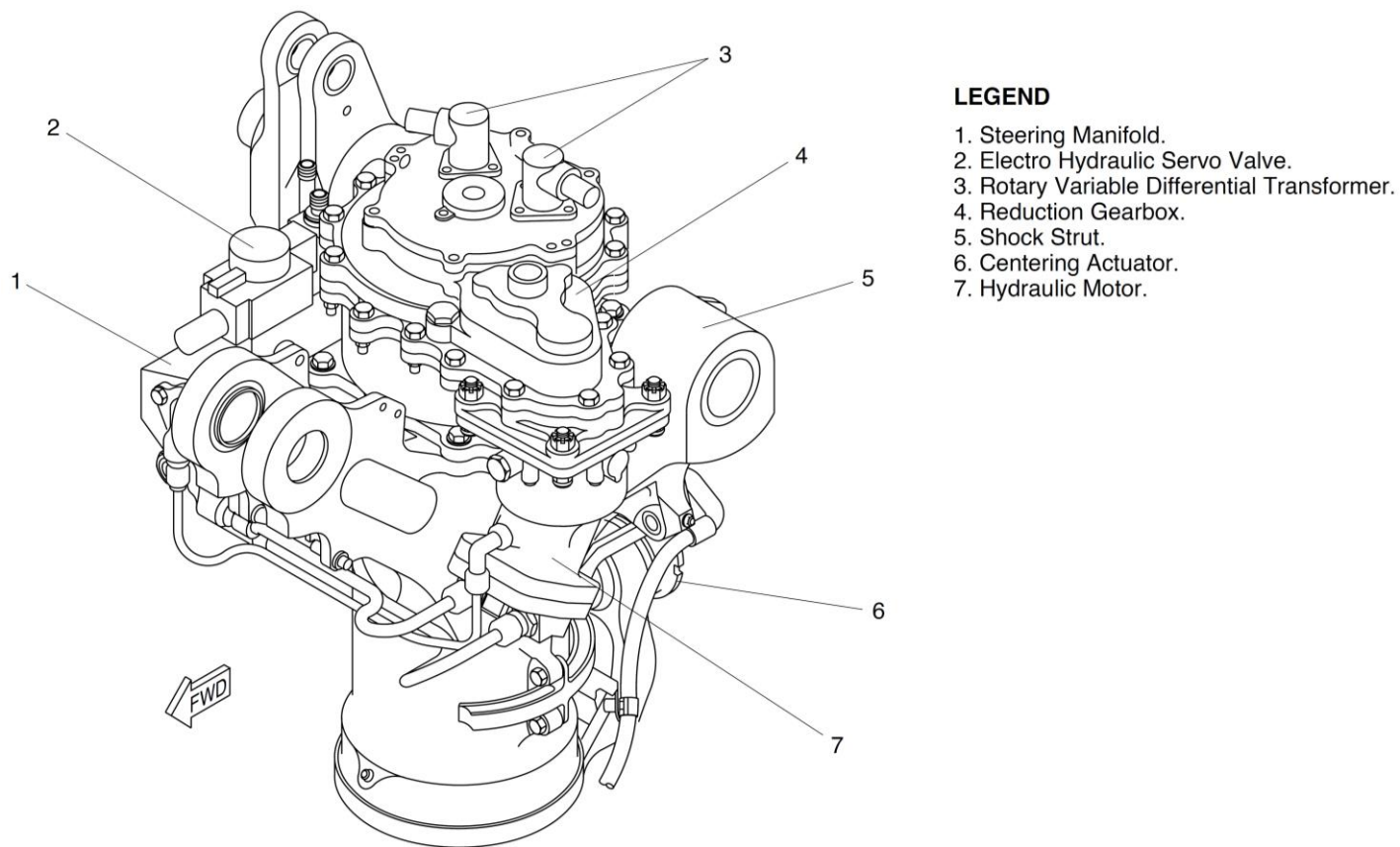
The NWS pressure switch is used to monitor hydraulic pressure status. When the SCU initially starts to power-up, the SCU PBIT momentarily commands the SOV to de-energize (the SOV closes). If the NWS pressure switch indicates that hydraulic pressure is available, the SCU will send a signal to the Caution and Warning panel. When the signal is received, the NOSE STEERING caution light comes on. This will be an indication that the SOV is stuck open. The pilot will then put the STEERING switch to the OFF position. If the SOV is stuck open, the NWS pressure switch gives a ground to the NOSE STEERING caution light to make it come on.

The mode selector valve stops hydraulic pressure to the EHSV when the NWS system is in the passive, free castor mode. The SOV is in the closed position until the NLG is compressed when the aircraft lands. The solenoid of the SOV is then energized by signals from the SCU to open the mode selector valve and to configure the steering system for powered steering. The EHSV gets signals from the SOV and converts them into proportional spool displacement. The spool displacement sends hydraulic pressure to the hydraulic motor to drive the reduction gearbox and to steer the nosewheels to the commanded position.

Nosewheel position signals are sent from the LVDT installed in the steering manifold, to the SCU to keep the nosewheels in the commanded steering position. The pressure switch senses any changes in hydraulic pressure that are caused by the failure of components in the steering manifold. The pressure switch sends a signal to the SCU. When the signal is received, a light on the Caution and Warning panel in the flight compartment comes on.

Reduction Gearbox

The reduction gearbox is a self-contained assembly that is installed on the top of the NLG outer cylinder. The feedback gear of the reduction gearbox connects with a spline on the inner cylinder of the NLG. The feedback gear of the reduction gearbox is operated by the hydraulic motor to transmit torque to the spline on the inner cylinder of the NLG. The reduction gearbox is lubricated with a mixture of grease and oil.



LEGEND

1. Steering Manifold.
2. Electro Hydraulic Servo Valve.
3. Rotary Variable Differential Transformer.
4. Reduction Gearbox.
5. Shock Strut.
6. Centering Actuator.
7. Hydraulic Motor.

Figure 32-102 Nose Wheel Steering Assembly

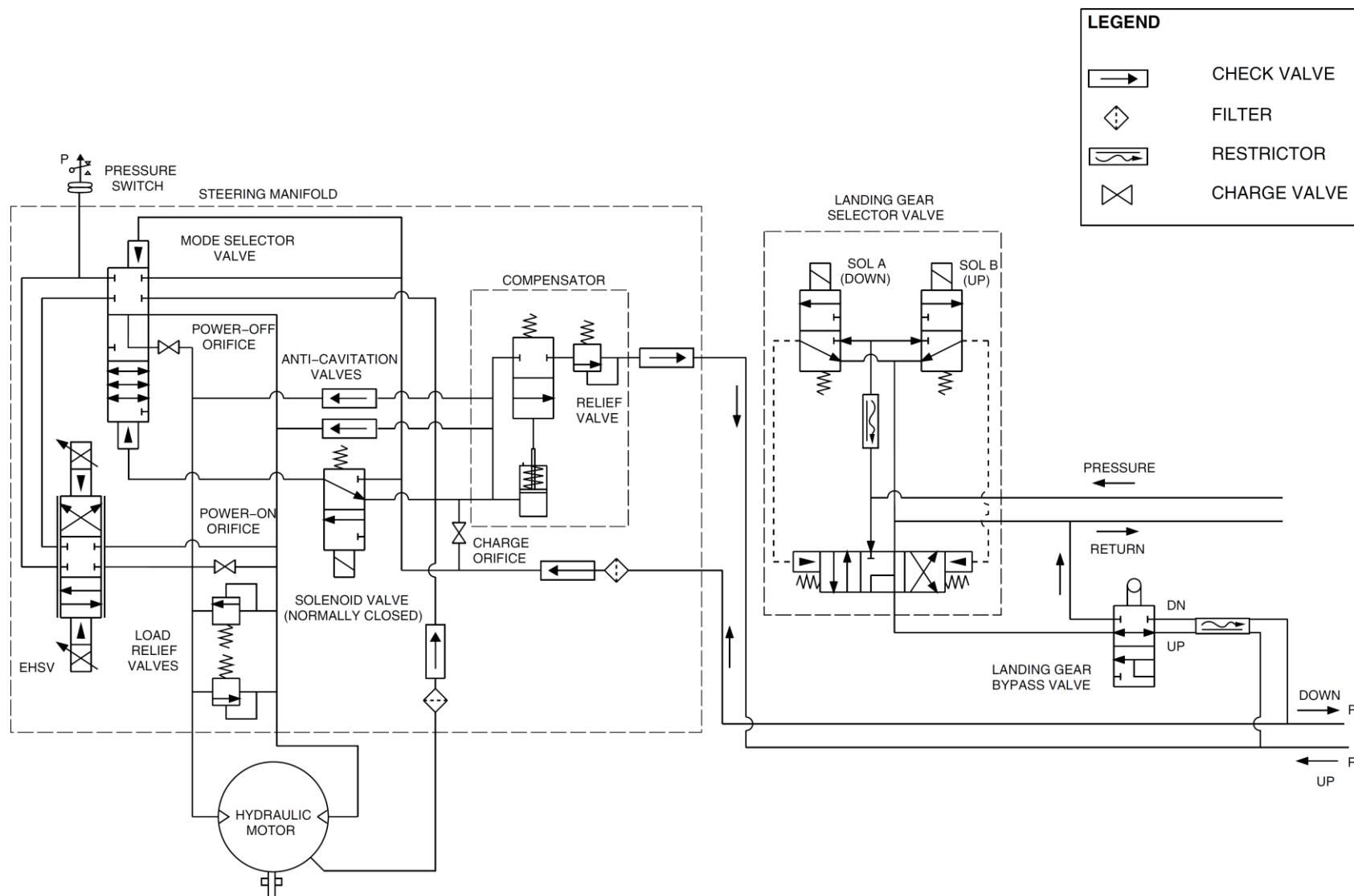


Figure 32-103 Nose Wheel Steering System Hydraulic Synoptic

Rotary Variable Differential Transformer (RVDT) Assembly

The RVDT assembly is a self-contained assembly bolted to the top of the reduction gearbox. The RVDT assembly connects with the feedback gear of the reduction gearbox. The SCU uses signals from the RVDT assembly to give the steered position of the nosewheels. The SCU uses one RVDT to give the position of the nosewheels. The SCU uses the second RVDT to check the accuracy of the first RVDT as a fault test.

Hydraulic Motor

The hydraulic motor is installed on the bottom, left forward side of the reduction gearbox. The hydraulic motor connects with the feedback gear of the reduction gearbox. The hydraulic motor receives hydraulic pressure for proportional steering control of the nosewheel from the EHSV of the steering manifold. The hydraulic motor operates a reduction gear assembly in the reduction gearbox to steer the nosewheels to the commanded position.

Nosewheel Centering Mechanism

[Refer Figure 32-104](#)

The centering mechanism center the nosewheels when the landing gear is retracted. The nosewheel centering mechanism is installed to a collar which is attached to the lower end of the NLG inner cylinder. The spring loaded centering actuator has a roller bearing that rolls on a machined race on the NLG outer cylinder. The machined race has a detent at the centered position. The detent returns the nosewheels to the straight ahead position. The centering actuator cannot move the nosewheels to the center position. It only holds them in the center position when the steering system aligns them. Attached to the centering actuator is the NWCENT proximity sensor.

The proximity sensor sends a signal to the PSEU when the centering actuator is in the straight ahead detent of the machined race. For the landing gear to retract, the PSEU must get an air mode signal and the NLG proximity sensor must read NEAR (NWCENT NEAR). After take-off the PSEU sends the "WOW OFF" signal to the SCU. The SCU ignores the pilot input and enters a "straight ahead mode". Then the SCU energizes the EHSV in the steering manifold and

actuates the hydraulic motor to center the nose gear. When the SCU senses the nose gear is at the neutral position through two RVDTs located at the top of the steering gearbox, the hydraulic motor stops. When the centering actuator rod is fully extended in the detent, the nosewheels are in the center position. The PSEU then lets the landing-gear retraction sequence start when the landing-gear handle is moved to the UP position.

NOTE: PSEU requires the NWCENT proximity sensor to read NEAR before retraction. Any discrepancy or miss alignment of the RVDT's will make the nose wheel steering hydraulics stop before the NLG has reached center position. This will give a NWCENT UNREASONABLE FAR condition.

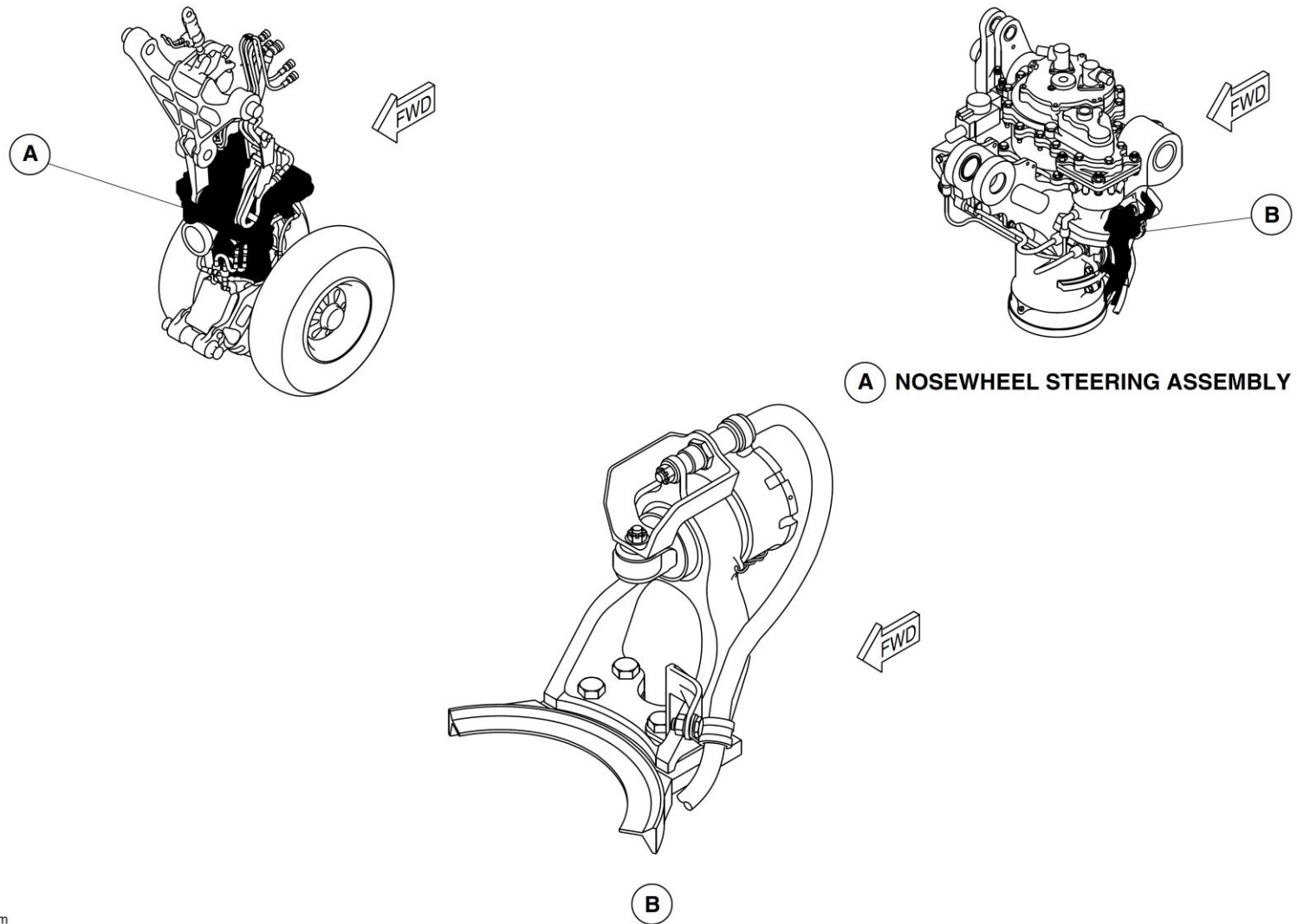


Figure 32-104 Nose Wheel Centering Mechanism

Pilot Hand Control

[Refer Figure 32-105](#)

The pilot hand control (PHC) is a self-centering unit and is located on the left side console in the flight compartment. The PHC lets the pilot steer the nosewheels. The PHC has a potentiometer that gives input signals to the SCU in proportion to the actual PHC position. The PHC has hard stops on either side of the center position. The centering spring returns the PHC to the neutral position when the input from the pilot is removed.

The PHC gives full authority steering control of the nosewheel for low speed, forward taxiing. For reverse taxiing, the PHC is in the center position. The PHC can steer the nosewheels ± 70 degrees to the left and to the right of the center position. The potentiometer of the PHC is monitored for failure on power up by the SCU PBIT function and is also monitored continuously for failure by the CBIT function.

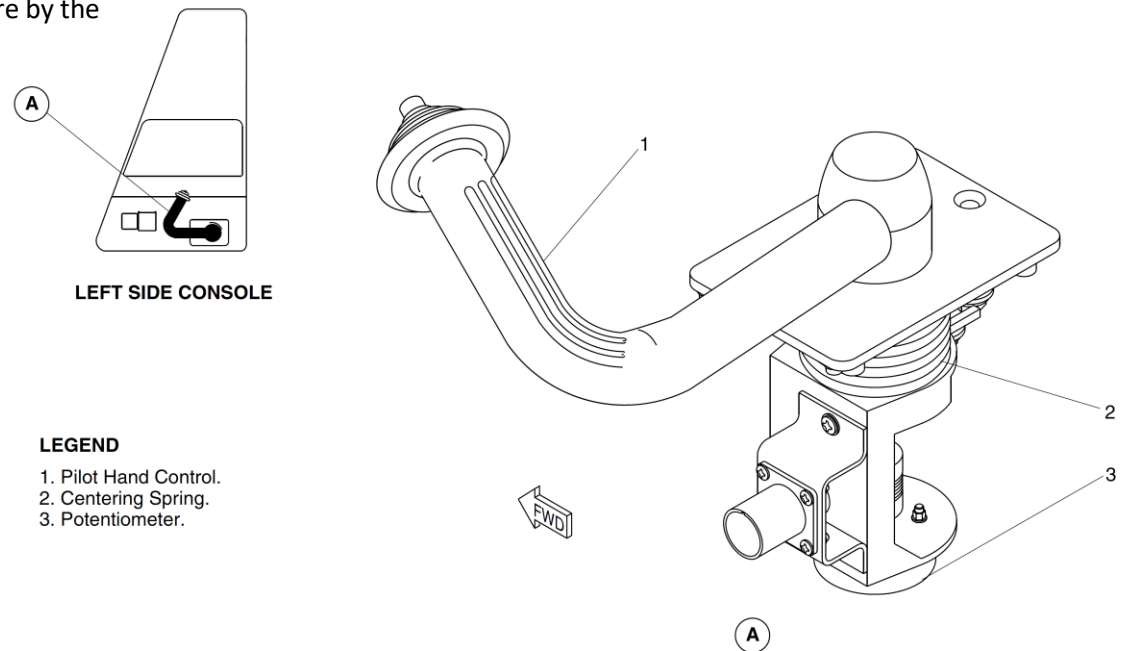


Figure 32-105 Pilot Hand Control

Rudder Pedal Potentiometer

[Refer Figure 32-106](#)

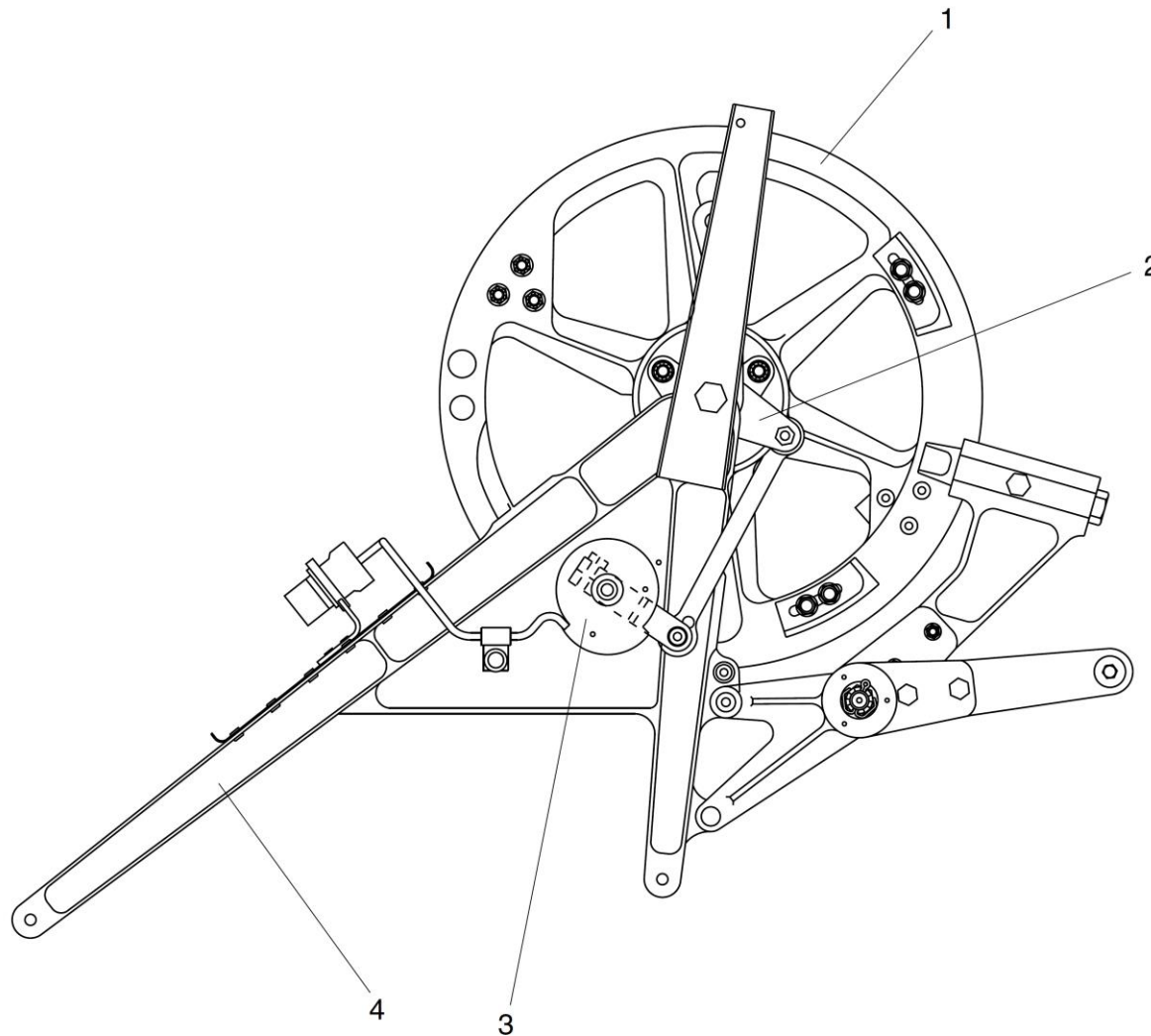
The rudder pedal potentiometer is a self-contained bolt-on assembly with electrical leads for remote airframe connections. The rudder pedal potentiometer is installed on the support structure of the rudder control quadrant in the equipment bay in the forward nose fuselage. The potentiometer is connected to the rudder quadrant with a linkage assembly for rotary inputs in proportion to rudder pedal movement. A panel located on the right side of the nose fuselage gives access to the potentiometer. Centering springs on the rudder pedals operate to return the pedals to the neutral position when the pilot inputs are removed.

The rudder pedals allow the crew to steer the nosewheels for high speed taxi, take-off and landing. For reverse taxiing, the rudder pedals are in the center position. The rudder pedals can steer the nosewheel ± 8 degrees to the left and to the right of the center position. The potentiometer gives input signals to the SCU. The signals are in proportion to the actual position of the rudder pedals. The potentiometer of the rudder pedal is monitored for failure on power up by the SCU PBIT function and is also monitored continuously for failure by the CBIT function.

Training Information Points

When you do maintenance on the nose landing gear make sure that the nose landing gear ground lock is engaged, the door lock pin is installed and that the hydraulic pressure is shut off. If you do not do this, you can cause injuries to persons and/or damage to the equipment.

Before you connect the tow bar to the aircraft make sure that the nosewheel steering switch is set to the off position and stays off during towing. After towing, make sure the nosewheel is centered before you disconnect the tow bar. If you do not do this, you can cause injuries to persons and/or damage to the equipment.



LEGEND

1. Rudder Quadrant.
2. Linkage Assembly.
3. Rudder Pedal Potentiometer.
4. Support Structure.

Figure 32-106 Rudder Pedal Potentiometer

Nose Wheel Steering Operation – Electrical Schematic

[Refer Figure 32-107](#)

When the nosewheel steering switch is selected ON, power from the left and right main 28 VDC buses is supplied from Pins H and J of the switch, to Pins 40 and 25 of the SCU.

With WOW, the steering control unit receives inputs from the PSEU as follows:

- Pin 11F to SCU Pin 7 - NLG Downlock
- Pin 2H to SCU Pin 14 - NLG WOW 2
- Pin 10D to SCU Pin 6 - NLG WOW 1.

The SCU outputs current to the following:

- Solenoid valve
- Servo valve torque motor.

The operator input is sensed through the rudder pedal or handwheel control potentiometer output. It is then sent to the SCU.

- The SCU positions the Electro Hydraulic servo valve to steer left or right. Feedback from the LVDT at Pins 9, 15 and 23, to the SCU confirms servo valve has moved to the correct position.
- Feedback from the pressure switch on Pins 1 and 4 at the SCU confirms that hydraulic pressure is available.
- Hydraulic pressure through the solenoid valve acts on the bottom of the Mode Selector Valve (MSV).
- The MSV sends pressure to the Electro Hydraulic servo valve to drive the steering motor left or right.

- As the steering motor rotates the NLG inner cylinder, the two RVDT's send the nosewheel position back to the control unit at Pins 49, 50 and 53 from RVDT 1 and Pins 46, 51 and 52 from RVDT 2.
- When the feedback signals from the RVDT's equal the input signals from the potentiometers, the SCU commands the Electro Hydraulic servo valve to the neutral position.

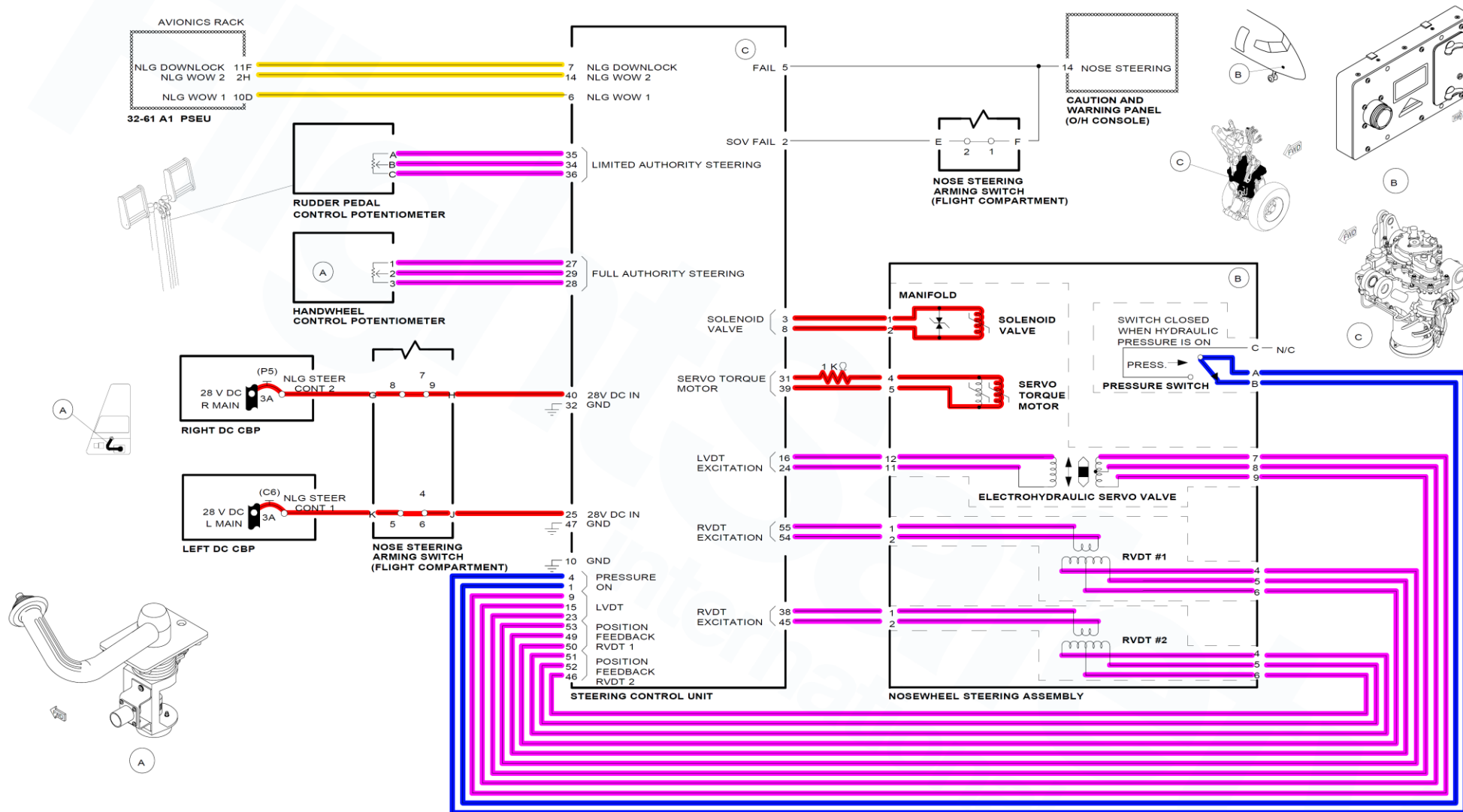


Figure 32-107 Nose Wheel Steering Operation – Electrical Schematic

Proximity Sensor System

Introduction

The Proximity Sensor System (PSS) is part of the landing gear system.

- Landing gear lamp test selected

General Description

[Refer Figure 32-108](#)

The PSEU receives signals from 20 landing gear and 11 fuselage door proximity sensors.

The PSEU then transmits discrete outputs to the landing gear and door indicators.

The Proximity Sensor System has the following components:

- Proximity sensor electronic unit
- Proximity sensors

System Description

The PSS is a 28 VDC energized, microprocessor based proximity system.

The system has signal conditioning, proximity sensing, discrete monitoring, control signals, interface wiring and a BITE keypad with display.

The PSS sends discrete outputs as follows:

- Landing gear control
- Landing gear status indication
- Alternate downlock indication
- WOW1
- Fuselage door indication
- WOW2.

The four discrete inputs to the PSS are:

- Landing gear extension command
- Landing gear retraction command
- INHIBIT switch set to INHIBIT

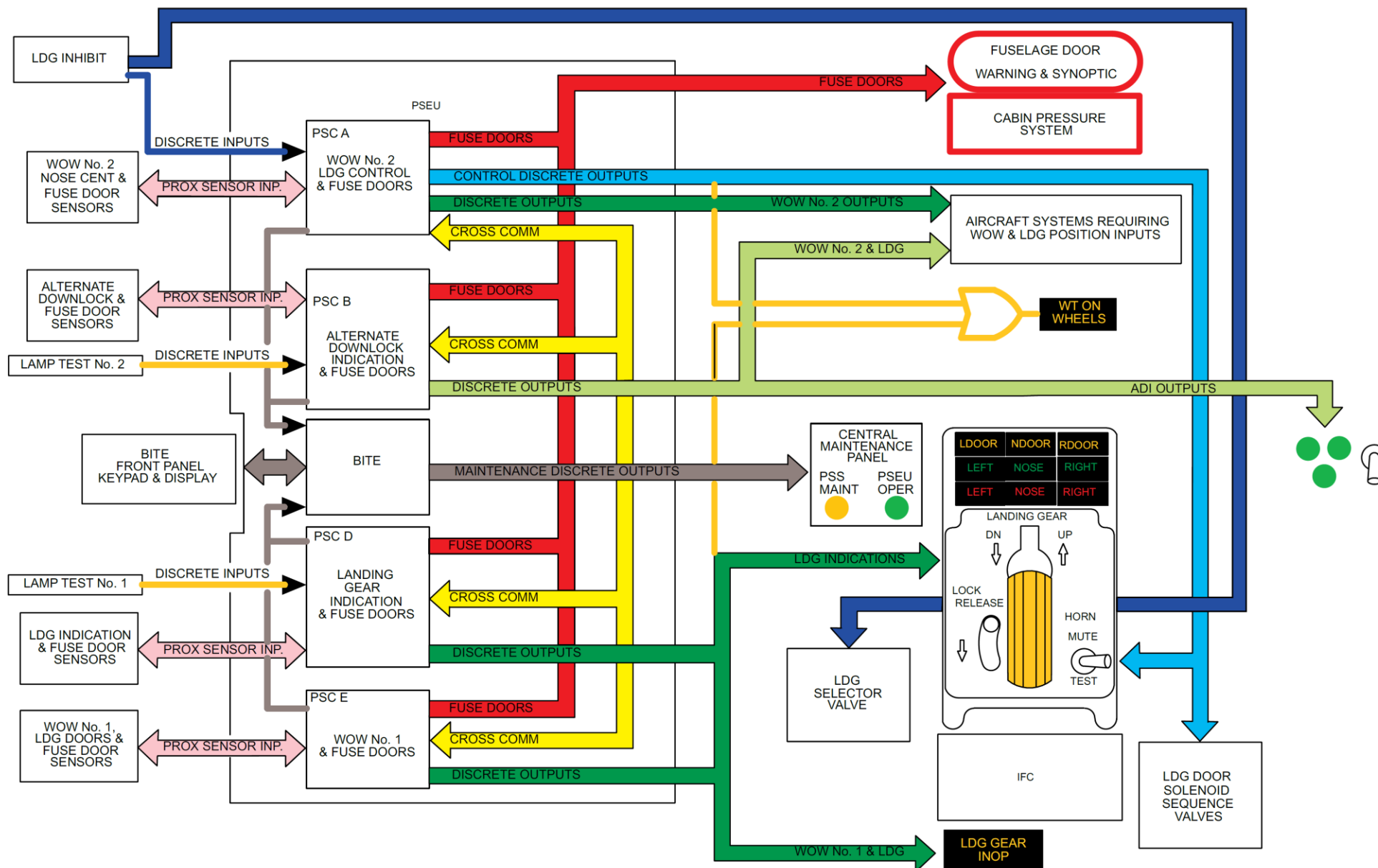


Figure 32-108 PSEU Input and Output Block Diagram

Proximity Sensor System – Component Description

Proximity Sensor Electronic Unit

[Refer Figure 32-109](#)

The PSEU is installed on the avionics rack.

The PSEU receives 28 VDC from the right and left main buses, and the right and left essential buses.

The PSEU provides visual indication in the flight compartment for:

- Gear position
- Gear doors position, and
- Fuselage doors status.

The primary functions of the PSEU are as follows:

- Monitor, indicate and operate the landing gear extension, and retraction system
- Monitor and supply the uplock, down lock, and WOW data to other systems
- Supply indication for the passenger and baggage doors.

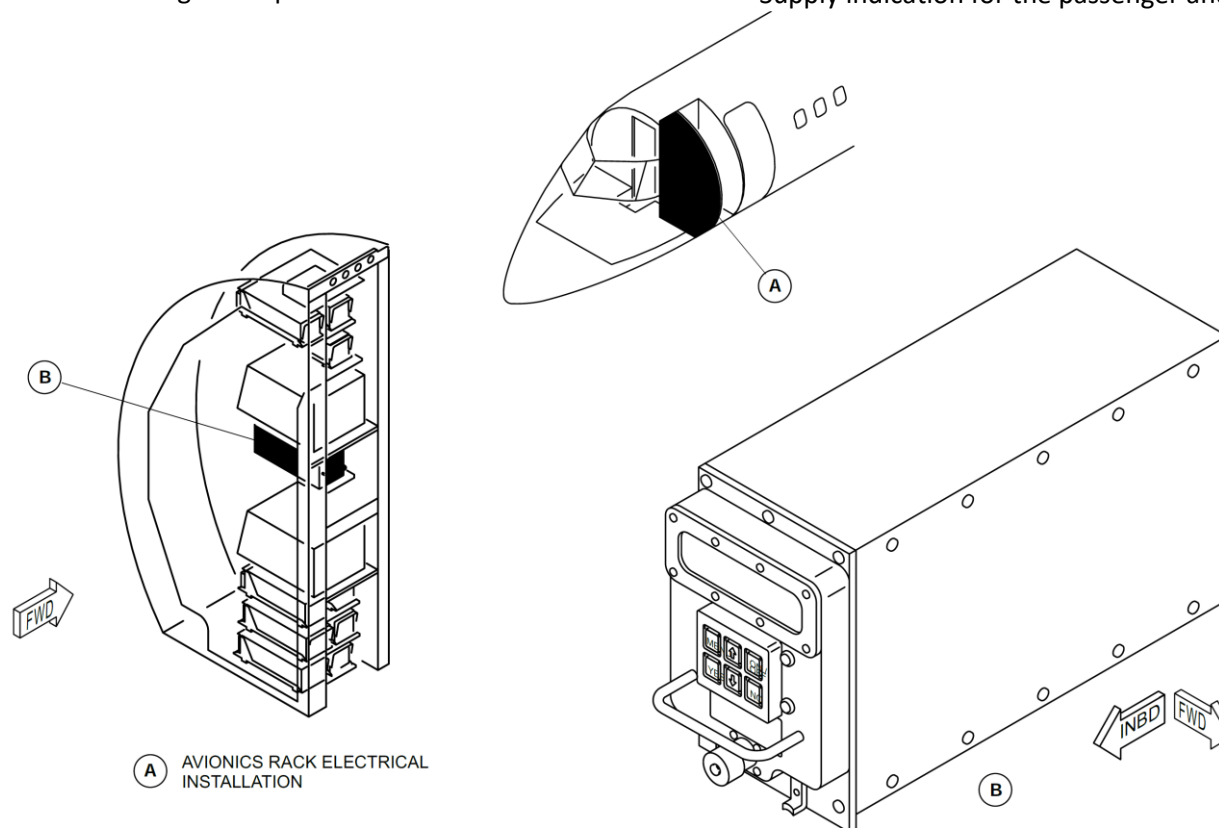


Figure 32-109 Proximity Sensor Electronic Unit

Proximity Sensors

[Refer Figure 32-110](#)

There are 31 proximity sensors. Each main landing gear has six proximity sensors. The nose landing gear has eight.

There are eleven sensors that monitor:

- Passenger entrance door
- The baggage compartment doors
- The emergency exit doors, and
- The service door

The proximity sensors are hermetically sealed, variable reluctance, two wire devices. The inductance at the sensor changes with the sensor's proximity to the target. As the target gets closer to the sensor face, the inductance increases.

The proximity sensors are not repairable

The function of the sensors is to supply position data to the PSEU.

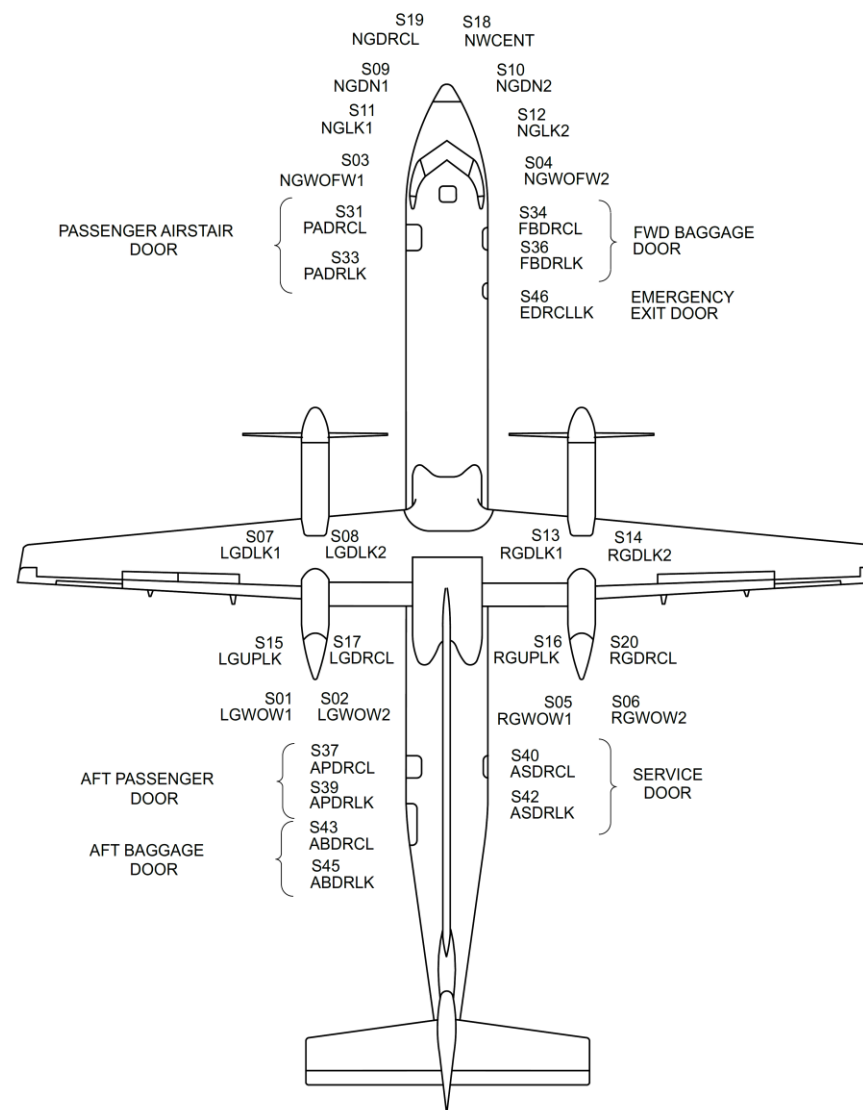


Figure 32-110 Proximity Sensors Location

Landing Gear Controls and Indications

Landing Gear Control

The PSEU provides discrete outputs that are used to control the movement of the landing gear (extend or retract), and landing gear doors (open or close). The PSEU generates seven discrete outputs that provide the landing gear control function. The outputs include the Flight Data Recorder, and the no smoking sign.

Landing Gear Indications

The following conditions are monitored by the landing gear indication system:

- Right, left, and nose gear door position
- Right, left, and nose gear position.

The outputs of the proximity sensors are processed by the PSEU.

The PSEU generates 16 discrete outputs which are sent to the two IOP in the IFC's, the landing gear control panel, the caution and warning panel, the Antiskid/Brake Control Unit, and the Nosewheel Steering Control Unit.

The landing gear control provides visual indication for each of the three landing gears:

- GEAR DOWN and LOCKED
- GEAR NOT IN COMMANDED POSITION
- GEAR DOOR NOT CLOSED.

Alternate Downlock Indications

The Alternate Down Lock Indication (ADI) provides a second DOWN and LOCKED indication of the three landing gears.

This function is used if the primary landing gear indication function shows an unsafe, or gear disagree condition. ADI is independent from the primary indication. Different sensors and PSC card are used to process inputs.

The following conditions are monitored by the ADI:

- Right and left gear down and locked
- Nose gear down and locked.

The left and right gears each have an alternate sensor that monitors down and locked. The nose gear has one sensor for the down position and one for locked.

The Advisory lamp test on the Passenger Warning panel will cause the ADI lights to come on. Incorrect Landing Gear Configuration/Landing Gear Warning Horn.

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WOW Functions

[Refer Figure 32-111](#)

The WOW function provides discrete outputs to other aircraft systems. Two separate channels WOW No.1 and WOW No.2 generate the weight-on-wheels discrete outputs.

The PSEU generates five types of WOW outputs, they are as follows:

- All three gear (WOW)
- Main gear (MGWOW) NEAR
- Main gear (MGWOFFW) FAR
- Nose gear (NGWOW) FAR
- Nose gear (NGWOFFW) NEAR

There are two discrete outputs supplied by the PSEU that are special to WOW1 and, WOW2 systems. These signals are WOW1 Operational, and WOW2 Operational respectively. These two signals are supplied in the same way.

The CAWP uses these two discrete outputs to show that the WOW functions operate correctly. If one of the two signals become deactivated on the CAWP, the WT ON WHEELS caution light illuminates. For this to occur, the WOW Operational discrete output will deactivate in the conditions as follows:

- There is a disagreement between two of the WOW sensors (in the same WOW system), that the aircraft is either on the ground, or in the air.
- A minimum of one proximity sensor becomes defective in each of the WOW1 and WOW2 systems at the same time.

PSEU Operating Mode	Condition
WOW	At least four of any of the six conditions that follow: LGWOW1 = True LGWOW2 = True RGWOW1 = True RGWOW2 = True NGWOFW1 = False NGWOFW2 = False

Table 2 – PSEU Operating Mode & Conditions

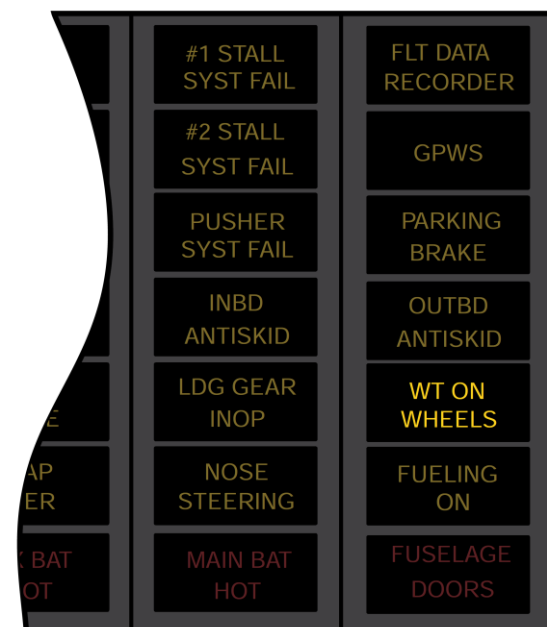


Figure 32-111 CAWP with WOW Caution

Landing Gear Configuration Warning System

Introduction

The landing gear configuration warning system supplies an aural warning to the flight crew if a landing is attempted with the aircraft not in the correct landing configuration (landing gears not down and locked).

General Description

The landing gear warning system logic is divided into three sets of conditions. Each condition depends on different flight phases and related parameters. In event of the aural warning the system logic provides sufficient time for the landing gears to extend down and locked or otherwise, the pilot can make a go-around. The system logic also prevents false warnings. A HORN MUTE and TEST switch is also installed on the landing gear selector panel.

Detailed Description

[Refer Figures 32-112 & 32-113](#)

The continuous 800 Hz landing gear configuration warning tone is generated by two Warning Tone Generator (WTG). The WTG's are the part of the Flight Data Processing System (FDPS) (Refer to Chapter 31-40-00). The WTG's are physically housed in the Input Output Processor (IOP) 1 and IOP 2 located in the two Integrated Flight Cabinets (IFC) 1 and IFC 2. The ARINC 429 digital data from different systems is directly input to IOP's. The analog and discrete data is input to the IOP's through the Input Output Module (IOM) 1 and IOM 2.

The IOP 1 and IOP 2 operate the WTG in a master slave configuration with IOP 1 as the master. The IOP 2 takes the command in case of a disagreement between generated warning tones. The WTG provides tone signals to the Remote Control Audio Unit (RCAU) through the IOM 1. The IOM 1 supplies the tone selection and modulation based on signals from the WTG. The IOM 2 monitors the output of the IOM 1.

The landing gear configuration warning is prioritized by the WTG with respect to other aural warnings. The RCAU amplifies the signal and sends the tone to the flight compartment speakers and to the pilot's and co-pilot's headsets.

The continuous 800 Hz landing gear configuration warning sounds in the flight deck, if the conditions that follow are sensed:

- If the HORN toggle switch, on the landing gear panel, is set to the TEST position
OR
- If all the landing gears are not down and locked.
AND

Condition 1 (Flaps and Engine torque):

- The flap lever is set to more than 8.5 degree
AND
- ENG1 torque OR ENG2 torque is less than 50%
AND
- PLA 1 AND PLA 2 are not in the rating power detent.

In this condition warning cannot be muted until the conditions are removed. This logic prevents false warning. As during a take-off or a go-around the PLA's are set to the rating power detent which disables the warning.

OR

Condition 2 (Airspeed, Radio Altitude and PLA setting):

- Indicated airspeed less than 156 Knots (288 km/h), AND RAD ALT 1 Invalid
OR
- Indicated airspeed less than 156 Knots (288 km/h), AND RAD ALT 1 Valid, AND Above Ground Level (AGL) altitude is less than 1053 feet
AND
- PLA1 AND PLA2 are less than FLIGHT IDLE plus 12 degrees.

In this condition warning cannot be muted until the conditions are removed. This logic prevents false warning. As during a take-off or a

go-around the PLA's are set to the rating power detent (above FLIGHT IDLE plus 12 degrees) which disables the warning.

OR

Condition 3 (Airspeed during single engine operation and PLA setting):

- Indicated airspeed less than 156 Knots (288 km/h), AND RAD ALT 1 Invalid
OR
- Indicated airspeed less than 156 Knots (288 km/h), AND RAD ALT 1 Valid, AND Above Ground Level (AGL) altitude is less than 1053 feet
AND
- PLA1 OR PLA2 is less than FLIGHT IDLE plus 12 degrees
AND
- HORN MUTE switch is not latched to MUTE
AND
- PLA 1 AND PLA 2 are less than the rating power detent.
In this condition warning can be muted in special simulated conditions like pilot training exercise. This logic prevents false warning. As during a take-off or a go-around the PLA's are set to the rating power detent, which disables the warning.

If the landing gear configuration warning is muted, the mute will reset in the conditions that follow:

- The down lock indication indicates all gear down and locked
OR
- The alternate down lock indication indicates all gear down and locked
OR
- PLA 1 AND PLA 2 are greater than OR less than FLIGHT IDLE plus 12 degrees
OR
- All the landing gears are not down and locked
AND
The Flap lever is set to more than 8.5 degree
AND
ENG1 torque OR ENG2 torque is less than 50%

AND

PLA 1 AND PLA 2 are not in the rating power detent.

[Refer Table 3](#)

The landing gear configuration warning has interfaces with other aircraft systems as follows:

Signal	Source
GEARS DOWN AND LOCKED 1	PSEU
GEARS DOWN AND LOCKED 2	PSEU
IAS < 156 ±5 Knots	ADU1, ADU2
AG L< 1053 feet	RA1
PLA 1 < FLIGHT IDLE + 12 degree	FADEC
PLA 2 < FLIGHT IDLE + 12 degree	FADEC
HORN MUTE	LANDING GEAR CONTROL PANEL
FLAP LEVER > 8.5°	FCU
ENG 1 TORQUE < 50%	FADEC
ENG 2 TORQUE < 50%	FADEC
PLA1 ≥ RATING POWER DETENT	FADEC
PLA2 ≥ RATING POWER DETENT	FADEC
LDG WARN TEST	LANDING GEAR CONTROL PANEL

Table 3 – Landing Gear Configuration Warning Interfaces

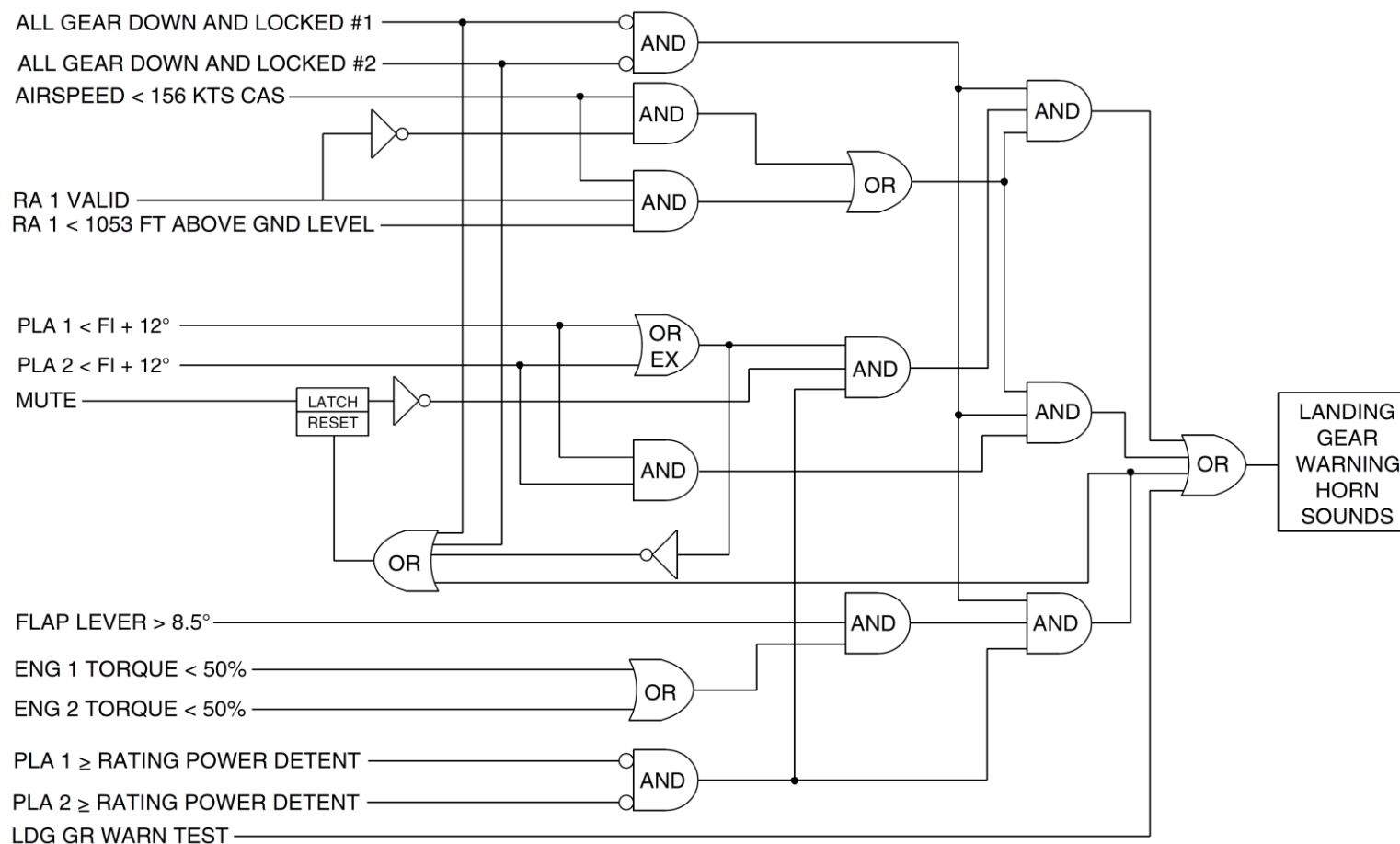


Figure 32-112 Landing Gear Configuration Warning System Logic

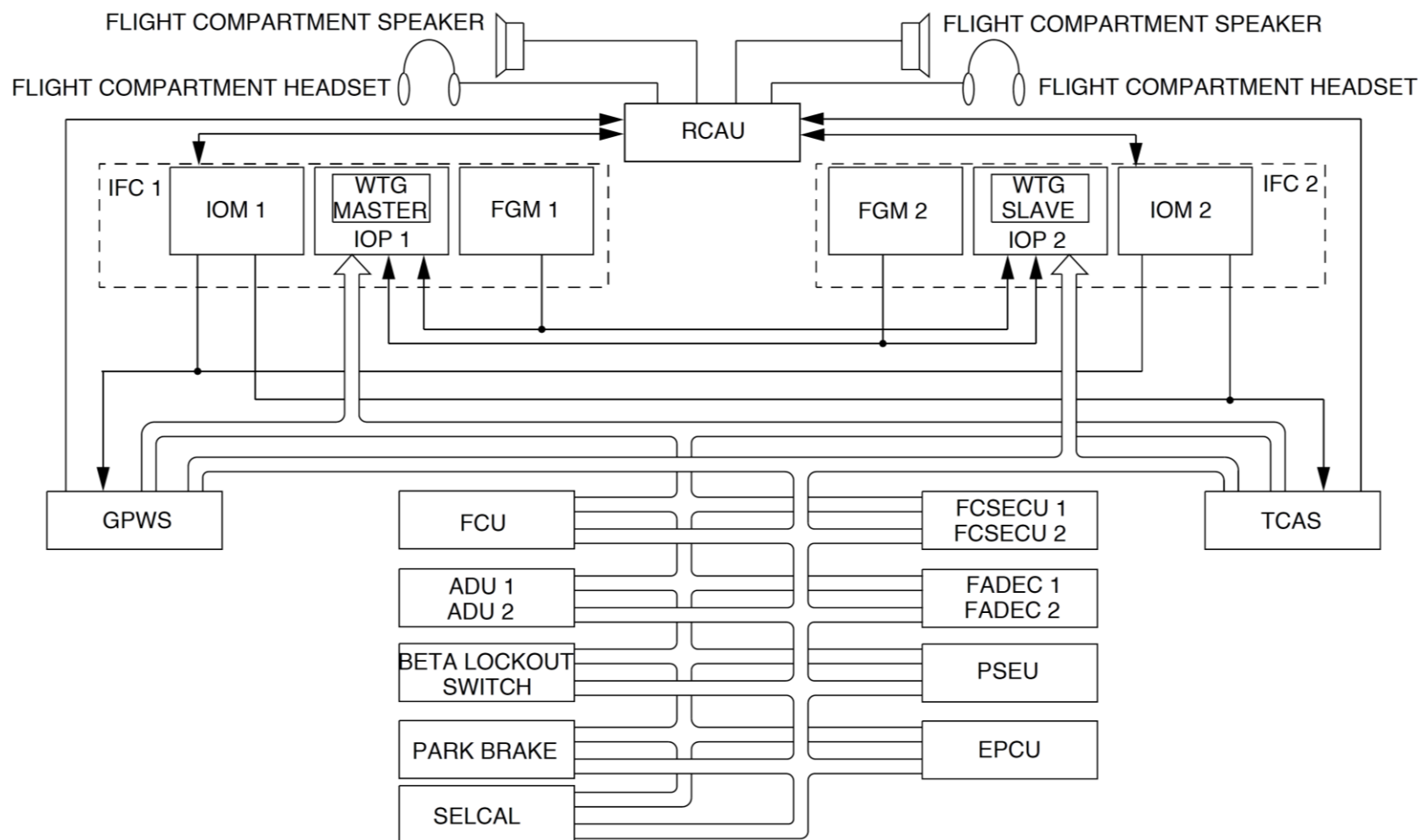


Figure 32-113 Landing Gear Warning System